PAROL6

Description	Documentation for the PAROL6 desktop robotic arm by Source Robotics.
Repository	https://github.com/Source-Robotics/PAROL-docs
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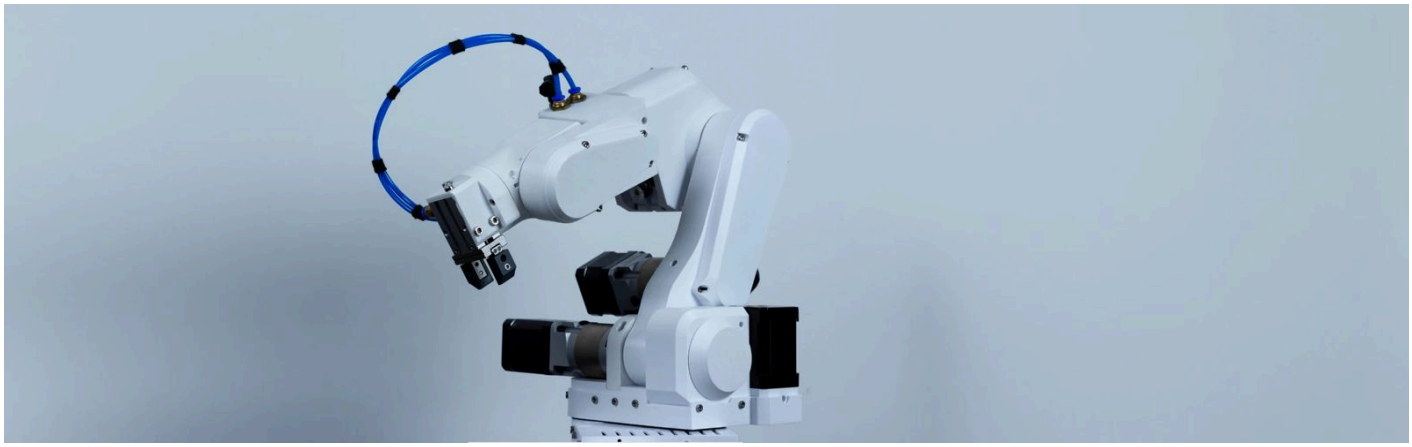
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PAROL6 Robot Manual

PAROL6 Robot Manual



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Original instructions

Quick start | About this manual

Source Robotics d.o.o. publishes this document as part of the open-source PAROL6 project. It is revised as the product changes; the document number, release, revision and date above identify the version you are reading, and the document history on the About page says what changed. Take care that the release named under *Applies to* matches the robot in front of you, and that the Python API version matches what `pip` installed: the version selector in the header switches between releases. The text and the images are licensed CC BY-SA 4.0, the code examples MIT.

Quick start

The robot manual

One page from an assembled robot to a homed one that moves: power, the commander software, the first jog, homing, with a link into the manual at every step.

Safety, Technical data, Installation, Operation, Peripherals, Maintenance, Troubleshooting.

Build guide

From the BOM to the first run: what to print and how, the parts to buy, the assembly manual and videos, the wiring, the firmware upload, the first power-on.

Control board

The board that drives the robot: specifications, connectors and pinout, the stepper drivers, the I/O and E-stop wiring, CAN, firmware upload, the test firmware, the serial protocol.

End effectors

What goes on the flange: the pneumatic and vacuum grippers, the SSG-48 and MSG electric grippers, your own tool; mounting, connections, the homing offset, driving them.

GUI interfaces

The graphical programs: the commander software, Waldo Commander in the browser, Motor GUI for the grippers.

{} Programming

The programming guide: the interfaces, the concepts behind them, the Python API, RBTscript, ROS 2; and the reference, read from the API's source.

Before you start

Quick start

From the PAROL6 you have just built to one that is homed and moves on command, on one page. It assumes the robot is assembled and wired as in the Build guide (Mechanical assembly, Electronics and wiring), and nothing more: no firmware uploaded yet, no software on your PC, and no experience with Python or a terminal. Every step ends with a link to the page that explains it; follow it when a step is not clear or something fails.

Read first

PAROL6 has no brakes: when the power goes, the arm falls. It has pinch points, it moves fast, and it moves by itself while homing. Anchor it to the table, keep the E-stop within reach, keep hands out of the arm's path, and read the Safety chapter: it is five minutes long.

1. What you need

	Where to get it
Your PAROL6, assembled and wired, with the control board inside the base	the Build guide; the parts in the BOM, or as kits from the shop
A 24 V power supply, 5 A	the BOM names one; any 24 V 5 A supply works. A closed one with a mains cable needs no mains wiring (The power supply)
A mains switch within reach, on the supply's mains side: a switched socket strip will do	it is the stop that still works when the PC software has frozen (Emergency stop)
An E-stop, a normally closed button in its printed box	the BOM names one; any NC switch works
A USB cable, type B, to the PC	
An ST-Link programmer and the programming adapter, to upload the firmware once	the programming adapter in the shop
A PC with Windows, Linux or macOS	see the minimum PC

Read in more detail: What you need.

2. Anchor the robot

Bolt the base to the table through its six holes, or clamp it to the edge. A free-standing PAROL6 tips over when it extends.

> ✓ **Check**
>
> Push the top of the base sideways, firmly: it must not lift or slide.

Read in more detail: Mounting the robot.

3. Install Python and Git, and open a terminal

The PC software is written in Python, and Git downloads it. A **terminal** is the window where you type commands: type one line, press Enter, wait until it finishes.

Windows

1. Download **Python 3.11.9** from python.org: the "Windows installer (64-bit)". 3.11.9 is the last 3.11 release with an installer.
2. Run it. On its first screen tick **Add python.exe to PATH**, then press **Install Now**.
3. Download **Git** from git-scm.com and install it with the settings it proposes.
4. Open a terminal: press the Windows key, type `cmd` and press Enter. A black window opens.
5. Type `python --version` and press Enter, then `git --version`.

The commands on this page start with `cd %USERPROFILE%\Documents`, which takes the terminal to your Documents folder. If Windows answers that it cannot find the path, your Documents folder is kept in OneDrive: type `cd %USERPROFILE%\OneDrive\Documents` instead. To open a terminal in any folder, open the folder in File Explorer, click the address bar, type `cmd` and press Enter.

Linux

1. Open a terminal: Ctrl+Alt+T on Ubuntu, or Terminal in the applications menu.
2. Install Git: `sudo apt install git` (it asks for your password).
3. The software needs **Python 3.11**: the commander runs on 3.10 or 3.11 only, the Python API on 3.11 or newer. Ubuntu's own Python is 3.10 on Ubuntu 22.04 and 3.12 on Ubuntu 24.04, so install 3.11 next to it, with its virtual environments and its window toolkit:

   ```
   sudo add-apt-repository ppa:deadsnakes/ppa
   sudo apt install python3.11 python3.11-venv python3.11-tk
   ```

4. Type `python3.11 --version` and press Enter, then `git --version`. Each program below gets its own virtual environment made with Python 3.11; inside it, `python` is that Python 3.11.

macOS

1. Download **Python 3.11.9** from python.org: the "macOS 64-bit universal2 installer". Run it.
2. Open a terminal: press Cmd+Space, type `Terminal` and press Enter.
3. Install Git with the Xcode command line tools: type `xcode-select --install`, press Enter and confirm the window that opens.
4. Type `python3.11 --version` and press Enter, then `git --version`. Each program below gets its own virtual environment made with Python 3.11; inside it, `python` is that Python 3.11.

Check

The terminal answers `Python 3.11` and a number, and `git version` and a number. If it says a command is not found, install that program again (on Windows, tick **Add python.exe to PATH**), then open a new terminal.

4. Upload the main firmware

A control board from the shop comes with the **test firmware**, which turns all six motors as soon as the board starts. Put the **main firmware** on it before you switch on the assembled robot. You do this once.

1. Download the robot repository into your Documents folder:

   ```
   cd %USERPROFILE%\Documents
   git clone https://github.com/Source-Robotics/PAROL6-Desktop-robot-arm.git
   ```

   On Linux and macOS, the first line is `cd ~/Documents`.

2. Install VS Code. Open it, click the **Extensions** icon on the left, search for **PlatformIO IDE** and install it. Restart VS Code when it asks.

3. On Windows, install the ST-Link driver: in File Explorer, open `PAROL6-Desktop-robot-arm` in your Documents folder, right-click `en.stsw-link009.zip`, choose **Extract All**, and run `dpinst_amd64.exe` from the extracted folder.

4. **Unplug the 24 V supply** from the robot. Plug the programming adapter onto the 2×5 header next to the USB port, and the ST-Link into the PC. The adapter powers the board's microcontroller, not the motors.

5. In VS Code: **File**, **Open Folder**, and choose `PAROL6 control board main software` inside `PAROL6-Desktop-robot-arm`. The first time, PlatformIO downloads its tools for a few minutes: wait until the bar at the bottom is quiet.

6. Click the → arrow (Upload) in the bar at the bottom of VS Code.

7. When it has finished, unplug the programming adapter.

Check

The terminal at the bottom of VS Code ends with `SUCCESS`. If not, see the Firmware page and Control board troubleshooting.

Read in more detail: Firmware · Firmware and first run.

5. Install the robot software

Choose one, or install both: the **commander software** is a window with buttons, a program editor and a 3D simulator; the **Python API** drives the robot from Python programs you write. Each one goes into its own folder with its own **virtual environment** (`.venv`), a private copy of Python for that program: the two need different versions of the same packages, so they cannot share one.

Commander software

In the terminal, download the commander software, make its virtual environment and install what it needs (a few minutes):

```
cd %USERPROFILE%\Documents
git clone https://github.com/Source-Robotics/PAROL-commander-software.git
cd PAROL-commander-software
python -m venv .venv
.venv\Scripts\activate
python -m pip install -r requirements.txt
```

On Linux and macOS:

```
cd ~/Documents
git clone https://github.com/Source-Robotics/PAROL-commander-software.git
cd PAROL-commander-software
python3.11 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.txt
```

After the activate line the prompt starts with `(.venv)`.

Python API

In the terminal, download the Python API, switch to version 0.9.1, make its virtual environment and install it (a few minutes):

```
cd %USERPROFILE%\Documents
git clone https://github.com/PCrnjak/PAROL6-python-API.git
cd PAROL6-python-API
git checkout v0.9.1
python -m venv .venv
.venv\Scripts\activate
python -m pip install .
```

On Linux and macOS:

```
cd ~/Documents
git clone https://github.com/PCrnjak/PAROL6-python-API.git
cd PAROL6-python-API
git checkout v0.9.1
python3.11 -m venv .venv
source .venv/bin/activate
python -m pip install .
```

After the activate line the prompt starts with `(.venv)`. Git reports a "detached HEAD" after the checkout: that is expected.

> ✓ **Check**
>
> The install ends without an `ERROR` line. If one appears, see Commander installation or Python API installation.

6. Connect

1. The E-stop to the two E-stop terminals on the base. Without it the robot will not run.
2. The USB cable to the PC.
3. The 24 V supply to the power connector, and the supply's mains plug into the switched socket strip, or behind the mains switch, within your reach. Nothing happens yet: the board is off until the button is pressed. A DIN-rail supply is wired and checked first (The power supply).

Read in more detail: Connecting.

7. Switch on

Hold the arm by the forearm and press the power button on the base. The motors energise, you hear six clicks, and the arm holds its position. Let go.

> ✓ **Check**
>
> Six drivers initialise and the arm supports itself before you let go. If it does not, use First power-on: if nothing happens.

Read in more detail: First power-on · Switching on and off.

8. Start the software

First find the robot's **serial port**, the name the PC gives the USB connection:

- **Windows**: open Device Manager (right-click the Start button), then **Ports (COM & LPT)**: the robot is a "USB Serial Device" with a number, for example `COM5`.
- **Linux**: `ls /dev/ttyACM*` in the terminal, usually `/dev/ttyACM0`. If the software later reports "permission denied", run `sudo usermod -aG dialout $USER`, then log out and in again.
- **macOS**: `ls /dev/cu.usbmodem*` in the terminal.

Commander software

For the first run, write the port into the program. Open `PAROL-commander-software/GUI/files/Serial_sender_good_latest.py` in a text editor (VS Code: **File**, **Open File**) and find these lines near the top:

```
if my_os == "Windows":
    STARTING_PORT = 58 # COM3
else:
    STARTING_PORT = 0
```

On Windows, replace `58` with the number of your port, `5` for `COM5`. On Linux, `0` is `/dev/ttyACM0`; change it if yours has another number. Save the file. Then start the program in the terminal:

```
cd %USERPROFILE%\Documents\PAROL-commander-software
.venv\Scripts\activate
cd GUI\files
python Serial_sender_good_latest.py
```

On Linux and macOS:

```
cd ~/Documents/PAROL-commander-software
source .venv/bin/activate
cd GUI/files
python Serial_sender_good_latest.py
```

Two windows open: the commander and the simulator. At the bottom of the commander, type the serial port in the port field (`COM5` on Windows, `ttyACM0` on Linux) and press **Connect**, more than once if the first try fails.

Python API

1. Start the **controller**, the program that talks to the robot, with your port, and leave its terminal open:

   ```
   cd %USERPROFILE%\Documents\PAROL6-python-API
   .venv\Scripts\activate
   parol6-server --serial COM5
   ```

   On Linux and macOS: `cd ~/Documents/PAROL6-python-API`, then `source .venv/bin/activate`, then `parol6-server --serial /dev/ttyACM0` (macOS: the `/dev/cu.usbmodem…` name).

2. Open a **second** terminal and activate the same virtual environment in it:

   ```
   cd %USERPROFILE%\Documents\PAROL6-python-API
   .venv\Scripts\activate
   python
   ```

   (Linux and macOS: `cd ~/Documents/PAROL6-python-API`, `source .venv/bin/activate`, `python`.) The prompt changes to `>>>`: Python now runs each line you type. Type these three lines:

```
from parol6 import RobotClient
rbt = RobotClient()
rbt.ping()
```

Keep this second terminal open: the next steps type into it.

Read in more detail: The commander software · The Python API.

Check

The commander's response log shows the robot's data, or `ping()` answers with `hardware_connected=True`. If it does not, use Communication troubleshooting.

9. Jog each joint

Before homing, check that every joint turns the way the software thinks it does: each one briefly, a few degrees. The figure shows each joint's positive direction, in colour, over the pose in grey.

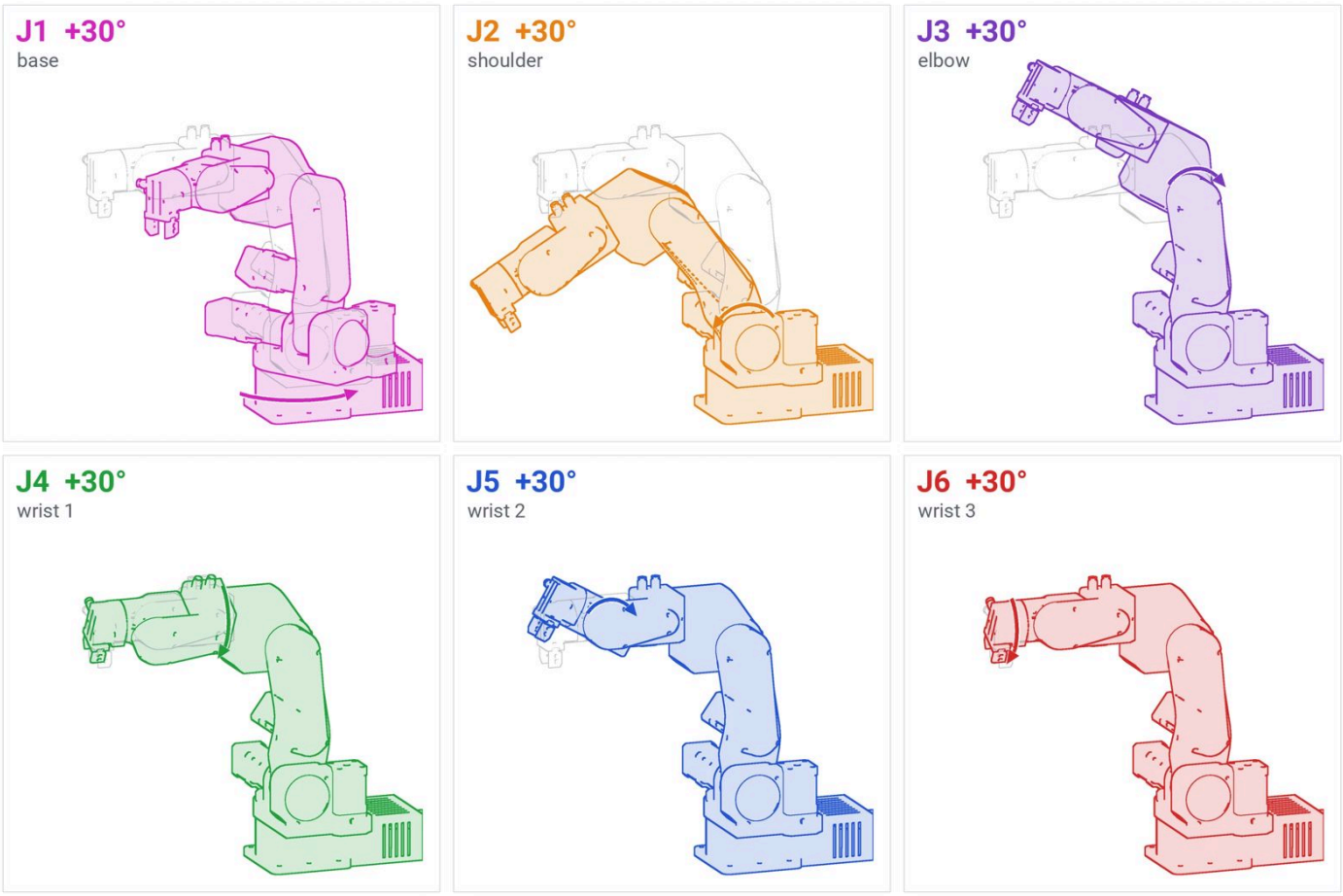


Figure 1: Each joint turned 30 degrees in its positive direction, in colour, over the pose in grey.

Before homing, both programs check each jog against the joint limits with a position count that starts at 0 when the board is switched on. That count puts joint 2 past its positive limit and joints 3 and 6 at or below their negative limits. So, until the robot is homed:

- **Joint 2** jogs only in its **negative** direction. Check that it turns opposite to the figure.
- **Joints 3 and 6** jog only in their positive direction, which is the one to check.
- **Joints 1, 4 and 5** jog both ways.

A refused jog does not move: the commander logs `Error: Robot jog -> Position out of range`, the Python API stops the joint before it starts.

Commander software

In the **Joint jog** menu, press the left arrow (positive) of joints 1 and 3 to 6 briefly, and the right arrow (negative) of joint 2.

Python API

```
rbt.jog_j(0, speed=0.2, duration=0.5)     # joint 1, positive, half a second
rbt.jog_j(1, speed=-0.2, duration=0.5)    # joint 2, negative
```

The first number is the joint: `0` is joint 1, `5` is joint 6. A negative `speed` jogs the negative way.

A joint that turns the wrong way is fixed in the firmware, not by trial: Firmware and first run.

Read in more detail: Jogging · Joint limits and directions.

Check

Joints 1 and 3 to 6 turned the way the figure shows, and joint 2 turned the opposite way on its negative jog. If one did not, fix it before you home: A joint turns the wrong way.

10. Home

Homing finds each joint's reference at its limit switch or sensor. Joints 1, 2 and 3 move to their switches together, then 4, then 6, then 5, and the arm ends in the after-homing pose: the standby pose turned 90° about joint 1. Stay at the E-stop: a joint that grinds at its end stop has a switch that does not trigger.

Before you press Home:

- **Joint 6** stands so that it finds its sensor within one turn in the negative direction, with the tool's cables and tubes not wound up. The figure shows the pose: the gripper as drawn, the sensor (blue) and the homing pin (red); joint 6 turns along the green arrow.
- **The tool** is mounted, and the firmware's joint 5 homing offset matches it. Joint 5 homes against a feature of the tool holder; `j5_homing_offset` in `main.cpp` is 8900 for the pneumatic gripper holder (as shipped) and 8035 for the SSG-48. If you change it, upload the firmware again as in step 4 (Joint 5 and the tool).

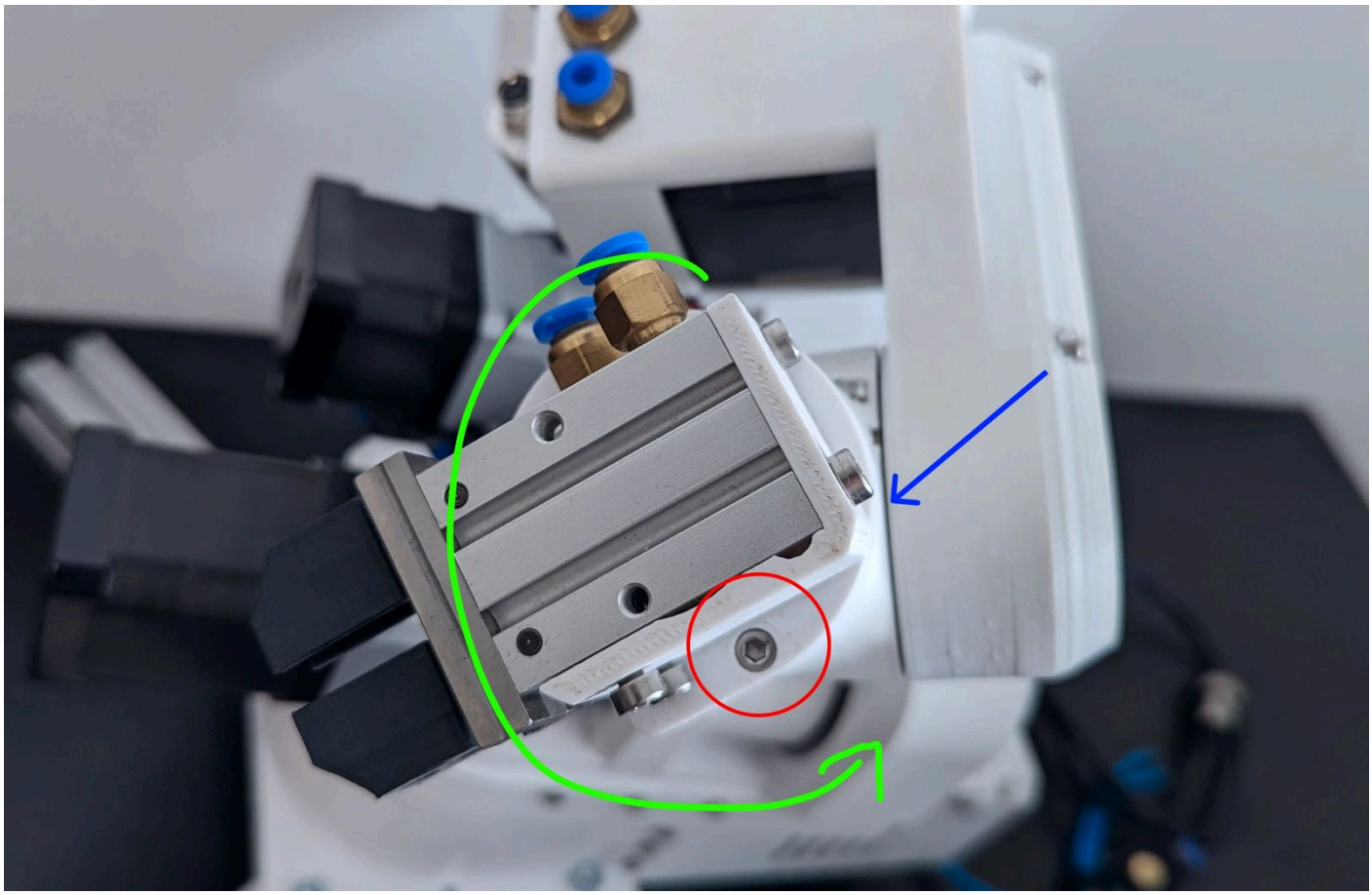


Figure 2: Joint 6 before homing: the sensor, the homing pin and the direction of rotation.

Commander software

Press **Home**.

Python API

```
rbt.home(wait=True)
```

Read in more detail: Homing · Homing failures.

Check

Homing finishes, the physical arm matches the simulator, and the pin passes through the witness marks of each joint, which line up in the after-homing pose (Witness marks). If a joint keeps travelling or the pose is wrong, stop and use Homing failures.

11. Move it

The robot now knows where it is. Three simple moves: bend the arm a little, turn the base, and come back. The Python API adds a short straight line down and up from the bent pose.

Commander software

Paste this program into the program window and press **Start**; **Stop** halts it.

```
Begin()
MoveJoint(90,-70,160,0,30,180,t=3)
Delay(1)
MoveJoint(60,-70,160,0,30,180,t=3)
Delay(1)
MoveJoint(90,-90,180,0,0,180,t=3)
End()
```

`MoveJoint` takes the six joint angles in degrees; `t=3` makes the move take three seconds. Jogging in the Cartesian jog menu moves the tool in straight lines instead.

Python API

Type these lines at the `>>>` prompt, one at a time; each waits until its move is done:

```
rbt.move_j([90, -70, 160, 0, 30, 180], speed=0.3)       # bend the arm: six joint angles in
degrees
rbt.move_l([0, 0, -30, 0, 0, 0], rel=True, speed=0.3)   # a straight line 30 mm down
rbt.move_l([0, 0, 30, 0, 0, 0], rel=True, speed=0.3)    # and 30 mm back up
rbt.move_j([60, -70, 160, 0, 30, 180], speed=0.3)       # turn the base
rbt.move_j([90, -90, 180, 0, 0, 180], speed=0.3)        # back to the after-homing pose
```

`speed` is a fraction of the maximum: 0.3 is gentle. With `rel=True`, `move_l` takes a change from where the tool is: x, y and z in mm in the world frame, then three rotations in degrees.

Read in more detail: RBTscript · The Python API.

12. Switch off

Bring the arm to the **parked pose** first, folded in over the base, so that it has the least distance to fall:

Commander software

Press **Park**: it writes a three-line program into the program window. Press **Start** to run it.

Python API

```
rbt.move_j([90, -144.683, 108.171, 2.222, 25.003, 180], speed=0.3)   # the commander's park pose
```

Type `exit()` to leave Python.

Then hold the forearm and press the power button. The arm goes limp the moment the motors lose power.

13. Where to go next

You want to	Read
attach a gripper	End effectors
write programs	Programming, RBTscript, The Python API
use inputs, outputs, pneumatics	Peripherals
understand what the robot does and does not protect against	Safety functions
fix something	Troubleshooting
look up a command	API reference

About this manual

Scope and audience

The PAROL6 Robot Manual covers the PAROL6 desktop robotic arm from the box, or the printer, to a robot that runs programs: what the robot is, how to install it safely, how it is switched on, homed, jogged and run, which tools and grippers it takes, its inputs, outputs and pneumatics, how it is maintained and what to do when something fails. The Build guide, part of the same document, covers building one from the published files. Four more documents live on this site: the Control Board Manual, for the electronics inside the base; the End Effectors Guide, for the grippers and tools on the flange; the GUI Guide, for the graphical programs; and the Programming Guide with its API reference, for everyone who writes programs for the robot. The list under *Related documents* gives their numbers.

The manual is written for someone who has a PAROL6 or is building one: a maker, a student, a teacher or an engineer who can run a Python script and tighten a belt. It does not assume robotics training; the concepts the robot's software uses are explained in the Theory corner, which anyone can read first. Read the Safety chapter before you power anything. The Quick start is the first day on one page; the chapters are in the order you need them and go deeper.

Finding your way

The header has seven tabs. **PAROL6 Robot Manual** is the document you read from the start: the quick start, then safety, technical data, installation, operation, peripherals, maintenance and troubleshooting, in numbered chapters, followed by the build guide. **Control Board** is the manual of the electronics inside the base: the connectors, the stepper drivers, the I/O wiring, the firmware, the serial protocol. **End effectors** is what goes on the flange: the pneumatic and vacuum grippers, the SSG-48 and MSG electric grippers, a tool of your own, and how each is mounted, connected and driven. **GUI interfaces** covers the graphical programs: the commander software, Waldo Commander, Motor GUI. **Programming** is for everyone who writes programs for the robot: the interfaces, the concepts, the Python API, RBTscript, ROS 2, and the reference where you look things up: every method of the Python API, read from its source, the RBTscript commands, the serial protocol byte by byte, the firmware parameters. **Theory corner** explains the robotics behind the software, from joint numbering and units to kinematics, singularities and motion profiles, for anyone who wants to read first. **Support** says where to get help and how to report a problem.

The whole site describes one PAROL6 release. The dropdown next to the site name switches between releases, and a page from an older release says so in a bar at the top. The repository card on the right of the header opens the repository the page belongs to; every page ends with a link to edit it and one to report a problem with it.

Warnings and notes

Three kinds of warnings and notes appear in this documentation. Read every Danger and Notice before you act on the step it belongs to. A green **Check** box after a setup step says what you should see when the step worked, and where to go when it did not.

Note

Points out something that is easy to miss or that changes what follows: a default, a limit, a side effect.

Notice

The robot, the tool, or data on the control board or the PC can be damaged or lost if this is ignored. Nobody is at risk.

Danger

Someone can be injured if this is ignored. Stop and read the whole box before acting on the step it belongs to.

Related documents

Document	Number	Covers
PAROL6 Robot Manual	SR-1002	the robot: quick start, safety, technical data, installation, operation, peripherals, maintenance, troubleshooting, transport; the build guide
PAROL6 Control Board Manual	SR-2002	the control board: features, technical data, connectors and pinout, stepper drivers, I/O and E-stop, CAN, firmware upload, the test firmware, the serial protocol
PAROL6 Programming Guide	SR-4004	the ways to program the robot, the concepts behind them, the Python API, RBTscript, ROS 2; the reference: every method of the Python API, the RBTscript commands, the serial protocol, the firmware parameters
PAROL6 End Effectors Guide	SR-3004	what goes on the flange: the pneumatic and vacuum grippers, the SSG-48 and MSG electric grippers, a tool of your own; mounting, connections, the joint 5 homing offset, driving them from the commander software and the Python API
PAROL6 GUI Guide	SR-4005	the graphical programs: the commander software (installation, the windows, the tabs), Waldo Commander in the browser, Motor GUI for the grippers

Every document describes PAROL6 release 1.1; the version selector in the header switches between releases. The compatibility table says which control board, firmware, Python API, commander and gripper versions work together.

The grippers made for PAROL6 are documented separately: the SSG-48 at docs.source-robotics.com/SSG48-gripper-docs, the MSG at docs.source-robotics.com/MSG-gripper-docs. PAR6, the larger arm, is documented at docs.source-robotics.com/par6.

Document history

The documents on this site describe one PAROL6 release; one revision is kept per release, and the version selector in the header switches between them. The Programming Guide and the API reference also name the Python API version they describe, because the API is released more often than the robot. The documentation as it was from 2023 to October 2026, before this site was rebuilt, stays in the version selector as *Old manual*; it is kept for reference and no longer updated.

The text and the images of this documentation are licensed under CC BY-SA 4.0, the code examples under the MIT licence. The robot's hardware files are published under CERN-OHL-S-2.0, its firmware and software under GPLv3, in their repositories. The Source Robotics name, the product names and the logos are not covered by any of these.

Document	Revision	Date	Changes
Robot Manual	2026.C	2026-11-05	The base screws, M5 × 16 mm with a 4 mm key; the E-stop in its own printed box on a cable; either of two E-stop buttons stops the robot; the GX16 polarity is the builder's choice; the robot model's link names; the commander's `STARTING_PORT` (58) in the quick start; the gripper socket's pins in the build guide (page 32 of the assembly manual has 1 and 3 the wrong way round)
Robot Manual	2026.B	2026-11-04	PAROL6 release 1.1, with the files as downloads on the release; the workspace side and top views per tool; the reference frames per tool and the flange frame (the Python API's reversed tool frame as KI-11); the mains side of the power supply; cleaning; the old manual as version 0 in the version selector; the hardware licence CERN-OHL-S-2.0
Robot Manual	2026.A	2026-11-03	First version in this form: the numbered chapters, the build guide, from the pages of the previous site
Control Board Manual	2026.C	2026-11-05	The E-stop in its own printed box; either of two E-stop buttons stops the robot
Control Board Manual	2026.B	2026-11-04	PAROL6 release 1.1: the same firmware as release 1

Document	Revision	Date	Changes
Control Board Manual	2026.A	2026-11-03	First version in this form, from the control board and peripherals pages of the previous site
End Effectors Guide	2026.B	2026-11-05	The M8's 24 V has no fuse of its own; the gripper socket's pins; the PAR6 cable reverses a gripper's supply, it does not put 24 V on the CAN pins
End Effectors Guide	2026.A	2026-11-03	First version: the flange, the pneumatic and vacuum grippers, the SSG-48 and MSG, your own tool
GUI Guide	2026.B	2026-11-05	The commander's `STARTING_PORT` is 58 in the code, not 8
GUI Guide	2026.A	2026-11-03	First version: the commander software, Waldo Commander, Motor GUI
Programming Guide	2026.A	2026-11-03	First version: the concepts, the Python API 0.9.1, the commander software and RBTscript, ROS 2; the generated API reference

Chapters

1. Overview

1. Overview

PAROL6 is a 3D-printed six-axis desktop robotic arm with a 408 mm reach and a 1 kg payload, built to work like an industrial arm in its mechanics, its control software and the way you use it. Everything about it is open source: the printable parts, the bill of materials, the control board's firmware, the commander software and the Python API. You can buy it assembled, as the full kit or as the partial kit of the hard-to-find parts, or print and build one from the published files: the Build guide compares them.

In this chapter

- The robot system: the arm, the control board, the PC software, and how they divide the work
- What you get: what is in the box of an assembled robot or a kit
- What you need: the power supply, the PC, the tools and the knowledge the robot asks of you

The robot system

A working PAROL6 is three things: the arm, the control board inside its base, and a PC running the high-level software. Each does a part of the job.

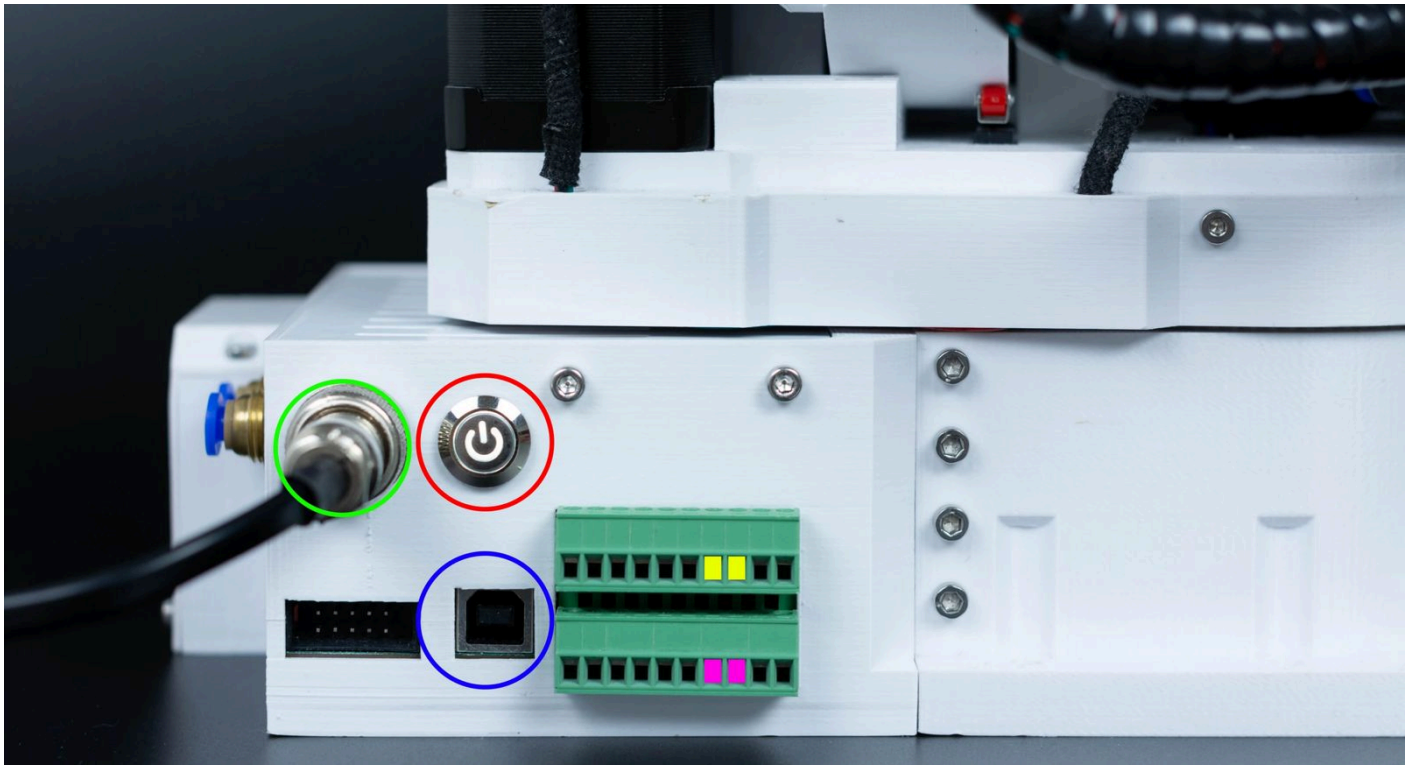


Figure 1.1: PAROL6 with the connections on its base.

The arm

Six revolute joints, driven by NEMA 17 stepper motors through planetary gearboxes and belts, in PETG printed links. Joints 1, 2 and 3 position the wrist; joints 4, 5 and 6 form a spherical wrist that orients the tool. Joint 6 ends in the flange the tool mounts on. The tool connections are on the forearm: an M8 connector for an electric gripper and two push-in fittings for pneumatic lines. Each joint has a limit switch or an inductive sensor that gives it a reference when the robot homes.

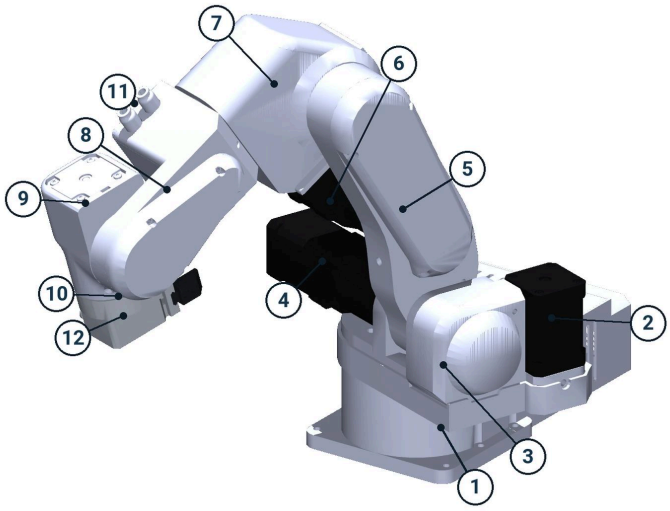


Figure 1.2: PAROL6 with the pneumatic gripper, its parts numbered.

1. Base, with the control board inside
2. Joint 1 motor
3. Shoulder, turned by joint 1
4. Joint 2 motor
5. Upper arm
6. Joint 3 motor
7. Elbow
8. Forearm, also called the lower arm, turned by joint 4
9. Wrist, turned by joint 5
10. Flange, turned by joint 6
11. Air fittings for a pneumatic tool
12. Gripper, here the standard pneumatic gripper

The robot model (the URDF and MuJoCo files in `robot_descriptions`) uses the standard ROS link names, which count the parts differently: the elbow is `forearm_link`, the forearm `wrist_1_link` and the wrist `wrist_2_link`.

The arm has no brakes and no joint encoders. The motors hold the arm while they are powered; the position is counted in steps from the homing reference. Both facts shape how the robot is used, and the Safety chapter returns to them.

The control board

The PAROL6 control board sits in the base: an STM32F446 microcontroller, six TMC5160 stepper drivers, two isolated inputs and two isolated outputs, the E-stop input, a CAN bus for electric grippers, and USB to the PC. Its firmware generates the step pulses, runs the homing sequence, reads the switches and the inputs, drives the outputs and the gripper, and exchanges a packet with the PC at 100 Hz. It does not plan motion: it executes the joint speeds and positions the PC sends it, 100 times a second.

The PC software

The PC does the thinking: kinematics, trajectory planning, the simulator, the program interpreter, the interface. There are two first-party programs, and others:

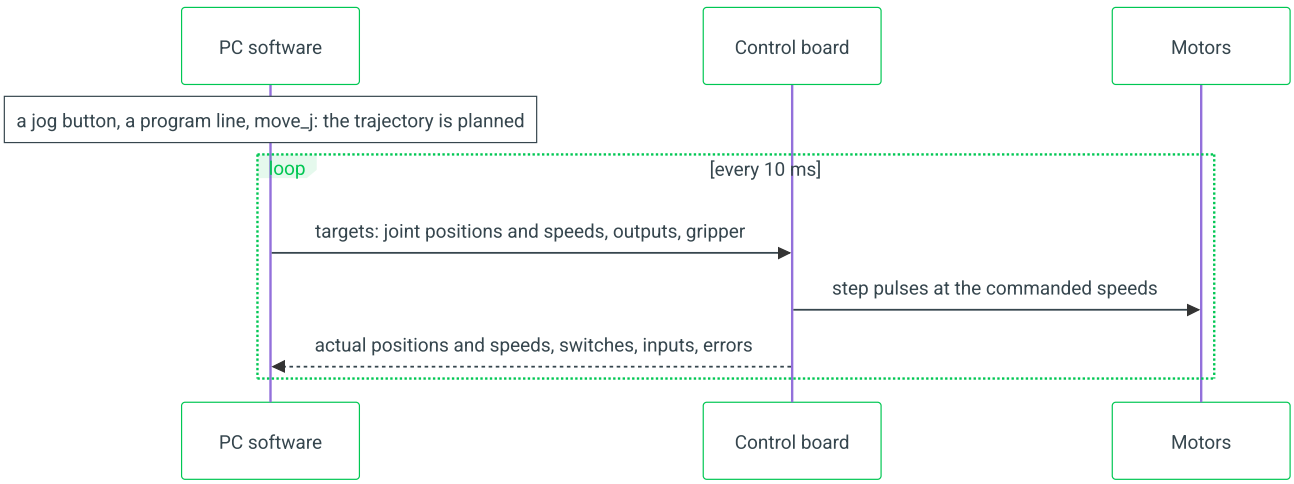
	What it is	For
Commander software	A desktop GUI with jogging, a 3D simulator, a program editor for RBTscript, I/O and gripper tabs	Using the robot by hand, teaching points, simple programs

	What it is	For
Python API	A controller process that talks to the board, and a Python client library with blocking and asynchronous calls, planned and streamed motion, tools, collision checking	Programs, integrations, research, teleoperation, AI agents
ROS 2 and MoveIt	A MoveIt configuration and a `ros2_control` hardware interface that drive the Python API's controller	ROS users
Community interfaces	Web GUIs, alternative firmware, vision demos	

All of them speak the same serial protocol to the control board, so a board with the main firmware works with any of them.

How a move happens

1. You press a jog button, run a program line or call `move_j`.
2. The PC software computes the trajectory: the joint angles over time, with the speed profile you chose, within the joint limits.
3. One hundred times a second it sends the control board a packet with the target position and speed of each joint, the output states and the gripper command.
4. The board sets the stepper speeds and runs the steps; the drivers turn the motors.
5. The board answers with the actual position and speed of each joint, the switch and input states, the errors and the gripper's status, which the PC software shows and uses for the next cycle.



Where everything is published

Mechanical files, BOM, building instructions, control board firmware, URDF	PAROL6-Desktop-robot-arm
Commander software	PAROL-commander-software
Python API	PAROL6-python-API
ROS 2 MoveIt configuration	PAROL6-ROS2-MOVEIT
This documentation	PAROL-docs
Shop, videos, community	source-robotics.com, YouTube, Discord

The full list is on Downloads and files.

What you get

An assembled robot

A PAROL6 from the shop comes assembled, wired and tested, with the control board in the base running the main firmware, the E-stop button, the pneumatic gripper holder, and the robot homed once on the bench. Check the contents against the order; the power supply, grippers and the programming adapter are separate items unless the order includes them.

A kit

The full kit contains every part of the BOM except the printed parts and the power supply. The partial kit contains the hard-to-find parts: the board, the drivers, the bearings, belts, pulleys, connectors, sensors, switches and the pneumatics; you add the gearboxes, the motors, the power supply and the printed parts. The Build guide lists every kit. You assemble a kit with the Build guide and the assembly manual. The control board in a kit carries the test firmware, as every control board from the shop does: it turns all six motors as soon as it starts. Upload the main firmware before you switch on the assembled robot (Firmware and first run).

A control board on its own

Every control board is tested end to end before it ships and comes with the **test firmware**, which exercises the drivers, the switches and the I/O from a serial terminal. Upload the main firmware before you use it with the commander software or the Python API: Firmware.

The files

Everything else you get is on GitHub: the STL and STEP files, the BOM with purchase links, the building instructions PDF and the video playlist, the firmware sources, the software. Downloads and files lists them.

What you need

Power

A 24 V DC supply of at least 5 A. The BOM names a Mean Well 120 W DIN-rail unit; any regulated 24 V 5 A supply with the robot's GX16 power connector wired to it does. The control board accepts 18 to 24 V; below 24 V the motors lose torque and speed. The robot draws about 40 W in typical use and more when all six motors accelerate at once. A closed supply with a mains cable needs no mains wiring; the DIN-rail unit does, in a closed box (The power supply).

Plug the supply in through a **mains switch within reach**, a switched socket strip for example. It is the stop that still works when the PC software has frozen (Emergency stop), and the switch you reach for when the E-stop has no effect.

The E-stop

A normally closed emergency stop button, wired to the two E-stop terminals of the control board. The robot does not run without it. The kit's button sits in its own 3D-printed box, on a cable as long as your setup needs; any NC switch works for a bench setup (Emergency stop).

The PC

The PC runs the kinematics and the planner and must send a packet to the robot every 10 ms without hiccups. A machine that falls behind makes the robot stutter. Tested systems and the minimum are in the commander repository's confirmed working systems: at least an Intel i5 six-core processor of 2017 or newer, or an AMD equivalent. The tested systems include two Windows 10 PCs with six-core AMD Ryzen processors, a Radxa ROCK 5B on Linux and a six-core MacBook Pro.

	Commander software	**Python API**
Operating system	Windows, Linux, macOS	Windows, Linux (x86-64 and ARM64), macOS (ARM64)
Python	3.10 or 3.11	3.11 or newer
Connection	USB (a serial port, 3 Mbaud)	USB on the PC that runs the controller; clients anywhere on the network

A USB type B cable connects the robot to the PC. USB does not power the board.

Tools

For an assembled robot: hex keys, 4 mm for the M5 mounting screws and 2, 2.5 and 3 mm for the gripper, a small screwdriver for the terminal blocks. For a build, the Build guide lists the rest, down to the blue threadlocker and the thermal cement.

Knowledge

You should be able to use a hex key. Python and the terminal you can learn on the way: the Quick start gives every command to type, for someone who has never used either. The Theory corner explains the robotics vocabulary the software uses, from joint numbering to singularities, in the order you meet it.

2. Safety

2. Safety

PAROL6 is a desktop robot, light enough to lift with one hand, and it can still hurt you: it pinches, it moves fast, it moves by itself when it homes, and it falls when it loses power. This chapter is short. Read all of it before the first power-on, and have everyone who works at the robot read it.

Danger

The project's Safety Warning and Disclaimer applies to everything on this site: the hardware, the firmware, the software and this documentation are under development, and you build, modify and operate the robot at your own risk.

In this chapter

- Intended use: what the robot is for, and what it is not
- Hazards: pinch points, the fall on power loss, induced voltage, hot motors, the tool
- Emergency stop: what the E-stop does, and what it does not
- Moving the robot by hand: when it is allowed and the two things never to do
- What the robot protects against, and what not
- Planning the cell: the person, the guarding, the program

Intended use

PAROL6 is made for education, research, hobby robotics and light automation on a bench: pick-and-place of light objects, camera and sensor experiments, teaching robot programming, prototyping a task before buying an industrial arm. It is an open-source, 3D-printed machine assembled by its owner or by a small workshop, and it is not a certified industrial robot.

Use it

- on a bench or frame it is bolted to, in a dry room
- with a payload within its rating: 1 kg near the base, 0.5 kg at full reach, tool included
- with the E-stop connected and within reach
- with the person who runs it able to see the whole arm

Do not use it

- around people who do not know it is a robot: children, visitors, a crowded desk
- for anything that must not fail: it has no brakes and no joint position feedback, and a lost step is silent
- with tools that cut, heat or spray without your own guarding and risk assessment
- in wet, dusty or hot places, or outdoors

A self-built robot is the builder's responsibility from the first screw: the currents, the wiring, the limit switches and the mechanical stops are all yours to get right, and the Build guide says where the mistakes are made.

Hazards

Pinch and crush points

Every joint is a pinch point, and the elbow and wrist fold against the upper arm and forearm. The motors are geared: a finger between two links when a joint closes is crushed, not pushed away. Keep hands, cables and tubes out of the arm's path while it is enabled, and never reach into the workspace of a running program. Under the elbow, the belt that drives joint 3 runs onto its pulley: a finger or a loose cable there is drawn in.

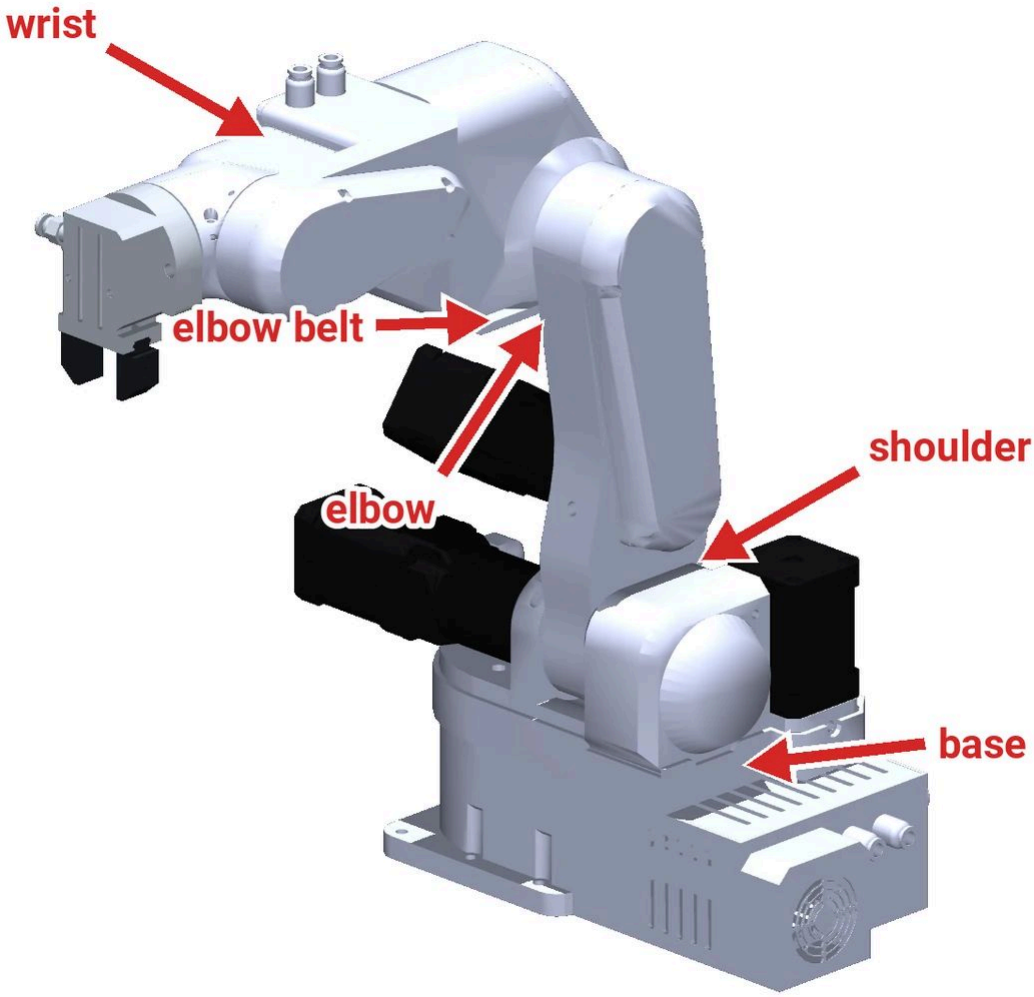


Figure 2.1: The pinch points of PAROL6, after homing: the wrist, the elbow, the belt and pulley under the elbow, the shoulder and the base.

Clothing, hair and eyes

A joint that closes catches more than fingers: its belts and pulleys pull in hair, loose sleeves, scarves, lanyards and jewellery. Tie back long hair and take those off before you work at a powered robot, and keep them out of the arm's reach while it moves. Wear safety glasses with a pneumatic gripper, with a sharp or pointed tool, and whenever the robot can drop or throw a part.

The arm falls when the power goes

PAROL6 has no brakes. The moment the motors lose power, from the button, the supply, a blown fuse or a pulled cable, the arm goes limp and falls under its own weight, with whatever it carries. The gearing slows the fall of joints 2 and 3 a little; it does not stop it.

- Never switch the robot off while it is extended or carrying something: bring it to the parked pose first, folded in over the base, where it has the least distance to fall (Switching on and off).
- Hold the forearm when you press the power button.
- Keep hands out from under the arm.
- Expect the fall during any power failure, and place nothing fragile under the arm.

Motion you did not command

- **Homing moves every joint**, at speed, toward its limit switch, as soon as Home is pressed. The arm must be free to move and the switches must work (Homing).
- **A program restarts** where you told it to, and `Loop()` restarts it forever.
- **A lost step** makes the arm move to a different place than the one on the screen. After a collision or a stall, stop and home again.
- **A wrong joint direction** on a new build sends a joint the wrong way on the first jog, into its stop. Jog slowly and briefly the first time (Firmware and first run).

Stepper-induced voltage

A stepper motor turned by hand is a generator. Moving the arm quickly while the supply is connected and the board off can power the board on and energise the motors, with the arm wherever you left it. Move an unpowered robot slowly, and only toward the parked pose or into its box.

Hot motors

After hours of motion the wrist motors reach 70 °C (Currents and temperatures). That is hot to the touch and close to the limit of the plastic around them.

Electricity

The robot runs on 24 V, which is safe to touch. The power supply you plug it into is connected to the mains: use a closed, approved supply, never an open-frame one on the desk. A DIN-rail supply such as the one in the BOM has open screw terminals: it belongs in a closed box, with an earthed cable, and if you are not sure how to wire it, an electrician wires it (The power supply). The control board has no reverse-polarity protection on its power input.

The tool

A pneumatic gripper closes with the force of the compressed air behind it, whatever is between its fingers; the electric grippers close with up to 80 N. A vacuum gripper drops its load the moment the air stops. Treat the tool as part of the robot in every rule above.

Emergency stop

The E-stop is the red button in its own 3D-printed box, a normally closed contact wired by its cable to the control board's E-stop input. It must be connected for the robot to run.

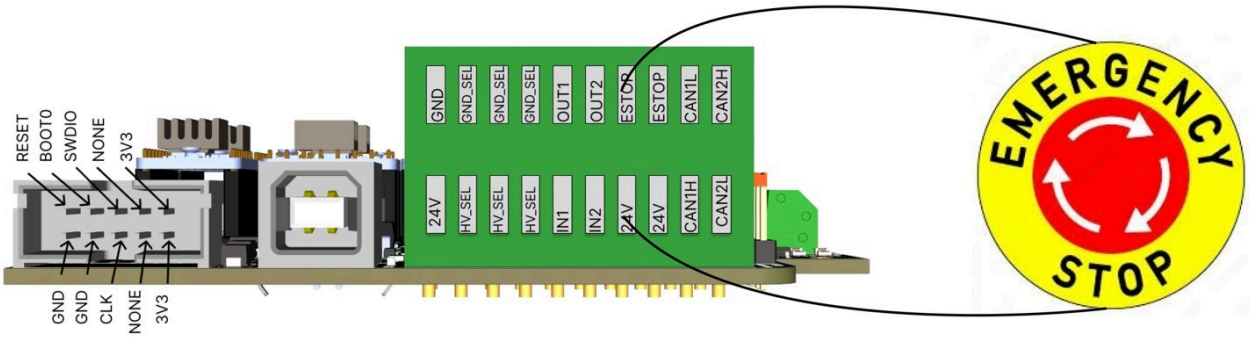


Figure 2.2: E-stop wiring to the control board.

What pressing it does

The control board reads the E-stop input in every command cycle and reports it to the PC in every packet. The PC software acts on it:

- the **commander software** disables the robot: it sends the disable command, the firmware stops accepting motion commands, and the log shows *Robot disable; button or estop*. A running program is stopped. The motors stay energised and hold the arm. Release the button and press **Enable** to continue; **Start** runs a program again from its first line.
- the **Python API's controller** cancels the move in progress and the queue, sends the disable command to the firmware, and refuses every motion command with `SYS_ESTOP_ACTIVE` while the button is pressed. When the button is released, the controller clears that error by itself and accepts commands again, but the firmware stays disabled and the arm does not move until `reset()` sends the enable command. Call `reset()` after every E-stop, before the next move.

The E-stop is a software stop

The control board's firmware (release 1) reports the E-stop and does not itself cut the motor power or the step pulses: the PC software stops the motion. A PC that has crashed, a program that has frozen, or a USB cable that has fallen out leaves the E-stop without effect, and the firmware keeps executing the last packet it received: a joint that was jogging in the commander keeps turning until it reaches its blocker. For those cases the hard stop is the **power supply**: a switch on the supply's mains side within reach, and the knowledge that the arm will fall when you use it.

What it does not do

- It does not cut power to the motors, so the arm does not fall when it is pressed.
- It does not stop the gripper: an electric gripper keeps its grip, a pneumatic one stays where the valve left it.
- It does not stop a Python script of your own: the script keeps running. Its commands are refused while the button is pressed and accepted again after the release, before the arm can move. Stop the script before you release the button, then call `reset()`.

If someone is caught by the arm

The E-stop does not free anyone

Pressed, the E-stop leaves the motors holding the arm where it is, and a joint cannot be pushed back by hand against a holding motor. To free a trapped hand or finger, either release the E-stop, enable the robot and jog that joint away slowly, the smallest step at a time; or support the arm with one hand and switch off the power supply with the other, knowing that the arm goes limp and falls. Decide which before you need it, and keep the supply's switch within reach (Planning the cell).

The contact

Normally closed, so that a cut wire, a loose terminal or an unplugged button is read as a stop. A normally open button does not work. The control board has two E-stop connectors on the same input: a second button can be wired in series, or a safety relay's contact. Inputs, outputs and E-stop has the wiring.

Moving the robot by hand

An unpowered PAROL6 can be moved by hand: the joints turn against the friction of their gearboxes and belts, and the arm has no brakes to release. This is how you pose it for packing and how you lift it. Two rules:

Never more than one turn, except joint 6

No joint may ever be turned more than one full turn, by hand or by software. Cables and pneumatic tubes run through the joints, and a second turn wraps and tears them. The one exception is **joint 6**: with the bare flange it can turn without limit; with a gripper or a tool, the tool's cables and tubes wrap as it turns, and they set how far it may go. Joints 1 to 5 have printed blockers that stop them; on a build without them, the rule is yours to keep.

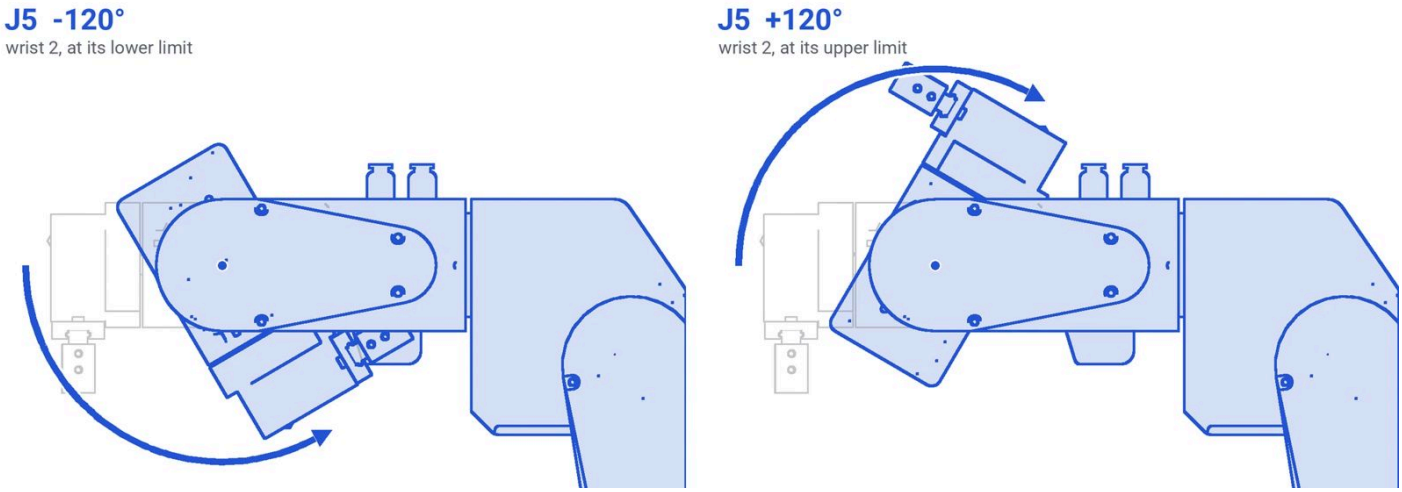


Figure 2.3: Joint 5 at its lower and upper limit, -120 and +120 degrees, over the standby pose in grey. A gripper can reach the forearm before that: check the clearance with a slow jog.

Not while the supply is connected

A stepper turned by hand generates voltage, and a quick move can switch the board on with the arm in your hands. Unplug the power supply, or move slowly, and only toward the parked pose.

When the robot is powered

Do not force a powered joint. The motor holds it with its full holding current; what gives is a belt, a set screw or a printed part, and afterwards the position the software reports is wrong. To move a powered robot, jog it.

After moving by hand

The robot does not know it was moved: the firmware counts steps and sees none. **Home again** before any program or Cartesian move. The commander's position display and the simulator are wrong until you do.

Lifting

Bring the arm to the parked pose, joints 2 and 3 toward their limit switches so that the arm folds over the base, then lift by the base, not by the arm. Transport and storage has the packing.

What the robot protects against, and what not

Know what the hardware and the software catch, and what they do not, before you decide how the robot is guarded.

What is protected

Against	How	Where
A target outside the joint limits	The planner refuses it; a jog stops at the limit. Firmware 1, the commander and the Python API check the earlier, partly wider limits (Joint limits)	Commander, Python API
A speed or acceleration above the limits	Clamped to the robot model's limits	Commander, Python API
A move through a singularity (Cartesian)	The commander stops the move; the API's IK fails and refuses the command	Commander, Python API
A pose the arm cannot reach	The IK fails and the command is refused	Commander, Python API
Self-collision and collision with declared obstacles	Collision checking of planned moves and jogs against the robot's meshes and your keep-out shapes	Python API (set_shapes)
The E-stop pressed	Reported by the board, acted on by the PC software (Emergency stop)	Commander, Python API
A driver overheating	The TMC5160 shuts its outputs down when it gets too hot; the fan keeps it far below that. The joint then goes limp, and firmware 1 does not tell the PC	Control board
A short on a motor output	The TMC5160 switches the output off. Firmware 1 reads the driver's short and open-load flags once at start-up and does not report them	Control board

What is not protected

Not against	Why	What you do instead
A collision with something the software does not know	There are no joint encoders and no force sensing: the stepper pushes until it stalls, then loses steps silently	Keep the workspace clear; check programs in the simulator first; declare obstacles to the Python API's collision checker
Lost steps	Open-loop motors; the position is counted, not measured	After a stall or a collision, home again; keep speeds and payloads within the limits
A fall on power loss	No brakes	Switch off only in the parked pose; keep hands and objects out from under the arm
A frozen PC	The E-stop acts through the PC software; the firmware does not stop on its own	A mains switch on the power supply within reach
Lost communication with the PC	The firmware has no timeout: it keeps executing the last packet until the next one arrives. After a move to a position (programs, the Python API) the joint stops near that target. After a jog in the commander the joint keeps turning at the jog speed until it reaches its blocker	Keep the mains switch within reach; a short, direct USB cable
Motor temperature	Only the drivers are measured, not the motors	Reduce the current for long runs (Currents and temperatures)
The tool	The robot does not know what the tool does	Guard the tool as you would the robot
A wrong joint direction on a new build	The firmware trusts its configuration	Jog each joint briefly before homing

The commander's safety functions are described in Errors and recovery; the Python API's in Errors.

Planning the cell

A robot cell is the robot, what it works on, and the people around it. For a PAROL6 on a bench the plan fits on a page, and it is worth writing.

The person

One person runs the robot and sees all of it. Nobody else reaches into the workspace while it is enabled. The E-stop is within that person's reach, and so is the power supply's switch.

The space

The space to keep free is everything the arm, the tool and the part in the gripper can sweep, not only where the tool centre point goes. The TCP reaches 434 mm from the axis of joint 1 with the pneumatic gripper and 494 mm with the MSG, and the elbow swings behind the base when the arm folds back (Workspace). Add the part and a margin:

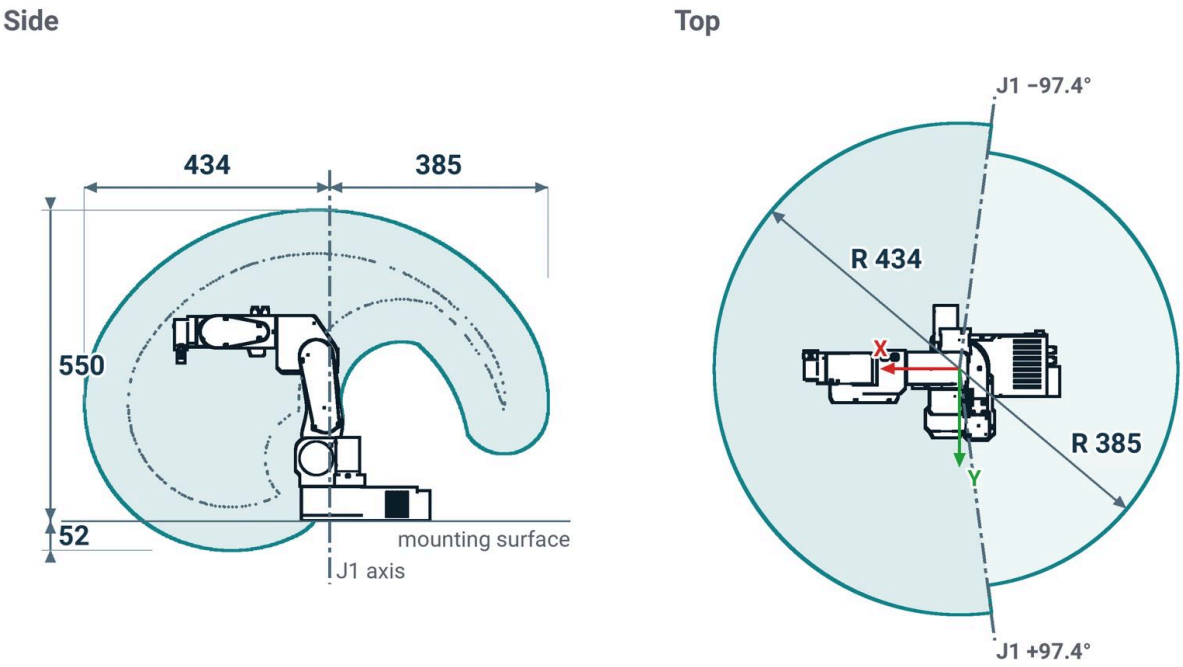


Figure 2.4: Where the pneumatic gripper's TCP can go, side and top view, in mm; the arm, the tool and the part need more room than this.

- A sphere of about half a metre around the base with the pneumatic gripper, about 0.6 m with the SSG-48 or the MSG, and more for a long part.
- Nothing fragile inside it, and nothing that the arm could knock onto the floor.

- The cables and tubes outside the sweep, or fixed so the arm cannot catch them.
- For a robot that runs unattended or near others, an enclosure: acrylic panels on the profile frame are enough to keep hands out; the robot cannot break them.

The program

- Run every new program in the simulator first, at full speed, and watch for the arm folding into the base or the table.
- Start on the real robot at low speed: in the commander, low `v=` or long `t=` in the moves (the speed slider only scales jogging); in the Python API, the `speed` fraction of each move.
- Make the first move of a program go from the after-homing pose to a known pose, so that a program started right after homing always begins safely.
- Use `Delay()` and the inputs to wait for the world, rather than assuming it.

The tool and the parts

- The payload includes the tool. A gripper and a camera together can be most of the 0.5 kg the arm carries at full reach.
- A pneumatic gripper needs 5 to 7 bar (check your gripper's rating) and a valve wired to an output; the valve's two tubes decide whether it is normally open or closed (Pneumatics).
- Know what the gripper does at an E-stop and at a power loss. The E-stop leaves the gripper as it is: an electric gripper keeps its grip, and a pneumatic or vacuum gripper stays where the valve left it. When the power or the air goes, the vacuum gripper drops its part.

When something changes

Changing the tool, the base, the table height or the fixture changes the kinematics and the clearances. Update the tool in the software, check the limits with a slow jog, and run the programs in the simulator again.

3. Technical data

3. Technical data

The numbers of PAROL6 V1 and V1.1, which share the mechanics: the specifications, the joint limits and their directions, the dimensions and the kinematic model, the drives with their ratios and resolutions, and the motor currents with the temperatures they lead to.

In this chapter

- Specifications
- Joint limits and directions
- Dimensions and kinematics
- Drives, ratios and resolution
- Currents and temperatures

The control board's own specifications are in the Control Board Manual.

Specifications

Parameter	Value
Degrees of freedom	6 revolute joints
Reach	408 mm to the flange, 434 mm to the standard pneumatic gripper's TCP, from the axis of joint 1 (Workspace)
Payload	1 kg near the base, 0.5 kg across the full workspace
Repeatability	0.1 mm
Weight	5.5 kg
Material	3D-printed PETG, steel shafts and bearings
Motors	NEMA 17 stepper motors, one per joint
Gearing	Planetary gearboxes (joints 2, 3, 6) and HTD3 belts (joints 1, 3, 4, 5)
Position reference	Limit switches (joints 2, 3, 5) and inductive sensors (joints 1, 4, 6); no joint encoders
Brakes	None: the arm falls when the motors lose power
Power supply	24 V DC, 5 A, recommended for optimal robot performance (18 V minimum)
Power consumption	About 40 W typical
Control	PAROL6 control board (STM32F446, six TMC5160 drivers) in the base
PC connection	USB, serial at 3 Mbaud, 100 Hz command cycle
Inputs and outputs	2 isolated inputs, 2 isolated outputs, E-stop input
Buses	1 CAN bus for electric grippers (a second is on the board, not populated)
Tool connections	M8 4-pin gripper connector (24 V, CAN) at the forearm; 2 pneumatic lines, 4 mm tube, base to forearm

Parameter	Value
Mounting	6 holes in the base, spaced for 30×30 mm aluminium profiles; clamps
Operating temperature	Room temperature; the printed parts limit the motor temperatures (Currents and temperatures)

Joint ranges

Joint	Range
Joint 1	194.8° (±97.4°)
Joint 2	113.4°
Joint 3	189.1°
Joint 4	229.2° (±114.6°)
Joint 5	240.6° (±120.3°)
Joint 6	360° in software; mechanically unlimited, but see Moving the robot by hand

The ranges are those of the robot's model (robot_descriptions 2.15.1); the exact limits are on Joint limits and directions.

Speeds

The speed limits are set in the PC software, not in the firmware, and the two programs differ: the commander allows less on joints 1, 2, 3 and 6 and on the tool, and more on joints 4 and 5.

Joint	Python API: maximum speed	Python API: acceleration	Commander: maximum speed
Joint 1	132 °/s (15000 steps/s)	395 °/s²	57 °/s (6500 steps/s)
Joint 2	70 °/s (25000 steps/s)	211 °/s²	51 °/s (18000 steps/s)
Joint 3	99 °/s (32000 steps/s)	298 °/s²	62 °/s (20000 steps/s)

Joint	Python API: maximum speed	Python API: acceleration	Commander: maximum speed
Joint 4	141 °/s (10000 steps/s)	422 °/s²	309 °/s (22000 steps/s)
Joint 5	141 °/s (10000 steps/s)	422 °/s²	309 °/s (22000 steps/s)
Joint 6	152 °/s (27000 steps/s)	456 °/s²	124 °/s (22000 steps/s)
Tool, linear	200 mm/s	550 mm/s²	60 mm/s
Tool, angular	100 °/s	275 °/s²	25 °/s

The steps per second are converted with the ratios of Drives, ratios and resolution. In the Python API a joint's acceleration is three times its speed, so a joint reaches full speed in a third of a second, and jogging is limited to 80 % of the maxima. The commander jogs within lower limits of its own, which its speed slider scales: 13 °/s on joint 1, 8 °/s on joint 2, 11 °/s on joint 3, 98 °/s on joints 4 and 5 and 101 °/s on joint 6 at 100 % (1500, 3000, 3600, 7000, 7000 and 18000 steps/s), and 60 mm/s for the tool. A joint driven at its maximum with a reduced motor current or a heavy payload loses steps: lower the speed before you lower the current.

Joint limits and directions

Joint numbering and positive directions

Joints are numbered from the base up. Joints 1, 2 and 3 are also called the base, shoulder and elbow joints, and joints 4, 5 and 6 wrist 1, wrist 2 and wrist 3, the names the robot model and ROS use. Each picture shows one joint turned 30° in its positive direction, in colour, over the pose in grey. The arrow at the joint points the same way, and the commander software's left arrows jog in that direction.



Figure 3.1: Each joint turned 30 degrees in its positive direction, in colour, over the pose in grey.

Joint	Also called	Positive direction
J1	base	counter-clockwise, seen from above

Joint	Also called	Positive direction
J2	shoulder	leans the upper arm forward, the way the forearm points in the standby pose
J3	elbow	raises the forearm
J4	wrist 1	clockwise, seen from the tool
J5	wrist 2	tips the tool up, in the standby pose
J6	wrist 3	clockwise, seen from the tool

Standby and after homing

The **standby pose** is the pose of the Denavit-Hartenberg table and the kinematics sheets, and the reference for the limits below: the upper arm vertical, the forearm horizontal, the arm pointing away from the electronics case at the back of the base. **After homing** the robot stands in the same pose turned 90° about joint 1, counter-clockwise seen from above: the firmware sets joint 1 to +90° when it finishes homing.

Pose	Joint 1	Joint 2	Joint 3	Joint 4	Joint 5	Joint 6
Standby	0°	−90°	180°	0°	0°	180°
After homing	90°	−90°	180°	0°	0°	180°

The Python API calls the after-homing pose its standby (`STANDBY_ANGLES_DEG`), and a later `home()` moves there; the commander software's standby is the standby pose above.

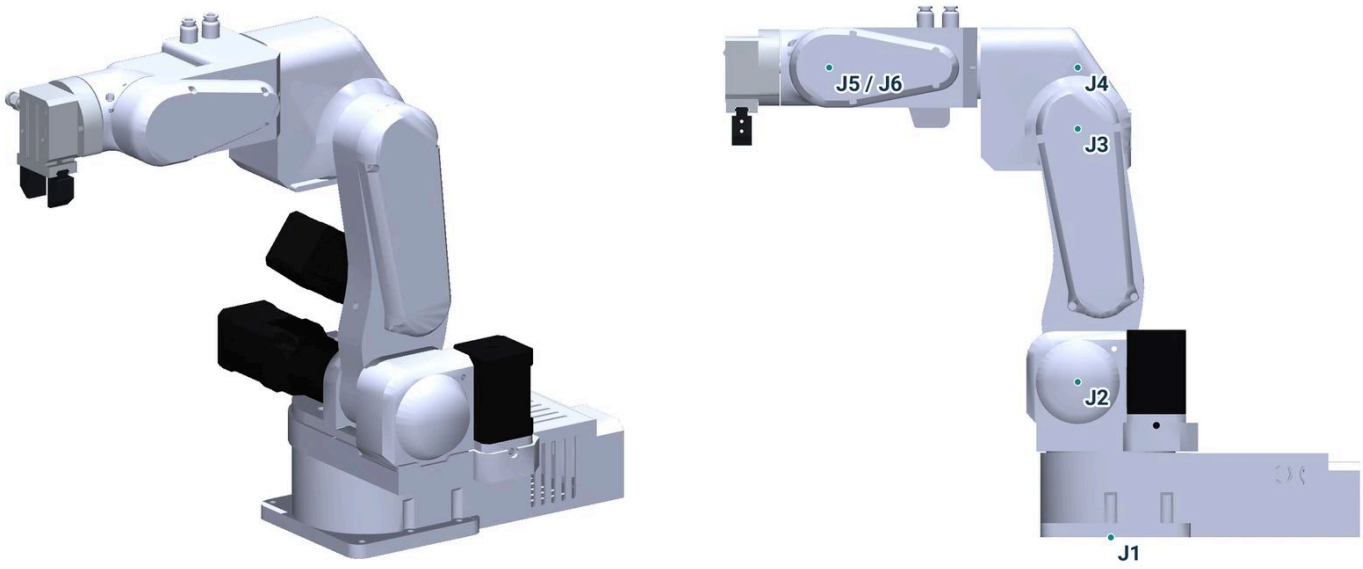


Figure 3.2: The standby pose, from the front and from the side, with the joints marked.

Limits

The joint limits of PAROL6's model in robot_descriptions (version 2.15.1), in degrees, in the angles the software shows (the standby pose above).

Joint	Negative limit	Standby	Positive limit	Range
J1	−97.40°	0°	+97.40°	194.8°
J2	−146.15°	−90°	−32.70°	113.4°
J3	105.52°	180°	294.59°	189.1°
J4	−114.59°	0°	+114.59°	229.2°
J5	−120.32°	0°	+120.32°	240.6°
J6	none	180°	none	no limit with the bare flange; with a tool, its cables and tubes set it

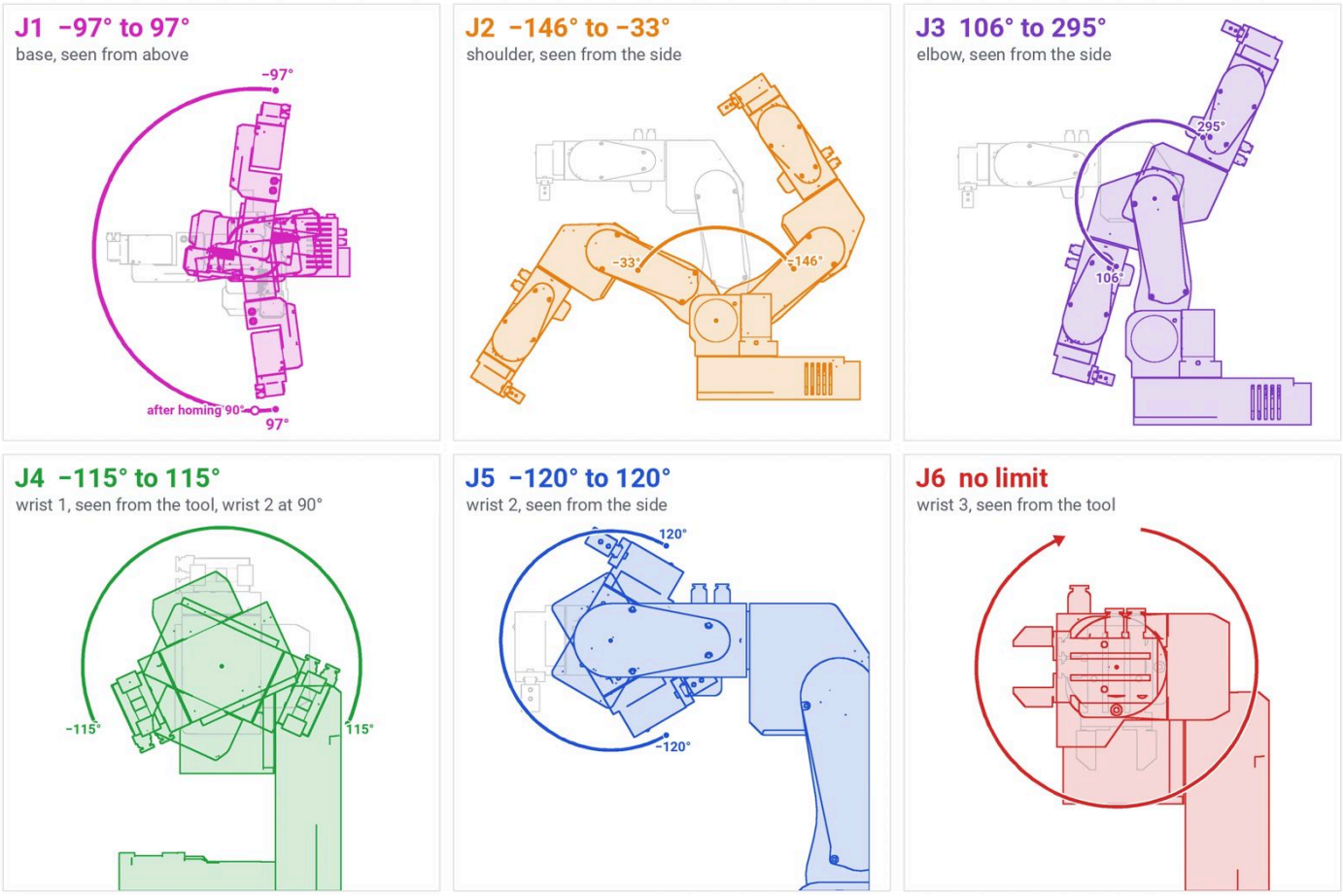


Figure 3.3: Each joint at both its limits, in its colour, over the standby pose in grey; the arc spans the range, in the angles the software shows. Joint 1 is seen from above, with the after-homing position marked; joint 4 is seen from the tool, with joint 5 at 90 degrees so that the tool's swing shows; joint 6 has no limit.

The firmware and the software still use the earlier limits

Firmware 1, the commander software and the Python API (0.9.1) enforce the limits of the earlier model until they are updated: J1 ±123.05°, J2 −145.01° to −3.38°, J3 107.87° to 287.87°, J4 ±105.47°, J5 ±90°, J6 0° to 360°. A target between the two is accepted by them; keep to the table above. Before homing they apply these limits to a position count that starts at 0 at power-up, so joint 2 cannot be jogged in its positive direction, nor joints 3 and 6 in their negative one, until the robot is homed (Jogging).

Joints 1 to 5 are stopped at both ends by printed blocker parts, the mechanical stops a little past the limits above. The limit switch or sensor that homes a joint sits near its blocker, so that it triggers just before the joint reaches it. Joint 6 has no blocker: with the bare flange nothing stops it, and with a tool only the tool's cables and tubes do.

Grippers and bases change the limits

A gripper changes where joint 5 can go without colliding with the forearm, and a riser under the base changes what joint 2 can reach. The software limits do not know about either: check the real clearances with a slow jog before you run a program near a limit.

The limits are not a safety function

The limits only stop the software from commanding a joint further. Inside them the arm can still hit a person, the table or itself, no safety-rated part checks them, and a wrong homing shifts them with it. They protect the robot's cables and parts, not the people near it (Safety).

Never more than one turn, except joint 6

No joint may ever be turned more than one full turn, by hand or by software: the cables and pneumatic tubes that pass through the joints wrap and tear. The one exception is **joint 6**: with the bare flange it can turn without limit; with a gripper or a tool, the tool's cables and tubes wrap as it turns, and they set how far it may go (Moving the robot by hand).

Dimensions and kinematics

Dimensions

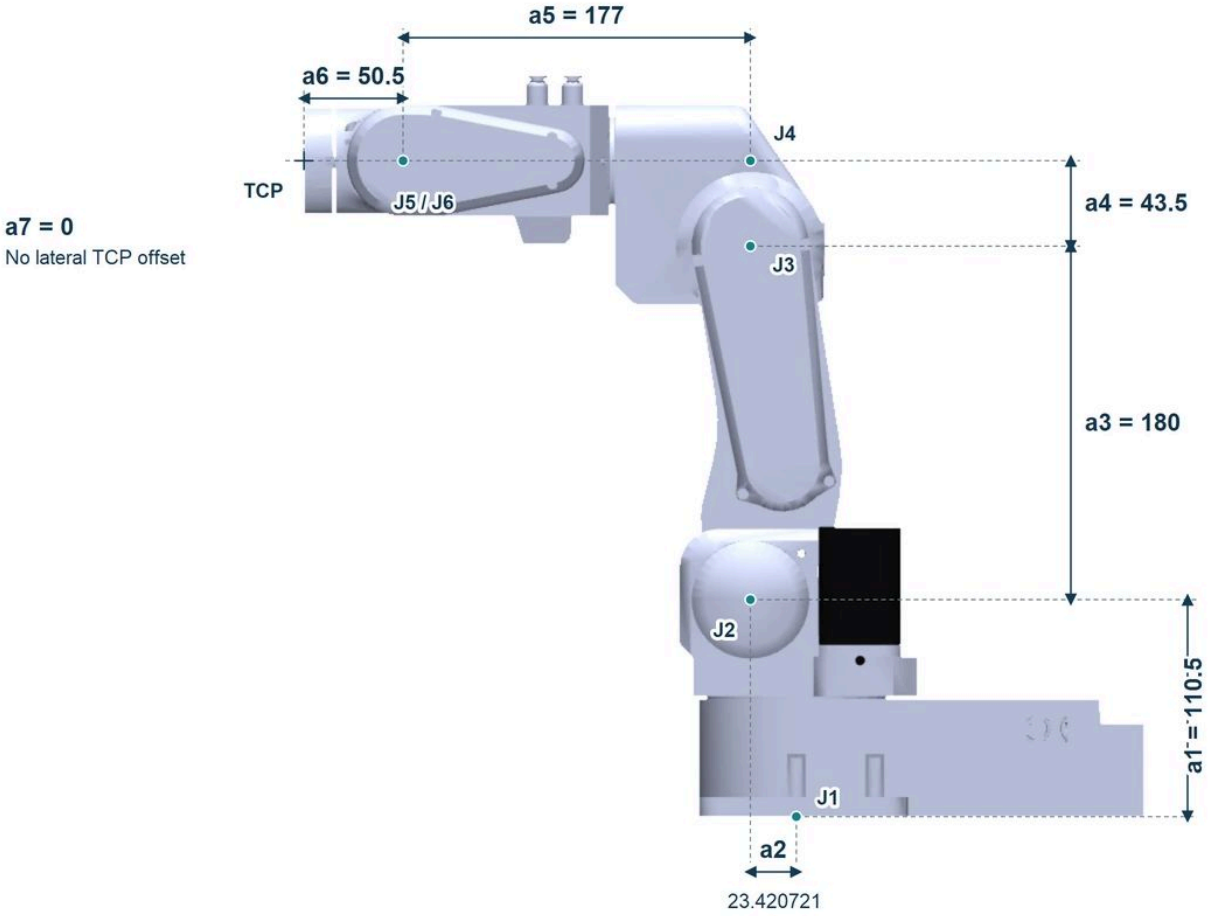
The sheets below come from PAROL6's models in robot_descriptions (version 2.15.1). Each shows the robot's own meshes in the standby pose (every joint of the model at 0), in one orthographic side view at one scale, with the dimensions taken from the URDF, not measured from the picture, and the standard DH table to the tool centre point (TCP) of that configuration. Click a sheet to enlarge it.

PAROL6 / Flange only

Kinematic review | Standard DH to the configured TCP

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Pose: q1 ... q6 = 0 rad; no jaws. Orthographic side view; dimensions in mm.



a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

Joint	Link	theta [rad]	alpha [deg]	a [mm]	d [mm]
1	Base	q1	-90	a2 = 23.420721	a1 = 110.5
2	Shoulder	q2 - pi/2	180	a3 = 180	0
3	Elbow	q3 + pi	90	-a4 = -43.5	0
4	Wrist 1	q4	-90	0	-a5 = -177
5	Wrist 2	q5	90	0	0
6	Wrist 3 / TCP	q6 + pi	180	-a7 = 0	-a6 = -50.5

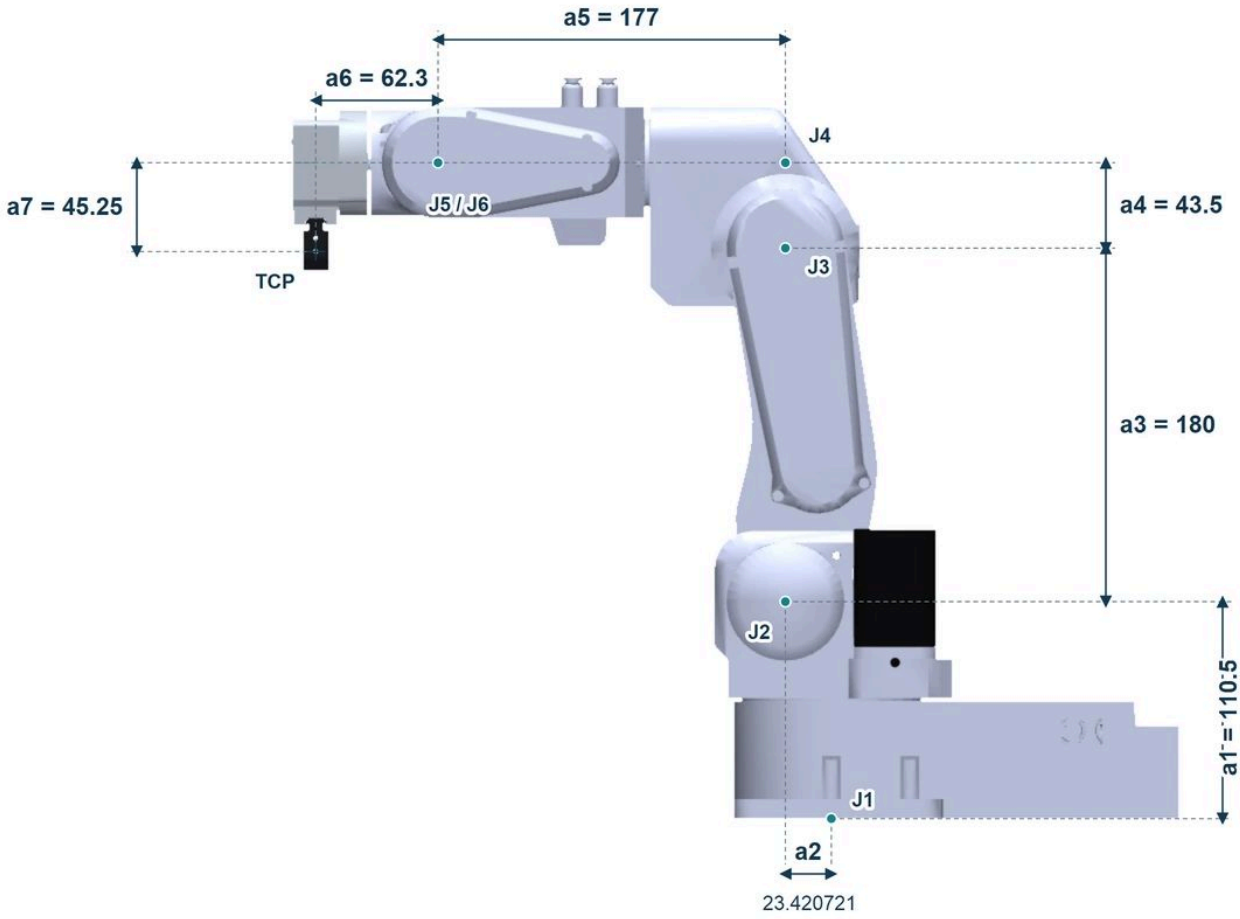
Figure 3.4: PAROL6 with nothing on the flange: dimensions and the DH table to the TCP.

PAROL6 / Pneumatic gripper

Kinematic review | Standard DH to the configured TCP

12 / 12

Pose: q1 ... q6 = 0 rad; jaws open. Orthographic side view; dimensions in mm.



a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

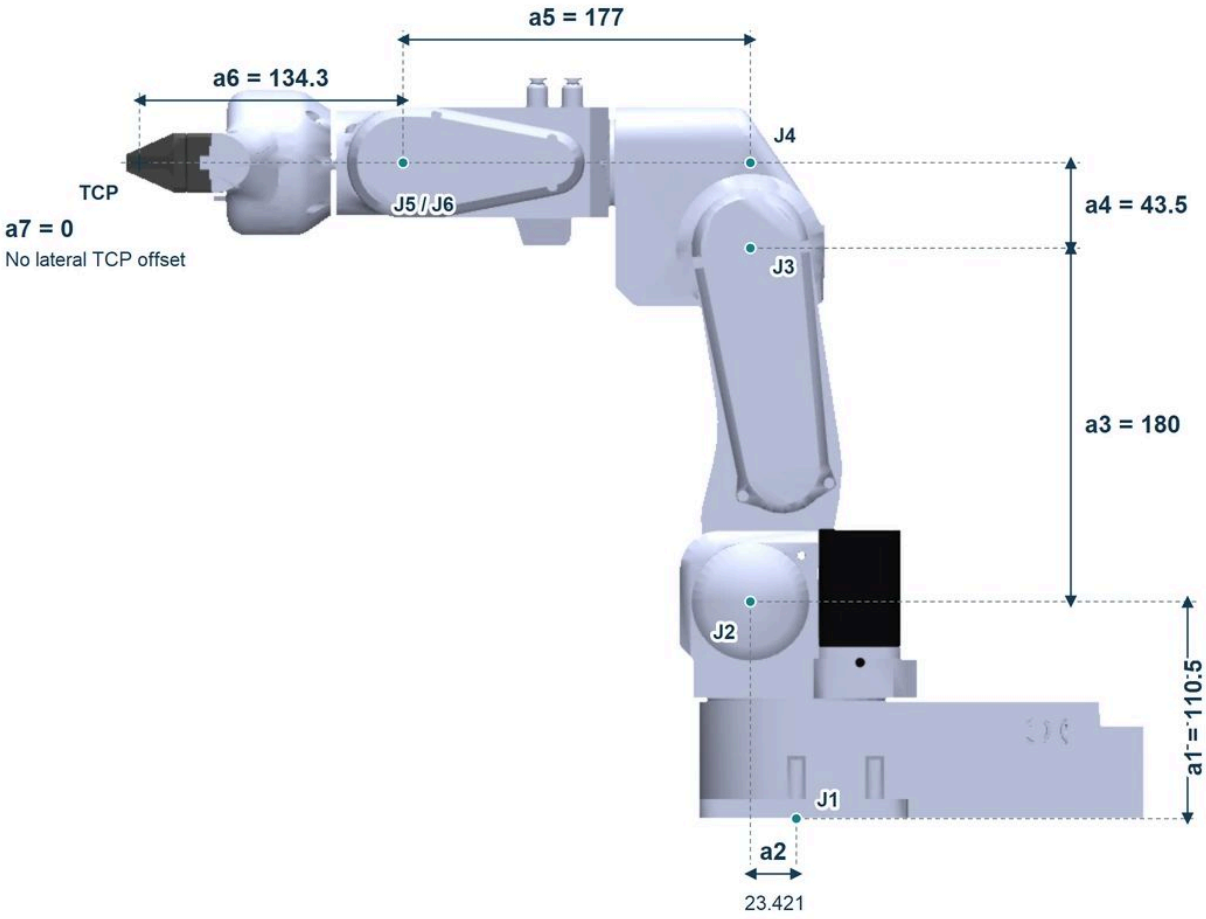
Joint	Link	theta [rad]	alpha [deg]	a [mm]	d [mm]
1	Base	q1	-90	a2 = 23.420721	a1 = 110.5
2	Shoulder	q2 - pi/2	180	a3 = 180	0
3	Elbow	q3 + pi	90	-a4 = -43.5	0
4	Wrist 1	q4	-90	0	-a5 = -177
5	Wrist 2	q5	90	0	0
6	Wrist 3 / TCP	q6 + pi	180	-a7 = -45.25	-a6 = -62.3

Figure 3.5: PAROL6 with the standard pneumatic gripper: dimensions and the DH table to the TCP.

PAROL6 / SSG48 / Solid jaws

Kinematic review | Standard DH to the configured TCP

Pose: q1 ... q6 = 0 rad; jaws open. Orthographic side view; dimensions in mm.



a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

Joint	Link	theta [rad]	alpha [deg]	a [mm]	d [mm]
1	Base	q1	-90	a2 = 23.421	a1 = 110.5
2	Shoulder	q2 - pi/2	180	a3 = 180	0
3	Elbow	q3 + pi	90	-a4 = -43.5	0
4	Wrist 1	q4	-90	0	-a5 = -177
5	Wrist 2	q5	90	0	0
6	Wrist 3 / TCP	q6 + pi	180	-a7 = 0	-a6 = -134.3

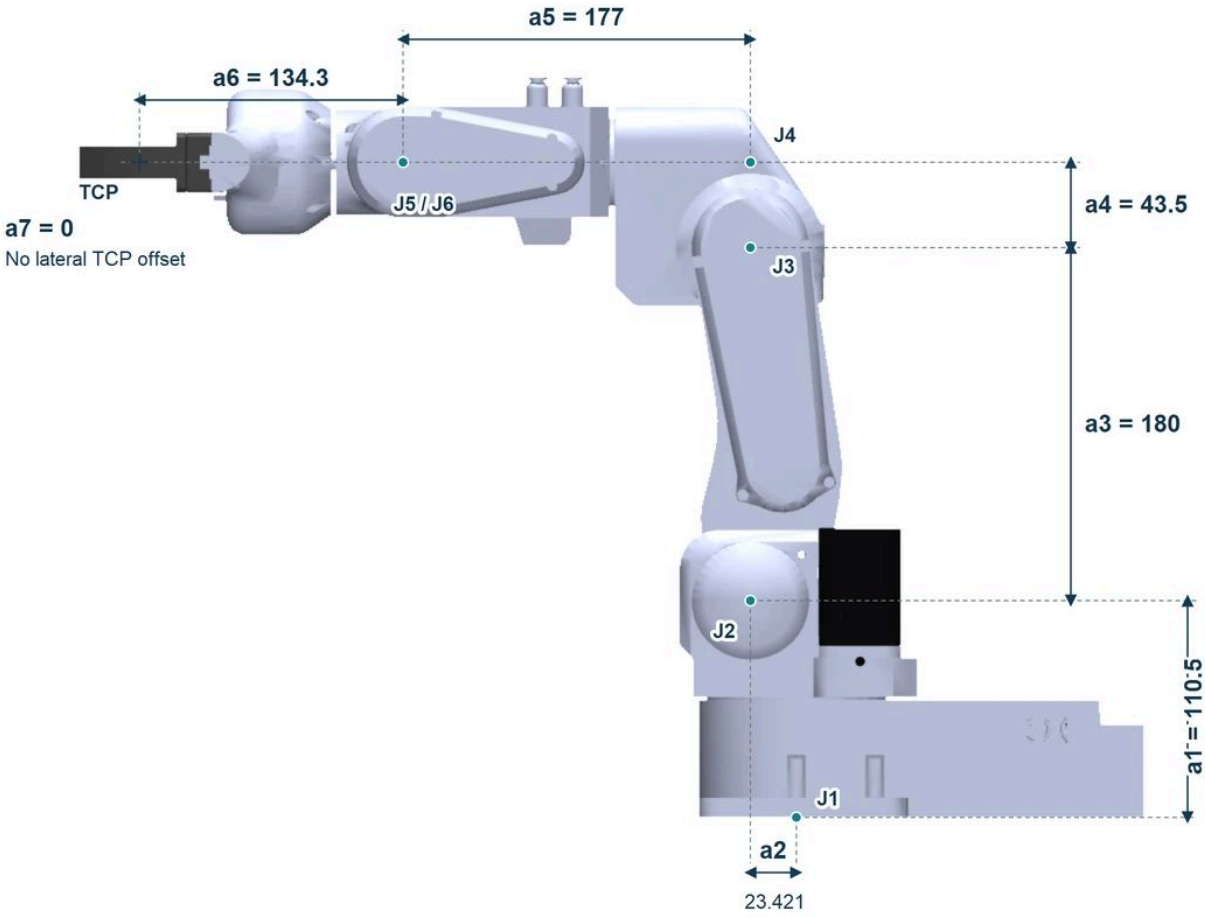
Figure 3.6: PAROL6 with the SSG-48 gripper, solid jaws: dimensions and the DH table to the TCP.

SSG-48 fin-ray

PAROL6 / SSG48 / Fin-ray jaws

Kinematic review | Standard DH to the configured TCP

Pose: q1 ... q6 = 0 rad; jaws open. Orthographic side view; dimensions in mm.



a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

Joint	Link	theta [rad]	alpha [deg]	a [mm]	d [mm]
1	Base	q1	-90	a2 = 23.421	a1 = 110.5
2	Shoulder	q2 - pi/2	180	a3 = 180	0
3	Elbow	q3 + pi	90	-a4 = -43.5	0
4	Wrist 1	q4	-90	0	-a5 = -177
5	Wrist 2	q5	90	0	0
6	Wrist 3 / TCP	q6 + pi	180	-a7 = 0	-a6 = -134.3

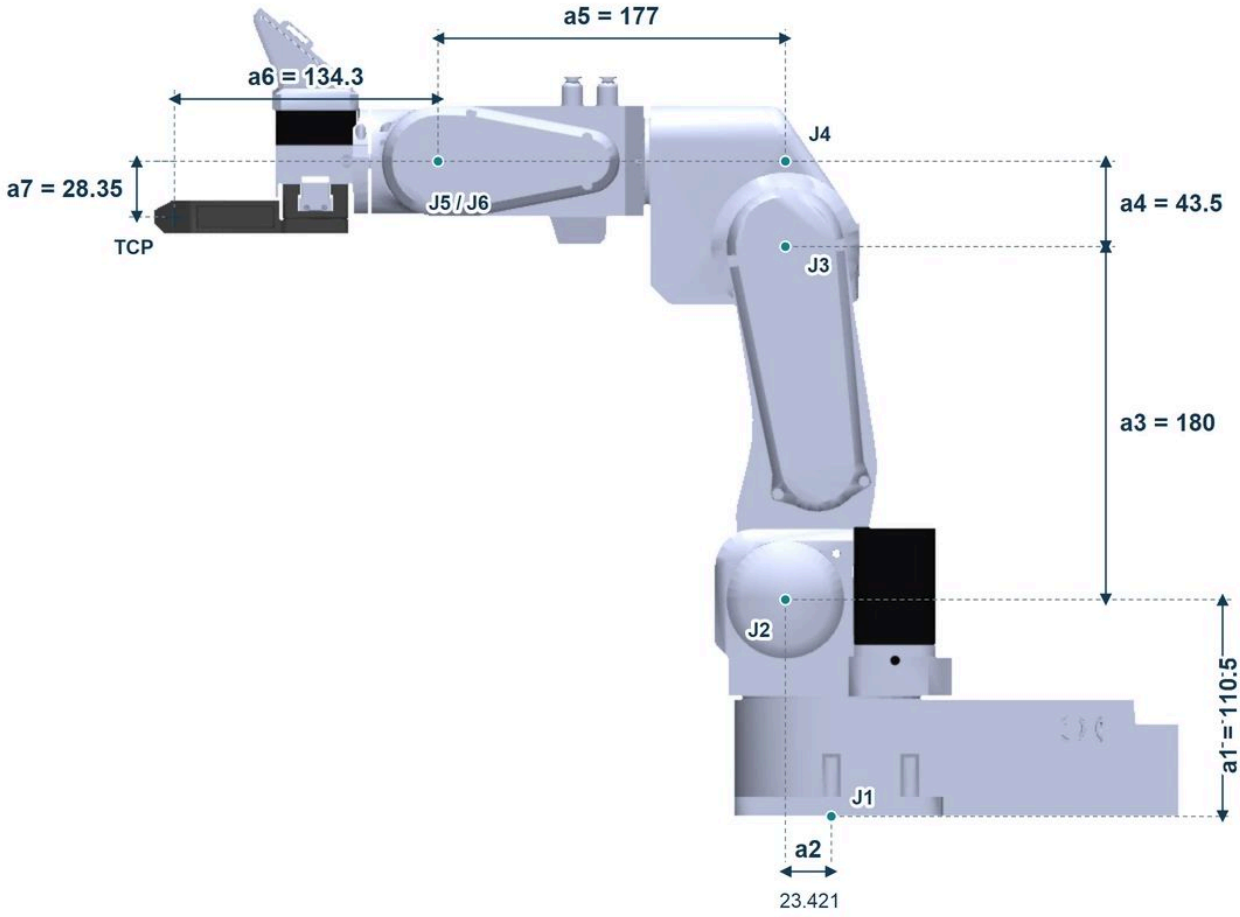
Figure 3.7: PAROL6 with the SSG-48 gripper, fin-ray jaws: dimensions and the DH table to the TCP.

PAROL6 / MSG / Solid jaws

Kinematic review | Standard DH to the configured TCP

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Pose: q1 ... q6 = 0 rad; jaws open. Orthographic side view; dimensions in mm.



a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

Joint	Link	theta [rad]	alpha [deg]	a [mm]	d [mm]
1	Base	q1	-90	a2 = 23.421	a1 = 110.5
2	Shoulder	q2 - pi/2	180	a3 = 180	0
3	Elbow	q3 + pi	90	-a4 = -43.5	0
4	Wrist 1	q4	-90	0	-a5 = -177
5	Wrist 2	q5	90	0	0
6	Wrist 3 / TCP	q6 + pi	180	-a7 = -28.35	-a6 = -134.3

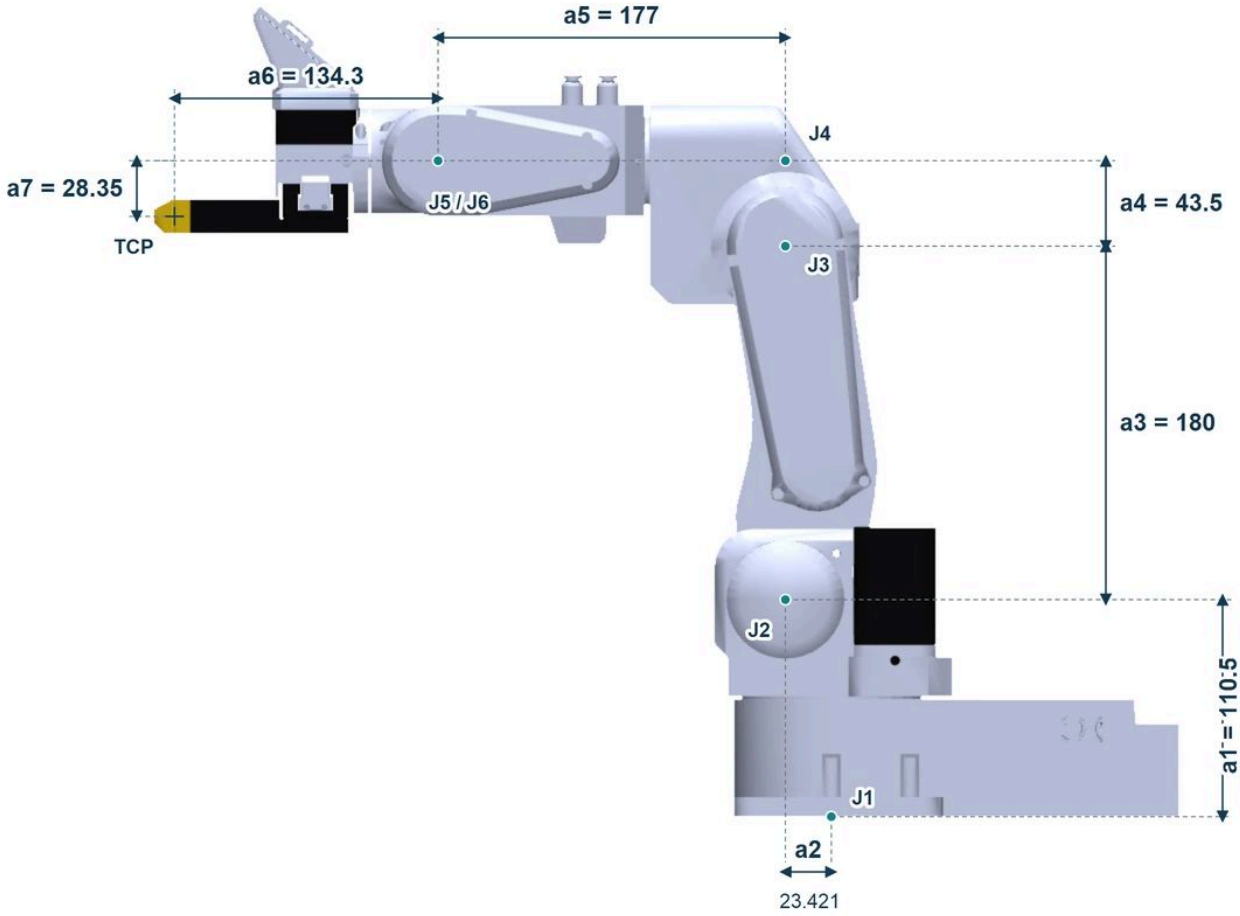
Figure 3.8: PAROL6 with the MSG gripper, solid jaws: dimensions and the DH table to the TCP.

PAROL6 / MSG / Soft TPU jaws

Kinematic review | Standard DH to the configured TCP

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Pose: q1 ... q6 = 0 rad; jaws open. Orthographic side view; dimensions in mm.



a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

Joint	Link	theta [rad]	alpha [deg]	a [mm]	d [mm]
1	Base	q1	-90	a2 = 23.421	a1 = 110.5
2	Shoulder	q2 - pi/2	180	a3 = 180	0
3	Elbow	q3 + pi	90	-a4 = -43.5	0
4	Wrist 1	q4	-90	0	-a5 = -177
5	Wrist 2	q5	90	0	0
6	Wrist 3 / TCP	q6 + pi	180	-a7 = -28.35	-a6 = -134.3

Figure 3.9: PAROL6 with the MSG gripper, soft TPU jaws: dimensions and the DH table to the TCP.

Dimension	What it is	Value
a1	Base height: the floor to the joint 2 axis	110.5 mm
a2	Shoulder offset: the joint 1 axis to the joint 2 axis	23.42 mm
a3	Upper arm: joint 2 to joint 3	180 mm
a4	Elbow offset	43.5 mm
a5	Forearm: the joint 3 axis to the wrist centre	177 mm
a6	The joint 6 centre to the TCP, along the tool axis	depends on the tool, below
a7	Sideways offset of the TCP	depends on the tool, below

Tool	a6	a7
Flange only (the TCP is the flange face)	50.5 mm	0
Pneumatic gripper	62.3 mm	45.25 mm
SSG-48, solid or fin-ray jaws	134.3 mm	0
MSG, solid or soft jaws	134.3 mm	28.35 mm

A riser under the base changes a1, and a tool of your own changes a6 and a7: tell the software about the tool (Tools and the TCP), or the poses it reports will be wrong.

The software still computes with the earlier model

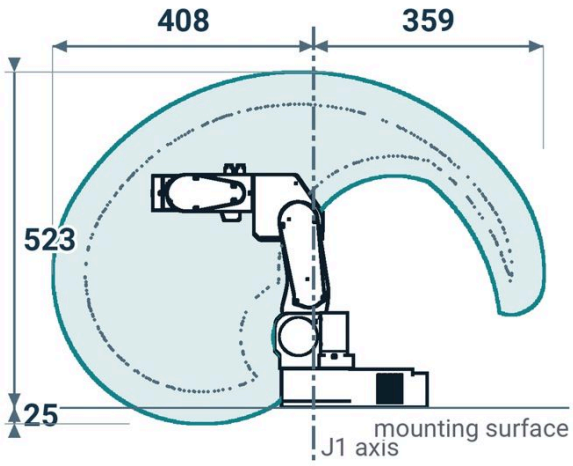
These pages give the model's dimensions. Until the software is updated, the Python API (its built-in URDF) and the commander software compute with a5 = 176.35 mm. The commander uses a6 = 62.8 mm for the pneumatic gripper, so a pose it reports can differ from these sheets by up to about 0.65 mm. The Python API has its own tool offsets, which put its TCP up to about 14 mm from the model's (The TCP of each tool).

Workspace

Where the tool centre point can go within the model's joint limits: on the left in the plane of the arm, on the right seen from above, as joint 1 turns through its range. Each tab is for one point: the flange centre, or the TCP of a gripper.

Flange only

Side



Top

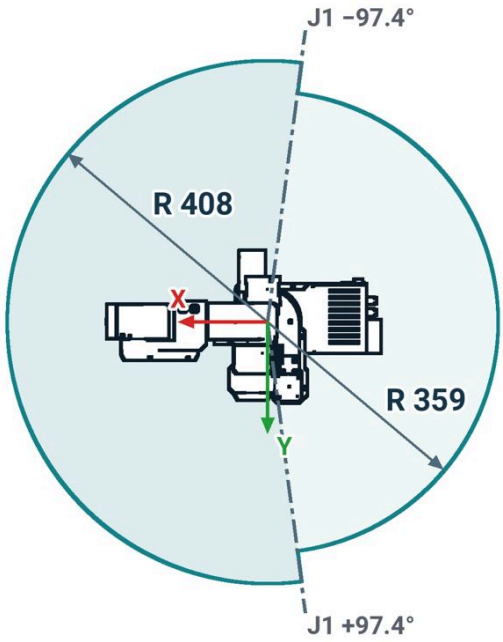


Figure 3.10: Workspace of the flange centre, side and top view, in mm.

Pneumatic gripper

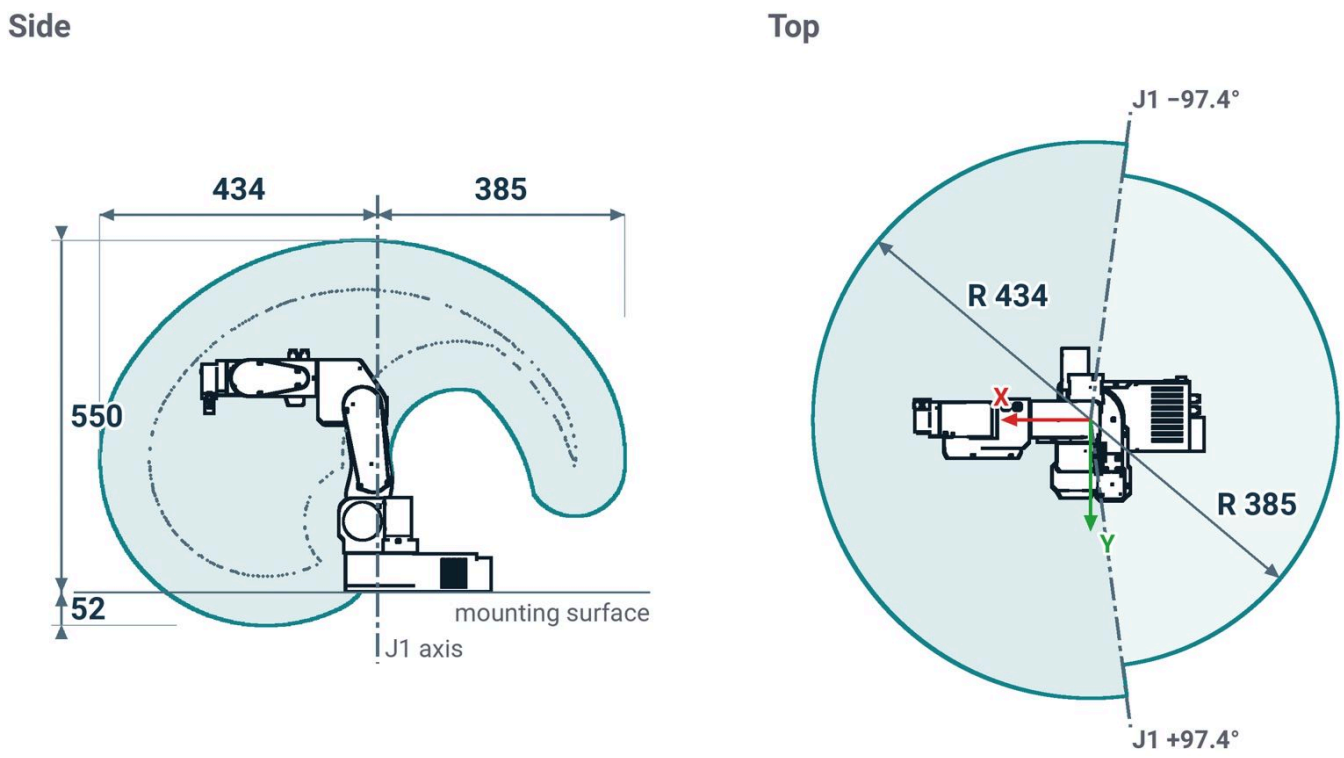


Figure 3.11: Workspace of the pneumatic gripper's TCP, side and top view, in mm.

SSG-48

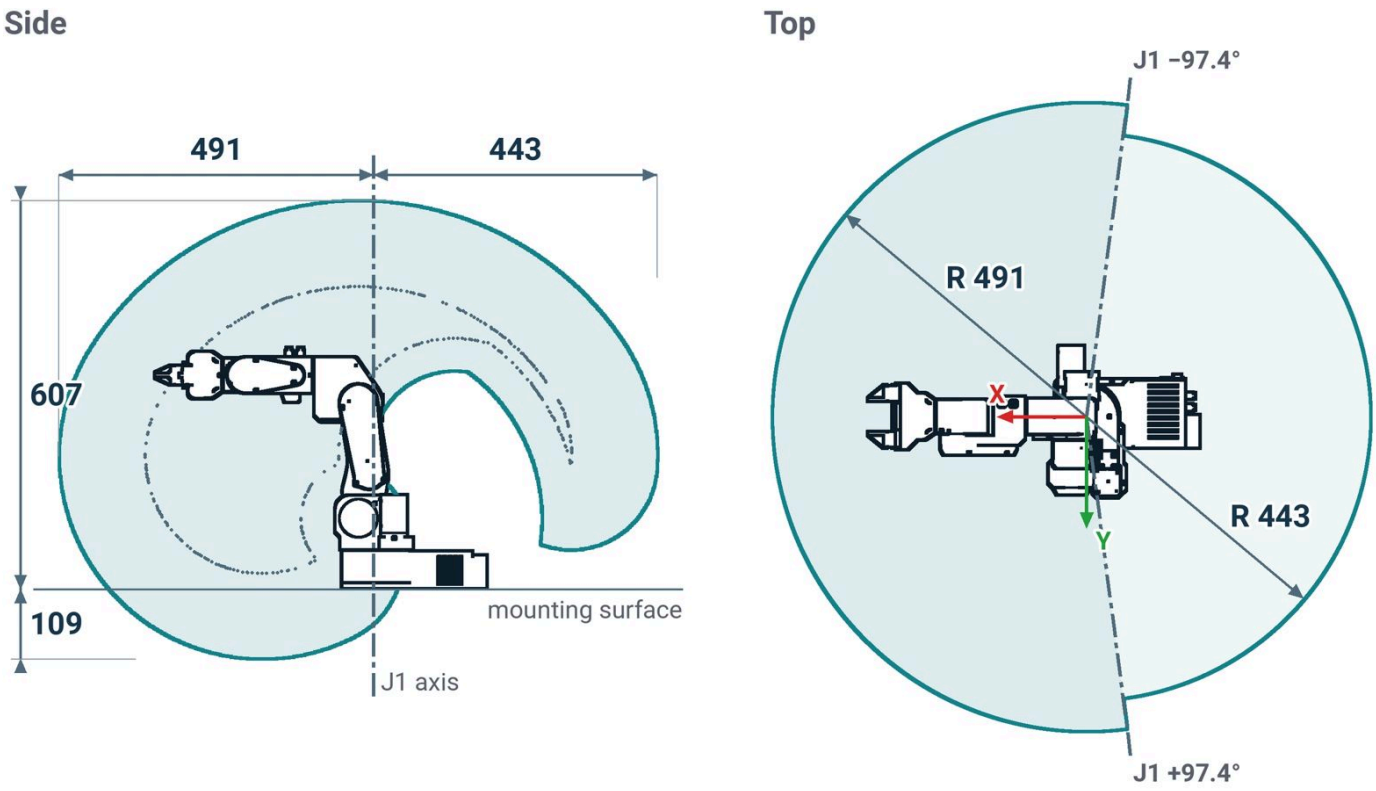


Figure 3.12: Workspace of the SSG-48's TCP, side and top view, in mm.

MSG

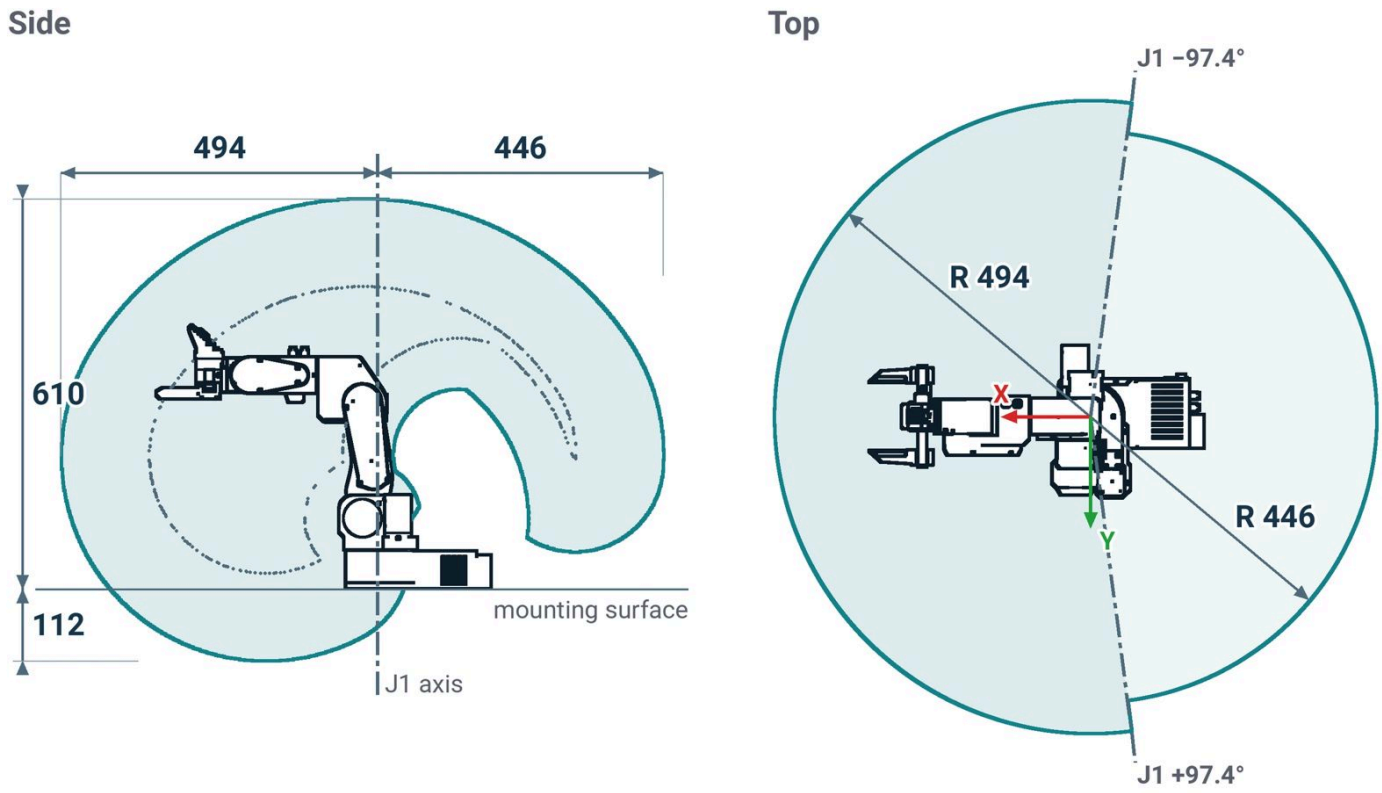


Figure 3.13: Workspace of the MSG's TCP, side and top view, in mm.

Point	In front of the joint 1 axis	Behind it	Highest, above the mounting surface	Lowest
Wrist centre	357 mm	308 mm	473 mm	26 mm above
Flange centre	408 mm	359 mm	523 mm	25 mm below
Pneumatic gripper TCP	434 mm	385 mm	550 mm	52 mm below
SSG-48 TCP	492 mm	443 mm	607 mm	109 mm below
MSG TCP	494 mm	446 mm	610 mm	112 mm below

In the side view the shaded region is where the point can go, with joints 2, 3 and 5 anywhere within their limits, joint 4 at 0° and joint 6 at 0° or 180°, so that a TCP beside the tool axis stays in the plane; the dotted line is the same for the wrist centre. The region shows where the point can go, not where the tool can take every orientation: at its edge the tool can only point outwards (Workspace and reach). It is computed from the joint limits only: the arm's own parts, the tool and the table can block some of it.

- **Below the mounting surface.** The region dips below the surface the robot stands on, in front of the base. On a table that part is solid; the arm reaches it only past the table's edge, or with the robot on a riser.
- **Behind the base.** Joint 1 turns ±97.4°, so seen from above the arm reaches a little more than the half circle in front of it. The lighter part behind the base it reaches only by folding back over the base, the way the side view's region curls behind the joint 1 axis.
- **Near the joint 1 axis.** Keep the work away from it. With the wrist centre on the joint 1 axis the arm is in its shoulder singularity, and close to it joint 1 has to turn fast for a small move of the tool (Configurations and singularities).
- **X and Y** in the top view are the axes of the world frame: X is the way the arm points in the standby pose, Y the way it points after homing (Reference frames).

Kinematic structure

The first three joints place the wrist centre; the axes of joints 4, 5 and 6 intersect in one point, a spherical wrist, which is what makes the inverse kinematics tractable. The Theory corner explains forward and inverse kinematics, and Configurations and singularities the eight ways the arm can reach one pose.

Denavit-Hartenberg parameters

Standard DH, each joint's transform being `A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)`, from the base to the TCP. `q1` to `q6` are the model's joint angles (the URDF's), all 0 in the standby pose; the theta column already contains the joint offsets, so do not add them twice. Theta is the angle the software shows: joint 2's software angle is q2 − 90°, joint 3's is q3 + 180° and joint 6's is q6 + 180°, the others are equal. The sheets above give the same table for each tool.

Joint	Link	theta	alpha	a	d
1	Base	q1	-90°	a2 = 23.42 mm	a1 = 110.5 mm
2	Shoulder	q2 - 90°	180°	a3 = 180 mm	0
3	Elbow	q3 + 180°	90°	-a4 = -43.5 mm	0
4	Wrist 1	q4	-90°	0	-a5 = -177 mm

Joint	Link	theta	alpha	a	d
5	Wrist 2	q5	90°	0	0
6	Wrist 3, to the TCP	q6 + 180°	180°	-a7	-a6

The models were checked at 128 poses including the joint limits: the DH table, the URDF and the MuJoCo model agree to within 0.006 mm and 0.002°. The Python API does not use the table directly: it loads its URDF and computes the kinematics numerically. The commander software uses a Robotics Toolbox DH model.

Mounting holes

Six holes in the base, on a pattern made for 30×30 mm aluminium profiles. The mounting plates folder of the repository has STEP drawings of a small and a big base plate: Mounting the robot.

Drives, ratios and resolution

Motors and gearing

Joint	Motor	Gearing	Ratio
Joint 1	NEMA 17, 65 N·cm (60 mm)	HTD3 belt, 15-tooth pulley to a 96-tooth printed pulley	6.4 : 1
Joint 2	NEMA 17, 65 N·cm (60 mm)	EG planetary gearbox	20 : 1
Joint 3	NEMA 17, 45 N·cm (40 mm)	EG planetary gearbox 20 : 1, then a belt 38 : 42	18.095 : 1
Joint 4	NEMA 17, 45 N·cm (40 mm)	HTD3 belt, 12-tooth pulley to 48 teeth	4 : 1
Joint 5	NEMA 17, 45 N·cm (40 mm)	HTD3 belt, 12-tooth pulley to 48 teeth	4 : 1
Joint 6	NEMA 17, 16 N·cm (20 mm)	EG planetary gearbox	10 : 1

The planetary gearboxes are the precision EG series with 15 to 20 arc-minutes of backlash; the cheaper MG series fits and works, with more backlash. The belts are 6 mm wide HTD3, tensioned by small bearings; their lengths are in the BOM.

Steps and microsteps

The stepper motors have 200 full steps per turn. The control board runs the TMC5160 drivers at 32 microsteps, so one motor turn is 6400 microsteps, and every position and speed the firmware and the PC exchange is in these steps. Through the gearing, one step at the joint is:

Joint	Steps per joint turn	Step size
Joint 1	40 960	0.00879° (0.000153 rad)
Joint 2	128 000	0.00281° (0.000049 rad)

Joint	Steps per joint turn	Step size
Joint 3	115 810	0.00311° (0.000054 rad)
Joint 4	25 600	0.01406° (0.000245 rad)
Joint 5	25 600	0.01406° (0.000245 rad)
Joint 6	64 000	0.00563° (0.000098 rad)

Conversions: degrees per step = 360 / (200 × 32) / ratio; the software's `DEG2STEPS` and `STEPS2DEG` do exactly this. A microstep is not a guaranteed position: the motor's holding torque at a fraction of a full step is small, so the real resolution under load is coarser than the table, which is why the repeatability is specified at 0.1 mm rather than at the step size.

Try it

The conversions of this page, with your numbers: 6400 microsteps per motor turn, through the joint's ratio.

Why open loop works here

The motors are not fitted with encoders: the position is the count of steps commanded since homing. This works because the planner never asks a motor for more than its torque at the commanded speed and acceleration, with margin; a stepper that is not overloaded does not lose steps. If one is overloaded (a collision, a payload over the limit, too high a speed with a reduced current) it loses steps silently, the arm is then not where the software thinks it is, and the only cure is to home again. Steppers, open loop and backlash goes deeper.

Currents and temperatures

Motor currents

The control board sets each motor's running current in the firmware. The values are chosen for the motors of the BOM and for the printed parts around them; joints 4 and 5 run below their motors' rating because their motors are enclosed in the forearm.

Joint	Maximum current (`constants.h`)	Current while moving	Holding current
Joint 1	2000 mA	85 % of the maximum: 1700 mA	70 % of that: 1190 mA
Joint 2	2000 mA	1700 mA	80 %: 1360 mA
Joint 3	1900 mA	1615 mA	80 %: 1292 mA
Joint 4	1700 mA	1445 mA	80 %: 1156 mA
Joint 5	1700 mA	1445 mA	80 %: 1156 mA
Joint 6	965 mA	820 mA	88 %: 722 mA

The firmware calls `rms_current(MOTORx_MAX_CURRENT * 0.85, hold_multiplier)`; the hold multiplier is 0.7 for joint 1, 0.8 for joints 2 to 5 and 0.88 for joint 6 (`motor_init.cpp`; Firmware parameters). Lower currents give less torque, which means lower reachable speeds and accelerations and a higher chance of lost steps; higher currents heat the motors and the plastic around them.

Temperatures

Stepper motors are rated for 100 °C and more, but PAROL6 is plastic, and PETG softens around 80 °C. These are the motor case temperatures measured on a robot at the default currents, at room temperature:

Joint	Holding position for 5 h	Moving for 2 h
Joint 1	51 °C	64 °C
Joint 2	54 °C	64 °C

Joint	Holding position for 5 h	Moving for 2 h
Joint 3	48 °C	52 °C
Joint 4	60 °C	73 °C
Joint 5	60 °C	72 °C
Joint 6	61 °C	65 °C

Through a thermal camera, a powered PAROL6 shows where the heat is: the motors of joints 2 and 3 beside the shoulder, the elbow (the hottest point in this picture, 50 °C) and the wrist motors in the forearm, and the control board at the back of the base. The scale runs from 29.6 to 50.0 °C.



Figure 3.14: A powered PAROL6 through a thermal camera: warm motors at the shoulder, the elbow and in the forearm, and the control board in the base.

Long runs

Joints 4 and 5 come within 10 °C of the plastic's limit after two hours of continuous motion. For a robot that runs for hours, reduce the currents in the firmware (`MOTORx_MAX_CURRENT` in `constants.h`, or the `hold_multiplier` in `motor_init.cpp`), lower the duty cycle, or let it rest in a pose that does not load the motors. A deformed forearm cannot be straightened: it has to be printed again.

The stepper drivers protect themselves: a TMC5160 that gets too hot shuts its outputs down, and its joint goes limp. The fan in the base keeps them well below that (Control board technical data). Firmware 1 does not tell the PC when a driver is hot: the joints' temperature error bits in its packet are always 0. The driver temperature is not the motor temperature.

4. Installation

4. Installation

Where to put the robot, how to fix it, what to connect, and the first power-on.

In this chapter

- Installation site
- Mounting the robot
- Connecting
- First power-on

Building the robot is the Build guide; wiring the control board inside the base is in the Control Board Manual.

Installation site

- **A rigid table or frame.** The arm accelerates its 5.5 kg quickly; a wobbly table shakes and spoils the repeatability. An aluminium profile frame or a solid bench is right.
- **Clearance.** The flange reaches 408 mm from the axis of joint 1 in front of the base and 359 mm behind it, plus the tool, and the elbow swings behind the base when the arm folds (Workspace). Keep a free sphere of about 500 mm around the base with the pneumatic gripper, about 600 mm with the SSG-48 or the MSG, and nothing fragile in it. The simulator shows the sweep of a program before the robot runs it.
- **Room temperature, dry, dust-free.** The base draws air through its fan over the stepper drivers.
- **The E-stop within reach** of the person at the PC, and reachable from where someone would stand to load the robot.
- **Cables.** The power, USB and E-stop cables leave the base at the back; route them so that nothing pulls on them and so that the arm cannot sweep over them.
- **Pneumatics.** If you use the pneumatic gripper, a valve and an air supply at 5 to 7 bar near the base; check your gripper's rating (Pneumatics).
- **A mains switch within reach** on the power supply's mains side, a switched socket strip for example: the stop for when the PC software has frozen (Emergency stop).

The Planning the cell page covers the safety side: who can reach the robot, guarding, and what the software can and cannot stop.

Mounting the robot

An unanchored robot falls over

When the arm extends, the centre of gravity leaves the base. Never run PAROL6 free-standing, not even for a test.

The base holes

Six holes in the base, spaced for 30×30 mm aluminium profiles. Bolt the base to a plate or a profile frame with six M5 × 16 mm socket head screws (DIN 912, in the kit), with T-nuts on a profile. They take a 4 mm hex key.



Figure 4.1: The six holes in the base.

Clamps

For a desk, clamp the base to the edge with two screw clamps over the base's rim. Check that the clamps do not interfere with the fan intake and the connectors at the back.

A profile frame

The repository's `STL/UTILITY` folder has `connectors_4_alu_profiles.STL`, which prints a set of corner connectors for a frame of 30×30 profiles, as in the figure. You need M5 T-nuts, M5 × 12 mm screws for the connectors and the six M5 × 16 mm screws for the base.

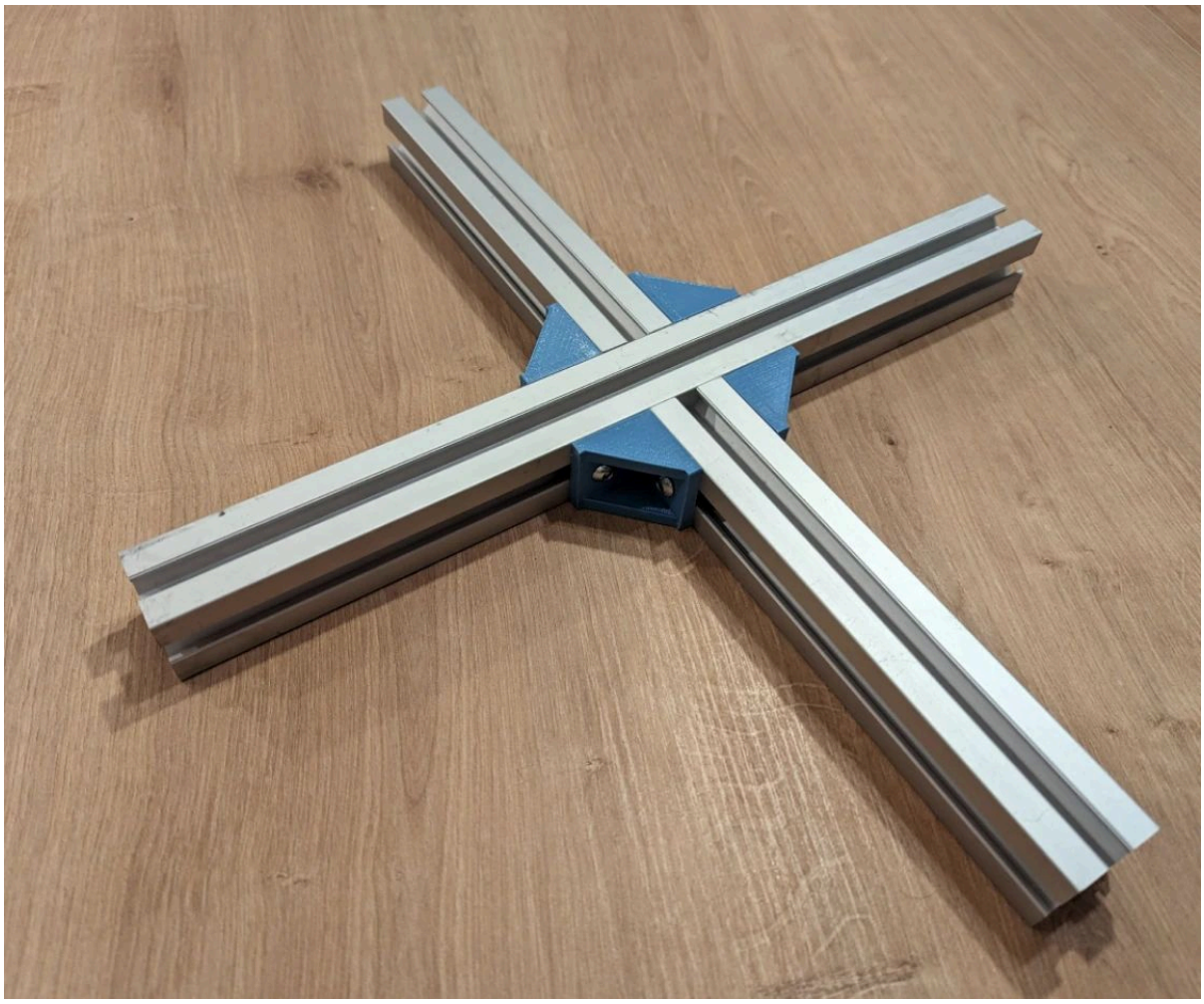


Figure 4.2: PAROL6 on a 30×30 aluminium profile frame.

Mounting plates

The `mounting plates` folder of the repository has STEP files of a small and a big base plate, with the hole pattern, for a plate you have machined or cut. The small plate has more holes than PAROL6 uses, two more M5 and six M4: they were made for other robots and designs, and stay empty under PAROL6.

Orientation

PAROL6 is designed to stand upright on its base. Wall and ceiling mounting are not supported: the payload, the motor currents and the speed limits are set for an upright arm, and with no brakes the arm would fall into the mount when powered off.

Connecting

The robot needs three connections to run. They are on the back of the base.

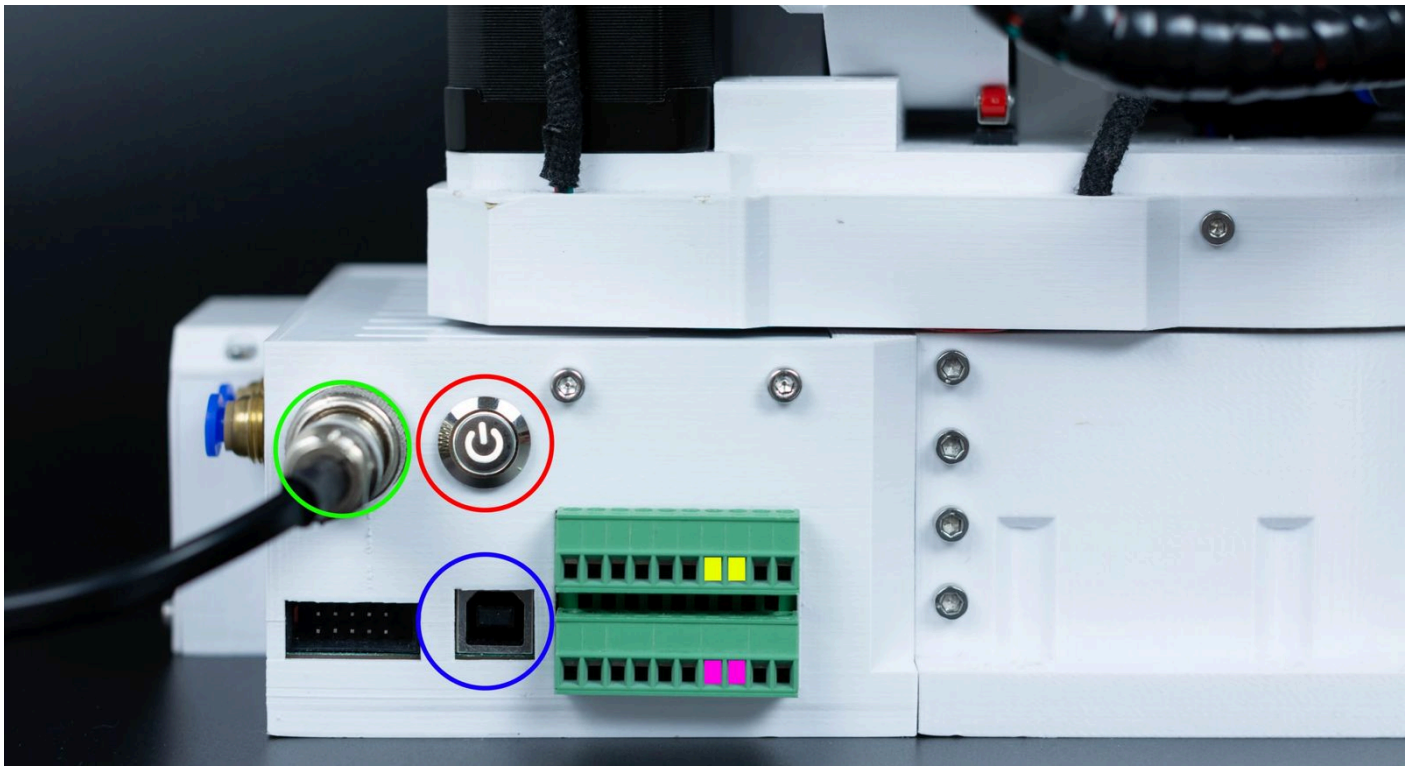


Figure 4.3: The power (green), USB (blue) and E-stop (yellow and pink) connections, and the power button (red).

Connection	Connector	Notes
24 V power	GX16 2-pin	From the 24 V 5 A supply (The power supply). Mind the polarity when you wire your own cable: the board has no reverse-polarity protection
USB	USB type B	To the PC. USB does not power the board
E-stop	Two screw terminals	A normally closed contact between them. Open means stop

Plug the power supply into the mains through a switch within your reach, a switched socket strip for example: it is the stop that still works when the PC software has frozen (Emergency stop).

The power supply

The robot needs 24 V DC at 5 A or more, into its GX16 socket. There are two good ways to get it: a closed supply with a mains cable, which needs no mains wiring, or the DIN-rail supply in the BOM, whose mains side you wire yourself.

> **Mains voltage kills**
>
> The mains side of a power supply carries 230 V or 120 V. Never open, wire or touch it while the plug is in the socket. If you are not sure how to wire it, use a closed supply (A), or have a qualified electrician wire the DIN-rail supply (B).

A. A closed supply with a mains cable

The simplest safe choice: a closed 24 V supply of at least 5 A (120 W), a desktop power brick with a moulded mains cable and plug, approved for your country. There is nothing to wire on its mains side. On its 24 V side:

- Its output plug will not fit the robot. Fit a GX16 2-pin plug to its output cable, or make a short adapter cable with the brick's mating socket on one end and a GX16 plug on the other.
- Find + and − on the output with a multimeter, and wire + to the GX16 pin that the robot's socket takes to the board's +. The board has no reverse-polarity protection.

B. The Mean Well EDR-120-24 from the BOM

A 120 W DIN-rail supply with screw terminals and no mains cable. From Mean Well's EDR-120 datasheet:

Input	90 to 264 V AC, 47 to 63 Hz
Input terminals	FG (earth), AC/L (live), AC/N (neutral)
Output terminals	Two −V and two +V, 24 V, 5 A: the robot's cable to one +V and one −V
Output adjustment	+V ADJ, 24 to 28 V. The board takes 24 V at most: leave it at 24 V
Room around it	40 mm above, 20 mm below, 5 mm at the sides, at full load

You need:

- **A mains cable with three wires, earth included, and a moulded plug.** Never a two-wire cable.
- **Wire ferrules** crimped on the stranded ends, so that no strand escapes the terminal.
- **Strain relief:** the cable clamped where it enters, so that a pull on the cable never reaches the terminals.

- **The earth wire always connected,** to FG.
- **An enclosure:** the supply on a DIN rail (TS35) in a closed box, or at least a cover over its terminals, so that no live screw can be touched. A plastic box is simplest; a metal one must be earthed as well.

Terminal	Europe and the UK	USA and Canada
AC/L, live	brown	black
AC/N, neutral	blue	white
FG, earth	green-yellow	green

Other countries can use other colours: check, and do not guess.

Check before the first power-on, in this order:

1. The plug is out of the socket.
2. Each wire is in the right terminal, by the table above, with no loose strands; a gentle pull on each wire moves nothing.
3. With a multimeter: the plug's earth pin to the FG terminal reads close to 0 Ω.
4. The box or the terminal cover is closed, and the cable is clamped.
5. The robot is not connected yet. Plug in: the supply's DC OK lamp lights. At the GX16 plug, measure 24 V, with + and − on the pins the robot's socket expects.
6. Unplug, connect the robot, plug in through the switch within reach.

If you are not sure about any step, have a qualified electrician wire the supply.

The E-stop

The E-stop must be a normally closed contact: pressing it, unplugging it or cutting its wire opens the circuit, and all of these stop the robot, which is what you want. A normally open button does not work. The control board has two E-stop connectors on the same input, so a second button can be wired in series: pressing either button stops the robot.

The kit's button sits in its own printed box with a PG7 cable gland, on a cable as long as you need. Emergency stop says what happens when it is pressed, and the Control Board Manual has the wiring diagram.

The gripper

An electric gripper (SSG-48, MSG) plugs into the M8 connector at the forearm, which carries 24 V and CAN: Electric grippers. Pneumatic tubes from the valve go to the two push-in fittings at the base: Pneumatics.

Inputs and outputs

The two inputs, two outputs and their supply terminals are on the side of the control board, reachable through the base: Inputs and outputs.

> **Nothing else on the programming port**
>
> The 2×5-pin programming header next to the USB port is for the ST-Link programming adapter only. An ST-Link connected with plain jumper wires without the adapter has destroyed boards. See Firmware.

First power-on

The first time, and every time after that:

1. **Check the E-stop is connected** and released.
2. **Hold the arm by the forearm.** The motors energise with a jolt, and until they do the arm is limp.
3. **Press the power button** on the base. The 24 V stage switches on, each driver initialises with a click, six clicks, and the arm holds its position.
4. Let go. The robot is now enabled, not homed, and waiting for the PC.

> ✓ **Check**
>
> Six clicks, the fan runs, and the arm holds when you let go. If not: If nothing happens.

Nothing moves until the PC software commands it. Start the commander software or the Python API's controller, connect, check the jog directions and home: the Quick start walks through it.

What the button does

The button is not a switch in the power line: it tells the firmware to turn the 24 V stage on, and the firmware keeps it on. A short press while the robot is on turns it off. The board stays powered, and turns itself on, as long as 3.3 V is present on the programming header, which is why a connected ST-Link keeps the robot on and the button useless: Switching on and off.

If nothing happens

- No clicks, no fan: no 24 V. Check the supply, the GX16 cable and its polarity.
- Clicks, but the PC software does not connect: the serial port name, or the board runs the test firmware. See Communication.
- The board turned on by itself when the supply was plugged in: something supplies 3.3 V to the programming header, or the arm was moved while unpowered (Hazards).

5. Operation

5. Operation

How the robot is switched on and off, homed, jogged and run, what happens when something goes wrong, and how it is mastered.

In this chapter

- Switching on and off
- Homing
- Jogging
- Running programs
- Errors and recovery
- Mastering

Each page says what to do in the commander software and in the Python API; the details of either are in the Programming Guide.

Switching on and off

Switching on

1. E-stop connected and released; power supply on; USB to the PC.
2. Hold the arm by the forearm.
3. Press the power button on the base. The fan starts, the six drivers initialise with a click each, and the motors hold the arm.
4. Start the PC software and connect. The robot is enabled and not homed; nothing moves until you command it.

The firmware energises the motors at the moment the 24 V stage comes up, whether or not a PC is connected, and holds them with the holding current from then on.

Switching off

> **Not while it moves**
>
> Never switch the robot off while it is running a program or carrying something. If it behaves unexpectedly, press the E-stop; if the E-stop has no effect, switch off the power supply. The power button is the last resort, and only with the arm held.

1. Stop the program and bring the arm to the **parked pose**, folded in over the base, where it has the least distance to fall. In the commander, **Park** writes a three-line program into the program window that moves there; press **Start** to run it. In the Python API, `rbt.move_j([90, -144.683, 108.171, 2.222, 25.003, 180], speed=0.3)` moves to the same pose.
2. Disconnect in the software, or stop the controller.
3. Hold the forearm.
4. Press the power button. The motors release at once and the arm goes limp in your hand.

The button and the programming adapter

The button does not switch the power line: it tells the firmware to turn the 24 V stage on and off. The microcontroller itself runs from 3.3 V derived from that stage, or from the programming header. **While an ST-Link or the programming adapter is connected, the board has 3.3 V, turns on by itself when the supply is plugged in, and cannot be switched off with the button.** Upload firmware with the 24 V supply disconnected, then unplug the adapter, then plug the supply in and press the button (Firmware).

After a power failure

The arm has fallen. Check that nothing is damaged and nothing is trapped under it, switch on as above, and home before anything else: the position is lost.

Homing

After power-up the robot does not know where its joints are: the motors count steps, and the count starts at zero wherever the arm happens to stand. Homing gives each joint a reference by driving it to its limit switch or sensor, then moving a known number of steps to the after-homing pose. It must be done after every power-up, and again after a collision, a stall or any move by hand.

Before you press Home

- The arm can move freely to every limit: nothing on the table in the way, no cable caught.
- The tool is mounted: joint 5 homes against the tool's feature on the SSG-48 and the pneumatic gripper, and the homing offset in the firmware matches the tool (Joint 5 and the tool, SSG-48).
- Joint 6 stands so that it finds its sensor within one turn in the negative direction, with the cables and tubes not already wound up. The figure shows the pose: the gripper as drawn, the sensor (blue) and the homing pin (red); joint 6 turns along the green arrow.
- You are at the E-stop.

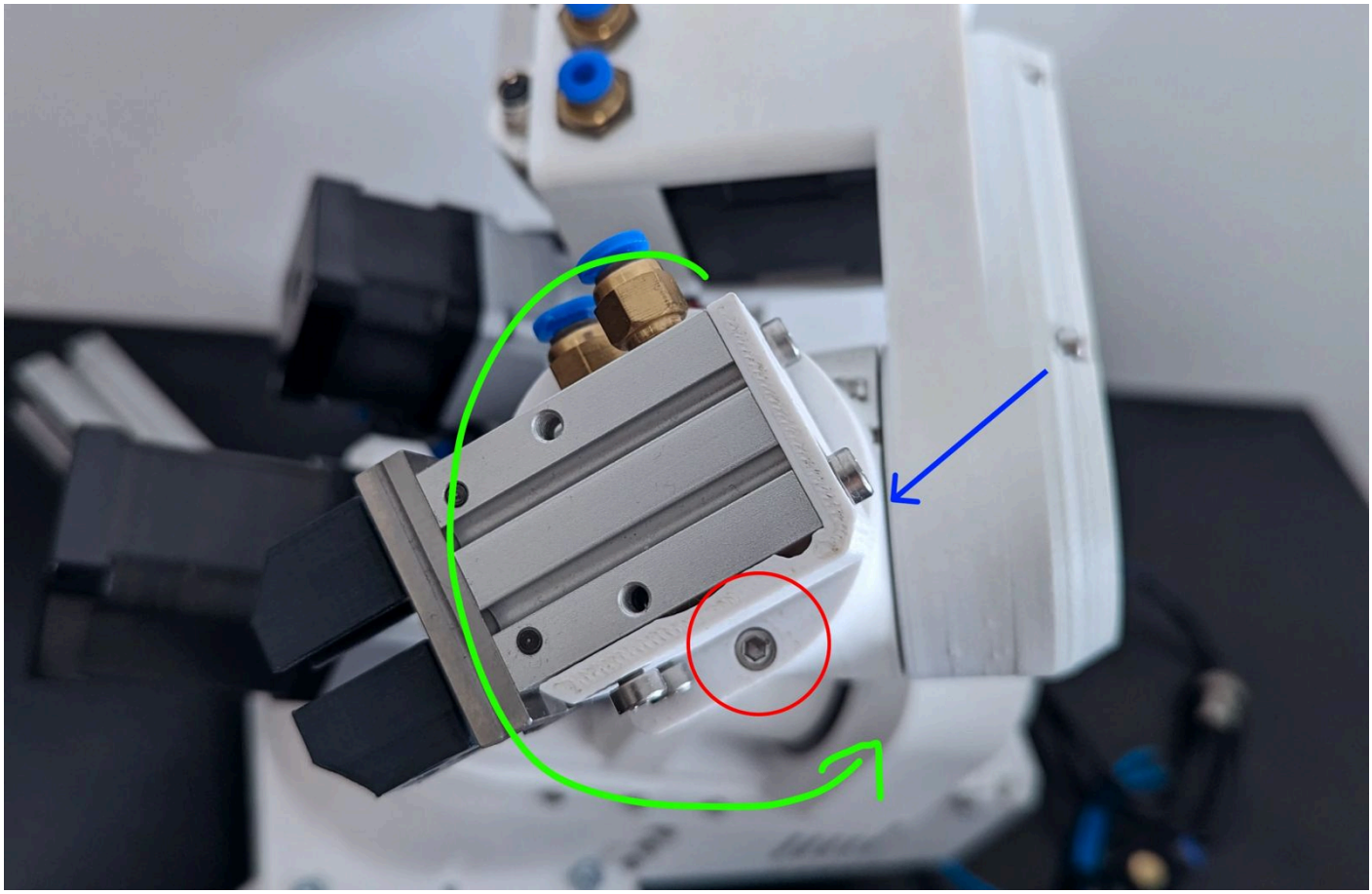
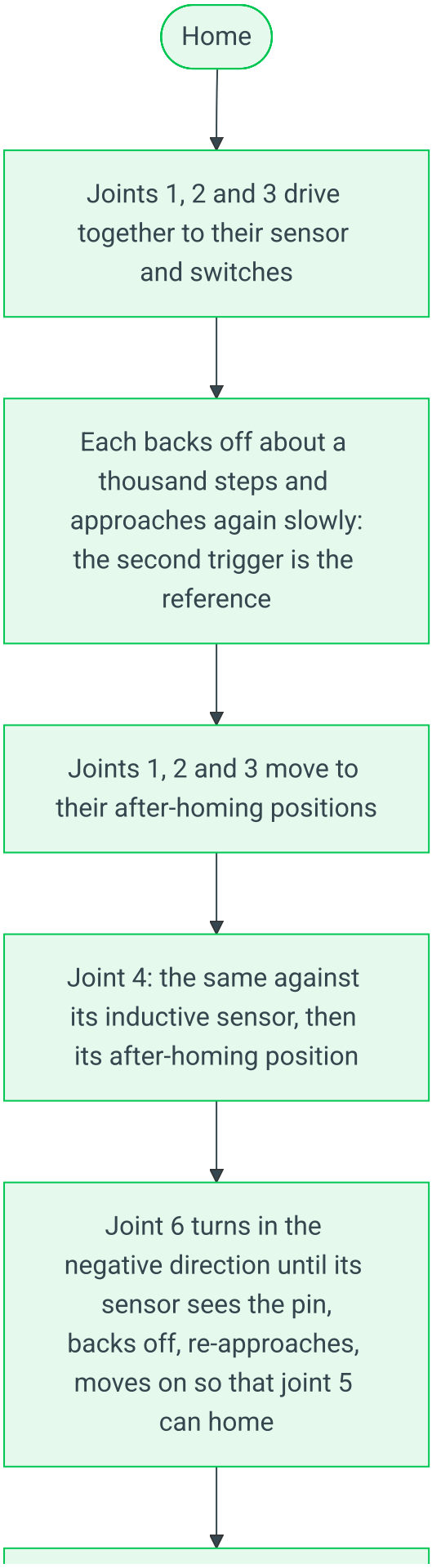
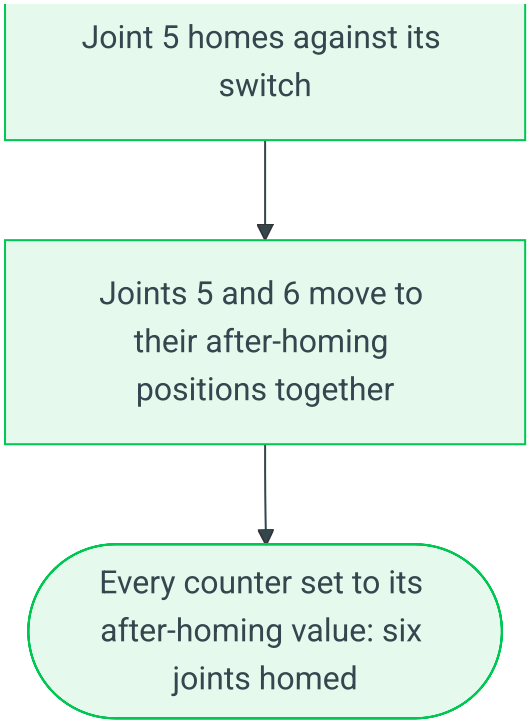


Figure 5.1: Joint 6 before homing: the sensor, the homing pin and the direction of rotation.

The sequence

The firmware runs the sequence; the PC only starts it and watches.


Home
Joints 1, 2 and 3 drive together to their sensor and switches
Each backs off about a thousand steps and approaches again slowly: the second trigger is the reference
Joints 1, 2 and 3 move to their after-homing positions
Joint 4: the same against its inductive sensor, then its after-homing position
Joint 6 turns in the negative direction until its sensor sees the pin, backs off, re-approaches, moves on so that joint 5 can home



1. **Joints 1, 2 and 3 move together** toward their switches: joint 1 to its inductive sensor, joints 2 and 3 to their limit switches. Each stops when it triggers.
2. Each backs off, 850 steps on joint 1 and 1050 on joints 2 and 3, and approaches again slowly, and the position where it triggers the second time is the reference: the first hit may have pressed the switch too far.
3. Joints 1, 2 and 3 move from the reference to their after-homing positions (13500, 19588 and 23020 steps away).
4. **Joint 4** does the same against its inductive sensor (backing off 1550 steps) and moves to its after-homing position.
5. **Joint 6** turns in the negative direction until its sensor sees the homing pin, backs off, re-approaches, and moves on to the position from which joint 5 can home.
6. **Joint 5** homes against its switch, and joints 5 and 6 move to their after-homing positions together: joint 5 by `j5_homing_offset`, joint 6 back to the point where its sensor triggered.
7. The firmware sets every joint's counter to its after-homing value (`standby_position` in the firmware), and reports all six joints as homed.

A video of a correct homing: Homing reference. It takes about half a minute. The result is the after-homing pose: the standby pose turned 90° about joint 1, with joint 1 at +90° (Joint limits and directions).

Commander software

Press **Home**. The response log reports the joints as they home; the position display and the simulator are right once all six are homed.

Python API

```
rbt.home(wait=True)
```

The first `home()` after power-up runs the referencing sequence. A later `home()` on a referenced robot is a planned, collision-checked joint move back to the after-homing pose; `home(calibrate=True)` runs the sequence again. Planned motion is refused with `MOTN_NOT_HOMED` until the robot is referenced.

If it goes wrong

Press the E-stop when a joint grinds at its end without stopping: its switch does not trigger. Joints 1, 4 and 6 use inductive sensors whose LED lights when they see metal; a sensor that does not light needs its trigger screw adjusted. If joint 6 keeps turning, its homing pin is not where the sensor looks. Homing failures has the cases.

Mastering

A homed robot stands within a few steps of the after-homing pose. The exact number of steps from each switch to the witness marks differs between builds; Mastering says how to measure and set it.

Jogging

Jogging moves one joint, or the tool along one axis, for as long as you hold a button or repeat a command. It is how you check a new robot, teach a pose and get out of trouble.

Joint jog

Each joint has a pair of arrows in the commander's **Joint jog** menu: the **left arrow is the positive** direction, the right arrow negative, as in Joint limits and directions. The speed slider sets the speed as a percentage of the commander's jog maximum, which is lower than the speed a program may use (Speeds). A jog stops at the joint limit.

Python API

```
rbt.jog_j(0, speed=0.3, duration=0.5)              # joint index 0 = J1, 30 % speed, 0.5 s
rbt.jog_j(joints=[1, 2], speeds=[0.2, -0.2], duration=1.0)
```

A jog command runs for `duration` and the arm brakes to rest when no newer jog arrives; a GUI repeats it while the button is held. Joints are indexed from 0 in the API.

Cartesian jog

The tool moves along X, Y or Z, or rotates about them, in the **world frame** (WRF, fixed to the base) or the **tool frame** (TRF, fixed to the tool). The commander's **Cartesian jog** menu has the six pairs of arrows and the WRF/TRF switch; a jog in the tool frame moves the tool along its own axes, which is the natural way to approach a part.

Python API

```
rbt.jog_l("WRF", "X", speed=0.2, duration=0.5)
rbt.jog_l("TRF", axes=["Z", "RZ"], speeds_list=[0.2, 0.1], duration=0.5)
```

A Cartesian jog needs a homed robot, and it can run into a singularity, where the planner cannot find joint speeds for the tool's motion: the commander stops the jog, and the Python API refuses it. Jog a joint away from the singular pose and continue (Configurations and singularities).

Jogging a robot that is not homed

Joint jog works before homing: it is how you check the directions on a new build. The positions shown are meaningless until the robot is homed, and a Cartesian jog is refused.

Both programs still check each jog against the joint limits, with a position count that starts at 0 when the board is switched on. That count puts joint 2 past its positive limit and joints 3 and 6 at or below their negative limits, so until the robot is homed:

- **joint 2** jogs only in its negative direction (the right arrow);
- **joints 3 and 6** jog only in their positive direction (the left arrow);
- **joints 1, 4 and 5** jog both ways.

A refused jog does not move: the commander logs `Error: Robot jog -> Position out of range`, the Python API stops the joint before it starts.

> **Warning**
>
> Before homing, the limits are checked against a count that does not match the arm, so they do not protect the joint: nothing stops it at its real limit except the printed blocker. Jog briefly and slowly.

Running programs

A program is a list of moves, delays, I/O and gripper commands. PAROL6 runs programs in two ways.

RBTscript in the commander software

The commander's **Program** window takes an RBTscript program: one command per line, `MoveJoint`, `MoveCart`, `MoveCartRelTRF`, `Delay`, `Output`, `Gripper`, with `Begin()` and `End()` or `Loop()` around it. The command list next to the window inserts a command with the robot's current pose as its arguments, which is how you teach points: jog to the pose, click the command. **Start** runs the program, **Stop** halts it.

```
Begin()
MoveJoint(90,-90,180,0,0,180,t=3)
MoveCart(0,263,242,90,0,90,t=2)
Output(1,HIGH)
Delay(1)
Output(1,LOW)
End()
```

RBTscript has the language and Examples has programs to start from. Programs are plain text files you can keep and share.

Python with the Python API

A Python script that calls the client is a program: every motion call queues a move on the controller, and the script waits for it or not, as you choose.

```python
from parol6 import RobotClient

with RobotClient() as rbt:
    rbt.home(wait=True)
    rbt.select_tool("SSG-48")
    rbt.tool.calibrate()
    rbt.delay(2)                  # the SSG-48 needs about 3 s; calibrate() waits only 2 s
    rbt.move_j([90, -90, 180, 0, 0, 180], speed=0.5)
    rbt.move_l([0, 263, 200, 90, 0, 90], speed=0.3)
    rbt.tool.close(speed=0.5, current=0.4)
    rbt.move_l([0, 263, 242, 90, 0, 90], speed=0.3)
    rbt.tool.open()
```

The Python API also runs programs written for other tools, and from other machines: the controller accepts commands over the network from any client, so a program can run on a laptop while the controller runs on the PC at the robot. The Python API is the guide.

Simulate first

Both programs have a simulator. The commander opens its 3D window with the GUI and mirrors the robot; the Python API's controller runs with `simulator(True)` or the `DryRunRobotClient` without hardware, through the same kinematics and collision checks. Run a new program in the simulator at full speed, then on the robot at low speed.

Speed

The commander's speed slider scales jogging only; a program's speed is in its moves, `v=` and `a=` or `t=`. The Python API takes a `speed` fraction per move and `set_execution_speed()` scales the queue while it runs; `pause()` and `resume()` hold and continue it.

Stopping a program

- **Stop** in the commander, or `stop()` in the API: the current move is cancelled, the queue cleared, the robot stays enabled and holding.
- The **E-stop**: the robot is disabled and the program or the queue is stopped; in the Python API, call `reset()` after releasing it (Emergency stop).
- Closing the software: stop the program first. The firmware has no timeout: it keeps executing the last packet it received until the next one arrives. After a move to a position the joints stop near that target and hold.

When the PC stops sending during a jog

If the commander closes, crashes or loses the USB connection while a joint jogs (in joint or Cartesian jog), the firmware keeps that joint turning at its jog speed until it reaches its blocker. Programs that use speed tracking (`speed` as the last argument of a move) behave the same way. Use the mains switch of the power supply (Emergency stop).

Errors and recovery

What the robot reports

Every 10 ms the control board sends the PC the joints' positions and speeds, which joints are homed, the state of the inputs, outputs and E-stop, the gripper's status, and two error bytes reserved for the drivers' temperature and position flags. The PC software adds its own: inverse kinematics failures, limit violations, timing problems.

The commander software

The **response log** shows what the robot and the program are doing and why a command was refused. The common entries:

Log	Meaning	What to do
Robot disable; button or estop	The E-stop is open, or Disable was pressed	Release the E-stop; press **Enable**
An inverse kinematics error	The target pose has no solution, or lies past a singularity	Change the pose or the approach; move in joint space to get near it
A joint limit error	The move would leave a joint's range	Choose a reachable target
Jogging stops in Cartesian mode	A singularity	Jog a joint away from it
The time between commands rises above 10 ms	The PC cannot keep the 100 Hz loop	Close other programs; a faster PC (Communication)

Enable clears the errors and enables the robot; **Clear error** resets the homing state but does not re-enable a disabled robot; **Disable** stops the firmware from accepting motion commands while the motors keep holding.

The Python API

A refused command raises `MotionError` with a code, a cause and a remedy, from the controller's error catalog: `IK_TARGET_UNREACHABLE`, `MOTN_NOT_HOMED`, `SYS_ESTOP_ACTIVE`, `SYS_SELF_COLLISION`, `MOTN_CANCELLED` and the others on Errors. `error()` returns the current error, `activity()` what the robot is doing. After an E-stop is released, call `reset()`: it enables the firmware again, and until then the arm does not move. After `stop()` nothing needs clearing.

After a collision or a stall

The motors have lost steps and the arm is not where the software says. Stop the program, move the arm clear if it is trapped (disable it, or switch off with the forearm held), check the belts and the printed parts, and **home again** before anything else.

After the arm fell

Power was lost with the arm extended. Check every joint by hand for free movement, look at the belts and the couplers, switch on with the forearm held, and home.

A joint moves the wrong way

On a new build or after work on the wiring: stop at once. The direction is set per joint in the firmware (Firmware and first run); do not try to home until every joint jogs in the right direction.

The gripper reports an error

The electric grippers report their own errors through the robot: not calibrated, E-stop, timeout. The commander's **Gripper** tab has **Clear gripper error** and **Calibrate gripper**: clear the error first, then calibrate, and wait until the calibration has finished before the first move. The Python API's `tool.calibrate()` calibrates but has no clear (SSG-48, MSG). The gripper manuals explain each error.

Mastering

Homing brings each joint to its switch and then a fixed number of steps to the after-homing pose. That number is in the firmware, measured on the reference robot. Printers and parts differ, so on your robot the homed pose may be off by a fraction of a degree per joint. Mastering measures the difference and corrects the number, so that a homed robot stands exactly at its witness marks.

Witness marks

Holes in neighbouring links that line up when the joint stands where homing should leave it, in the after-homing pose (90°, −90°, 180°, 0°, 0°, 180°): a pin pushed through them proves the pose. PAROL6 has them on every joint.

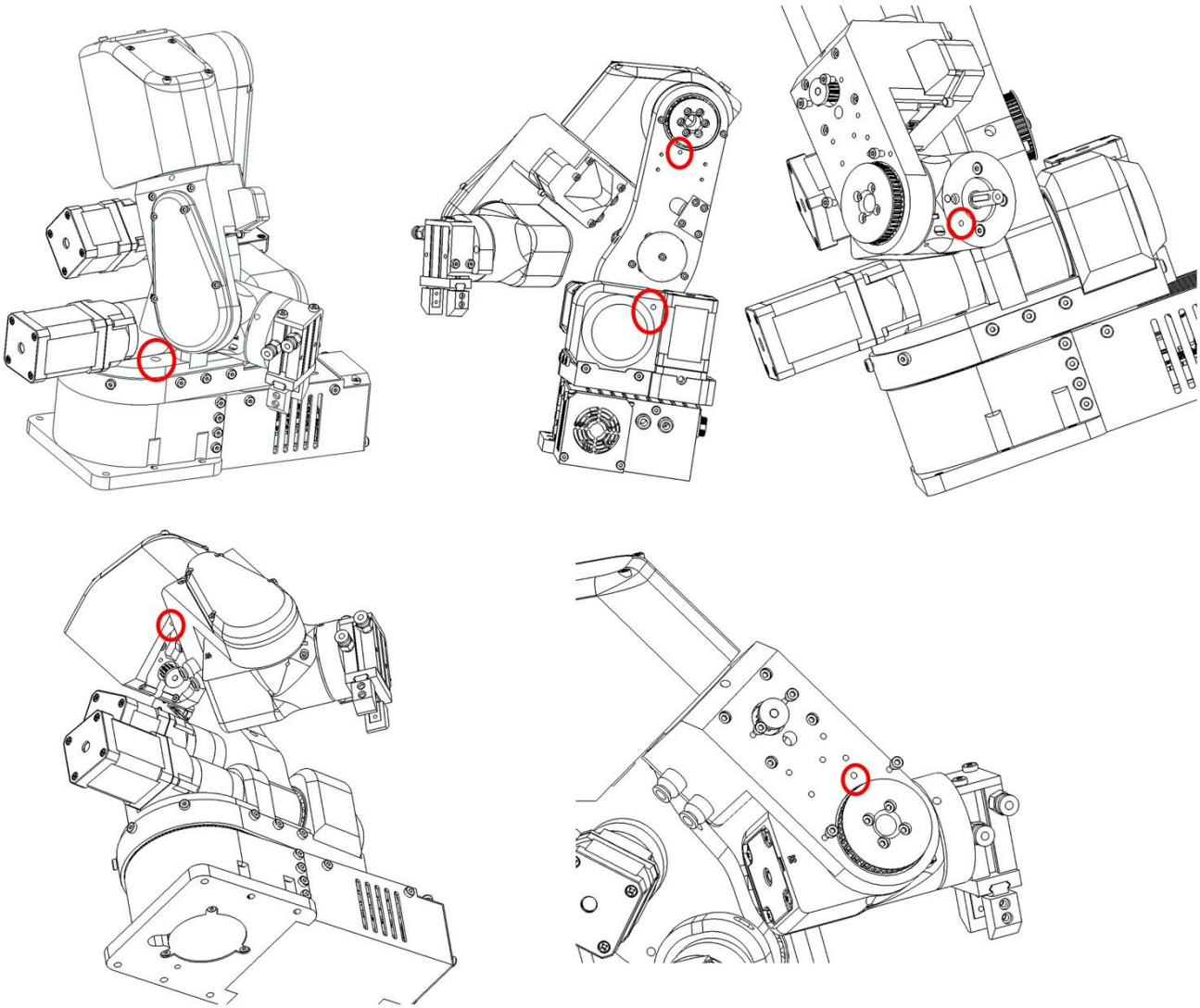


Figure 5.2: Witness marks of PAROL6.

The procedure

1. Home the robot.
2. For each joint, jog it in small steps until the pin fits through its witness marks. Jog at the lowest speed and read the joint's position from the commander's position display or `rbt.angles()`.
3. Convert the angle you jogged by to motor steps with the joint's ratio (Drives, ratios and resolution), or read the step count from the firmware's telemetry.
4. For joints 1 to 4: in `motor_init.cpp`, add that number of steps to the joint's `homed_position` (subtract if you jogged in the negative direction), upload the firmware, and home again. The pin should now fit without jogging. Joints 5 and 6 work differently, below.

```
/// @brief Number of steps from when the limit switch is hit to the standby position
Joint__->homed_position = 13500;   // joint 1: raise or lower by the steps you measured
```

The values of the reference robot are on Firmware parameters.

Joint 5 and the tool

Joint 5 homes against a feature of the tool, so its number depends on the tool: 8900 steps with the pneumatic gripper holder and 8035 with the SSG-48, set as `j5_homing_offset` in `main.cpp`. Master joint 5 by changing that value, not its `homed_position`. A custom tool needs its own value, found the same way.

Joint 6

Joint 6 has no offset to master in firmware 1. Its `homed_position` (15900) is only the position from which joint 5 homes: when joint 5 is done, joint 6 turns back to the point where its sensor saw the homing pin, and that point is its after-homing position. If joint 6 does not line up with its witness marks after homing, move its sensor or its homing pin.

When to master again

After replacing a belt, a switch, a sensor or a printed part of a joint, or when the robot's absolute accuracy matters more than its repeatability. The repeatability does not depend on mastering: a homed robot returns to the same place every time, whether or not that place is exactly the witness mark.

6. Peripherals

6. Peripherals

Everything the robot talks to besides the PC and the tool: switches and sensors on its inputs, valves and lamps on its outputs, the air for the pneumatic gripper, and the CAN bus for the electric grippers.

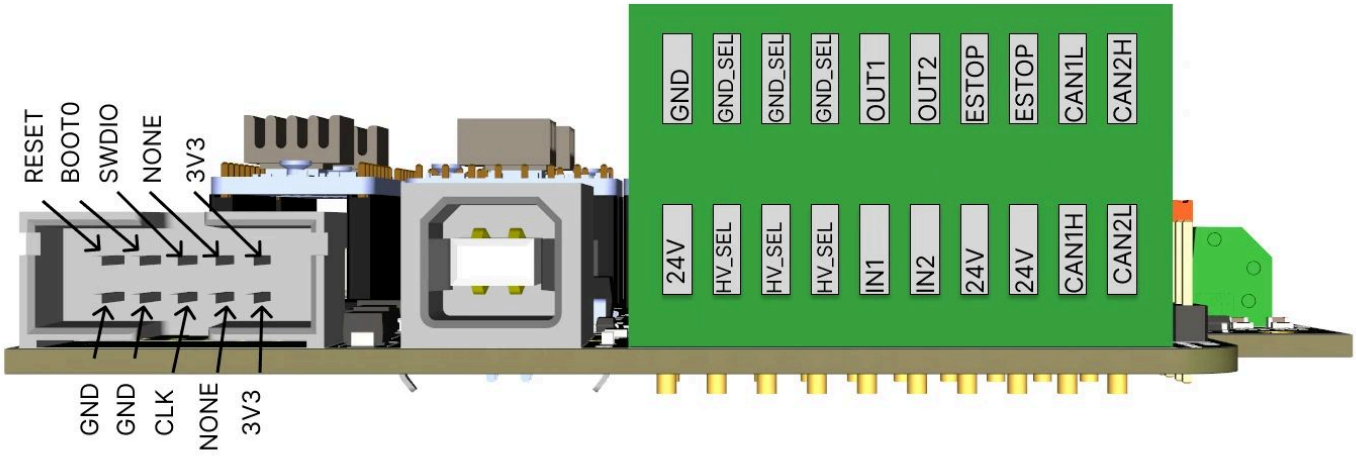


Figure 6.1: The control board's connections to the outside world.

In this chapter

- Inputs and outputs
- Pneumatics
- CAN bus

The electrical details of the connectors are in the Control Board Manual.

Inputs and outputs

The control board has two isolated inputs and two isolated outputs on screw terminals at the side of the board, reachable through the base. They are isolated from the robot's electronics through optocouplers, and they share a supply that you choose.

The supply

- **Isolated**: connect an external supply's negative to **GND_SEL** and positive to **HV_SEL**. The inputs and outputs then work at that voltage, 12 or 24 V (the inputs need at least 12 V), and a fault on them cannot reach the robot.
- **Not isolated**: link the robot's own 24 V and GND to HV_SEL and GND_SEL. Simpler, and the valve of the pneumatic gripper is usually wired this way. A 2 A fuse protects the outputs in this mode.

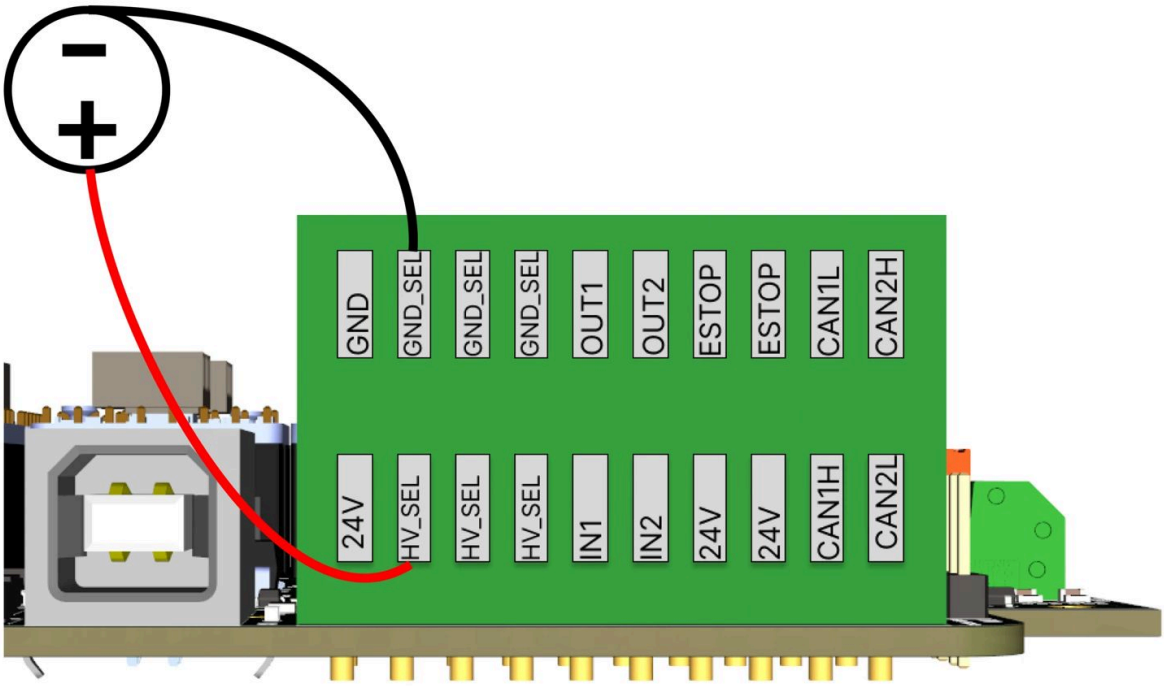


Figure 6.2: Isolated supply wiring.

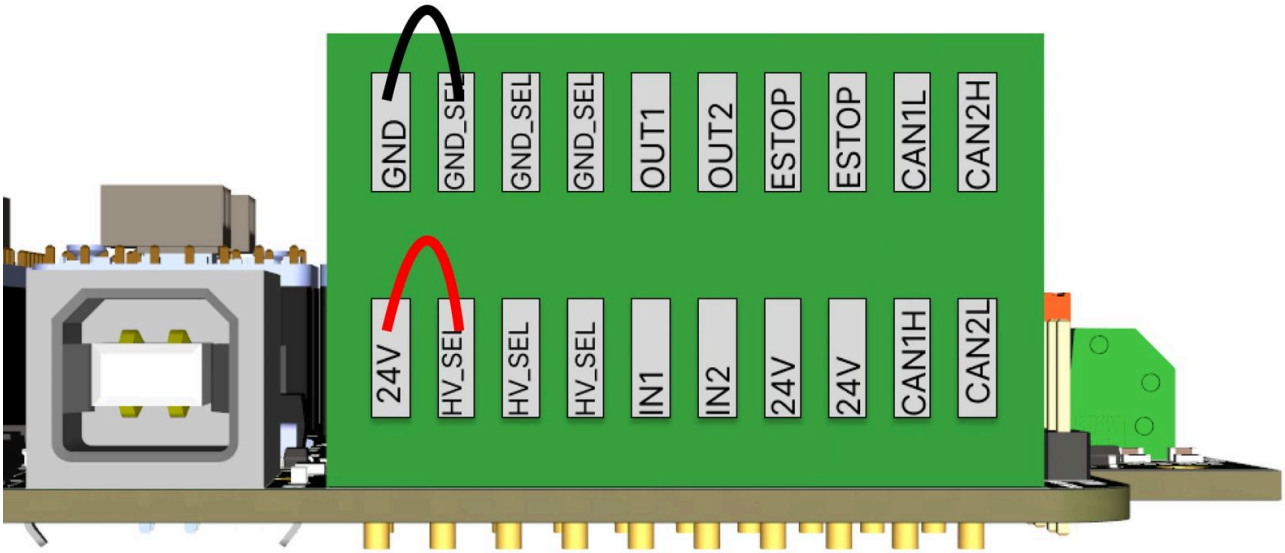


Figure 6.3: Non-isolated wiring from the robot's 24 V.

The E-stop is not part of this: it is wired to the robot's own supply whichever option you choose.

Inputs

Switches, push buttons, light barriers and sensors. A switch goes between **HV_SEL** and **IN1** or **IN2**; a three-wire sensor takes HV_SEL and GND_SEL for its supply and gives its signal to the input. Input voltage 12 to 50 V, 24 V nominal.

The robot reports the inputs in every packet: the commander's **IO** tab shows them, and the Python API's `io()` returns `[in1, in2, out1, out2, estop]`. RBTscript's `Input()` has no action yet: a program stops on it (Known issues).

Outputs

Relays, valves, lamps, small actuators, up to 0.5 A each (1 A peak), 48 V maximum. A load goes between **GND_SEL** and **OUT1** or **OUT2**; the output switches it to the supply. An inductive load, a valve or a relay coil, gets a flyback diode across it.

RBTscript

```
Output(1,HIGH)
Output(1,LOW)
```

Only output 1: in the current commander version `Output(2,...)` is refused by a bug in its check and stops the program (RBTscript).

Python API

```
rbt.write_io(0, 1)       # output 1 high (logical index 0)
rbt.write_io(1, 0)
```

The commander's **IO** tab switches the outputs by hand.

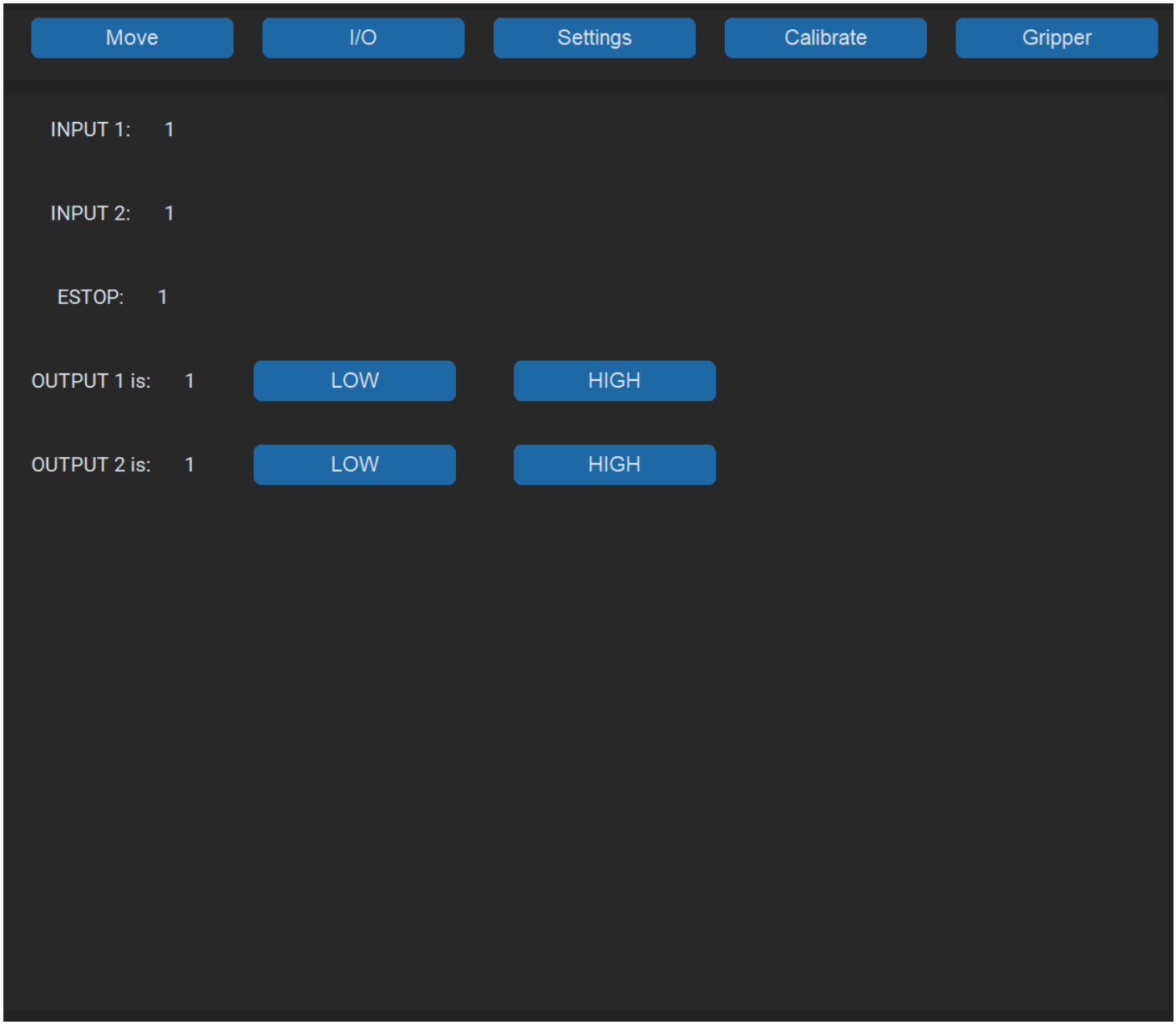


Figure 6.4: The IO tab of the commander software.

Pneumatics

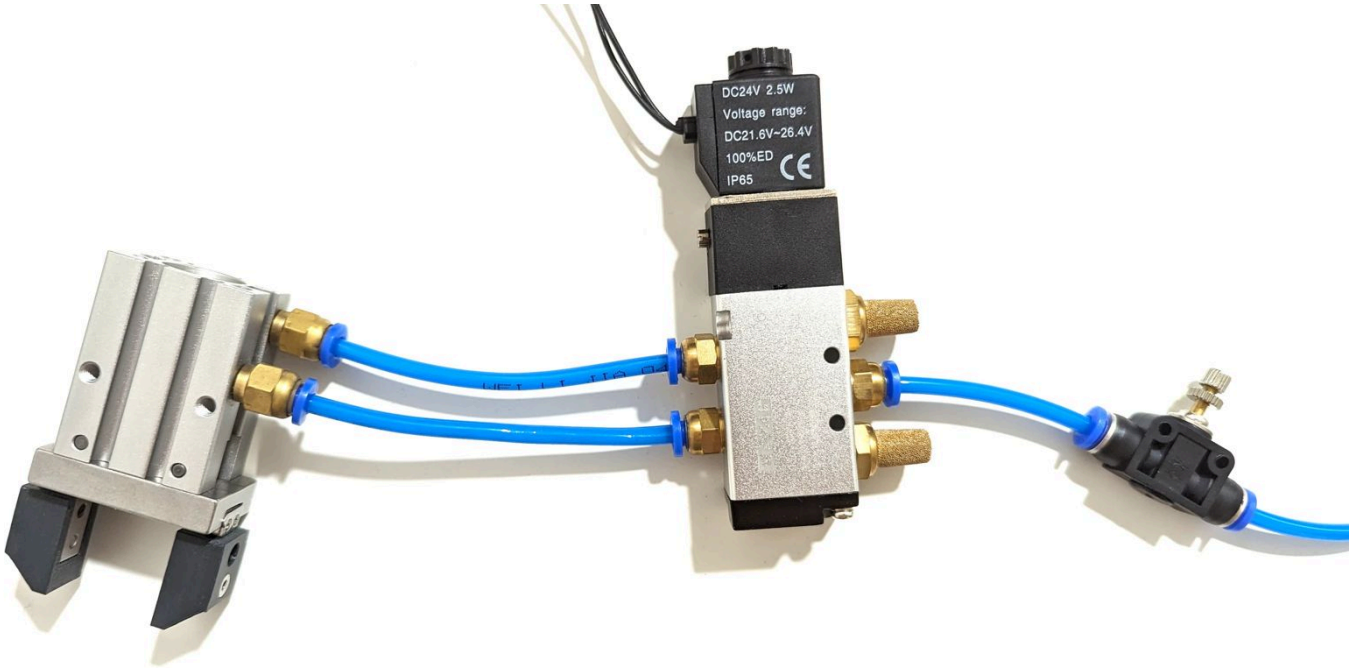
Two 4 mm tubes run inside the arm from the push-in fittings at the base to the forearm, where the pneumatic or vacuum gripper connects. A 5/2 solenoid valve at the base, driven by an output, switches the air.

Pressure

5 to 7 bar; check your gripper's rating, and stay below the lowest rating of the gripper, the valve and the fittings. The tubes and fittings are standard 4 mm pneumatic parts.

The valve

A 5/2 valve, 24 V coil, 4 mm ports: one port from the compressor, two to the base fittings, the exhausts open. The coil goes between **GND_SEL** and **OUTPUT1** (Inputs and outputs). Which of the two outputs is pressurised when the coil is off decides whether the gripper is normally open or normally closed: swap the two tubes to change it.



Routing

The arm's tubes run inside it from the base to the push-in fittings on the forearm; they are one reason for the one-turn rule of the joints. From the forearm fittings, the gripper's own tubes cross joints 5 and 6 outside the arm, like an electric gripper's cable: give them enough slack for the range you use, since they, not the flange, limit joint 6. Outside the base, give the tubes a loop so the arm cannot pull them.

A vacuum gripper

Needs one line only: block the valve's second port (Vacuum gripper).

In the Python API, a pneumatic or vacuum tool's `open()` switches output 1 high and `close()` switches it low, whatever the tubing does: lay the tubes so the two match (Pneumatic gripper).

CAN bus

The control board has a CAN bus for the electric grippers, wired to the M8 connector at the forearm. The board is the master: it sends the gripper the command from the PC's packet every cycle (a move, a calibrate, a clear error, or an empty status poll when nothing changed) and relays the gripper's answer back to the PC.

- **Bit rate**: 1 Mbit/s, the grippers' rate.
- **Node ID**: the gripper's CAN ID, 0 as shipped; the PC packet carries the ID the board addresses, so a gripper with another ID works if the software is told.
- **Termination**: 120 Ω at each end of the bus. At the board end, the **CAN1 JUMPER** connects the board's 120 Ω resistor when it is on; at the other end, the gripper has its own termination switch, on as shipped (CAN bus).

A second CAN controller is on the board, but its transceiver is not populated: CAN 2 is for your own experiments. More axes on the bus, a seventh axis or a second gripper, are not supported by the firmware or the software yet; the protocol the grippers speak is documented in the STEPFOC and Spectral Micro references for anyone who wants to extend it.

The electrical side is in CAN bus of the Control Board Manual.

7. Maintenance

7. Maintenance

PAROL6 needs little: belts that stay tight, set screws that stay put, switches that trigger. Check them at the intervals below and after any collision.

Interval	What
Before each session	The E-stop works (press it with the robot enabled); nothing in the workspace; the cables and tubes free
Monthly, or every 50 hours	Inspection: belts, couplers, switches, fan, printed parts
After a collision, a fall or a lost step	Inspection, then Checking the homing
When a belt slips or a joint is loose	Belts, Couplers and threadlocker

Before any work inside the base, unplug the supply and touch earthed metal: static electricity damages the control board and its drivers.

In this chapter

- Inspection
- Belts
- Couplers and threadlocker
- Checking the homing

Inspection

With the robot switched off and the supply unplugged:

- **Belts** (joints 1, 3, 4, 5): press each with a finger through its cover opening; it should give a little and spring back. A belt that sags or that you can push against the pulley needs tensioning (Belts). Look for frayed edges and missing teeth.
- **Couplers and set screws**: hold the output of each joint and try to turn the motor side; any play that is not the gearbox's backlash is a loose set screw (Couplers and threadlocker).
- **Limit switches and sensors**: press each switch lever; it should click. The inductive sensors' LEDs light when a screw head comes near.
- **Printed parts**: cracks at screw bosses and bearing seats, especially on the elbow and the upper arm, and any warping of the forearm from heat.
- **Cables and tubes**: chafing where the arm's cables pass through the joints, and where the gripper's cable and tubes cross joints 5 and 6 from the forearm; the gripper cable's connector tight.
- **The fan**: turning when the robot is on, and the intake free of dust. A stopped fan overheats the drivers.
- **The base**: the mounting bolts tight, the E-stop button firm in its housing.

Then switch on, home, and watch the homing: each joint should back off and re-approach its switch cleanly, and the pose at the end should match the witness marks (Checking the homing).

Cleaning

Clean the robot switched off and unplugged. Wipe the printed parts with a dry or slightly damp cloth, or with a little isopropyl alcohol on the cloth for grease; use no other solvents on them, and never spray anything into the base, where the control board and the fan are open to it. Clear dust from the fan intake with a soft brush or a vacuum cleaner, not with compressed air, which drives the dust into the board.

Belts

Belts drive joints 1, 3, 4 and 5. Over time they stretch a little and the joint loses stiffness; a belt that slips a tooth loses the robot's position.

Tension

The reference is the belt tension video: in it, maximum hand force is applied to the belt, and that is the tension to aim for on joints 3 and 5. Joints 1 and 4 take slightly less. A belt is tensioned by adding or moving the small tension bearings (3 × 8 × 4 mm and 3 × 10 × 4 mm) that press on its back; if a belt is not tight enough, add another bearing as in the photos. You may or may not need a tensioner on a given joint: it depends on your belts and printed parts, so fit one only where the belt is not tight enough without it.

Joint 5 belt

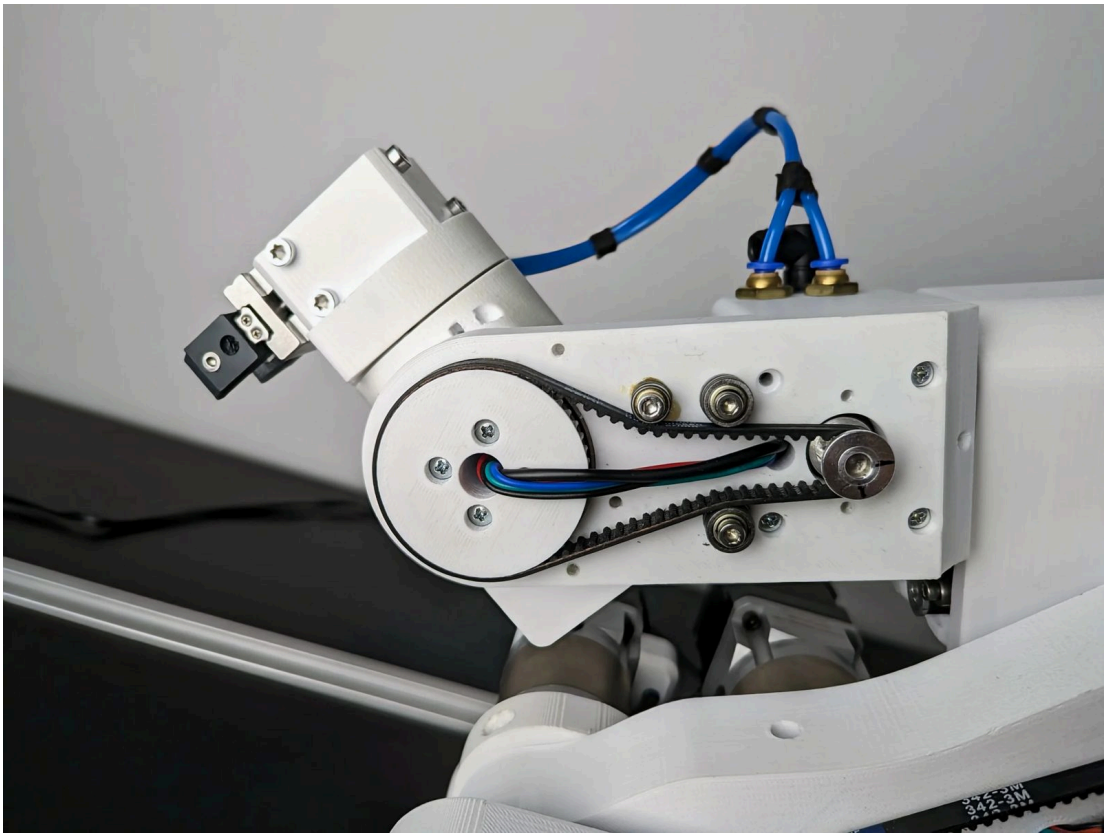


Figure 7.1: Joint 5 belt and tensioner.

Joint 4 belt



Figure 7.2: Joint 4 belt and tensioner.

Joint 3 belt

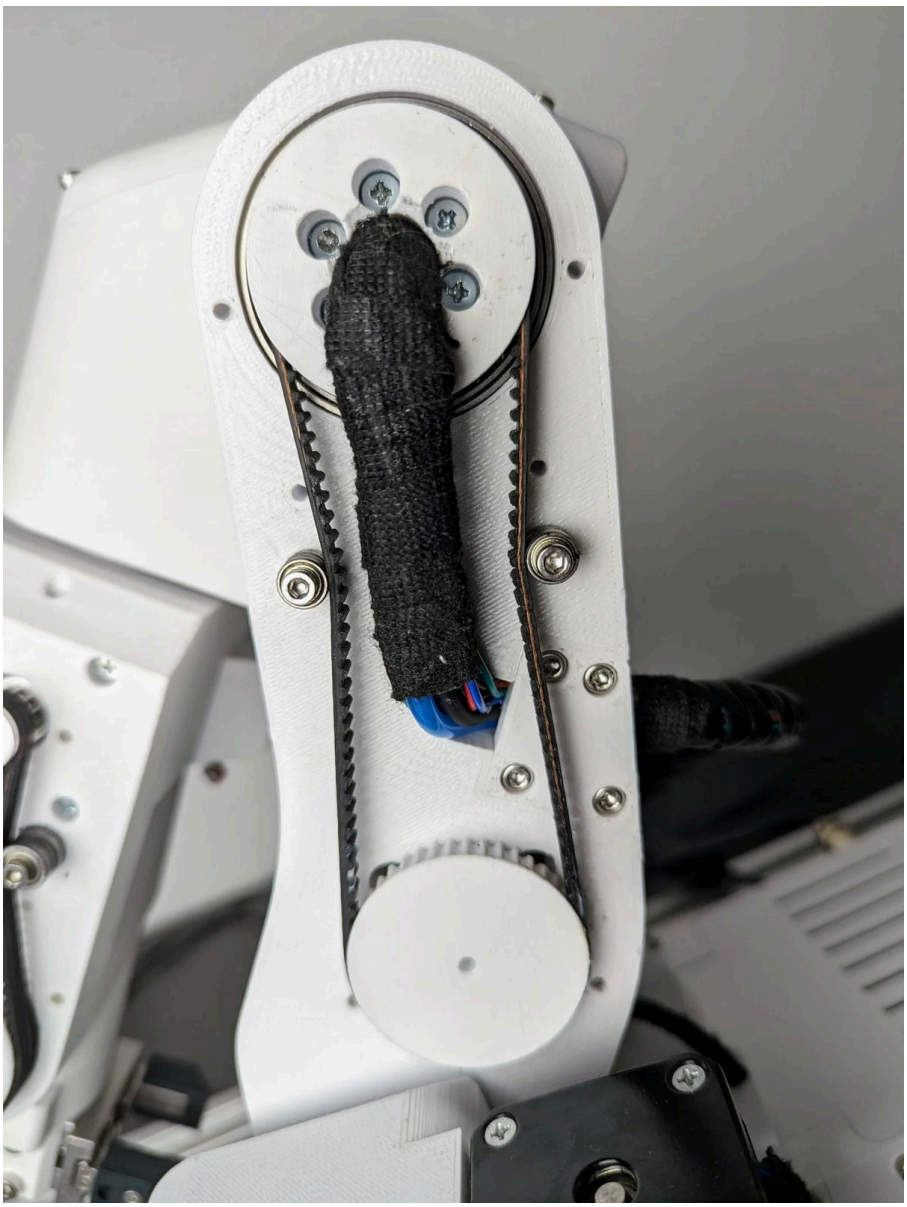


Figure 7.3: Joint 3 belt and tensioner.

Joint 1 belt

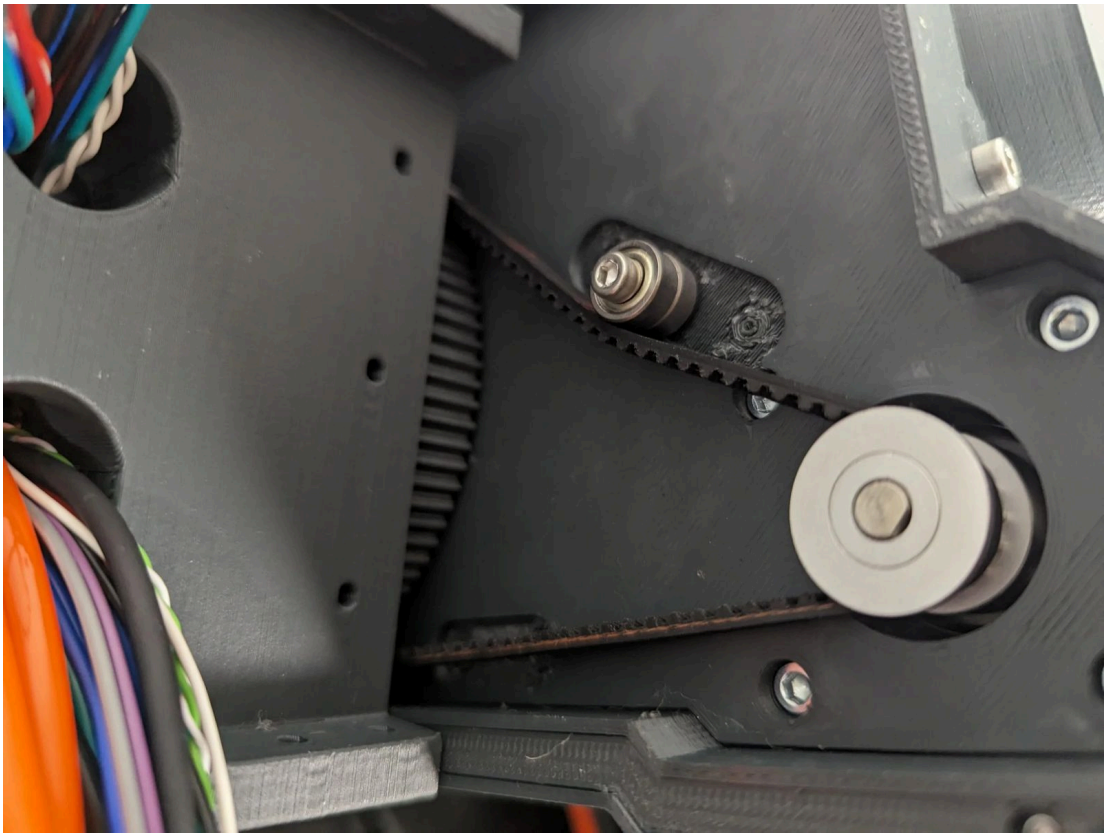


Figure 7.4: Joint 1 belt and tensioner.

Grease

A thin film of grease on the belts' teeth reduces noise and wear. Reapply when it dries out.

Replacement

Joint	Belt
Joint 1	HTD3 396 mm, 6 mm wide
Joint 3	HTD3 342 mm
Joint 4	HTD3 201 mm
Joint 5	HTD3 246 mm

Replacing a belt means opening the joint as in the assembly manual; afterwards home the robot and check the witness marks, and master if they do not line up.

Couplers and threadlocker

The motor shafts, the gearbox outputs and the joint 6 shaft are joined by couplers and pulleys with set screws on flats or keys. Vibration loosens set screws; a loose one lets the motor turn without the joint, and the robot loses its position without noticing.

- If a joint starts to slip, or its position drifts between homings, find the set screw, tighten it, and apply **blue (medium) threadlocker**. Do not move the robot for 24 hours while it cures.
- Put a screw on the key as well as on the flat where the part has both.
- Apply threadlocker to every metal-to-metal screw joint on the robot: the motor mounts, the gearbox flanges, the bearing clamps. Printed parts and brass inserts do not need it, and the solvent in some threadlockers attacks plastic: keep it on the metal.
- Never use red (permanent) threadlocker; the parts cannot be taken apart again without heat.

After tightening a coupler, home the robot and check the witness marks.

Checking the homing

Homing is the robot's only position reference, so check it after any mechanical work, after a collision, and whenever the robot's accuracy seems off.

1. Home the robot and watch each joint: it must reach its switch, back off, re-approach slowly, and move on. A joint that grinds at a stop, or re-approaches at full speed, has a switch problem (Homing failures).
2. At the end, try the pin at the witness marks of each joint (Mastering). It should slide in, or nearly. A joint that is off by more than a degree was knocked, slipped, or its switch moved.
3. Home three times in a row and compare the witness marks each time. The pose should repeat exactly: if it does not, the switch or the sensor triggers inconsistently, or something in the joint moves.
4. Jog to a taught pose from a program and check that the tool is where it was taught.

A joint that homes repeatably but to the wrong place is mastered; a joint that does not repeat is repaired.

8. Troubleshooting

8. Troubleshooting

- By symptom: the robot does not turn on, does not connect, moves wrong, stops, makes noise
- Homing failures
- Communication: serial ports, the 100 Hz loop, slow PCs
- Known issues: the faults in the firmware and software that are known and not yet fixed, and what to do until they are

For the control board itself, see the Control Board Manual's troubleshooting; for the software, Programming troubleshooting. When you ask for help, Report a problem says what to include.

By symptom

Symptom	Likely cause	What to do
Nothing happens at the power button	No 24 V; polarity; the button cable	Check the supply's LED, the GX16 cable, the JST button cable on the board
The board turns on by itself when the supply is plugged in	3.3 V on the programming header (an ST-Link or the adapter is connected); the arm was moved while unpowered	Unplug the adapter; move an unpowered robot slowly
The robot cannot be switched off with the button	The same 3.3 V	Unplug the programming adapter
Clicks on power-up, but the PC does not connect	Wrong serial port; the test firmware is on the board; Linux permissions	Communication
A joint jogs the wrong way	Direction not set for that joint in the firmware; a motor's wires swapped	A joint turns the wrong way
A joint does not move at all	Driver not seated or faulty; motor connector; the joint is at its blocker	The test firmware: its driver report at power-on, and whether the motor turns with the others (The test firmware)
A joint grinds at its end during homing	Its switch or sensor does not trigger	E-stop; Homing failures
Joint 6 keeps spinning during homing	The homing pin is not where the sensor looks	E-stop; set the joint 6 pose as in Homing
The pose after homing is off by a few degrees on one joint	A loose coupler, a slipped belt, a moved switch	Couplers, Belts, then Mastering
The robot's position drifts during a program	Lost steps: too fast, too heavy, a collision, a reduced current	Lower the speed; check the payload; home again
A joint hums or squeals and does not hold	A driver in error, a lost phase	The test firmware's driver report at power-on (its fault flags); check the motor cable

Symptom	Likely cause	What to do
Cartesian jogging stops by itself	A singularity	Jog a joint away (Configurations and singularities)
A pose from the GUI cannot be reached by MoveCart	The inverse kinematics finds no solution along the straight line	Move in joint space; choose another orientation
The robot stutters while moving	The PC misses the 10 ms cycle	Communication
The forearm is hot	Hours of motion at the default currents	Currents and temperatures
The gripper does nothing	Not calibrated; an error latched; the cable	Calibrate, clear the error; the gripper's manual
The E-stop has no effect	The PC software is frozen	Switch off the power supply; see Emergency stop

Homing failures

Stop with the E-stop first; a joint driven against its stop strips a belt or cracks a part.

A joint does not stop at its switch

- **Joints 2, 3, 5** use mechanical limit switches. Check that the lever is pressed by the moving part, that the switch clicks, and that it is wired to 24 V and its signal terminal (polarity does not matter). The test firmware's `#LIMIT` shows the state live.
- **Joints 1, 4, 6** use inductive sensors that detect a screw head. The sensor's LED lights when it sees metal. If it does not light as the joint passes, adjust the trigger screw closer; the 4 mm sensors see about 1 to 2 mm. Wiring: brown 24 V, blue GND, black signal.
- A switch on the wrong joint's terminal: wire each switch and sensor to the terminal the wiring diagram shows for its joint; the test firmware's `#LIMIT` shows which input changes when you press a switch.

A joint stops too early

The switch triggers before the joint reaches it: a wire shorted, a sensor too close to a screw it should not see, or a switch lever caught. Also the reason a homed pose is far off on one joint.

Joint 6 turns more than a turn

Its sensor never sees the homing pin. Before homing, turn joint 6 by hand so that the pin is just before the sensor in the direction of rotation (Homing), and check the sensor's gap.

Joint 5 does not reach its switch

The tool's homing feature is missing or in the wrong place, or the firmware's `j5_homing_offset` is for another tool (Your own tool).

The homed pose does not match the witness marks

By a fraction of a degree: master the joint. By more: the joint slipped, a belt or coupler, or the switch moved. Inspect, then master.

Homing never finishes in the software

The PC lost the packet stream, or the firmware is waiting for a joint that does not report. Check the log, press Clear error or `reset()`, and home again.

Communication

The serial port

The control board appears on the PC as a serial port over USB: `COMx` on Windows, `/dev/ttyACMx` on Linux and `/dev/cu.usbmodem…` on macOS. The commander takes `COM5` or `ttyACM0` in its connection field; the Python API's controller takes `--serial COM5` or `--serial /dev/ttyACM0`, or finds the port from `PAROL6_COM_PORT`.

- **Windows**: Device Manager, Ports, shows the number. If the board is missing, try another cable or port; the board enumerates only when it has power.
- **Linux**: `ls /dev/ttyACM*`. Permission denied: `sudo chmod 666 /dev/ttyACM0`, or add yourself to the `dialout` group and log in again.
- **Connect fails the first time** in the commander: press Connect again; two or three tries are normal.
- **The port is busy**: another program has it, often a previous instance of the controller.

The test firmware answers, the main firmware does not

A board from the shop carries the test firmware, which talks in text. Upload the main firmware (Firmware).

The 100 Hz loop

The PC must send a packet every 10 ms. The commander prints `Time between 2 commands in ms` in its terminal output; it should sit near 0.010. The number is in seconds, despite the label: 0.010 is 10 ms. If it is often higher, the robot stutters and the position tracking suffers.

- Close browsers and other heavy programs; disable power saving on the USB port and the CPU.
- A faster PC: six cores from 2017 or newer (What you need).
- The Python API's controller logs `loop overbudget by +N%` when its loop runs late, and shows the figures in `loop_stats()`; it also accepts a lower `PAROL6_CONTROL_RATE_HZ`, at the cost of smoothness, and runs worse with TRACE or DEBUG logging on.

The connection drops

A USB cable without ferrite near the motors, a hub, or a long cable: use a short, shielded cable straight into the PC. When packets stop, the firmware keeps executing the last one: after a move to a position the joints stop near that target and hold, but a joint that was jogging in the commander keeps turning until it reaches its blocker. Keep the mains switch of the power supply within reach (Emergency stop).

Known issues

The faults in PAROL6's firmware and software that are known and not yet fixed, and what to do until they are. The pages that describe each part say the same where it matters; this page collects them in one place. Which versions work together is in Versions and compatibility.

Control board firmware

Seen in: release 1.

	What happens	What to do	More
KI-1	Enforces the joint limits of the earlier robot model, which differ from those in these pages; a target between the two is accepted.	Keep to the limits in the Joint limits table.	Joint limits

Control board test firmware

Seen in: release 1.

	What happens	What to do	More
KI-8	Enables all six drivers when it boots and turns every motor slowly, without a command, with no limits and no homing, until the power is switched off; no command stops them.	Run it with the arm free to move or the motors unmounted, never on a robot folded against its stops.	The test firmware
KI-9	Most command names the parser knows (`#STATUS`, `#STEP1` to `#STEP6`, `#FLASH`, `#CAN`) are answered with `Unknown command`: only `#IO`, `#LIMIT`, `#STOP` and `#start` do something.	Use the four commands that work; CAN has its own test program.	Test firmware commands

Commander software

Seen in: main branch, 2025-09-23.

	What happens	What to do	More
KI-2	Enforces the joint limits of the earlier robot model, and computes with its dimensions (a5 = 176.35 mm, the pneumatic gripper's a6 = 62.8 mm), so its Cartesian positions are off by up to about 0.65 mm.	Keep to the limits in the Joint limits table; teach the poses that need accuracy on the robot.	Kinematics
KI-3	`Output(2,HIGH)` and `Output(2,LOW)` are refused by a bug in the program check (`Invalid Output() command`), and the program stops.	Use output 1 in commander programs, or switch output 2 from the Python API.	RBTscript, I/O
KI-4	`SpeedJoint()`, `Input()` and `Home()` pass the program check but have no action: the program stops on that line until you press **Stop**.	Leave them out of programs; home with the **Home** button before you start a program.	RBTscript, SpeedJoint and SpeedCart
KI-5	`SpeedCart()`, `Get_data` and `Timeouts` are in the command menu but fail the program check: a program that contains one is refused.	Leave them out of programs.	RBTscript, SpeedJoint and SpeedCart
KI-6	A move's profile written in quotes (`"trap"`) or as `func=trap` is ignored without a message, and the move gets the default profile.	Write the bare word, `trap` or `poly`, as the last argument.	RBTscript
KI-7	The Cartesian rotation jog buttons do nothing in the world frame (WRF).	Rotate in the tool frame (TRF), the default.	The GUI

Python API (parol6)

Seen in: 0.9.1.

	What happens	What to do	More
KI-10	Enforces the joint limits of the earlier robot model, and computes with its dimensions (a5 = 176.35 mm), so its Cartesian positions are off by up to about 0.65 mm from the model these pages use; its tool offsets put the TCP up to about 14 mm from the model's.	Keep to the limits in the Joint limits table; teach the poses that need accuracy on the robot.	Kinematics
KI-11	The tool frame (TRF) and the flange frame have Y and Z the other way from the commander's and these pages': the frame is turned 180° about X, so its Z points back into the wrist. A TRF move or jog along +Z, or a TCP offset with a positive z, goes towards the robot instead of out of the tool, and +Y goes the other way.	In the Python API, use negative z for out of the tool: `move_l([0, 0, -20, 0, 0, 0], frame="TRF", rel=True)`, `set_tcp_offset(0, 0, -120)`. Jog slowly in the TRF before you run a TRF move written for the commander.	Reference frames

Something that is not here

Check By symptom first. If it is still not explained, Report a problem says what to include so it can be reproduced. A fixed issue stays on this page, under "Fixed", with the version that fixes it.

9. Transport and storage

Moving the robot

1. Home it, then bring it to the **parked pose**: joints 2 and 3 toward their limit switches so the arm folds over the base. The commander's **Park** button writes a program that moves there (Switching off). Or move it by hand once it is switched off (Moving the robot by hand).
2. Switch off with the forearm held; unplug the supply, the USB and the E-stop.
3. Carry it by the base, not by the arm.

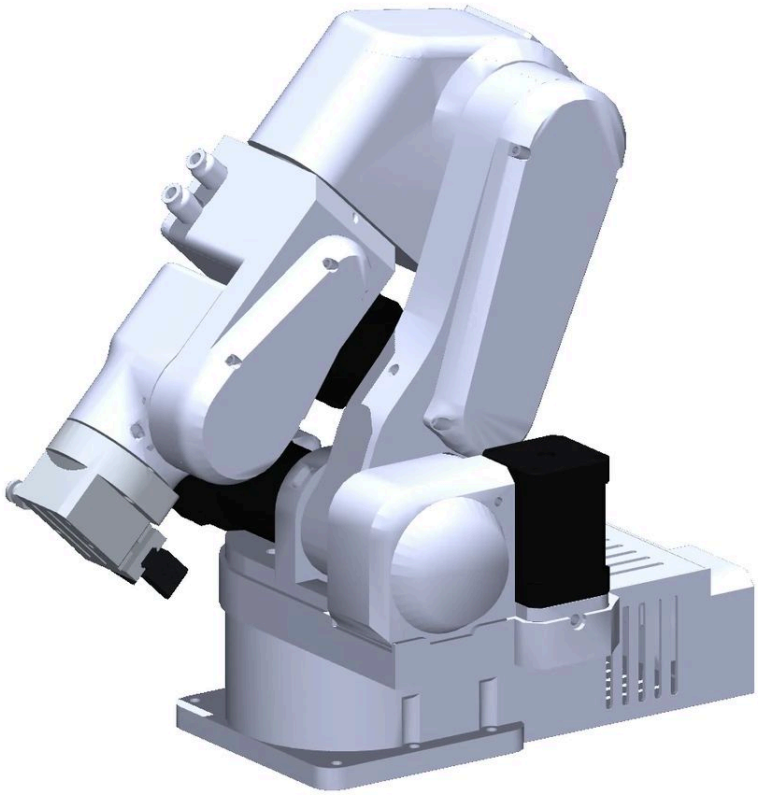


Figure 9.1: The parked pose: the arm folded in over the base.

Packing

Fold the arm as above, put foam under the joint 3 limit switch so that the forearm rests on it and not on the switch, and tape the forearm to the upper arm so that the joints cannot swing. Pack the robot upright in a box with foam around the base; the gripper and the E-stop separately.

Storage

Dry, at room temperature, folded. The stepper motors and the drivers have no shelf-life issues; the PETG parts should not be stored where it gets hot, in a car or near a radiator. Before the first use after storage, run the Inspection.

Build guide

Build guide

PAROL6 is open hardware: you can print the parts, buy the rest from the bill of materials and build the robot yourself. This guide is the order to do it in, with the facts that are not in the assembly manual; the step by step assembly is in the building instructions PDF and the video playlist.

> **Warning**
>
> A self-built robot is yours to make safe: the motor currents, the wiring, the switches and the stops. Read the Safety chapter before the first power-on, as you would for a bought one.

The steps

1. Bill of materials: what to buy
2. Printing the parts: the material, the settings, the print table
3. Mechanical assembly: the manual, the videos, the things to get right
4. Electronics and wiring: the control board, the drivers, the motors, the switches
5. Firmware and first run: the test firmware, the main firmware, directions, homing

Kits

You do not have to source everything yourself. The shop sells the robot in every state between a bag of parts and a working arm:

	What is in it	You add
Assembled robot	the robot, built, tested and ready for the commander software	nothing
Full kit	every part of the BOM except the printed parts and the power supply	the printed parts, a 24 V 5 A supply
Partial kit	the hard-to-find parts: the board, the drivers, the bearings, belts, pulleys, connectors, sensors, switches, the pneumatics	the gearboxes, the stepper motors and the power supply, which are heavy to ship and cheaper from a supplier near you; the printed parts
Printed parts kit	every printed part, in PETG, in several colours	everything else

	What is in it	You add
Screw kit	every screw and nut of the BOM, counted	everything else
Control board, programming adapter	the electronics alone	everything else

The partial kit is the best value for a builder with a printer: it carries the parts that are hard to find or easy to get wrong, and leaves out the three items a local supplier sells cheaper than we can ship them. A full kit skips the bill of materials; a printed parts kit skips the printing.

Bill of materials

This is a copy

The original is BOM/README.md in the robot's repository; this page is that file as of 2026-11-05 (commit `38d0b1a`), with the reference images shown inline. If a link is broken or a part is wrong, open an issue there, and the fix lands here with the next sync. If you would rather not source the parts yourself, the shop's kits cover every part or the hard-to-find ones.

Most items have links where you can buy the item, and some of them are affiliate links. Items like screws, belts and bearings can be bought locally or from many other vendors.

Note

If any product links are broken, unavailable, or no longer point to the correct item, please open an issue. You can also use the reference images to find the same or an equivalent part from another supplier.

Screws and Nuts

Diameter / Type	Length	Quantity	Description
M4	10 mm	20	DIN 912 / ISO 4762 (hexagon socket head cap)
M3	8 mm	10	2 mm head height
M3	8 mm	50	DIN 912 / ISO 4762 (hexagon socket head cap)
M3	14 mm	30	DIN 912 / ISO 4762 (hexagon socket head cap)
M3 nut	—	10	DIN 934 / ISO 4032 (hexagon nut)
M3	25 mm	10	DIN 912 / ISO 4762 (hexagon socket head cap)
M2	10 mm	10	DIN 912 / ISO 4762 (hexagon socket head cap)
M3	16 mm	10	DIN 912 / ISO 4762 (hexagon socket head cap)
M3	6 mm	10	Pan head phillips

Diameter / Type	Length	Quantity	Description
M3	12 mm	30	DIN 912 / ISO 4762 (hexagon socket head cap)
M3	35 mm	10	DIN 912 / ISO 4762 (hexagon socket head cap)
M4	16 mm	30	DIN 912 / ISO 4762 (hexagon socket head cap)
M4	14 mm	10	DIN 912 / ISO 4762 (hexagon socket head cap)
M4	50 mm	10	DIN 912 / ISO 4762 (hexagon socket head cap)
M3	15 mm	2	Tapered screw head

Gearboxes

Note

For cheaper gearbox options you can use MG series planetary gearboxes. They have bigger backlash but are cheaper. Link — Aliexpress Link

Name	Type	Quantity	Description	Link	ALT link	Reference Image
Gearbox 20:1	Nema 17 20:1	2	EG precision planetary	Link	Alt Link	
Gearbox 10:1	Nema 17 10:1	1	EG precision planetary	Link	Alt Link	

Motors

Name	Type	Quantity	Description	Link	ALT link	Reference Image
Stepper 1	Nema 17	1	16Ncm, 42x42x20mm	Link	Alt Link	
Stepper 2	Nema 17	3	45Ncm, 42x42x40mm	Link	Alt Link	
Stepper 3	Nema 17	2	65Ncm, 42x42x60mm	Link	Alt Link	

Belts

Name	Type	Quantity	Description	Link	ALT link	Reference Image
Joint 1 belt	HTD3 396	1	6 mm width	Link		
Joint 3 belt	HTD3 342	1	6 mm width	Link		
Joint 4 belt	HTD3 201	1	6 mm width	Link		
Joint 5 belt	HTD3 246	1	6 mm width	Link		

Pulleys

Name	Type	Quantity	Description	Link	ALT link	Reference Image
J4 pulley, J5 pulley	HTD3M 12 tooth	2	5mm bore, 10mm width	Link		
J1 pulley	HTD3M 15 tooth	1	5mm bore, 10mm width	Link		

Shaft couplers

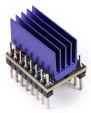
Name	Type	Quantity	Description	Link	ALT link	Reference Image
Shaft coupler	Shaft coupler	3	8 mm hole shaft, M4 screw holes	Link		

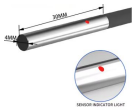
Bearings

Name	Type	Quantity	Description	Link	ALT link	Reference Image
Bearing 1	AXK3552	1	35X52X4, needs to have top and bottom plate	Link		
Bearing 2	NSK HR32906J	5	Joint 5, 4 and 3	Link		
Bearing 3	NSK HR32907J	4	Joint 1 and 2	Link		

Name	Type	Quantity	Description	Link	ALT link	Reference Image
Tension bearing	Ball bearing 3x8x4	20	Used for belt tension	Link		
Tension bearing	Ball bearing 3x10x4	20	Used for belt tension	Link		

Electronics

Name	Type	Quantity	Description	Link	ALT link	Reference Image
PAROL control board		1	source-robotics.com	Link		
TMC5160		6	Drivers from the links are tested. Use any other at own risk	Link	Alt Link	
Thermal paste / cement		2	Needed only if your control board has no drivers!	Link		
On/Off button		1	12mm 3-6V	Link		
JST PH 2.0mm 200mm 4PIN		1	For power button	Link		

Name	Type	Quantity	Description	Link	ALT link	Reference Image
Noctua NF-A4x20 5V 40mm		1	or any 40mm 5V fan			
Power connector - male + female		1	GX16 2PIN	Link		
Gripper connector (electric)		1	M8 sensor Female 4 pin	Link		 Female 4Pin
Power supply		1	Any 24V 5A supply	Link	Alt Link	
Limit switches		3	Type ZW12-3	Link		
Inductive sensor 1		1	4mm NPN NO	Link	Alt Link	
Inductive sensor 2		1	GX-F8A	Link		
Inductive sensor 3		1	M5, NPN NO	Link	Alt Link	
Estop		1	Any NC estop	Link		

Name	Type	Quantity	Description	Link	ALT link	Reference Image
Cable gland for estop		1	PG7 type	Link		
M3 brass insert		4	M3, 6mm lenght	Link		
Pneumatic connector - robot		4	4 mm type PM	Link		
Pneumatic Gripper		1	MHZ2-16D	Link		
Pneumatic gripper connector		2	PC4 - M5	Link		
Pneumatic tube		5 meters	1 meter, 4x2.5mm	Link		
xt30 connector female		1	Used to deliver power to PAROL6 PCB	Link		

Name	Type	Quantity	Description	Link	ALT link	Reference Image
Stlink / programming adapter		1	Use our Official programming adapter for ease of use! Pin order needs to be: SWCLK, SWDIO, GND, 3V3, 5V	Link		
USB B cable		1	Comms to the PC			
5/2 solenoid air valve		1	4mm tube connectors, 5/2, 24V	Link		

Vacuum gripper

Name	Type	Quantity	Description	Link	ALT link	Reference Image
Suction cup gripper		1		Link		
Vacuum generator		1		Link		
4mm to 6mm coupler		1		Link		

Small print bed modification

Note

These parts are only needed if your printer bed is too small for `J1_turret_rework_blocker` and `Upper_arm`. Both parts are split in two and joined with dowel pins and epoxy after printing. STL files are in STL/SMALL_PRINT_BED. Nothing else in the build changes.

Name	Type	Quantity	Description	Link	ALT link	Reference Image
J1 turret dowel pin	Dowel pin 4 x 18 mm	3	Solid cylindrical, stainless or hardened steel	Link		
Upper arm dowel pin	Dowel pin 6 x 22 mm	4	Solid cylindrical, stainless or hardened steel	Link		
Two part epoxy	Epoxy adhesive	1	1:1 mix, slow setting	Link		

Printing the parts

Material

PETG. The motors run warm and the parts around them see 60 to 70 °C in long use, which PLA does not survive; PETG softens around 80 °C, prints without an enclosure, and bonds layers well. ABS or ASA work with an enclosure. The repository's PETG notes give the printing basics: hot end 220 to 250 °C, bed 60 to 90 °C with glue, a layer fan, and the warping is small.

The files

All parts are in the repository's `STL` folder, grouped by link: `BASE`, `SHOULDER`, `UPPER_ARM`, `ELBOW`, `FOREARM`, `WRIST`, `ESTOP`, `GRIPPER_ATTACHMENTS`, `UTILITY`, `mounting plates`, and `SMALL_PRINT_BED` for the split versions of the two largest parts. The print table below says how each was printed on the reference robots; it is a tested starting point, not a rule.

Print table

The print table from the robot's repository, as it is. Every printed part with the settings it was printed with on the reference robots: supports, mass, infill, print time, cooling, layer height, top and bottom layers, perimeters. Click the table to enlarge it. The same table is in the repository as a PDF and as an Excel file.

THIS TABLE HAS SUGGESTIONS ON WHAT SETTINGS TO USE!
IT IS NOT CRITICAL TO FOLLOW IT BUT NOTE THAT THESE SETTINGS HAVE BEEN TESTED ON MUTLIPLE ROBOTS!

Part name	Position in arm	Supports	mass (g)	Infill type	Infill %	Time	Cooling %	Layer h	Top layers	Bot layers	Perimeters
Wrist	J6	Minimal	77,65	honey	40	7h51m	15	0.2 LA	6	4	3
Pneumatic gripper holder	J6	None	31	honey	40	2h52m	15	0.2 LA	6	4	3
J5 pulley	J5	None	20	honey	40	2h	15	0.2 LA	6	4	3
J5 part	J5	Minimal	114	honey	30	10h	15	0.2 LA	6	4	3
J5 belt lid	J5	None	20	honey	15		15	0.2 LA	6	4	3
J5 electronics lid	J5	None	20	honey	15		15	0.2 LA	6	4	3
Elbow part	J4	alot	215	honey	25	16h	15	0.2 LA	6	4	3
J4 limiter	J4	None	1	honey	80	2m	15	0.2 LA	6	4	3
J4 bearing backplate	J4	None		honey			15	0.2 LA	6	4	3
	J4	None		honey			15	0.2 LA	6	4	3
Elbow lid 1	J4	None	20	honey	20	1h46m	15	0.2 LA	6	4	3
Elbow lid 2	J4	None	4	honey	20	15m	15	0.2 LA	6	4	3
Upper arm	J3	Minimal	220	honey	35	18h	15	0.2 LA	6	4	3
J3 limit switch lid	J3	None	4	honey	50	21m	15	0.2 LA	6	4	3
J3 pulley	J3	None	25	honey	50	2h42m	15	0.2 LA	6	4	3
J3 pulley input	J3	None	14	honey	50	1h16m	15	0.2 LA	6	4	3
Upper arm cover	J3	minimal	34	honey	15	2h	15	0.2 LA	6	4	3
J2_cover	J2	None	10	honey	20	42m	15	0.2 LA	6	4	3
J2_bering backplate	J2	None	10	honey	20	42m	15	0.2 LA	6	4	3
J1_belt_cover	J2	None	37	honey	20	2h	15	0.2 LA	6	4	3
J2_stepper_blocker	J2	None	7	honey	20	15m	15	0.2 LA	6	4	3
J1_turret_motor_holder	J2	Minimal	53	honey	80	6h	15	0.2 LA	6	4	3
J1_turret	J1	Minimal	210	honey	25	18h	15	0.2 LA	6	4	3
J1_lid_motor	J1	Minimal	9	honey	25	51m	15	0.2 LA	6	4	3
J1_rotation_shaft	J1	none	31	honey	50	3h34m	15	0.2 LA	6	4	3
J2_shaft	J2	none	43	honey	50	4h41m	15	0.2 LA	6	4	3
Electronics_cover	J1	Minimal	80	honey	20	6h	15	0.2 LA	6	4	3
J2 limit switch cover	J2	Minimal	3	honey	25	13m	15	0.2 LA	6	4	3
Electronics_case	J1	Minimal		honey	50		15	0.2 LA	6	4	3
Fan_cover	J1	Minimal		honey	20		15	0.2 LA	6	4	3
J1_bottom_lid	J1	Minimal		honey	25		15	0.2 LA	6	4	3
Main_base	J1	Minimal		honey	50		15	0.2 LA	6	4	3
HTD_48_J5_PULLEY	J5	Minimal		honey	60		15	0.2 LA	6	4	3
J4_output_pulley	J1	Minimal		honey	50		15	0.2 LA	6	4	3
Pneumatic_gripper_holder	Gripper	Minimal		honey	30		15	0.2 LA	6	4	3
J1_backplate	J1	Minimal		honey	50		15	0.2 LA	6	4	3
J1_turret	J1	Minimal		honey	45		15	0.2 LA	6	4	3
J2_limit_switch_cover		Minimal		honey	50		15	0.2 LA	6	4	3
J1_stepper_block	J1	Minimal		honey	45		15	0.2 LA	6	4	3
Upper_arm_wires_cover	J3	Minimal		honey	45		15	0.2 LA	6	4	3
38_pulley_J3	J3	Minimal		honey	70		15	0.2 LA	6	4	3
42_pulley	J3	Minimal		honey	70		15	0.2 LA	6	4	3

Figure 1: The PAROL6 print table: part, position in the arm, supports, mass, infill type and percentage, print time, cooling, layer height, top and bottom layers, perimeters.

Print the pulleys with high infill and slow outer walls: their teeth carry the load. Check the first printed pulley against a belt before printing the rest.

Tolerances

Bearing seats and shafts are designed for a well-calibrated printer; a seat that is too tight is reamed, one that is too loose is shimmed with tape or a drop of epoxy. Test the fit of the tapered bearings in their seats before assembling a joint.

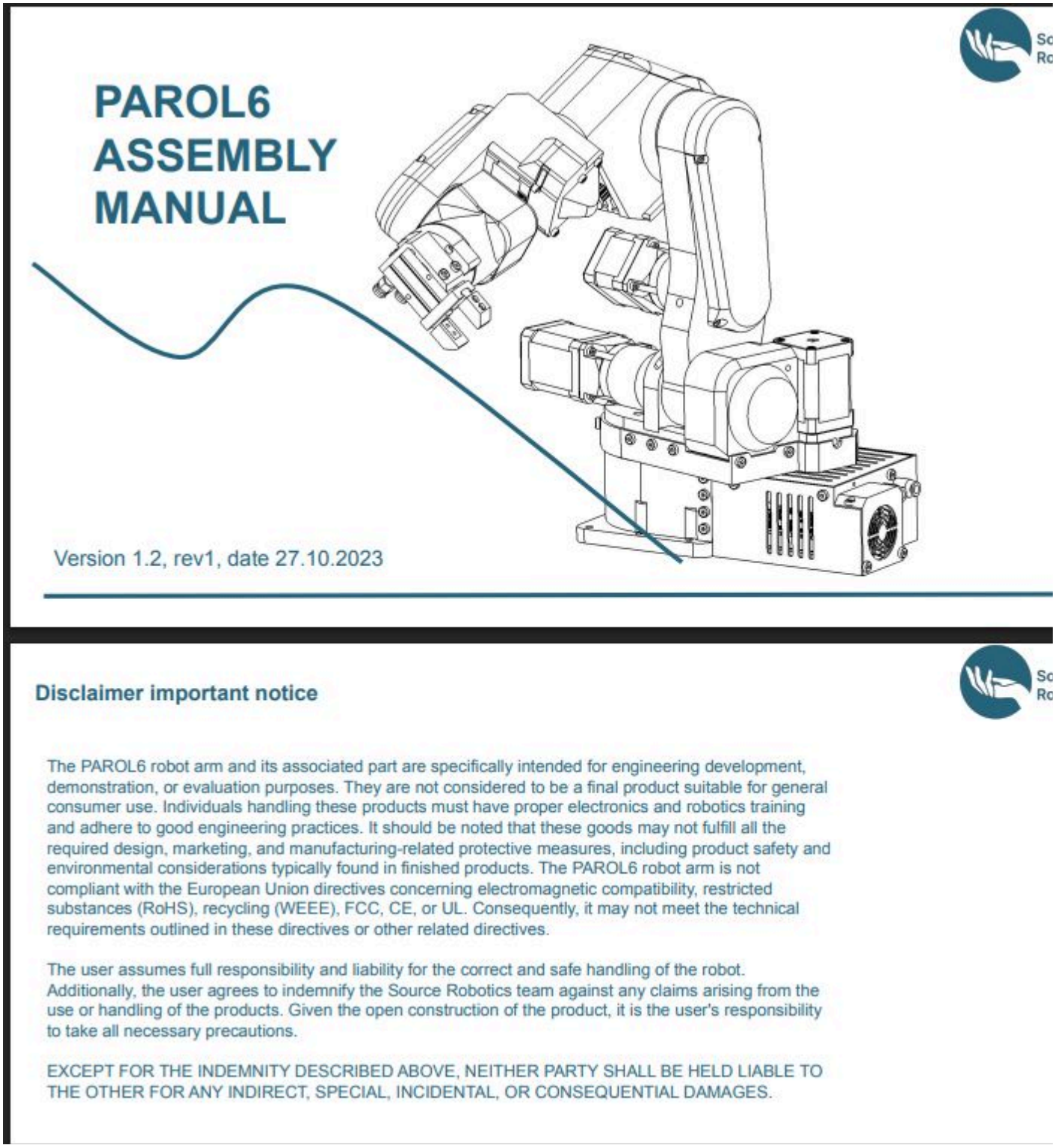
Inserts

Four M3 brass inserts are melted into their holes with a soldering iron; everything else screws directly into the plastic.

Mechanical assembly

The assembly is documented step by step in the Parol building instructions PDF and shown in the video playlist. Read the whole manual once before you start: some steps cannot be undone without taking the joint apart again. This page lists what to get right.

The building instructions



Download the building instructions (PDF, 8 MB)

Things to get right

- **Bearings seated square**: the tapered roller bearings carry the joint; a bearing that sits crooked makes the joint stiff in one direction and loose in the other.
- **Belt tension** as in the belt tension video: maximum hand force for joints 3 and 5, slightly less for 1 and 4, with the tension bearings (Belts).
- **Threadlocker** on every metal-to-metal screw, blue, and on all set screws (Couplers and threadlocker). The couplers between motors, gearboxes and shafts are where a loose screw costs the most.
- **The limit switches and sensors** positioned so that the moving part presses the lever, or passes the sensor within a millimetre or two, before the joint reaches its mechanical stop.
- **Joint 1's blockers** fitted: without them joint 1 can spin a full turn and tear its wires.
- **Wires and tubes** with enough slack through each joint for its full range, and fixed so that they cannot be pinched by the belts.
- **Wire markings**: the print table spreadsheet's *Wire_markings* sheet lists the colour code of the motor and sensor cables used on the reference robots, which helps when the base is wired.

Small print beds

The J1 turret blocker and the upper arm come split in `STL/SMALL_PRINT_BED`, with a README. Join the halves with the dowel pins and slow epoxy, clamp them straight, and let them cure before the joint is assembled.

Custom couplers and flanges

The repository's `custom_couplers.md` is a work in progress on machined alternatives to the shaft couplers, with a press-fit flange; use the BOM's couplers until it is finished.

Before the electronics

Turn every joint by hand through its range. Everything should move smoothly against the gearbox's resistance, with no clicks and no tight spots, and every switch should click as its joint reaches it.

Electronics and wiring

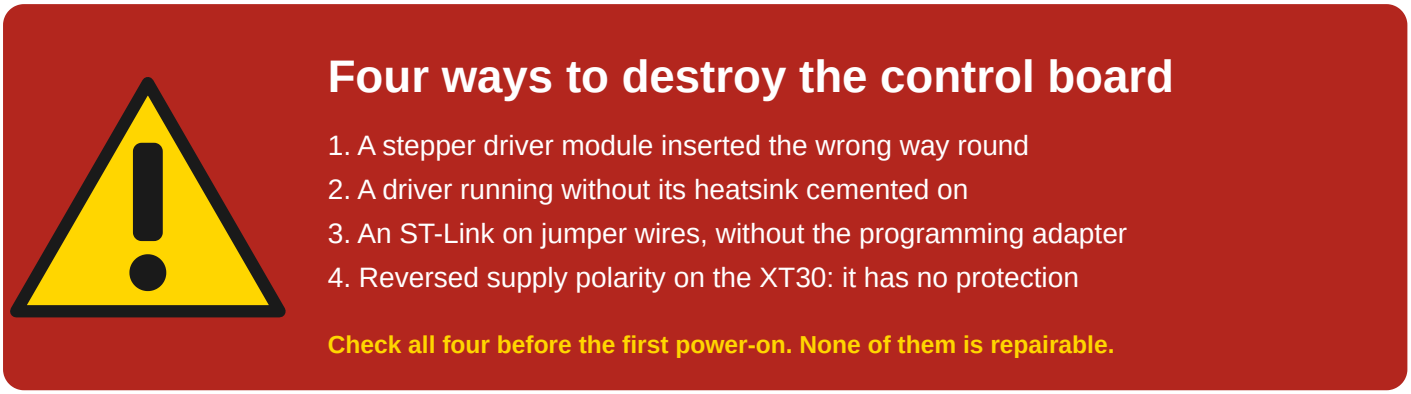


Figure 1: Four ways to destroy the control board.

The control board goes into the base with its fan and its six driver modules; the motors, switches, sensors, the gripper connector, the E-stop and the power connector are wired to it.

The control board

Everything about the board itself is in the Control Board Manual: the connectors, the pin map, the drivers. The two things that destroy boards, both on the Stepper drivers page: a driver module inserted the wrong way round, and a driver without its heatsink cemented on. Read that page before you plug anything in.

Wiring the arm to the board

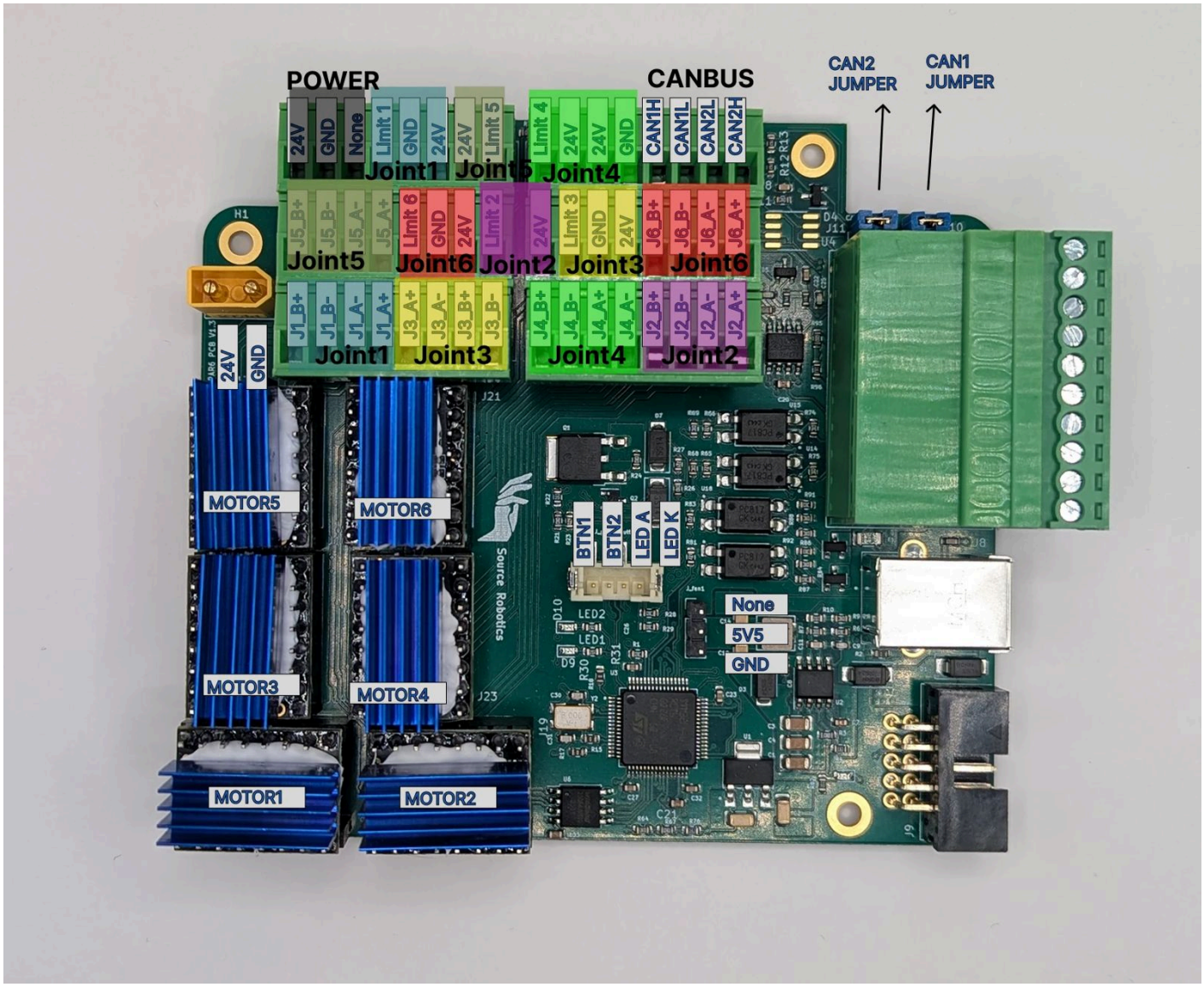


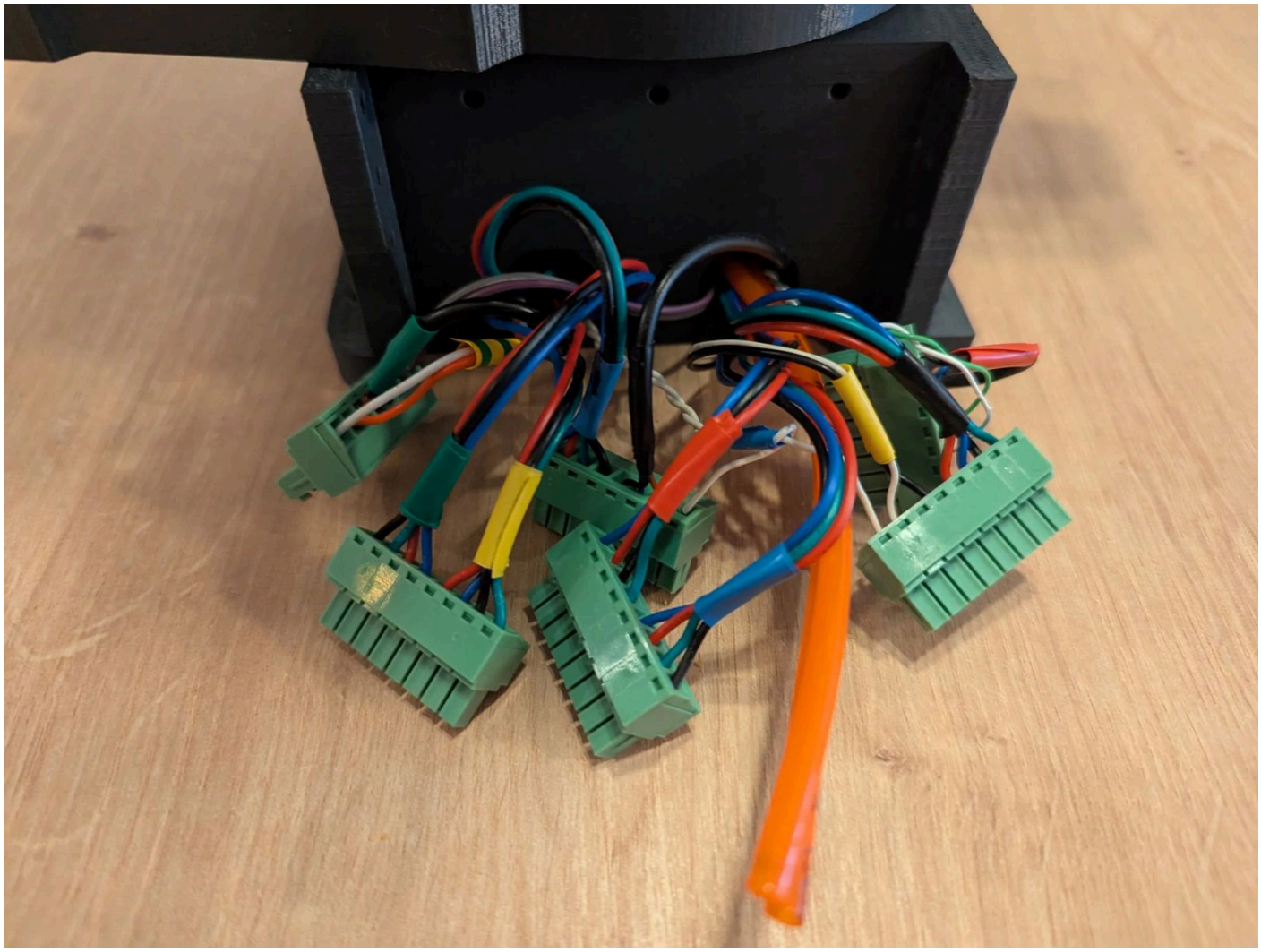
Figure 2: Wiring diagram: motors, switches, sensors and connectors to the control board.

- **Motors**: one four-wire cable per joint, to the terminals marked with its joint number (`J1_A+` to `J6_B-`).
- **Limit switches** (joints 2, 3, 5): one lead to 24 V, the other to the LIMIT terminal the wiring diagram shows in that joint's group; polarity does not matter.
- **Inductive sensors** (joints 1, 4, 6): brown to 24 V, blue to GND, black (signal) to the LIMIT terminal the wiring diagram shows in that joint's group.
- **The gripper connector**: 24 V, GND, CAN H and CAN L from the board to the female M8 socket at the forearm: pin 1 GND, 2 CAN H, 3 24 V, 4 CAN L, by the numbers moulded into it (The gripper socket). Solder it as the photo on page 32 of the assembly manual shows, the view from the back where the wires go. The numbered list next to that photo has pins 1 and 3 the wrong way round for this socket: wired by the list, the PAROL6 gripper cable reverses the gripper's supply.

- **The E-stop**: the NC contact to the two E-stop terminals.
- **Power**: the GX16 socket on the base to the XT30 on the board, polarity checked twice. The GX16 has no fixed +24 V pin: choose one, wire the supply's plug the same way, and mark it.
- **The power button**: the JST PH cable to the button connector.
- **The fan**: 5 V, three-pin, on the fan header; not a PWM fan.

Wire each switch and sensor to the terminal the diagram shows for its joint, joints 1 and 6 included, and check each one with the test firmware's `#LIMIT` before you close the base.

The finished base looks like this:



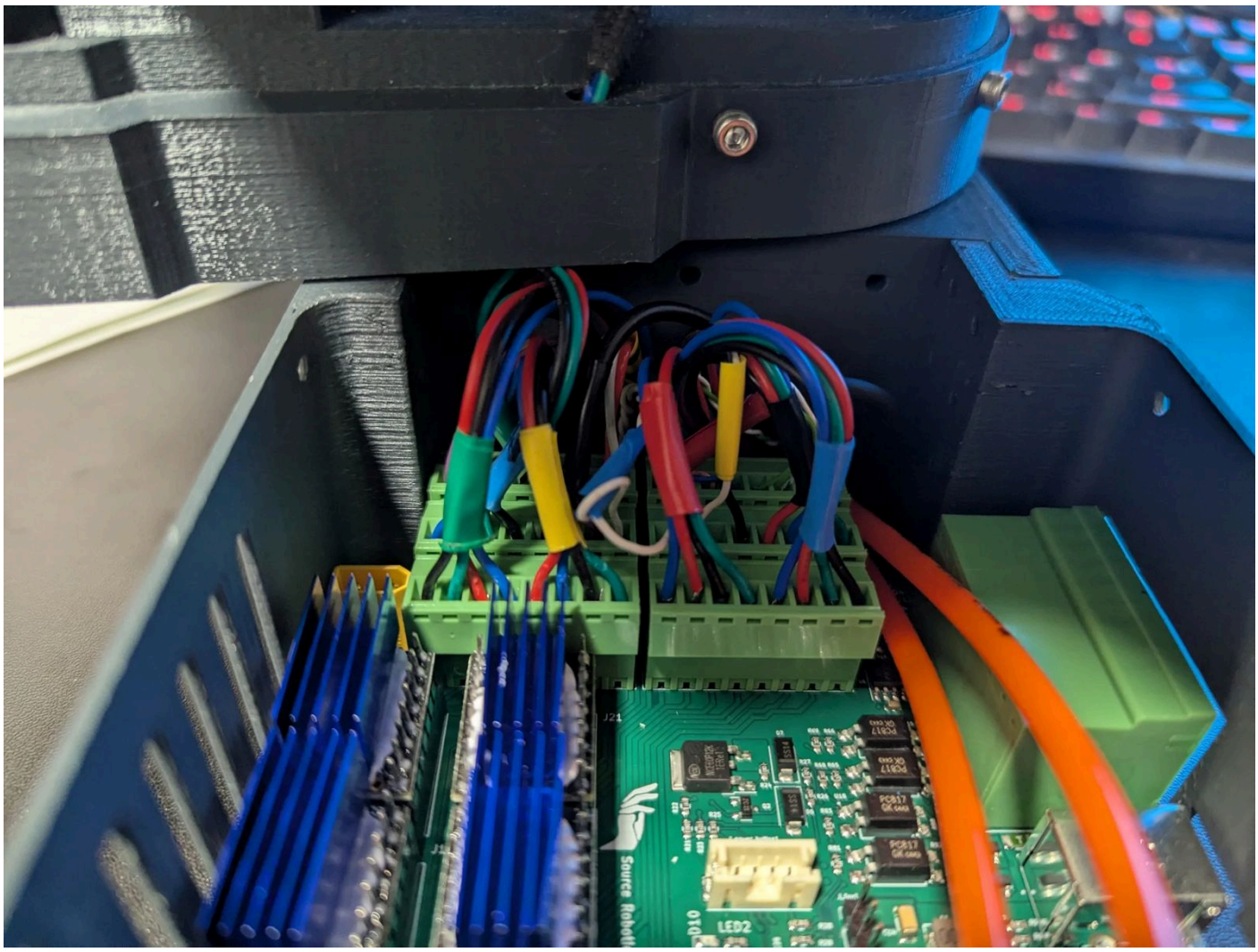


Figure 3: The connectors on the base.

Test before you close the base

Power the board with the test firmware and check every driver, switch, sensor, input and output from the serial terminal (The test firmware). It is far easier now than with the base closed and the arm on top. You can also test your motors here, before the arm is assembled, to see that each one works.

Every control board sold by Source Robotics comes with the test firmware preloaded. It is the firmware every board was tested with: each board is fully tested, every part of it, at the end of the production line.

Firmware and first run

1. The test firmware

A control board from the shop ships with the test firmware. Use it, with the base open and the motors not yet driving the arm, to check every driver, every switch and sensor, and the I/O from a serial terminal: on the bench before the arm is assembled, as in Electronics and wiring. At power-on it prints a report on each driver (`joint N Connection is: GOOD` or `BAD`) and turns every motor slowly; `#LIMIT` then shows the switches and `#IO` the E-stop, the inputs and the supply voltage. The test firmware has the commands and what good output looks like.

Danger

The test firmware turns all six motors slowly from the moment it boots until the power is switched off; no command stops them. On an assembled arm joints 1 to 5 reach their blockers within about a minute and keep pushing. Use it with the motors free to turn, never on an assembled robot.

Check

Six `Connection is: GOOD` lines in the report; `#LIMIT` names each switch while you press it; `#IO` follows the E-stop and the inputs and shows the supply near 24000 mV. If not: Control board troubleshooting.

2. Upload the main firmware

The main firmware is in the repository under `PAROL6 control board main software` ; it is a PlatformIO project for VS Code. Install VS Code and the PlatformIO extension, open the folder, and upload with an ST-Link through the programming adapter, with the 24 V supply disconnected: Firmware has the procedure and the warnings.

Check

PlatformIO ends the upload with `SUCCESS` . If not: Control board troubleshooting.

3. First power-on

Unplug the programming adapter, plug in the supply, hold the forearm and press the button. Six clicks, and the arm holds. Start the commander software, connect, and open the joint jog menu.

Check

Six clicks, the fan runs, the arm holds itself, and the commander shows the robot's data. If not: First power-on: if nothing happens.

4. A joint turns the wrong way

Jog each joint briefly. The joint must turn as in the figure, which shows each joint's positive direction (Joint limits and directions). Before homing, the commander checks jogs against the joint limits with a position count that starts at 0, so some directions are refused with `Error: Robot jog -> Position out of range`:

- **Joint 2**: press its **right** arrow (negative). It must turn opposite to the figure; its left arrow does nothing until the robot is homed.
- **Joints 1, 3, 4, 5 and 6**: press the **left** arrow (positive). It must turn as in the figure.



Figure 1: Each joint turned 30 degrees in its positive direction, in colour, over the pose in grey.

A joint that turns the other way is the most common first-run problem, and it is fixed in the firmware, not by rewiring:

```
// motor_init.cpp, in Init_Joint_N: switch between 0 and 1
Joint__->direction_reversed = 1;
```

Upload again and check again. Swapping one motor phase's wires has the same effect, if you prefer to open the base.

> ✓ **Check**
>
> The left arrows of joints 1 and 3 to 6 turn them as in the figure, and joint 2's right arrow turns it the opposite way.

5. Home

With every direction right, set joint 6 and check the tool's homing offset as in Before you press Home, then press **Home** and stay at the E-stop. The sequence is on Homing; if a joint grinds at a stop, press the E-stop and see Homing failures.

> ✓ **Check**
>
> Homing finishes, the simulator matches the robot, and the pin passes through the witness marks of each joint, which line up in the after-homing pose (Witness marks).

The commander's terminal output

While it runs, the commander prints the robot's data to its terminal: the positions, the homed and I/O bytes, the errors, the gripper data, and `Time between 2 commands in ms`, which should be near 0.010 (the number is in seconds, despite the label: 0.010 is 10 ms). Watch it for a few minutes while jogging: if it is often much higher, the PC is too slow (Communication).

Appendix

Release notes

PAROL6

V1.1 — 2026-11-04

The PAROL6 files as of November 2026, with downloads on the release page: the STL files, the building instructions, the BOM with a reference image for every part, the print table and checksums. Since V1: the revised building instructions, the small print bed modification for the upper arm and the J1 turret, the holders for a horizontal and a custom pneumatic gripper, the BOM's current links and images, the URDF and MuJoCo models from robot_descriptions, and the hardware licence CERN-OHL-S-2.0. The firmware is the same as in V1.

V1 — 2025-03-08

Stable version of the PAROL6 mechanical files, code, documentation and BOM: the first tagged release of the robot repository. The repository's main branch continued with BOM updates, the small print bed variants and the community projects list, which release V1.1 collects; the firmware is unchanged since.

Python API

The releases of PCrnjak/PAROL6-python-API, newest first, as published on GitHub. The tags `v0.9.0` and `v0.9.1` have no release notes on GitHub; the reference on this site describes v0.9.1.

v0.8.1 — 2026-09-20

A patch on **v0.8.0**: PR #51, the Cartesian streaming executor. No API changes.

Fixes

- **MOVECART and SERVOL hold their straight line.** Ruckig's synchronization was never set, so it defaulted to `Time` — the six tangent components only *finish* together, each taking its own time-optimal route there, and the tangent bows. The executor now runs under `Phase`, one shared profile for all six.
- **A repeated target no longer costs that synchronization.** A servo stream sets the same target every tick, and each re-plan drops to time synchronization unless the current velocity and acceleration line up exactly with the fresh profile; once out of phase the state drifts further out and the next tick fails too. Setting a target Ruckig is already tracking is now a no-op. On a 160 mm diagonal: 4.7 mm off the line before, 0.000 mm after.
- **The speed ceilings are TCP speeds again.** Ruckig bounds each component on its own, so an isotropic envelope let a diagonal run the resultant up to sqrt(3) times the configured limit — 1.35x as measured. The per-component ceilings are now scaled by the motion direction's largest component over its norm, and the same move peaks at exactly the 0.160 m/s jog ceiling.

v0.8.0 — 2026-09-19

Everything since **v0.7.0**: PRs #41–#48, the PAROL6 side of waldoctl v0.14.0's skill runtime, full TCP calibration, bounded I/O, execution speed, demonstration metadata, held-object geometry and supervised restart. Pre-1.0, so breaking changes bump the minor.

Breaking changes

- **The waldoctl pin moves v0.12.0 → v0.14.0.** `set_tcp_transform` / `tcp_transform` and `set_execution_speed` / `pause` / `resume` are required of every client there and implemented here.
- **Completion waits query the exact command result** (#47, #48), and client and controller must both speak the new query. A controller session change during a wait raises `ConnectionError`; unknown, cancelled and expired results stay unconfirmed instead of passing for success, and replies to timed-out queries are filtered by response type.
- **TCP queries fail explicitly when the controller does not answer** (#43) rather than returning a stale value.
- **Unconfirmed blocking motion completion raises** `TimeoutError` (#45); the queued motion stays until it completes or is stopped.

Skills

- **Python skills run on the connected client** (#41). Decorated skills go through the existing synchronous connection and event loop, can use the typed PAROL6 async client internally, compose nested calls, and run unchanged under Commander's stepping wrapper. Live and dry-run clients advertise their skill capabilities. Stopping a planned trajectory now clears the segment player's activity, so status no longer sits at EXECUTING after a cancel.
- **Gripper skills discover PAROL6 gripper support** on the live client and native preview (#42); the preview tool exposes its key and type as metadata, so a requirement check cannot dispatch a command. Streaming updates preserve unread configuration, query and stop packets across a polling batch boundary.

Setup and calibration

- **Full TCP transforms through planning and readback** (#43). Translation and intrinsic XYZ orientation are queued, applied and confirmed before use; earlier queued motion keeps its frame, cancelled corrections do not leak into planning, and a tool/variant change clears the correction. The XYZ API stays and clears user rotation. Tool status reports its applied variant, so a calibration cannot silently bind to different jaws.
- **Bounded digital I/O** (#44). Reads and writes take a per-call deadline covering endpoint setup and retries; a missing read returns `None`, an unconfirmed write raises `TimeoutError`, and neither is mistaken for cancellation. Both facades advertise `io.digital`; dry-run output writes use logical indices and report dispatch errors.

Motion

- **Queued execution speed and explicit pause/resume** (#45). `set_execution_speed(0.1..1.0)` retimes the planned joint path; `pause()` decelerates to a hold with the queue retained and `resume()` continues at the selected speed, and changing speed while paused keeps the hold. Rate changes are checked against acceleration derived from the emitted positions, with preallocated buffers in the control loop; queued dwell pauses, while jog and servo-stream timing stay separate. Preview retimes durations and reports paused operations as unresolved. Physical transition tuning is unverified.

- **Held-object geometry under the flange** (#47). Only declared contact pairs are exempt; attach/detach requires confirmed whole-world application and invalid contact names keep the previous geometry. Reference or controller-context changes invalidate declarations and gate arm motion until explicit reconciliation or removal. The controller retains 1024 successful indices, including every command consumed by a blend.

Status surface

- **Session and timing metadata on every status frame** (#46): a nonzero session id, publication sequence and monotonic snapshot timestamp, so a client can record observation cadence and detect a controller restart. Missing metadata is explicitly unavailable. The offline client's no-tool identity matches the native controller.
- **Queue readback includes accepted commands** still waiting for planning or paused (#48), and clears completed or cancelled indices, so a supervised restart can tell whether the controller still has pending work. Restart detection also watches status broadcasts when the command socket goes quiet after a peer reset.

Dependencies

- **interpolatepy is gone.** Both profiles it supplied were only ever asked for rest-to-rest motion, so the trapezoidal and quintic curves are now evaluated here in closed form, across a numpy array rather than one Python call per sample. Positions match the previous output to within a thousandth of a motor step. This also removes a compiled dependency that, as of its 3.3.0, ships no Windows or Python 3.14 wheels.

v0.7.0 — 2026-09-04

Everything since **0.3.2**, the last release. Four tags landed in between without notes, so this collects them. Pre-1.0, so breaking changes bump the minor.

Breaking changes

- `RobotError` **comes from waldoctl** rather than being defined here. A frontend represents a refusal the same way whichever backend raised it. It is an exception now, not a frozen dataclass, so a client can raise it as-is; anything that built one positionally or called `dataclasses.replace` on it needs updating. Tool-error attribution rebuilds the error from its wire form for exactly that reason.
- `stop`, `estop` **and** `reset` **replace** `halt` **and** `resume`, and a stop cancels the whole motion pipeline rather than just the active command.
- `home(calibrate=)` **replaces the separate CALIBRATE command.** Unhomed, or with `calibrate=True`, the arm runs the full end-stop referencing sequence; already homed, it takes a planned collision-checked move back to standby.
- **Planned motion is refused while unhomed.** Jogging stays available.
- **The waldoctl pin moves v0.7.0 → v0.12.0**, which removes the `Plane` keep-out shape. Persisted worlds containing a plane now raise on load.

Status surface

- **The control loop broadcasts its own health.** The p99 period and the deadline-miss count ride every STATUS frame in a `loop_health` mapping, so a display no longer polls the loop-statistics query for them. `drive_health` sits

beside it and stays empty on this backend: the drives report per-joint error flags, not analog registers, and empty distinguishes "no such sensor" from "all zero".

- `StatusBuffer`**, homing progress and loop statistics conform to waldoctl**, and homing progress now survives a cancel.
- **A hardware e-stop disables the controller**, rather than only halting motion.

Motion

- **Trajectory settling is gated on progress, not elapsed ticks.**
- **Fast-path home**: an already-referenced arm takes a planned return move instead of re-running the firmware seek.
- **Command indices stay monotonic across a reset.**
- **The dry-run snapshot carries the pending blend chain**, so a preview mid-blend matches what the runtime will execute.

Fixes and hygiene

- A system command's `Command_out` was clobbered in the same tick by the executor fallback.
- The MSG gripper is seated on the flange rather than inside the wrist, and its built-in camera mount is declared.
- Test fixes surfaced by loaded runners: the unreachable-client test takes a free port instead of adding to the live one, which overflowed past 65535; the loop-health comparison waits for a frame that has caught up instead of demanding two samples of a climbing counter be equal.
- Typing: scipy stubs pinned per Python version, stale suppressions dropped, sync tool wrappers typed as `SyncTool`.

0.3.2 — 2026-07-14

What's Changed

- Fix stale error poisoning unrelated wait_command calls by @Jepson2k in https://github.com/PCrnjak/PAROL6-python-API/pull/26

Full Changelog: https://github.com/PCrnjak/PAROL6-python-API/compare/v0.3.1...0.3.2

v0.3.1 — 2026-07-09

What's Changed

- Bump waldoctl pin to v0.6.2 by @Jepson2k in https://github.com/PCrnjak/PAROL6-python-API/pull/25

Full Changelog: https://github.com/PCrnjak/PAROL6-python-API/compare/v0.3.0...v0.3.1

v0.3.0 — 2026-07-09

What's Changed

- parol6: camera_spec in tool registry + StrEnum ToolType membership (commander-public-api base) by @Jepson2k in https://github.com/PCrnjak/PAROL6-python-API/pull/18
- parol6: plugin tools in every process registry + variant TCP rpy (panel-extensions stack) by @Jepson2k in https://github.com/PCrnjak/PAROL6-python-API/pull/19
- Panels → main: plugin tools + planner/status fixes by @Jepson2k in https://github.com/PCrnjak/PAROL6-python-API/pull/20
- fix: bump waldoctl pin to v0.4.0 (repairs red main) by @Jepson2k in https://github.com/PCrnjak/PAROL6-python-API/pull/22
- ci: scope push trigger to main (kill duplicate PR-branch runs) by @Jepson2k in https://github.com/PCrnjak/PAROL6-python-API/pull/23
- Collision checking: keep-out shapes, enablement gates, MoveIt-style invalidation by @Jepson2k in https://github.com/PCrnjak/PAROL6-python-API/pull/24
- chore: comment sweep (why-only comments) by @Jepson2k in https://github.com/PCrnjak/PAROL6-python-API/pull/21

Full Changelog: https://github.com/PCrnjak/PAROL6-python-API/compare/v0.2.7...v0.3.0

v0.2.7 — 2026-05-14

First release with `setuptools_scm`-managed versioning. `pip install git+...@v0.2.7` now resolves to the matching version directly (no auto-bump dance, no tag/metadata drift).

Changes since v0.2.6

- Switch to `setuptools_scm` for dynamic versioning. The git tag is now the source of truth for the package version; `pyproject.toml` no longer carries a static `version = "..."` line. Removes `[tool.tbump]` and the `release.yml` workflow that auto-bumped on tag publish.

Release notes written by Claude on the author's behalf.

v0.2.6 — 2026-05-14

Bug fixes

- Electric-gripper stall detection: terminate on stall when grasping, ignore PID hunting on clamped jaws, keep `move_active` set so subsequent commands don't short-circuit.
- `SetTcpOffset` now routes through the motion-planner subprocess (was applied controller-side only, leaving the planner with stale IK seeds).
- Failed tool actions clear stale error state and `command_index` so subsequent commands aren't blocked.
- `RobotClient.async_client` property removed (cross-loop foot-gun). Async users construct `AsyncRobotClient` directly. Minor API break.

Kinematics / motion

- TOPP-RA trajectory shake: njit'd IK-outlier smoother (median-filters IK output), denser IK sampling, sim-dt jitter fix.
- IK `max_iter` bumped 10 → 20 to help wrist-near-singular targets converge.
- Deterministic wrist-flip restart for IK branches that converge outside joint limits.

Dependencies

- pinokin pin bumped v0.1.5 → v0.1.6.

Release notes written by Claude on the author's behalf.

Versions and compatibility

PAROL6 release	Released	Control board	Control board firmware	Python API (parol6)	Commander software	Grippers
1.1	2026-11-04	v2.0 (pluggable terminal blocks); v1.2 (spring contacts) is functionally identical	1 (VERSION 1 in constants.h), unchanged since release 1	0.2.6 to 0.9.1 (the reference describes 0.9.1); Python 3.11 or newer	main branch (Python 3.10 or 3.11)	SSG-48 (Spectral Micro firmware 101 or newer), MSG (STEPFOC firmware 103 or newer), pneumatic MHZ2-16D, vacuum
1	2025-03-08	v2.0 (pluggable terminal blocks); v1.2 (spring contacts) is functionally identical	1 (VERSION 1 in constants.h), main branch	0.2.6 to 0.9.1 (the reference describes 0.9.1); Python 3.11 or newer	main branch (Python 3.10 or 3.11)	SSG-48 (Spectral Micro firmware 101 or newer), MSG (STEPFOC firmware 103 or newer), pneumatic MHZ2-16D, vacuum

Reading the table

- **Control board** v2.0 has pluggable terminal blocks; v1.2 had spring-loaded contacts. They are functionally identical and run the same firmware.
- **Control board firmware** is released with the robot (`#define VERSION 1`). The serial protocol it speaks is the one every PC program on this site uses.

- **Python API**: the reference on this site is read from the version named on the Programming landing page. Older 0.x versions have fewer commands; the API is pre-1.0, so a minor version can change an interface, as the release notes say.
- **Commander software** has no releases; the main branch is what you clone. The `experimental_kinematics` branch fixes most Cartesian jogging errors but can make dangerous moves: main is the safe one.
- **Grippers**: the firmware's gripper protocol works with every released firmware of both grippers.

Community projects

Software, hardware modifications and experiments built around PAROL6 by its community. They are third-party projects, not maintained by Source Robotics: use them at your own risk and check each one's licence and documentation. The list is kept in the robot repository's community_projects.md, where a pull request adds yours.

Control software and GUIs

Project	What it is
Waldo Commander	A browser-based commander: Python programs with an editor and debugger, 3D simulation, teach by demonstration, and an MCP server so that an LLM can drive the arm. Built on the Python API through the waldoctl abstraction
Parol6_Better_Command	The commander software refactored for easier editing, with vision
PAROL6 Pilot	A web GUI
PAROL6_GUI	An alternative desktop GUI
parol6_commander	A web GUI on NiceGUI, ROS 2 and MoveIt 2, for calibrating, jogging and programming
Parol6-firmware	Alternative motion firmware for a Teensy 4.1 with ROS 2 over micro-ROS
parol6_ros2_control	The `ros2_control` hardware interface and bridge used by the MoveIt configuration's real-robot launch

Vision and AI

Project	What it is
PAROL6-Python-API-Gemini-Vision	Controlling PAROL6 with the Gemini LLM and a camera
Tic-Tac-Toe-Robot-Project-Parol6	A vision-guided tic-tac-toe player

Hardware

Project	What it is
parol6-printed-eg17	A printed gearbox alternative

The legacy Python API

PAROL6-python-API-Alvar is the first Python API, a headless commander with a text-over-UDP client, kept as it is for the projects built on it. New work should use the current Python API.

Resources

Glossary

After-homing pose

Where homing leaves the robot: the standby pose turned 90° about joint 1, joints at 90°, −90°, 180°, 0°, 0°, 180°. The firmware (`standby_position`) and the Python API (`STANDBY_ANGLES_DEG`) call this pose standby. The witness marks line up in it.

Blending (blend radius)

Passing through a waypoint without stopping: the path cuts the corner within the given radius. The Python API's `r` parameter.

Commander software

Source Robotics' desktop program for PAROL6: a GUI with jogging, a simulator, a program editor for RBTscript, I/O and gripper control, talking to the control board over USB.

Control board

The electronics inside the base: an STM32 microcontroller, six TMC5160 stepper drivers, the inputs, outputs, E-stop input, CAN and USB. It runs the low-level firmware; the PC runs the high-level software.

Controller (Python API)

The `parol6-server` process: it talks to the control board over serial at 100 Hz, plans the motion and accepts commands over UDP from any number of clients.

Direction, positive

The direction in which a joint's angle increases, shown in the figure on *Joint limits and directions*. The commander's left arrows jog positive.

E-stop

The normally closed button in its own printed box, on a cable to the robot. Pressing it, unplugging it or cutting its wire opens the circuit, and the robot stops. It must be connected for the robot to run.

Firmware

The program on the control board. The *main* firmware runs the robot with the PC software; the *test* firmware checks the board and the wiring from a serial terminal.

Flange

The mounting face at the end of joint 6, where the tool is screwed on. With nothing on it, the TCP is the flange face.

Forearm (lower arm)

The link from the elbow (joint 3) to the wrist. It carries joint 4. Forearm and lower arm mean the same link.

Homing (referencing)

Finding each joint's position after power-up by driving it to its limit switch or sensor, then moving a known number of steps to the after-homing pose. Needed after every power-up; the robot does not know where it is before.

Jogging

Moving a joint, or the tool along an axis, at a chosen speed for as long as a button is held or a command repeats.

Joint names

Joints 1 to 6 by number, from the base up. Other names for the same joints: base, shoulder and elbow for 1 to 3, and wrist 1, wrist 2 and wrist 3 for 4 to 6, as in the robot model and ROS.

Limit switch, inductive sensor

The references for homing: mechanical switches on joints 2, 3 and 5, inductive sensors on joints 1, 4 and 6.

Mastering

Fine-tuning the number of steps from a joint's limit switch to its witness mark, so that a homed robot stands exactly in the after-homing pose.

Motion profile

The shape of the speed curve of a move: trapezoidal, polynomial, time-optimal, and so on. The Python API offers several; the commander uses trapezoidal and quintic polynomial.

Parked pose

The arm folded in over the base, joints 2 and 3 toward their limit switches: how the robot rests when it is off, and the pose to switch off in, since the arm has the least distance to fall from it. The commander's **Park** button writes a program that moves there (90°, −144.683°, 108.171°, 2.222°, 25.003°, 180°).

Pose

The position and the orientation of the tool together: six numbers, x, y, z in millimetres and three angles in degrees. The standby, after-homing and parked poses are poses of the whole arm, given as six joint angles.

RBTscript

The commander software's program language: one command per line, `MoveJoint`, `MoveCart`, `Delay`, `Gripper` and so on.

Standby pose

The pose of the Denavit-Hartenberg table and the reference of the joint angles: joints at 0°, −90°, 180°, 0°, 0°, 180°, the upper arm vertical, the forearm horizontal, the arm pointing away from the electronics case at the back of the base. After homing the robot stands in it turned 90° about joint 1 (joint 1 at 90°): the after-homing pose, which the Python API calls its standby.

Stepper motor

The motors of PAROL6: open-loop motors that move in steps, 200 full steps per turn, 6400 with the 32 microsteps the control board uses. The position is counted, not measured.

Test firmware

The firmware every control board from the shop carries when it ships. It reports each driver at power-on and answers a few text commands (`#LIMIT`, `#IO`) from a serial terminal, and it turns all six motors slowly from the moment it starts until the power is switched off. Replace it with the main firmware before the robot is used (The test firmware).

TCP (tool center point)

The point on the tool the robot positions and the origin of the tool frame. Selecting a tool in the software sets it; an offset moves it.

TRF, WRF

The tool reference frame, fixed to the tool, and the world reference frame, fixed to the base. Cartesian jogs and relative moves can be made in either.

Upper arm

The link from the shoulder (joint 2) to the elbow (joint 3).

Witness marks

Holes in the printed parts of neighbouring links that line up, so that a pin passes through, when the joint stands exactly in the after-homing pose.

Wrist 1, wrist 2, wrist 3

Other names for joints 4, 5 and 6. See *Joint names*.

Downloads and files

Everything to download for PAROL6, at the versions these pages describe. The versions come from one file, `versions.yml`, so this page changes with them; which versions work together is in Versions and compatibility.

Firmware and software

What	Version	Download	How to install
Control board firmware, a PlatformIO project	V1.1	PAROL6 control board main software	Firmware
Control board test firmware, and the CAN test	the same release	PAROL6 control board test code	The test firmware
ST-Link USB driver, for uploading from Windows		STSW-LINK009	Firmware: tools
Python API (`parol6`)	0.9.1 (2026-10-03)	PAROL6-python-API	Installation
Commander software	main (2025-09-23)	PAROL-commander-software	Installation
Robot model: URDF and MuJoCo, with the meshes and every tool	2.15.1	robot_descriptions: PAROL6	
ROS 2 MoveIt configuration	main branch	PAROL6-ROS2-MOVEIT	ROS 2

The version links open the files at exactly that release. The commander software has no releases: its main branch, as of the date given, is the one these pages describe. Known faults in each are on Known issues.

CAD and build files

The latest files, on the main branch of the robot repository.

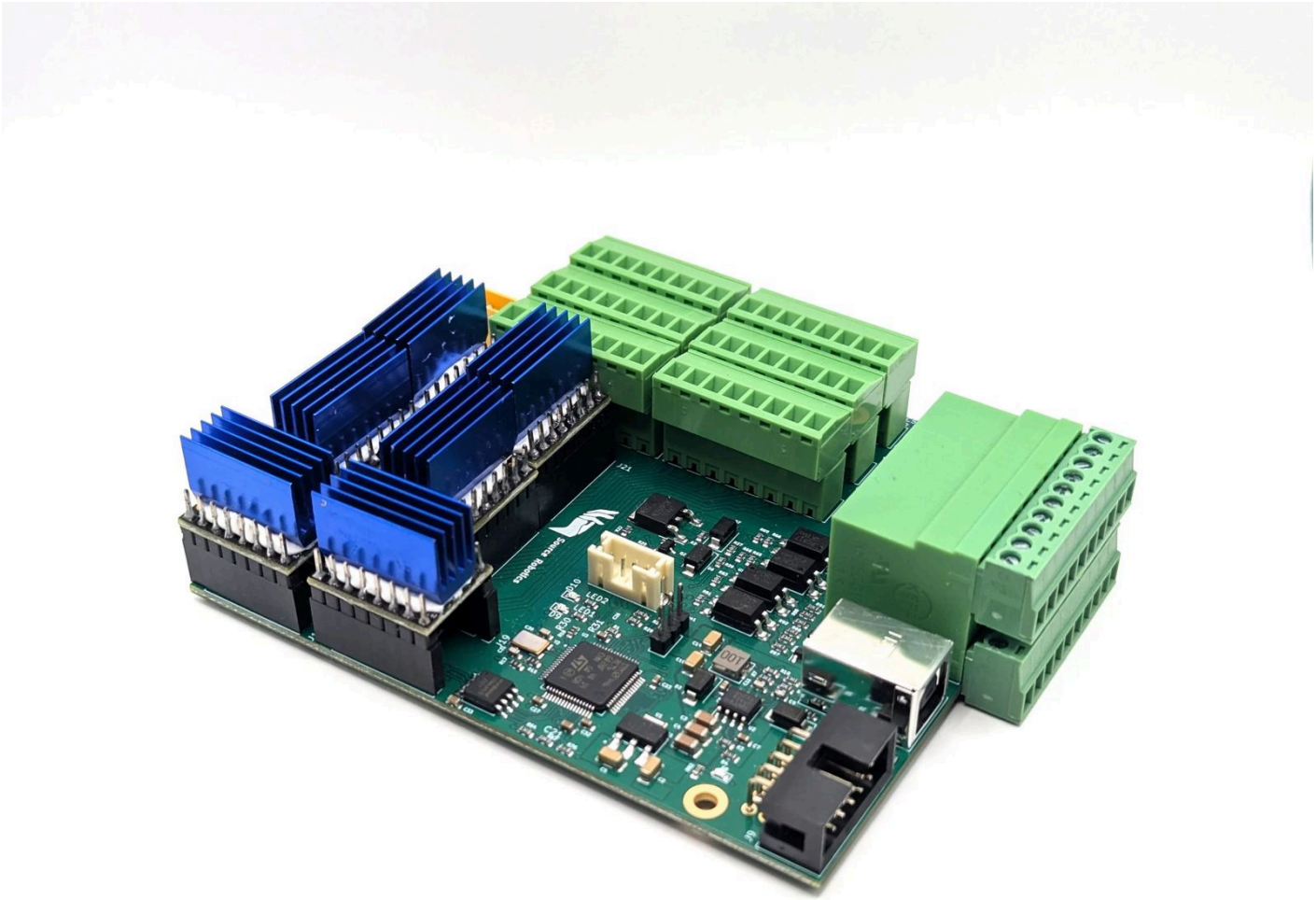
What	Where
Printable parts (STL), by link, with the small print bed variants	STL
Mounting plates (STEP)	STL/mounting plates
Bill of materials with purchase links and reference images	BOM, and on this site: Bill of materials
Building instructions (PDF), small print bed modification, PETG notes	Building instructions
Print settings table	Print table
Assembly and setup videos	YouTube playlist

More

What	Where
Safety warning and disclaimer	SAFETY_WARNING_AND_DISCLAIMER.md
Community projects	community_projects.md
Gripper documentation	SSG-48, MSG
This documentation	PAROL-docs
The old manual (2023 to October 2026), kept for reference	Old manual
Shop	source-robotics.com
Community	Discord, Hackaday, blog

Control Board

PAROL6 Control Board Manual



Document number: SR-2002

PAROL6 release: 1.1

Revision: 2026.C

Published: 2026-11-05

Applies to: PAROL6 control board v2.0 and v1.2, PAROL6 release 1.1

Original instructions

The PAROL6 control board is the controller of the PAROL6 arm: a 32-bit microcontroller, six stepper drivers, the inputs and outputs, the E-stop input, CAN and USB on a board the size of a pack of playing cards, which is what lets PAROL6 run without a control cabinet. This manual covers the board itself: its specifications, connectors and pinout, the stepper drivers and their installation, the I/O and E-stop wiring, the CAN bus, uploading the firmware, the test firmware, and the serial protocol the PC software uses. How the robot is used is in the Robot Manual; how it is programmed in the Programming Guide.

The board

Overview, Safety, Technical data, Connectors and pinout, Stepper drivers.

Wiring

Inputs, outputs and E-stop, CAN bus.

Firmware

Uploading the firmware, the test firmware, the serial protocol, troubleshooting.

1. Overview

The control board sits in the base of PAROL6 and does the low-level work: it generates the step pulses for six stepper motors, runs the homing sequence, reads the limit switches, the inputs and the E-stop, switches the outputs, drives an electric gripper over CAN, and exchanges a data packet with the PC over USB 100 times a second. It plans nothing: the PC computes every trajectory and sends the board the joint speeds and positions to execute, cycle by cycle.

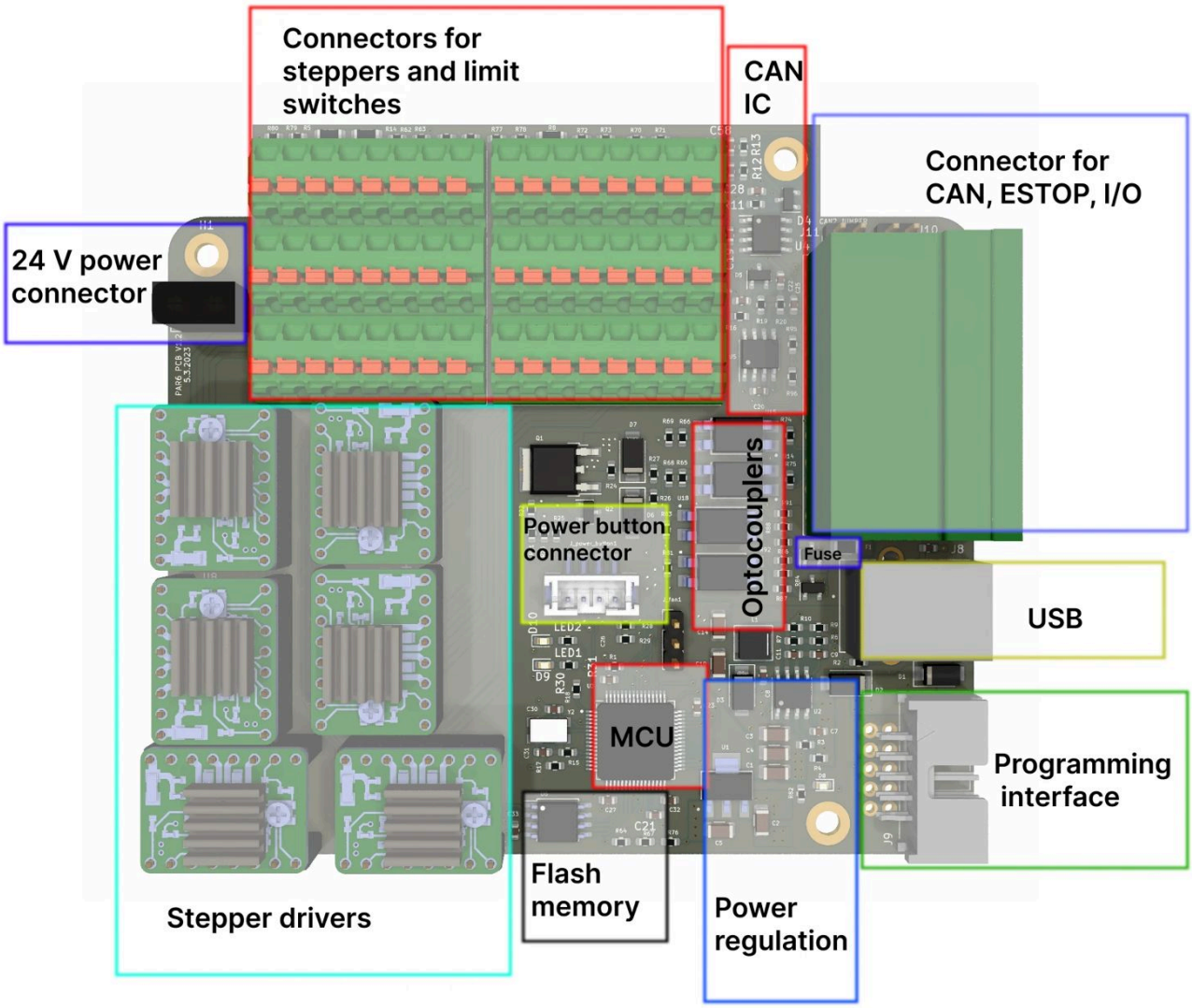


Figure 1.1: Features of the control board.

It works out of the box with PAROL6 and the commander software or the Python API, and it drives any six-axis stepper arm that is wired like PAROL6.

Versions

Version	Status	Difference
v2.0	Current, in the shop	Pluggable terminal block connectors
v1.2	Discontinued	Spring-loaded Phoenix contacts

The two are functionally identical and run the same firmware.

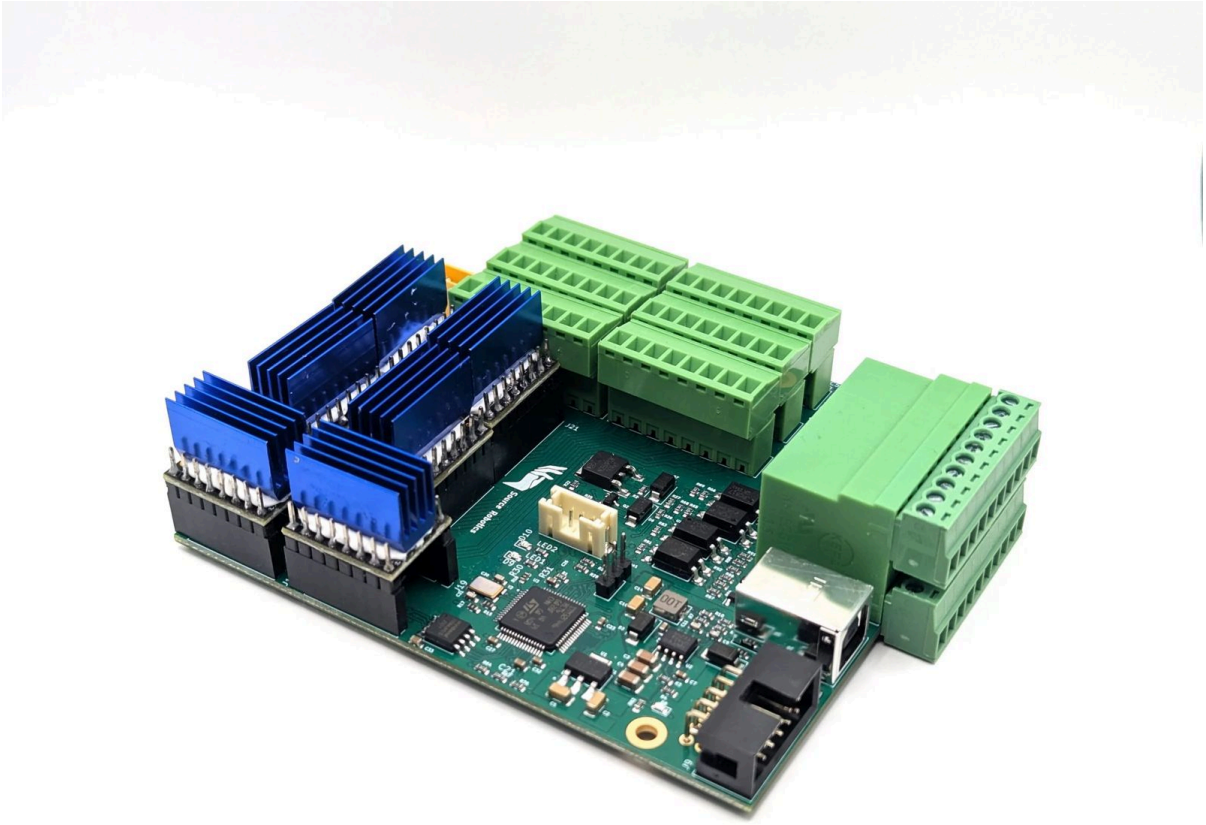


Figure 1.2: Control board v2.0.

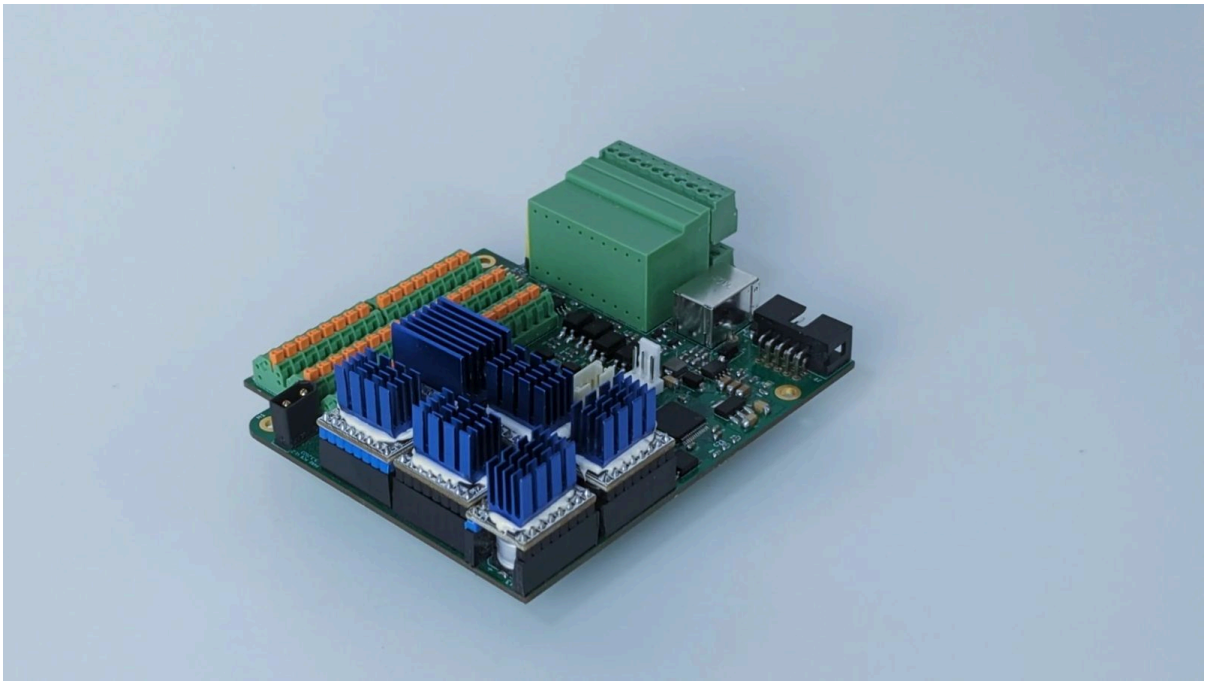


Figure 1.3: Control board v1.2 (discontinued).

Testing

Every board is tested end to end before it ships, and comes with the test firmware preloaded, so that you can check the wiring of your robot before the main firmware goes on (The test firmware).

Where it is published

The firmware, main and test, is in the robot repository: PAROL6 control board main software and PAROL6 control board test code. The board is sold in the shop.

The board schematics and PCB design are not available as open source materials due to licensing agreements and confidentiality terms established with our development partners. These proprietary design elements are subject to contractual obligations that restrict their public distribution.

2. Safety

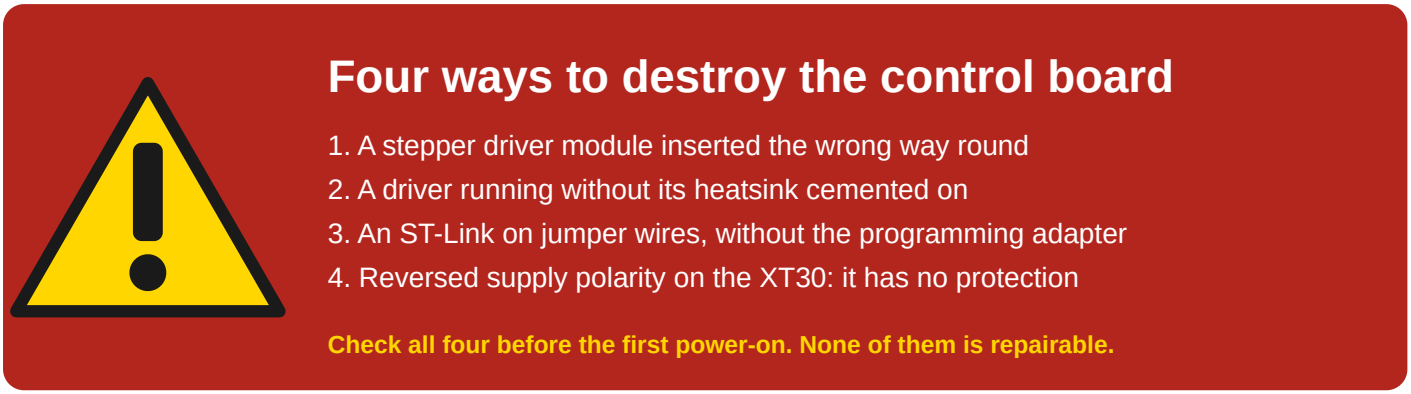


Figure 2.1: Four ways to destroy the control board.

Four ways to destroy the board

1. **A stepper driver module inserted the wrong way round.** Check the orientation against the DIAG pins before the first power-on (Stepper drivers).
2. **A driver without its heatsink cemented on.** The TMC5160 modules need the heatsink and the fan.
3. **An ST-Link connected with jumper wires without the programming adapter.** Use the adapter, or only the four pins named on Firmware.
4. **Reversed supply polarity.** The XT30 power input has no reverse-polarity protection.

- The board runs on 24 V, which is safe to touch; the power supply you connect it to is the mains device.
- The outputs switch up to 1 A at up to 48 V; the inputs take up to 50 V. The isolation between them and the robot holds only if you power them from an isolated supply (Inputs, outputs and E-stop).
- The E-stop input is read by the firmware and acted on by the PC software; it does not cut the motor power (Emergency stop in the Robot Manual).
- With 3.3 V on the programming header the board turns on when the supply is connected and cannot be switched off with the button. Disconnect the 24 V supply before you connect the programmer.
- The drivers are hot after use; the fan must run whenever the motors are energised.
- Static electricity damages the microcontroller and the drivers. Unplug the supply and touch earthed metal before you handle the board or a driver module, and hold them by the edges.

3. Technical data

Hardware

Microcontroller	STM32F446RE: Arm Cortex-M4 with DSP and FPU, 180 MHz, 512 KB flash
Stepper drivers	6 × TMC5160 modules, SPI configured, 10 to 35 V, with stall, short and temperature protection; currents below
Microstepping	32 (set by the firmware)
Inputs	2, isolated, 12 to 50 V, 24 V nominal
Outputs	2, isolated, 48 V maximum, 0.5 A continuous, 1 A peak; 2 A fuse for the non-isolated mode
E-stop	One input on a dedicated pin, two connectors in parallel, normally closed contact
CAN	2 controllers; CAN 1 with its transceiver, CAN 2 transceiver not populated
USB	USB 2.0 full speed, a virtual serial port; the PC software opens it at 3 000 000 baud, a setting USB ignores
Additional memory	W25Q64FV SPI flash, 64 Mbit
Programming	SWD through the 2×5-pin header (ST-Link with the programming adapter)
Fan	5 V, 3-pin header, 0.3 A maximum, not PWM
Power button	Soft switch: the button tells the firmware to switch the 24 V stage on and off
Supply voltage	18 V minimum, 24 V maximum; 24 V is recommended for optimal robot performance
Supply voltage sense	The firmware reads the supply voltage (VBUS)
Size	Three M3 mounting holes; dimensions in the drawing below

Operating limits

Stepper driver rated current	2.5 A per motor
Stepper driver maximum current	3.6 A, short bursts or with extreme cooling only
Driver overtemperature	The driver shuts its outputs down and its joint goes limp; firmware 1 does not tell the PC
Isolated input voltage	12 V minimum, 24 V nominal, 50 V maximum
Isolated output voltage	48 V maximum
Isolated output current	1 A maximum
Fan current	0.3 A maximum; around 0.1 A recommended

The PAROL6 firmware runs the motors well below the drivers' ratings: 0.8 to 1.7 A (Currents and temperatures).

Dimensions

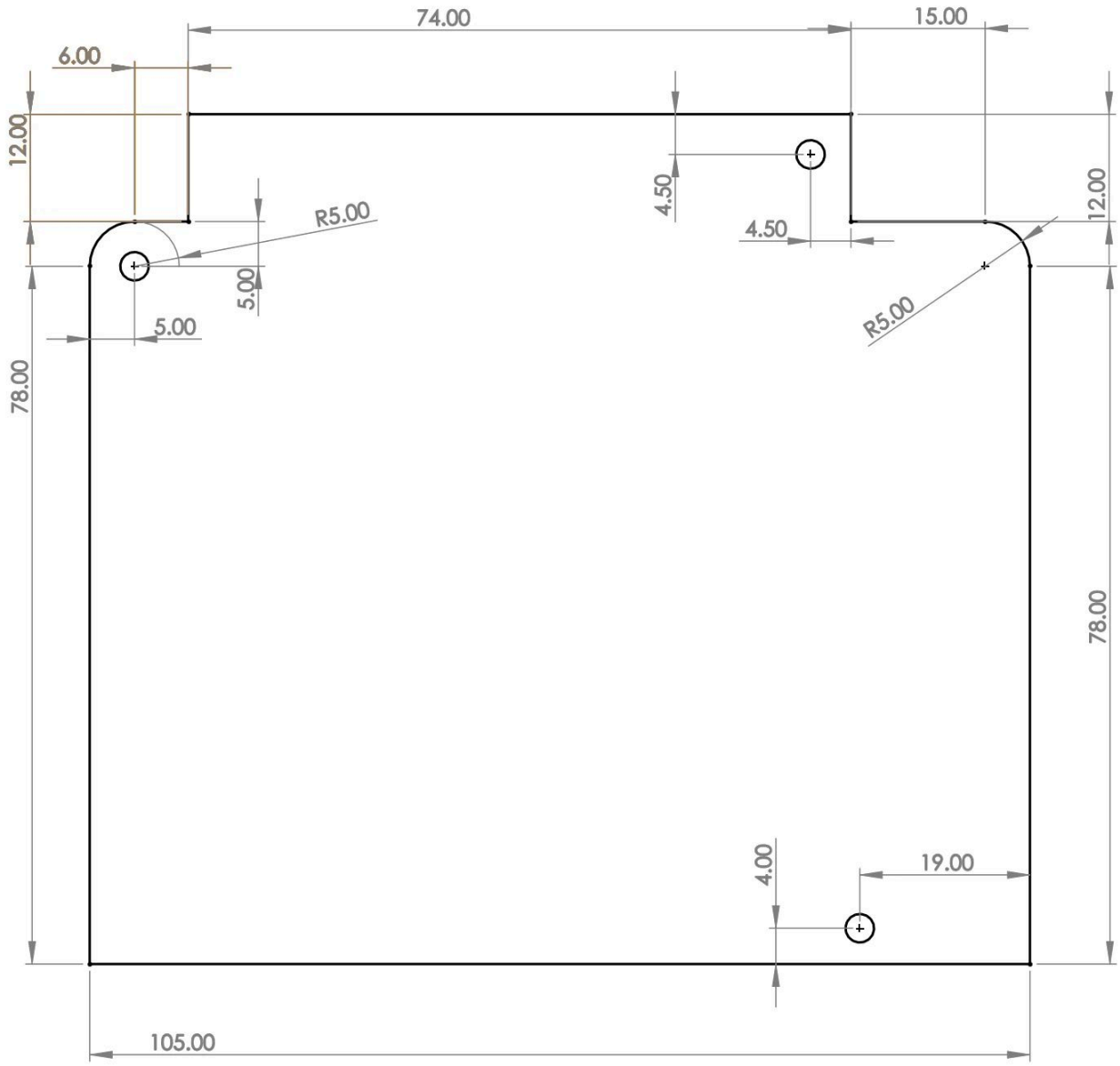
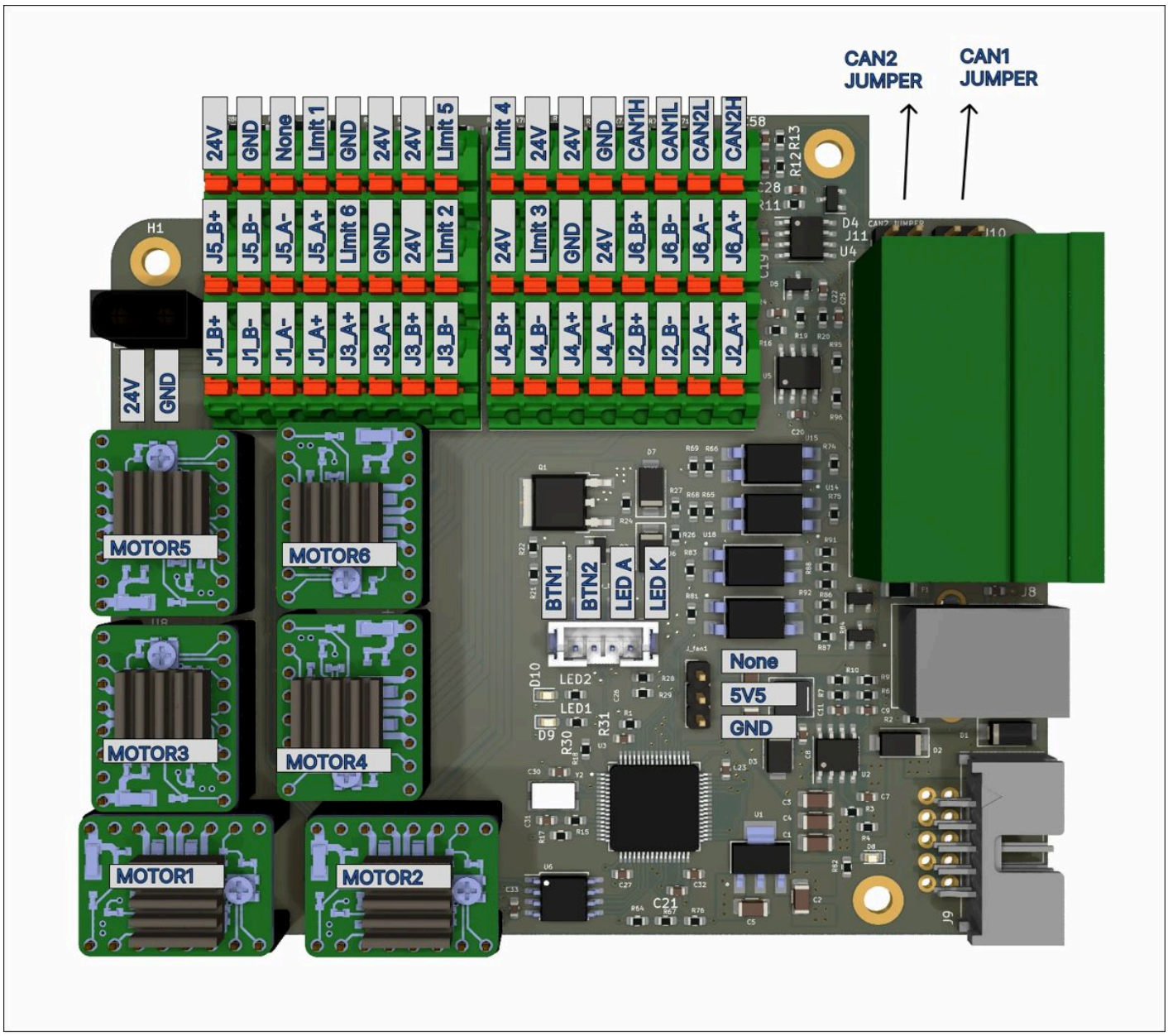


Figure 3.1: PCB dimensions in millimetres.

Mounting and cooling

Three M3 holes. The drivers need active cooling: a 40 × 40 × 20 mm 5 V fan, a Noctua NF-A4x20 or any 5 V fan of that size that is not PWM controlled, blowing over the heatsinks. In PAROL6 the fan sits in the base's fan cover.

4. Connectors and pinout



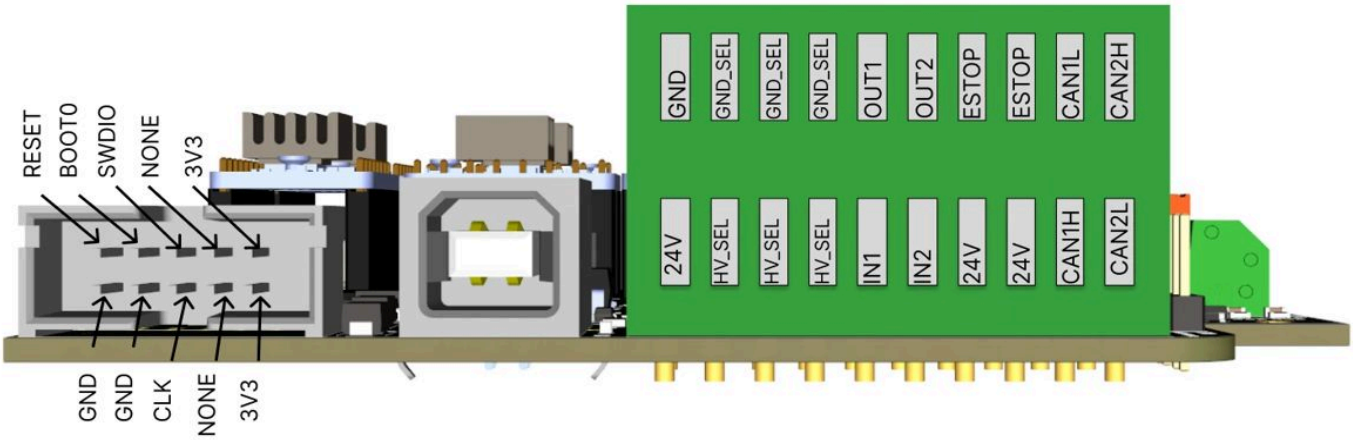


Figure 4.1: The connections of the control board.

Connectors

Connection	Connector
24 V power	XT30 male
Cooling fan	3-pin, 2.54 mm pitch, 5 V
Power button	JST PH 4-pin (B4B-PH-SM4-TB)
USB	USB type B
Programming	2×5-pin, 2.54 mm pitch (SWD)
Motors	6 × 4-pin terminal blocks, one per driver
Limit switches, inputs, outputs, E-stop, CAN, 24 V out, GND, HV_SEL, GND_SEL	Pluggable terminal blocks along the edges
CAN termination	Two jumpers, CAN1 JUMPER and CAN2 JUMPER: on, each connects a 120 Ω resistor across its bus (CAN bus)

Pin definitions

The microcontroller pins as the firmware defines them (`iodefs.h`), for anyone who writes firmware or probes the board:

```
#define PUL1 PC6     // step, joint 1          #define DIR1 PB15    // direction, joint 1
#define PUL2 PA10    // step, joint 5          #define DIR2 PA1     // direction, joint 5
#define PUL3 PC0     // step, joint 6          #define DIR3 PC1     // direction, joint 6
#define PUL4 PC3     // step, joint 4          #define DIR4 PA0     // direction, joint 4
#define PUL5 PC9     // step, joint 3          #define DIR5 PA8     // direction, joint 3
#define PUL6 PC5     // step, joint 2          #define DIR6 PB1     // direction, joint 2

#define SELECT1 PC7  // SPI chip select of driver 1 (joint 1)
#define SELECT2 PA9  // driver 2 (joint 5)
#define SELECT3 PC15 // driver 3 (joint 6)
#define SELECT4 PC2  // driver 4 (joint 4)
#define SELECT5 PC8  // driver 5 (joint 3)
#define SELECT6 PC4  // driver 6 (joint 2)
#define GLOBAL_ENABLE PA3   // ENABLE of all drivers

#define LIMIT1 PC12  // limit input of joint 6      #define LIMIT4 PD2   // limit input of joint
4
#define LIMIT2 PB3   // limit input of joint 2      #define LIMIT5 PB4   // limit input of joint
5
#define LIMIT3 PA15  // limit input of joint 3      #define LIMIT6 PC11  // limit input of joint
1

#define MISO PA6, MOSI PA7, SCK PA5   // SPI to the six drivers and the flash
#define FLASH_SELECT PA4              // chip select of the flash memory

#define LED1 PB2, LED2 PB10           // the two LEDs
#define SUPPLY_ON_OFF PC10            // switches the 24 V power stage
#define SUPPLY_BUTTON_STATE PC14      // reads the power button
#define INPUT1 PB6, INPUT2 PB5        // IN1, IN2 terminals
#define OUTPUT1 PC13, OUTPUT2 PB7     // OUT1, OUT2 terminals
#define ESTOP PB14                    // E-stop terminals
#define VBUS PB0                      // supply voltage sense
#define USB_D_PLUS PA12, USB_D_MINUS PA11
#define CAN1TX PB9, CAN1RX PB8        // CAN 1 (the gripper bus)
#define CAN1TX PB13, CAN1RX PB12      // CAN 2's pins (transceiver not populated)
```

`iodefs.h` defines `CAN1TX` and `CAN1RX` twice, the second time with CAN 2's pins, PB13 and PB12. The CAN driver does not use these names: `CANInit()` in `CAN.cpp` picks the pins itself.

Wiring the robot

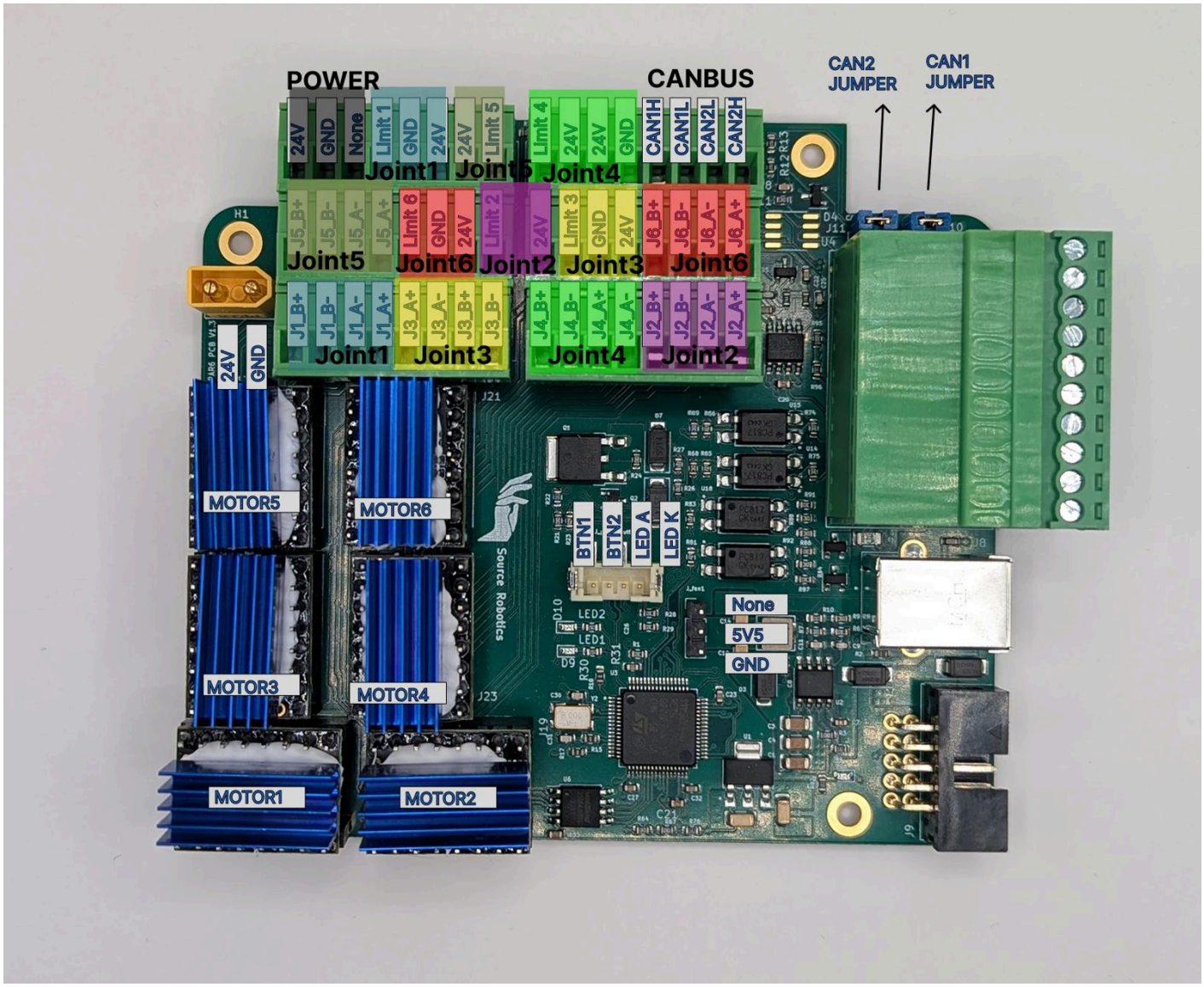
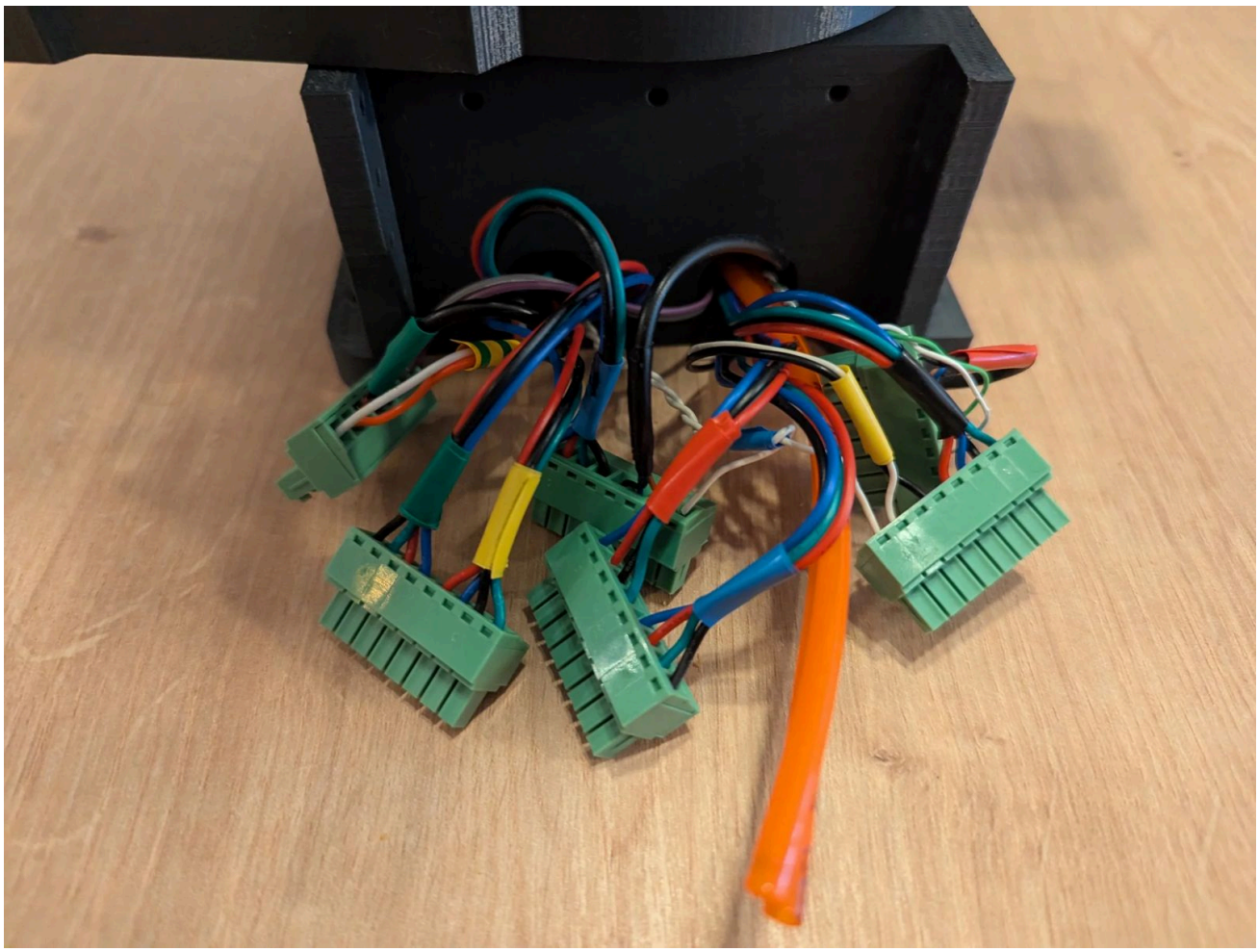


Figure 4.2: Wiring diagram of the robot to the control board.

Limit switches to 24 V and their LIMIT terminal; inductive sensors to 24 V, GND and the LIMIT terminal (brown, blue, black). The gripper's M8 socket to 24 V, GND, CAN H and CAN L. The finished wiring in PAROL6's base:



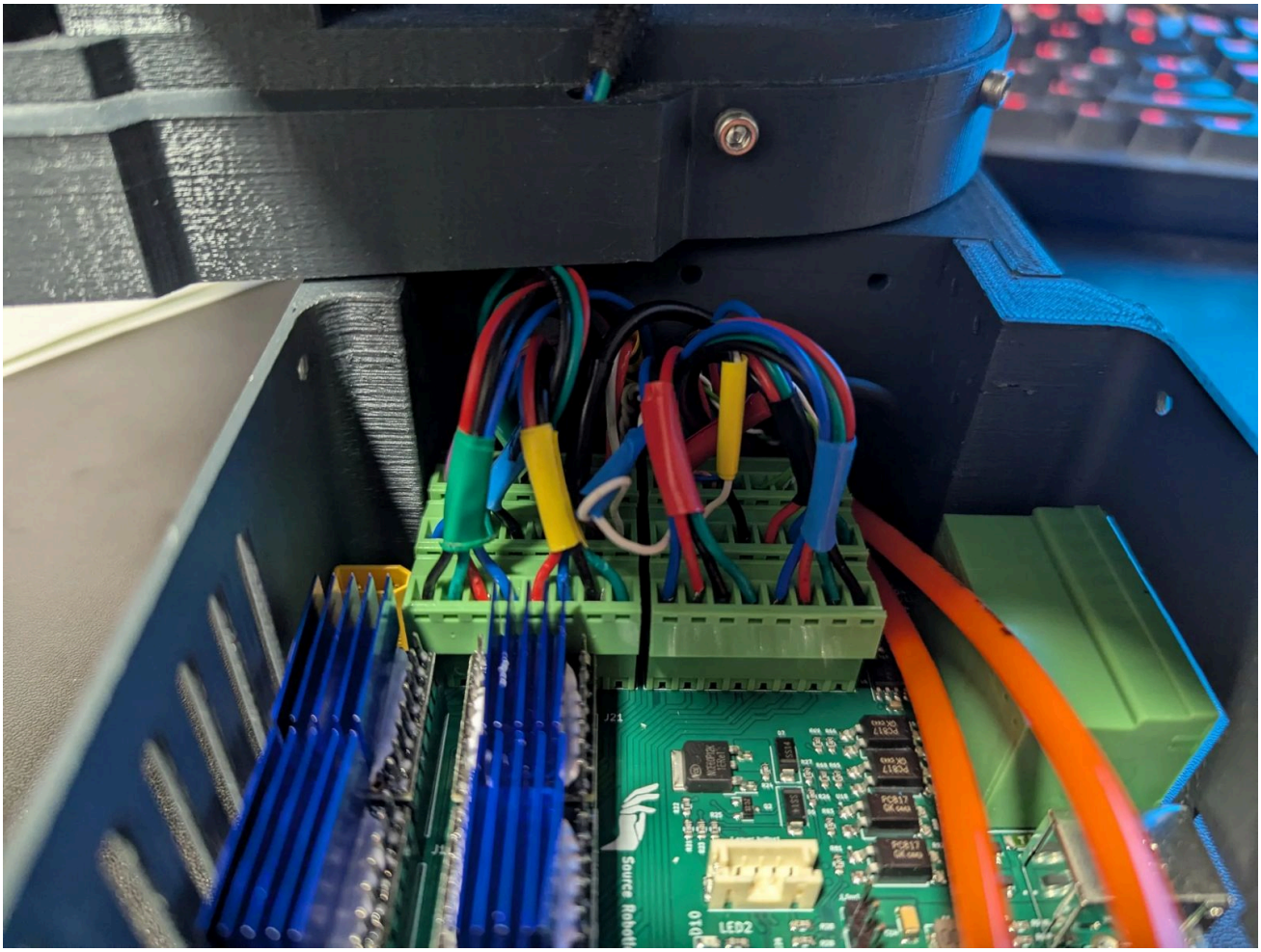


Figure 4.3: The connectors at the base.

5. Stepper drivers

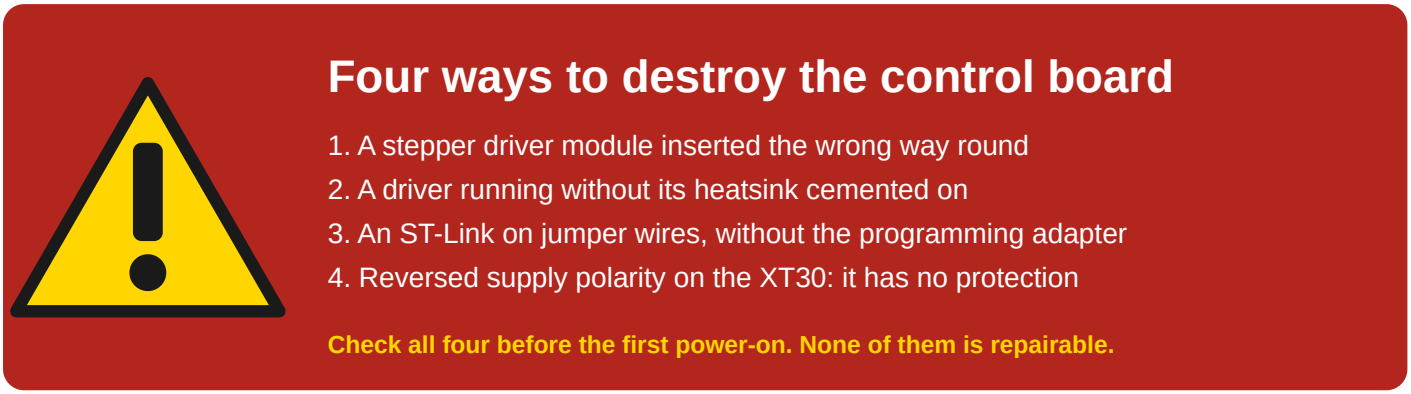


Figure 5.1: Four ways to destroy the control board.

The board takes six TMC5160 driver modules in the StepStick format, configured over SPI by the firmware.

Orientation

A driver module inserted the wrong way round destroys the board the moment power is applied, permanently. The orientation is given by the two DIAG pins: they must point at the red arrows in the figure. Check all six before every power-on after the drivers were touched.

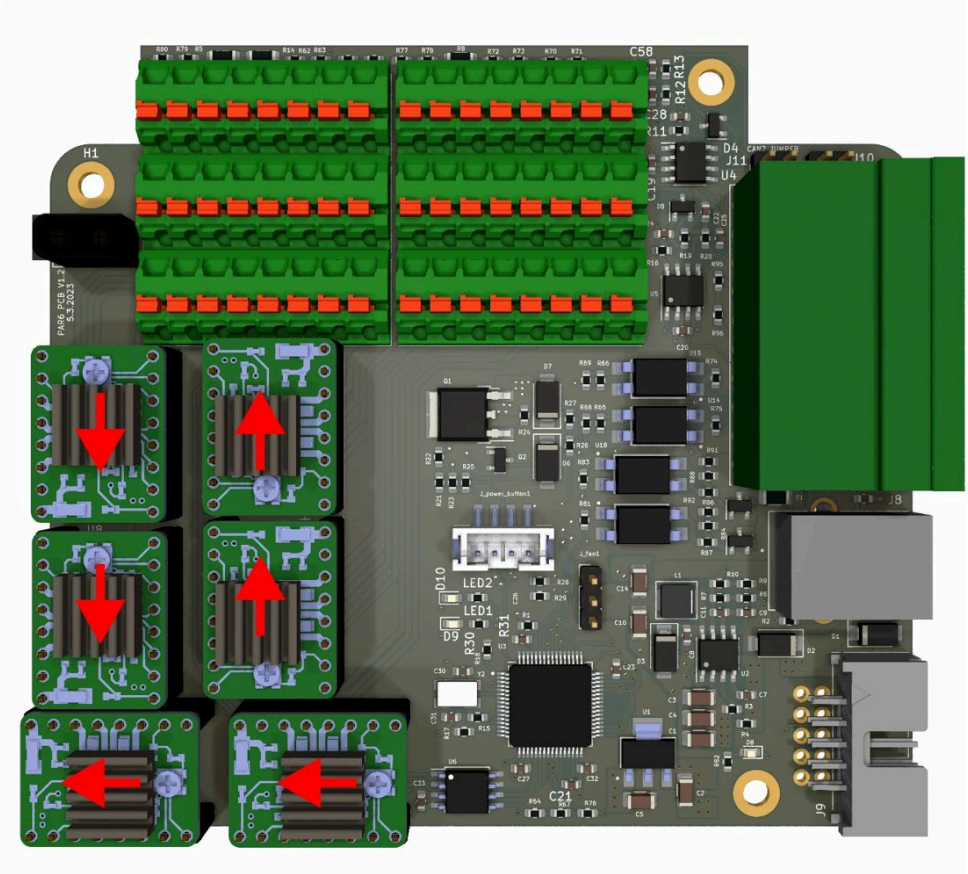


Figure 5.2: Driver orientation: the DIAG pins at the red arrows.

Which drivers

Only the TMC5160 modules listed in the BOM are tested; a board bought with drivers has them fitted. A module from elsewhere must have the pinout in the figure, and:

- **Remove the two DIAG pins** if they are soldered: the capacitors on the board are in their way and the module will not seat.
- **Cement the heatsink** to the driver with thermal cement (not paste, not tape); a driver without a bonded heatsink overheats and fails, and takes the board with it.

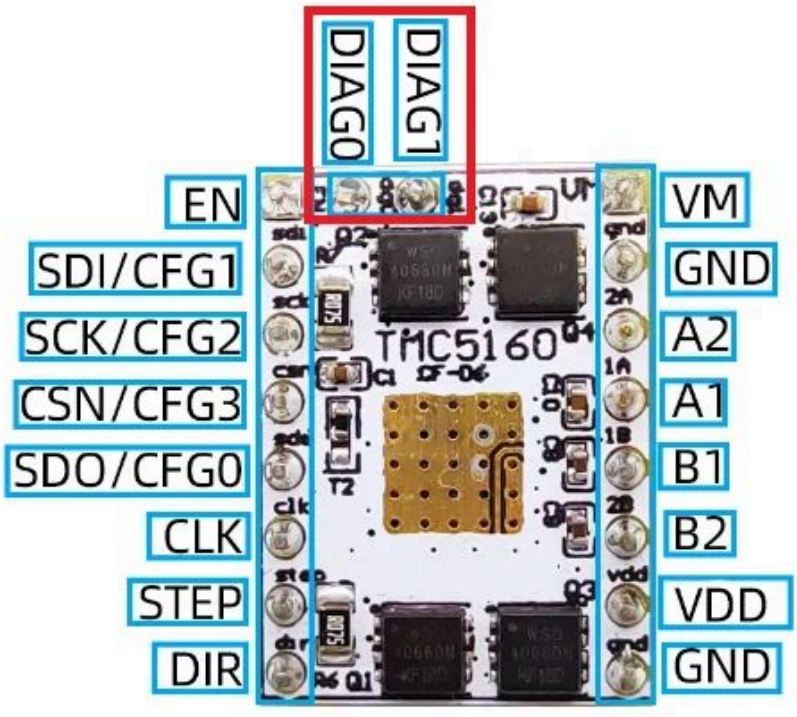


Figure 5.3: The TMC5160 module pinout.

Current

The firmware sets each driver's running and holding current for its joint (Currents and temperatures and Firmware parameters). The modules' potentiometer, if they have one, is not used.

Cooling

The fan over the heatsinks must run whenever the motors are energised. A driver that gets too hot shuts its outputs down and its joint goes limp, and firmware 1 does not tell the PC; in PAROL6's base with the fan they stay far below that.

Checking a driver

The test firmware reads every driver over SPI about three seconds after power-on and prints whether it answers (`joint N Connection is: GOOD` or `BAD`) with its currents and fault flags, then turns every motor slowly (The test firmware). A driver that does not answer is not seated, dead, or in the wrong slot.

6. Inputs, outputs and E-stop

The isolated I/O

Two inputs and two outputs on optocouplers, with their own supply rails **HV_SEL** and **GND_SEL** on the terminal blocks. You decide what feeds those rails:

Isolated supply

An external supply's negative to GND_SEL, positive to HV_SEL. The I/O then runs at that voltage, 24 V nominal and at least 12 V for the inputs, and a short or a surge on a sensor cannot reach the robot's electronics.

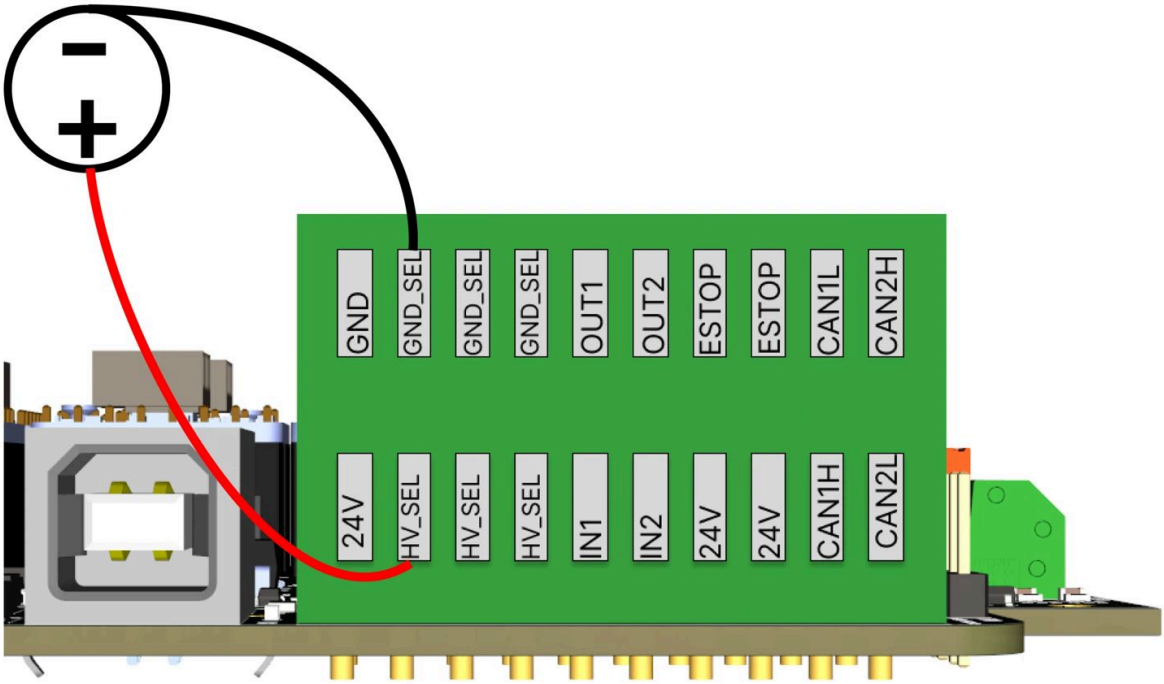


Figure 6.1: Isolated supply wiring.

The robot's 24 V

Link the board's 24 V and GND terminals to HV_SEL and GND_SEL. The I/O shares the robot's supply, with a 2 A fuse protecting the outputs. Simpler, and enough for a pneumatic valve.

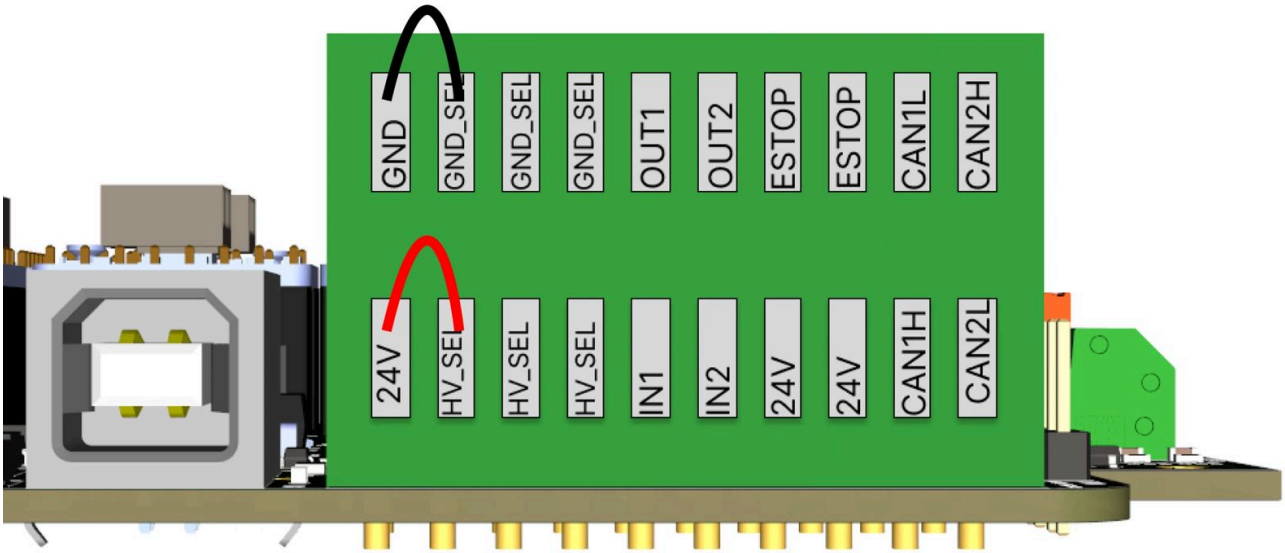


Figure 6.2: Non-isolated wiring.

The E-stop is on the robot's own supply in both cases.

Inputs

Between HV_SEL and IN1 or IN2: a switch, a push button, a relay contact, or the signal of a sensor fed from HV_SEL and GND_SEL. 12 to 50 V. The firmware reads them every cycle and reports them in the I/O byte of the packet.

Outputs

Between GND_SEL and OUT1 or OUT2: a relay coil, a valve, a lamp. 0.5 A continuous, 1 A peak, 48 V maximum, with a flyback diode across any coil. The firmware sets them from the I/O byte of the PC's packet.

The E-stop

Two connectors on the same input, for a normally closed contact. The firmware reads the input every cycle and reports it in the I/O byte; the PC software stops the robot when it sees the contact open (Emergency stop).

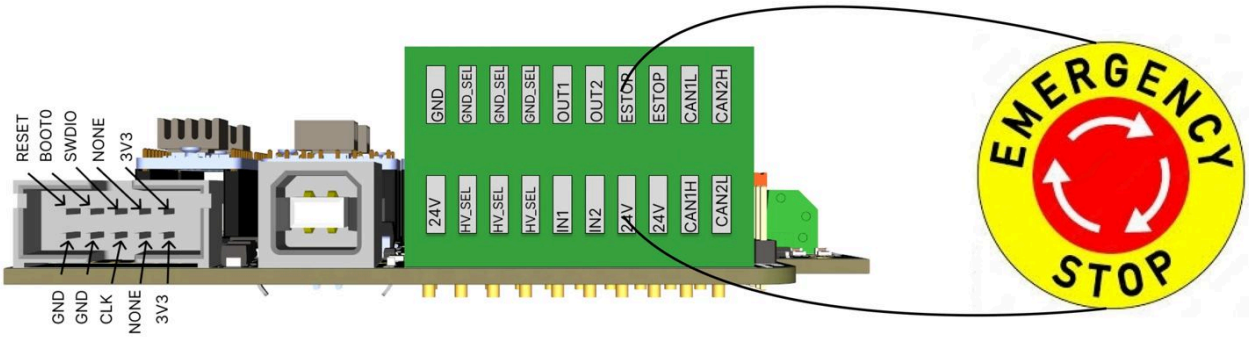


Figure 6.3: E-stop wiring.

- The contact must be **normally closed**: an open circuit means stop, so an unplugged button or a cut wire stops the robot too. A normally open button does not work.
- Both connectors share one microcontroller pin: a second button goes in series on the other connector, so pressing either button stops the robot.
- Any NC switch works for a bench setup. The BOM's 19 mm E-stop sits in its own printed box (`STL/ESTOP`), on a cable as long as the setup needs.

The power button

A momentary push button with an LED, on the JST PH connector. A press switches the 24 V stage on; a press while on switches it off. The LED is fed from the button connector. The firmware reads the button on every pass of its main loop.

7. CAN bus

CAN 1	Transceiver fitted; terminals CAN H and CAN L; the gripper bus in PAROL6
CAN 2	Controller on the microcontroller, transceiver not populated
Bit rate	1 Mbit/s, set by the firmware
Termination	One jumper per bus, **CAN1 JUMPER** and **CAN2 JUMPER**: with the jumper on, a 120 Ω resistor is connected between CAN H and CAN L; with it off, the bus is not terminated on the board

In PAROL6, CAN 1 runs with 24 V and GND to the M8 socket at the forearm. The firmware addresses one gripper at the CAN ID the PC's packet names (0 as the grippers ship), sends it a move, a calibrate, a clear-error or an empty status frame every cycle, and relays its answer to the PC: CAN bus in the Robot Manual, and the gripper manuals for the messages themselves.

A CAN bus is terminated at both of its ends, 120 Ω at each. In PAROL6 the ends are the board, with its CAN1 jumper, and the gripper, with its own termination switch. Nodes in the middle of a longer bus stay unterminated. The jumpers are on the board figure in Connectors and pinout.

8. Firmware

Two firmware programs exist for the board, both in the robot repository as PlatformIO projects:

	Folder	For
Main firmware	`PAROL6 control board main software`	Running the robot with the commander software or the Python API
Test firmware	`PAROL6 control board test code`	Checking the board and the wiring from a serial terminal (The test firmware)

A board from the shop carries the test firmware. Upload the main firmware before you use the robot.

Tools

1. VS Code with the **PlatformIO** extension from its marketplace.
2. A clone of the robot repository.
3. An **ST-Link** programmer and the PAROL6 programming adapter, or an ST-Link whose cable carries exactly SWCLK, SWDIO, GND and 3V3 to the pins of the 2×5 header in the table below.
4. On Windows, the ST-Link USB driver (STSW-LINK009) if PlatformIO cannot find the programmer.

Danger

Use the programming adapter, or the four named pins only. An ST-Link wired to the header with jumper wires without the adapter has destroyed boards: other pins of the header are not for it.

The 2×5 programming header, as labelled in the side view of the connections figure, left to right, the header to the left of the USB port:

Row					
Top	RESET	BOOT0	**SWDIO**	none	**3V3**
Bottom	**GND**	GND	**CLK** (SWCLK)	none	3V3

An ST-Link needs four of them: SWDIO, CLK, GND and 3V3. Leave RESET and BOOT0 unconnected.

Uploading

1. **Disconnect the 24 V supply** from the robot.
2. Connect the adapter to the 2×5 header next to the USB port, and the ST-Link to the PC. The adapter supplies 3.3 V, which powers the microcontroller (not the motors) and turns the board on.
3. In VS Code, **Open Folder** on the firmware's folder. PlatformIO downloads the toolchain the first time.
4. Build and **Upload** (the arrow in PlatformIO's toolbar). The upload takes a few seconds.
5. Disconnect the adapter. With it connected, the board cannot be switched off with the button and turns on by itself when the supply is connected.
6. Reconnect the 24 V supply and press the power button.

> ✓ **Check**
>
> PlatformIO ends the upload with `SUCCESS`, and after step 6 the drivers click and the PC software connects. If not: Troubleshooting.

Version

`#define VERSION 1` in `constants.h`. The firmware is released with the robot (tag V1) and has not changed since: release V1.1 ships the same firmware; the protocol it speaks is on The serial protocol.

9. The test firmware

The test firmware turns the board into a bench instrument. At power-on it checks the six stepper drivers over SPI and prints the result, then turns all six motors slowly, blinks the LEDs and toggles the two outputs; on request it prints the E-stop, the inputs, the supply voltage and the limit switches over USB. It is what every board ships with, and the right tool for a freshly wired robot before the base is closed.

The motors turn as soon as it starts

The test firmware enables all six drivers when it boots and turns every motor slowly, at 1000 steps per second, without being asked, with no limits and no homing, until the power is switched off. No command stops them: `#STOP` only ends a printout. Use it with the arm free to move or the motors unmounted, never on a robot folded against its stops.

Connecting

Upload the firmware if the board does not have it (Firmware), then open a serial terminal on the board's USB port: PlatformIO's monitor, PuTTY or `screen`. The board is a USB device, so the baud rate does not matter; PlatformIO's monitor uses the 3000000 set in the project's `platformio.ini`. Make the terminal send a line feed at the end of each line: the firmware acts on a command only when the line feed arrives, so a terminal that sends only a carriage return on Enter gets no answer. Two ways to power the board for the test:

- the programming adapter's 3.3 V, with the 24 V supply disconnected: the board talks, but the drivers have no power and report errors;
- the 24 V supply: the board is off until you press the power button, or supply 3.3 V from the adapter as well.

What it does by itself

- **About three seconds after power-on** it reads every driver over SPI and prints a report per joint: `joint N Connection is: GOOD` or `BAD`, then the driver's current, hold and run current settings and its fault flags.
- **From then on** all six motors turn slowly and keep turning.
- **Every second** the two LEDs blink and outputs 1 and 2 toggle.

Commands

`#` followed by the name, case as written, and the line feed. Four commands do something:

Command	What it does
`#IO`	Prints, once a second, the E-stop input, input 1, input 2 and the supply voltage in millivolts
`#LIMIT`	Prints, once a second, `Joint N is pressed` for every limit switch that is triggered, numbered by joint
`#STOP`	Ends the `#IO` or `#LIMIT` printout; the motors keep turning
`#start`	Toggles the two LEDs

Every other name, including `#STATUS`, `#STEP1` to `#STEP6`, `#FLASH` and `#CAN`, is answered with `Unknown command`: the parser knows the names, but the firmware does nothing with them. Test firmware commands is the reference list.

Testing CAN

CAN has its own test, next to the test firmware in the repository's `PAROL6 control board test code` folder and not part of its build: the sketch `can_12_12` sends a frame with ID 0x446 every 500 ms at 1 Mbit/s and prints every frame it receives, and `can_parol6_test.py` is the PC side, a python-can script for a USB-CAN adapter (slcan, 1 Mbit/s; set its port, `COM44` in the file) that prints the frames it receives with their node and command IDs.

What good looks like

All six drivers answering, in the report at power-on:

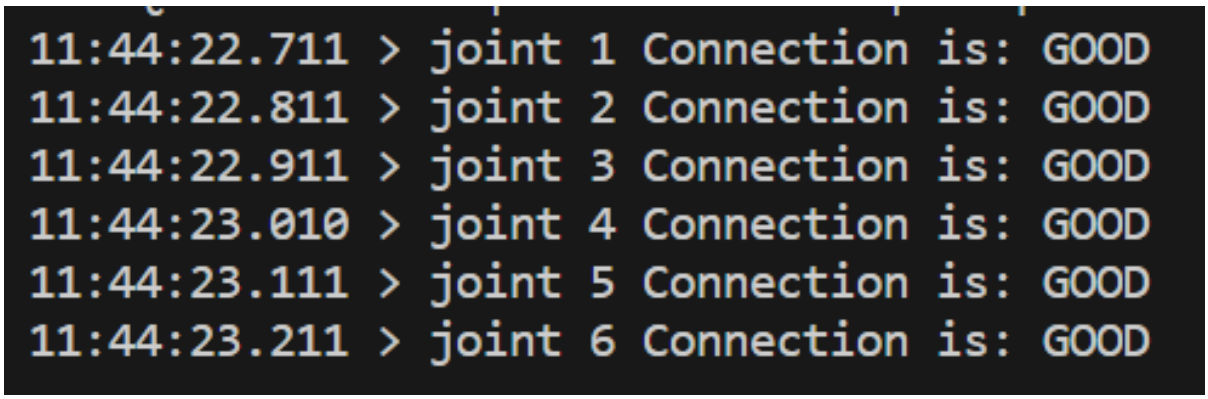


Figure 9.1: Serial output with all drivers responding.

A driver missing or faulty:

```
--- Terminal on COM13 | 1000000 8-N-1
--- Available filters and text transformations: colorize, debug, d
--- More details at https://bit.ly/pio-monitor-filters
--- Quit: Ctrl+C | Menu: Ctrl+T | Help: Ctrl+T followed by Ctrl+H
11:39:59.765 > joint 1 Connection is: BAD
11:39:59.765 > Overtemp. Pre Warning
11:39:59.765 > Overtemp.
11:39:59.864 > joint 2 Connection is: BAD
11:39:59.864 > Overtemp. Pre Warning
11:39:59.864 > Overtemp.
11:39:59.964 > joint 3 Connection is: BAD
11:39:59.964 > Overtemp. Pre Warning
11:39:59.964 > Overtemp.
11:40:00.064 > joint 4 Connection is: BAD
11:40:00.064 > Overtemp. Pre Warning
11:40:00.064 > Overtemp.
11:40:00.164 > joint 5 Connection is: BAD
11:40:00.164 > Overtemp. Pre Warning
11:40:00.164 > Overtemp.
11:40:00.264 > joint 6 Connection is: BAD
11:40:00.264 > Overtemp. Pre Warning
11:40:00.264 > Overtemp.
```

Figure 9.2: Serial output with a faulty driver.

With good drivers and the 24 V on, the motors turn slowly at 200 to 300 mA. A triggered limit switch, with `#LIMIT`:

```
11:41:28.617 > Received 'LIMIT' command
11:41:28.618 > Received command: LIMIT
11:41:28.618 > Received argument:
11:41:58.021 > Joint 2 is pressed
11:42:11.864 > Joint 5 is pressed
11:42:12.853 > Joint 5 is pressed
11:42:13.841 > Joint 5 is pressed
11:42:16.806 > Joint 1 is pressed
```

Figure 9.3: A limit switch triggered.

The I/O report, with `#IO`; change the E-stop or an input and the values follow:

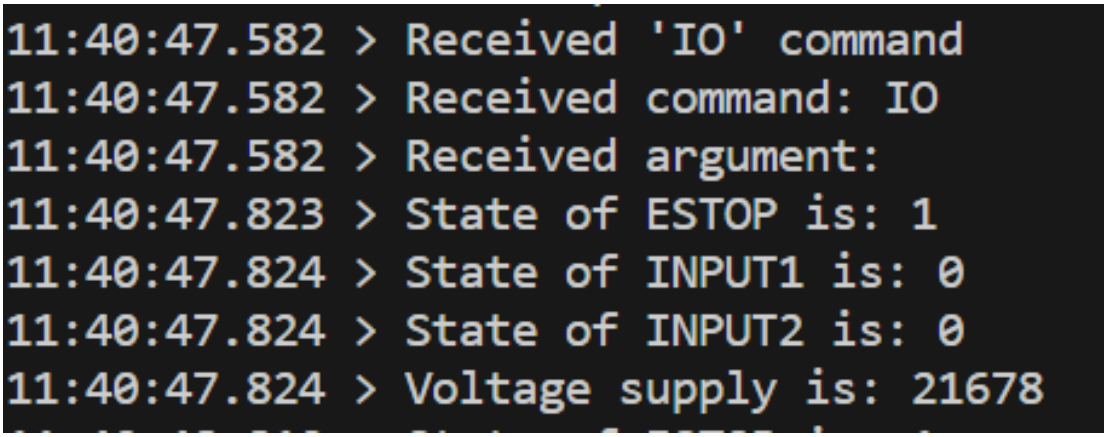


Figure 9.4: The IO report.

Outputs 1 and 2 toggle every second, and the two LEDs blink.

10. The serial protocol

The PC and the board exchange one packet each per control cycle over USB, a virtual serial port (the PC software opens it at 3 000 000 baud, a setting USB ignores). The PC sends a command packet at 100 Hz; the board answers each valid packet with a telemetry packet. Both are binary, framed with start bytes, a length byte, two end bytes and a CRC byte, and together under 120 bytes. The byte-exact layout is on The serial protocol in the API reference; this page explains how it works.

Robot -> PC	
Start bytes	3
Package length	1
Position out	18
Speed out	18
Homed	1
I/O	1
Temp error	1
Position error	1
Timing data	2
Timeout error	1
Additional 2	1
Gripper ID	1
Gripper position	2
Gripper speed	2
Gripper current	2
Gripper status	1
Object detection	1
CRC	1
End bytes	2
Total bytes	60

PC -> Robot	
Start bytes	3
Package length	1
Position	18
Speed	18
Command	1
Affected joint	1
I/O	1
Timeout	1
Gripper data	9
CRC	1
End bytes	2
Total bytes	52

Figure 10.1: Packet structure.

The cycle

```
loop:
    read the power button; run the active command (jog, go-to-position, home, idle)
    while serial data is available:
        read a byte; look for the three start bytes 0xFF 0xFF 0xFF
        the next byte is the length: that many bytes follow, including the CRC and the end bytes
        when the end bytes are 0x01 0x02: unpack the packet into the commanded positions, speeds,
command,
            outputs, gripper fields; read the E-stop and inputs; write the outputs; send the
gripper its frame;
            pack and send the telemetry packet
```

A valid packet starts one robot cycle: the firmware does nothing on its own between packets except run the speeds it was last given. The PC must keep the packets coming; the time between the last two packets is measured with a 703 125 Hz timer (90 MHz / 128, so 1.422 µs per tick: 10 ms is 7031 ticks) and returned in the telemetry as `timing_data`, which is how the PC software checks that it keeps the 100 Hz loop.

Commands

The command byte selects what the board does with the carrier data (the positions and speeds) in the same packet:

Command	Name	What the board does
255	Idle (dummy)	The joints stand still: it sets their speeds but steps none; it only continues a homing in progress. The default between moves
123	Jog	Runs each joint at its commanded speed (`runSpeed`)
156	Go to position with speed	Runs each joint at the average of its commanded speed and the speed that would close the position error in one cycle: the tracking the planners use
100	Home	Starts the homing sequence; idle packets continue it until done
101	Enable	Enables the robot and resets the homing state
102	Disable	Disables motion commands; the motors keep holding
103	Clear error	Resets the homing state
69	Repeatability test	A built-in two-pose cycle, for testing

Motion commands (100, 123, 156) are ignored while the robot is disabled. A jog or go-to packet ends a homing in progress; the homed bits in the telemetry change only by homing. The firmware does not check them before a move: the PC decides when a move is allowed.

Passive data

The I/O byte carries the two output states in every packet, and the gripper fields (ID, position, speed, current, command bits, mode) are forwarded to the gripper over CAN every cycle: a move when they change, a calibrate for mode 1, a clear-error for mode 2, and an empty status poll when nothing changed. They ride along with any command.

Telemetry

The board returns the six positions and speeds in motor steps, which joints are homed, the inputs, outputs and E-stop, two bytes of driver flags (temperature and position, reserved: the joint bits are cleared and the two spare bits set), the timing, the last command byte it received, and the gripper's data. Of that, the gripper reports only its position, current and status byte; the ID, speed and object-detection bytes are fixed values, and the object detection is read from two bits of the status byte (The serial protocol).

Writing your own host

Anything that frames the packets as on the reference page and keeps the 100 Hz cadence can drive the board: the commander software and the Python API are two such hosts, in Python. Start from the Python API's serial transport rather than from scratch; the Control ownership page says why two hosts must not talk to the board at once.

11. Troubleshooting

Symptom	Cause	Remedy
No life at all with 24 V connected	The button; the supply; a reversed XT30	Check the supply's output, the JST button cable, the XT30 polarity. A reversed supply has usually destroyed the board
The board is on but cannot be switched off	3.3 V on the programming header	Unplug the ST-Link or adapter
The board turns on when the supply is connected	The same, or a motor was turned by hand	Unplug the programmer; move an unpowered robot slowly
The test firmware's report at power-on says `Connection is: BAD` for a joint	Not seated; dead; wrong orientation	Power off, check the orientation against the DIAG pins, reseat. A driver that was powered the wrong way round is dead, and the board may be too
A motor hums but does not turn	A phase open; a motor wire loose in the terminal	Check the four wires of that joint's cable
A motor turns the wrong way	Normal for a new build	Flip `direction_reversed` in the main firmware
A switch does not show in `#LIMIT`	Wiring; the switch; the wrong terminal	Check 24 V to the switch and the signal wire; try the other terminals to find which input changes
An inductive sensor never triggers	Gap; no 24 V; the sensor is PNP	NPN NO sensors only; brown 24 V, blue GND, black signal; gap of a millimetre or two
The E-stop reads pressed with the button released	Wiring open; the button is NO	NC contact between the E-stop terminals
The drivers are hot and shut down	The fan	The fan must run; 5 V, not PWM
No serial port on the PC	The board has no power; the cable; the driver (Windows)	The board enumerates only when powered; try another cable and port

Symptom	Cause	Remedy
The main firmware does not upload	ST-Link driver; the adapter; the target not powered	Install STSW-LINK009; check the adapter's 3.3 V; disconnect the 24 V supply

For anything the firmware reports to the PC, the robot-side pages apply: Troubleshooting in the Robot Manual.

End effectors

PAROL6 End Effectors Guide



Document number: SR-3004

PAROL6 release: 1.1

Python API (parol6): v0.9.1

PAROL6 commander software: main

Revision: 2026.B

Published: 2026-11-05

Applies to: PAROL6 release 1.1, Python API (parol6) v0.9.1, PAROL6 commander software main

Original instructions

What goes on PAROL6's flange and how to set it up: the pneumatic and vacuum grippers from the robot's own files, Source Robotics' SSG-48 and MSG electric grippers, and a tool of your own. Each page says how the end effector mounts, how it connects, what to tell the firmware and the software, and how the commander software and the Python API drive it. The electric grippers have manuals of their own; here is only what is PAROL6-specific.

End effector	Driven by	Mounts with	Commander	Python API	Manual
Pneumatic gripper, SMC MHZ2-16D	a 5/2 valve on OUTPUT1, 5 to 7 bar	a printed holder on the joint 6 shaft coupler	`Output(1,HIGH)`, `Output(1,LOW)`	`select_tool("PNEUMATIC")`	this guide
Vacuum gripper	the same valve and an ejector	a printed holder	`Output(1,HIGH)`, `Output(1,LOW)`	`select_tool("VACUUM")`	this guide
SSG-48 adaptive electric gripper	CAN and 24 V through the M8 connector	`coupler_connector_PAROL6`	the Gripper tab, `Gripper()`	`select_tool("SSG-48")`	SSG-48 manual
MSG compliant AI stepper gripper	CAN and 24 V through the M8 connector	`ring_cover-PAROL6`	the Gripper tab, `Gripper()`	`select_tool("MSG")`	MSG manual
Your own tool	24 V and CAN, two air lines, two inputs and two outputs	the shaft coupler, or the coupler template	`Output(1,HIGH)`, `Output(1,LOW)`; the inputs in the IO tab	`set_tcp_offset()`, a `ToolConfig`	this guide

In this guide

- The flange: the 8 mm shaft of joint 6, the connections at the forearm, the payload, the tool in the software

- Pneumatic gripper
- Vacuum gripper
- SSG-48
- MSG
- Your own tool

Get the robot working before you attach a gripper: troubleshooting both at once is hard.

The flange

Joint 6 ends in an 8 mm output shaft. Tools clamp onto it with a shaft coupler (8 mm bore, M4 set screws, three in the BOM) that bolts to the tool's holder; the SSG-48 and MSG grippers use a printed coupler connector made for this shaft. The coupler's set screws take blue threadlocker: joint 6 slipping on its shaft is the first thing to check when a tool's orientation drifts (Couplers and threadlocker).

Connections at the forearm

M8 4-pin connector	24 V, GND, CAN H, CAN L for an electric gripper; wired from the control board's CAN 1 and 24 V. The 24 V has no fuse of its own: the gripper draws what it needs, and a short in the gripper or its cable is limited only by the power supply
Two pneumatic tubes, 4 mm	From the two push-in fittings at the base, through the arm, to the forearm

The tubes and the gripper cable run through the arm and come out at the forearm. Joint 5 is held inside its limits by its stops. Joint 6 has no limit with the bare flange; with a tool, the tool's cable and tubes run from the forearm to the tool and wrap as joint 6 turns, so keep joint 6 within the turn they allow (Joint limits and directions).

The gripper socket

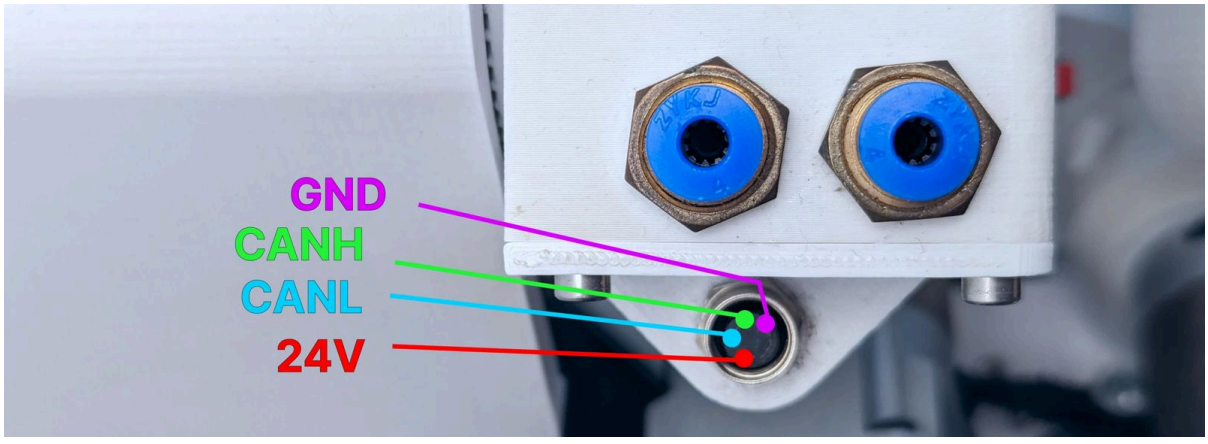


Figure 1: The gripper socket at the forearm, seen from the front, with its four signals.

The M8 socket at the forearm is a female 4-pin. Its pins, by the numbers moulded into it:

Pin	Signal
1	GND
2	CAN H
3	+24 V
4	CAN L

The grippers have +24 V on pin 1 and GND on pin 3, so the PAROL6 gripper cable crosses pin 1 with 3 and 2 with 4 between its ends. A straight M8 cable, the PAR6 gripper cable among them, reverses a gripper's supply here and destroys its driver. A tool of your own takes its pins from the table above.

The tool in the software

The software must know the tool to compute the pose of its tip. The commander's kinematic model is built for the standard pneumatic gripper; the Python API has tool definitions with variants and a settable TCP offset (Tools and the TCP):

```
rbt.select_tool("PNEUMATIC")                      # default variant: vertical
rbt.select_tool("PNEUMATIC", variant_key="horizontal")
rbt.select_tool("SSG-48", variant_key="pinch")
rbt.select_tool("MSG", variant_key="150mm")
rbt.select_tool("VACUUM")
rbt.select_tool("NONE")                           # the bare flange
```

Tool centre point

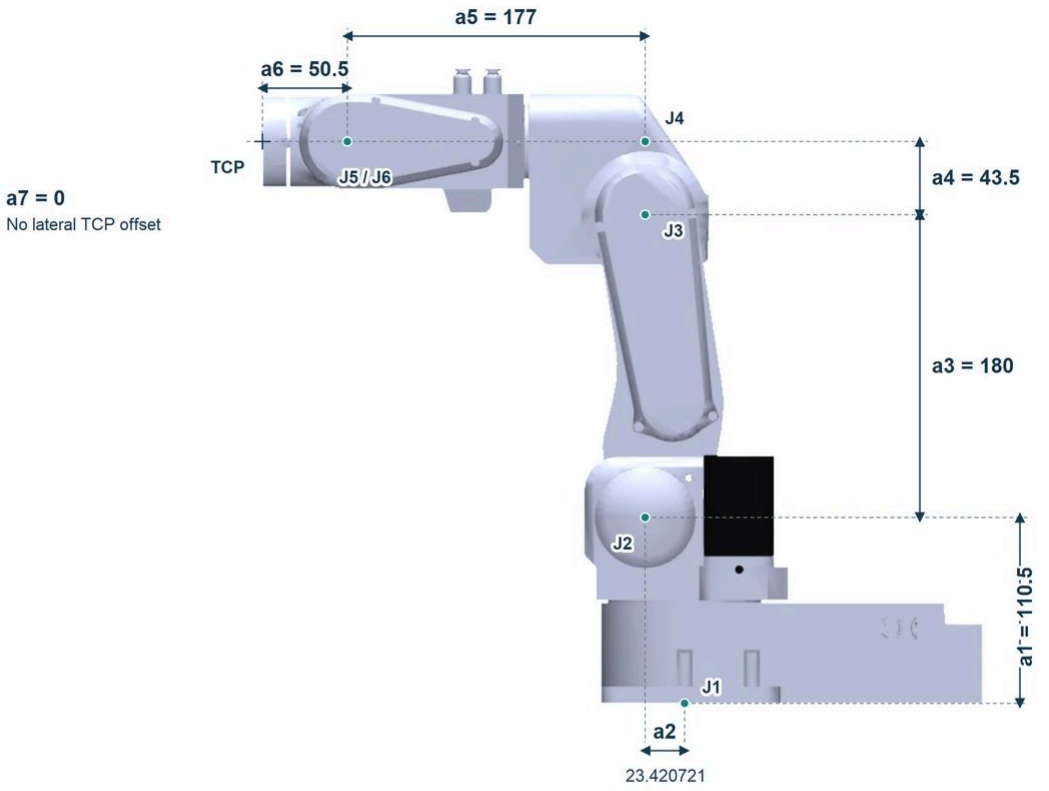
The tool centre point (TCP) is a6 = 50.5 mm from the joint 6 centre along the tool axis and a7 = 0 mm to the side. With nothing on the flange, the TCP is the flange's reference point, the origin of the flange frame: on the joint 6 axis at the flange face, Z out of the flange, X up in the standby pose. The Python API has the frame's Y and Z the other way (Reference frames). The sheet is the robot's model from robot_descriptions (version 2.15.1) with every joint at 0. Tell the software about the tool (Tools and the TCP); every tool is compared on Dimensions and kinematics.

PAROL6 / Flange only

Kinematic review | Standard DH to the configured TCP

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Pose: q1 ... q6 = 0 rad; no jaws. Orthographic side view; dimensions in mm.



a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

Joint	Link	theta [rad]	alpha [deg]	a [mm]	d [mm]
1	Base	q1	-90	a2 = 23.420721	a1 = 110.5
2	Shoulder	q2 - pi/2	180	a3 = 180	0
3	Elbow	q3 + pi	90	-a4 = -43.5	0
4	Wrist 1	q4	-90	0	-a5 = -177
5	Wrist 2	q5	90	0	0
6	Wrist 3 / TCP	q6 + pi	180	-a7 = 0	-a6 = -50.5

Figure 2: PAROL6 with the flange only: dimensions to the TCP and the DH table.

Homing and the tool

Joint 5 homes against a feature of the tool holder. The pneumatic gripper holder and the SSG-48's coupler connector have it; a tool of your own needs it too, and the firmware's joint 5 homing offset must match the tool (Your own tool).

Payload

1 kg near the base, 0.5 kg at full reach, the tool included. A gripper with a camera and a cable can be most of that. A short video shows the arm with a 1 kg payload: PAROL6 robot arm 1kg payload.

Pneumatic gripper

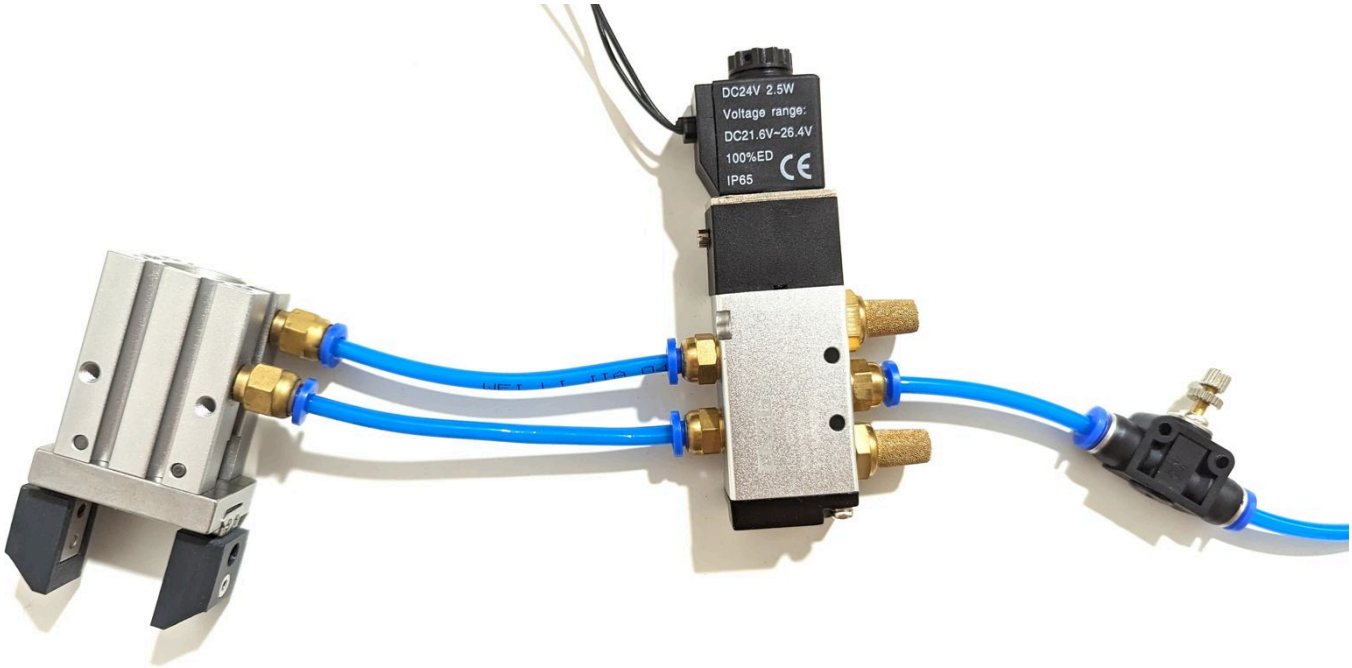
The standard tool: an SMC MHZ2-16D parallel gripper in a printed holder, driven by a 5/2 solenoid valve from one of the control board's outputs.

Parts

Gripper	MHZ2-16D, two PC4-M5 push-in fittings
Holder	`GRIPPER_ATTACHMENTS/Pneumatic_gripper_holder.STL` (vertical) or `Pneumatic_gripper_holder_horizontal.stl`; `Gripper_ARMS.STL` for the fingers
Valve	5/2 solenoid valve, 24 V, 4 mm tube ports
Tube	4 × 2.5 mm pneumatic tube, 5 m
Air	5 to 7 bar; stay below the lowest rating of the gripper, the valve and the fittings (Pneumatics)

Connections

The two tubes from the gripper leave the forearm and are joined to the two tubes that run through the arm to the base; the valve's two outputs go to the two fittings at the base. **Swapping the two tubes swaps normally open and normally closed**: choose the state the gripper should take without power.



The valve's coil goes between **GND_SEL** and **OUTPUT1** on the control board, with the output supply set to the robot's 24 V or an isolated one (Inputs and outputs).

Tool centre point

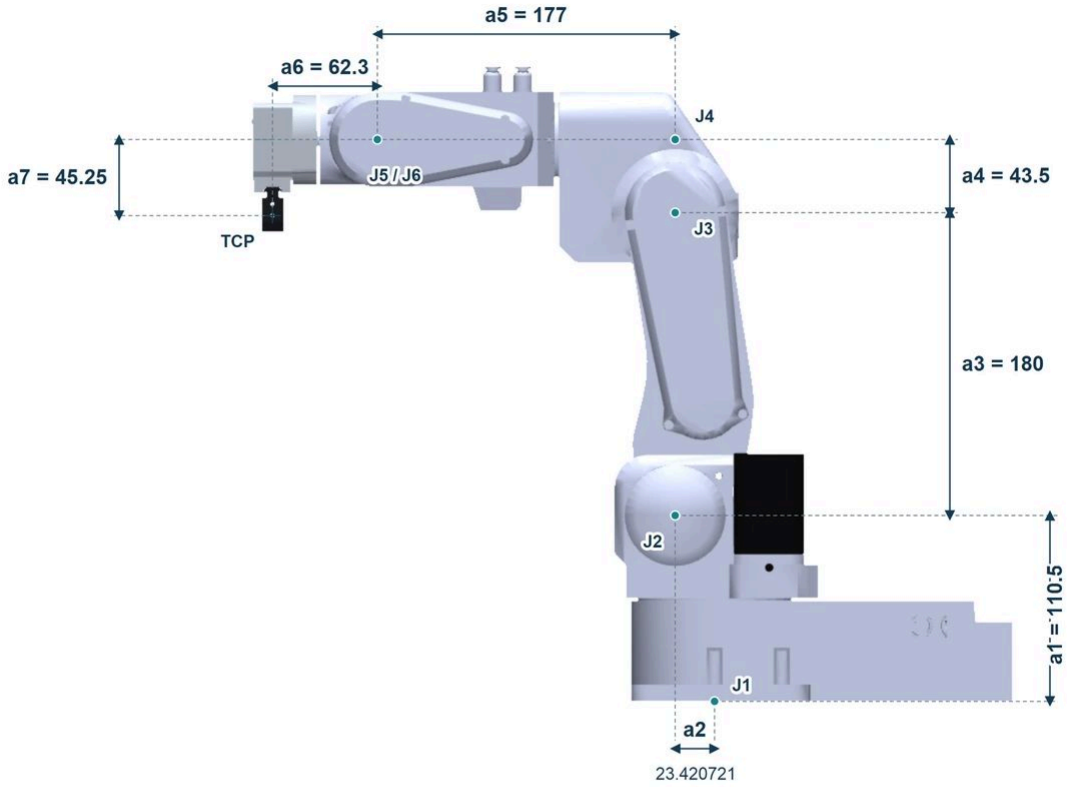
The tool centre point (TCP) is a6 = 62.3 mm from the joint 6 centre along the tool axis and a7 = 45.25 mm to the side. The TCP is the reference point of the original MHZ2-10D gripper, not a newly measured grasp centre. The sheet is the robot's model from robot_descriptions (version 2.15.1) with every joint at 0. Tell the software about the tool (Tools and the TCP); every tool is compared on Dimensions and kinematics.

PAROL6 / Pneumatic gripper

Kinematic review | Standard DH to the configured TCP

12 / 12

Pose: q1 ... q6 = 0 rad; jaws open. Orthographic side view; dimensions in mm.



a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

Joint	Link	theta [rad]	alpha [deg]	a [mm]	d [mm]
1	Base	q1	-90	a2 = 23.420721	a1 = 110.5
2	Shoulder	q2 - pi/2	180	a3 = 180	0
3	Elbow	q3 + pi	90	-a4 = -43.5	0
4	Wrist 1	q4	-90	0	-a5 = -177
5	Wrist 2	q5	90	0	0
6	Wrist 3 / TCP	q6 + pi	180	-a7 = -45.25	-a6 = -62.3

Figure 1: PAROL6 with the pneumatic gripper: dimensions to the TCP and the DH table.

Driving it

RBTscript

```
Output(1,HIGH)    // the valve switches
Delay(0.5)
Output(1,LOW)     // and back
```

Python API

```
rbt.select_tool("PNEUMATIC")
rbt.tool.close()              # output 1 low
rbt.tool.open()               # output 1 high
rbt.write_io(0, 1)            # the same as open(), as a plain output
```

The tool's action lasts its estimated stroke, so a move queued after `close()` waits for the gripper. The API's `open()` switches output 1 high and `close()` switches it low: lay the tubes so that the gripper is open with the valve switched, or `open()` and `close()` come out the other way round.

Orientation

The vertical holder puts the gripper along the axis of joint 6; the horizontal holder turns it 90°, which reaches a table from the side. The Python API has a variant for each (`vertical` , `horizontal`); with the commander, remember that the DH dimensions a6 and a7 assume the vertical holder.

Vacuum gripper

A suction cup on a vacuum generator, fed by the same valve and tubes as the pneumatic gripper.

Parts

The BOM's vacuum section: a suction cup gripper, a vacuum generator (ejector) and a 4 mm to 6 mm coupler. The holder is `GRIPPER_ATTACHMENTS/vacuum_gripper_holder.STL`.

Mounting

Print the holder and bolt the gripper to it. The gripper spins freely in the holder by default; to fix its orientation, tie it with a zip tie to the screw shown in the figure.

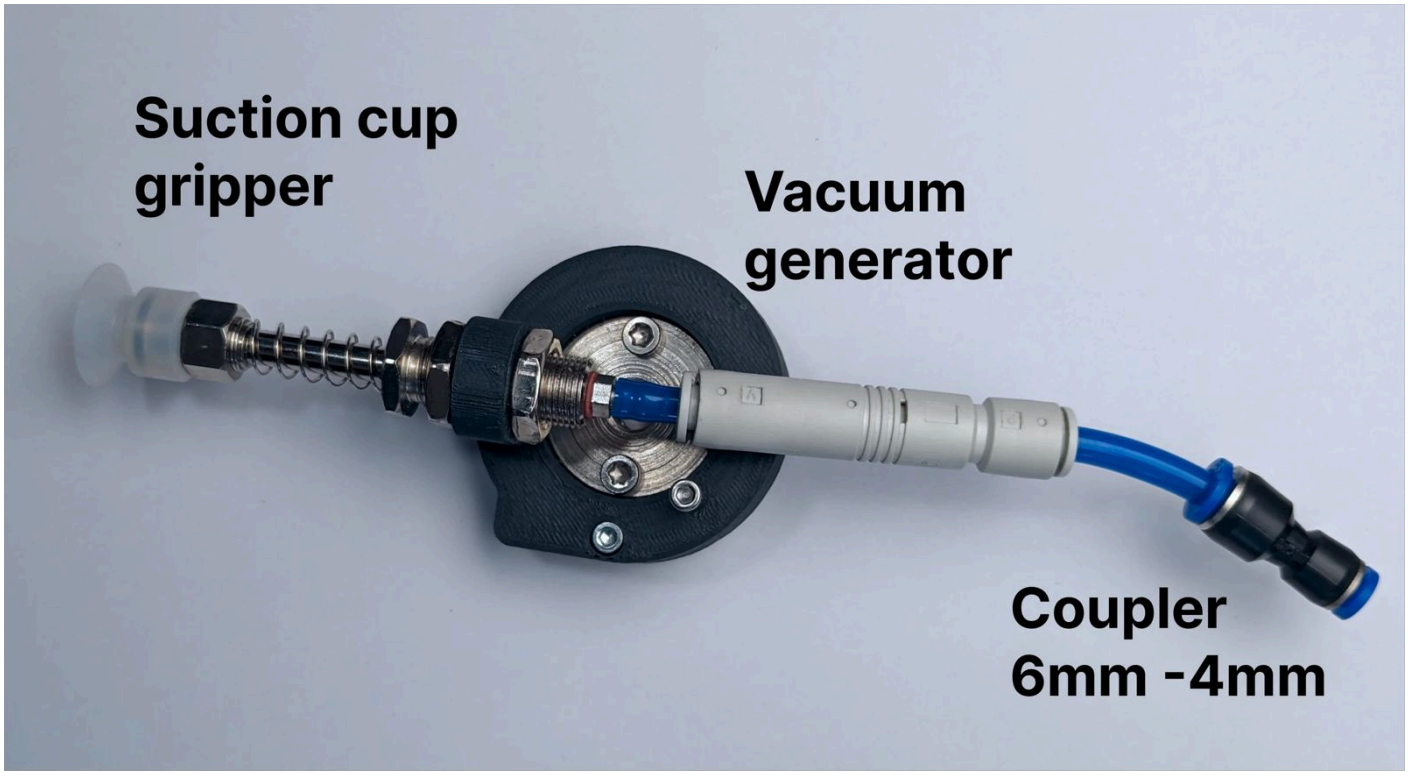


Figure 1: The vacuum gripper on the forearm.

Air

A vacuum generator needs one supply line, so one of the valve's two ports is blocked: a short piece of tube with an M3 screw and threadlocker screwed into it does the job.

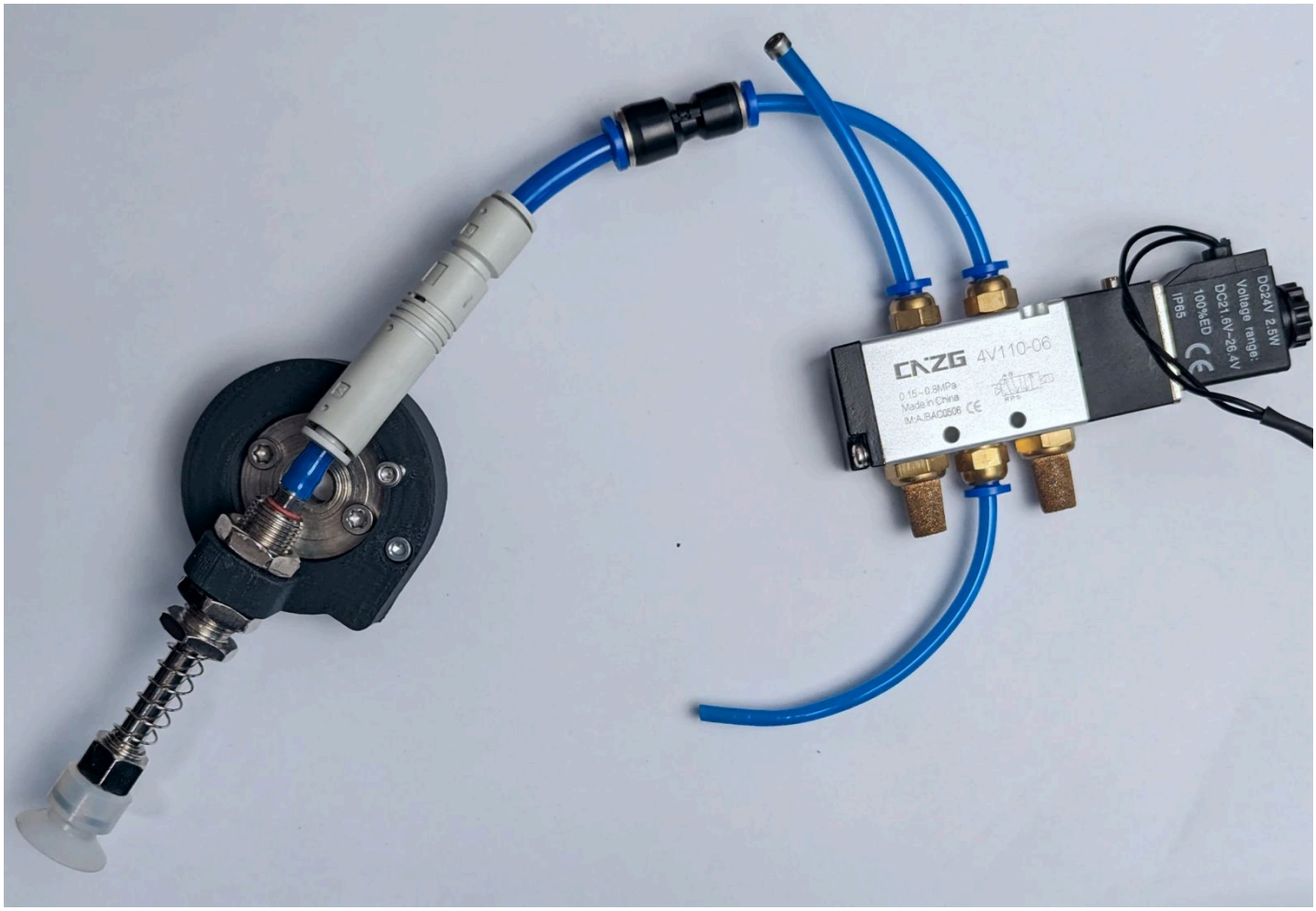


Figure 2: One valve port blocked with a tube and an M3 screw.

Driving it

The valve switches the air to the ejector, which pulls the vacuum: `Output(1,HIGH)` holds, `Output(1,LOW)` drops. In the Python API, `select_tool("VACUUM")` gives `tool.open()`, which switches output 1 high, and `tool.close()`, which switches it low. With the tubing above, `open()` holds the part and `close()` drops it; `write_io(0, 1)` and `write_io(0, 0)` do the same.

> **Warning**
>
> The part drops the moment the air stops, with the E-stop, a valve fault or a power failure. Do not carry anything over something it can damage.

SSG-48 adaptive electric gripper



Source Robotics' 48 mm electric gripper: a Spectral Micro BLDC driver in gripper mode. It plugs into the M8 connector at the forearm and the control board drives it over CAN; the commander software and the Python API both know it. Its own manual, SSG48-gripper-docs, covers the gripper; this page covers PAROL6.

Stroke	48 mm
Grip force	about 54 N per ampere, 5 to 80 N; on PAROL6 up to about 54 N from the commander, 70 N from the Python API
Fingers	fin-ray or pinch, printed: `finger` and `pinch` in the Python API
Shop	SSG-48

Mounting

1. Fit `coupler_connector_PAROL6` from the gripper's repository to the gripper. It carries a shaft coupler with an 8 mm bore and two M4 clamp screws for the joint 6 shaft, and the homing tab that joint 5 homes against: the tab goes on the same side as the gripper's cable connector, with an M3 × 12 screw in the marked hole, as the gripper's assembly manual shows on its last pages.
2. Slide the gripper onto the joint 6 shaft, orient it, and tighten the two clamp screws, one on each side, with blue threadlocker (Couplers and threadlocker).
3. Connect the PAROL6 gripper cable between the gripper and the M8 connector at the forearm.

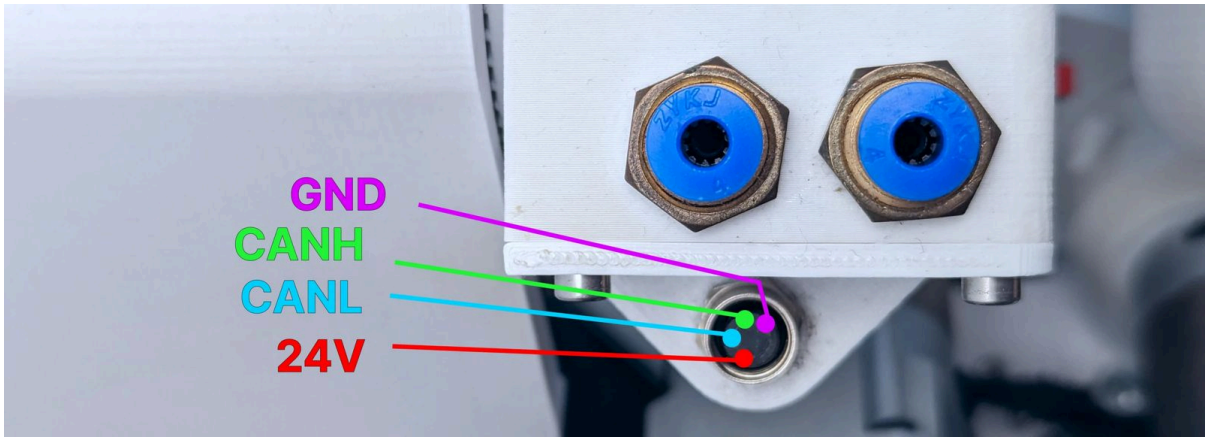


Figure 1: The gripper connector at the forearm.

Only the PAROL6 cable

PAROL6 and PAR6 put the four signals on different pins. The PAR6 cable reverses the gripper's supply, and a self-made cable with a wrong pin can put 24 V on a CAN pin; either destroys the gripper's driver. The socket's pins are in The gripper socket, the cable's in the gripper manual's cable page.

Leave the gripper's CAN node ID at 0 and its termination switch on: it is the last node on the robot's bus (CAN bus).

Tool centre point

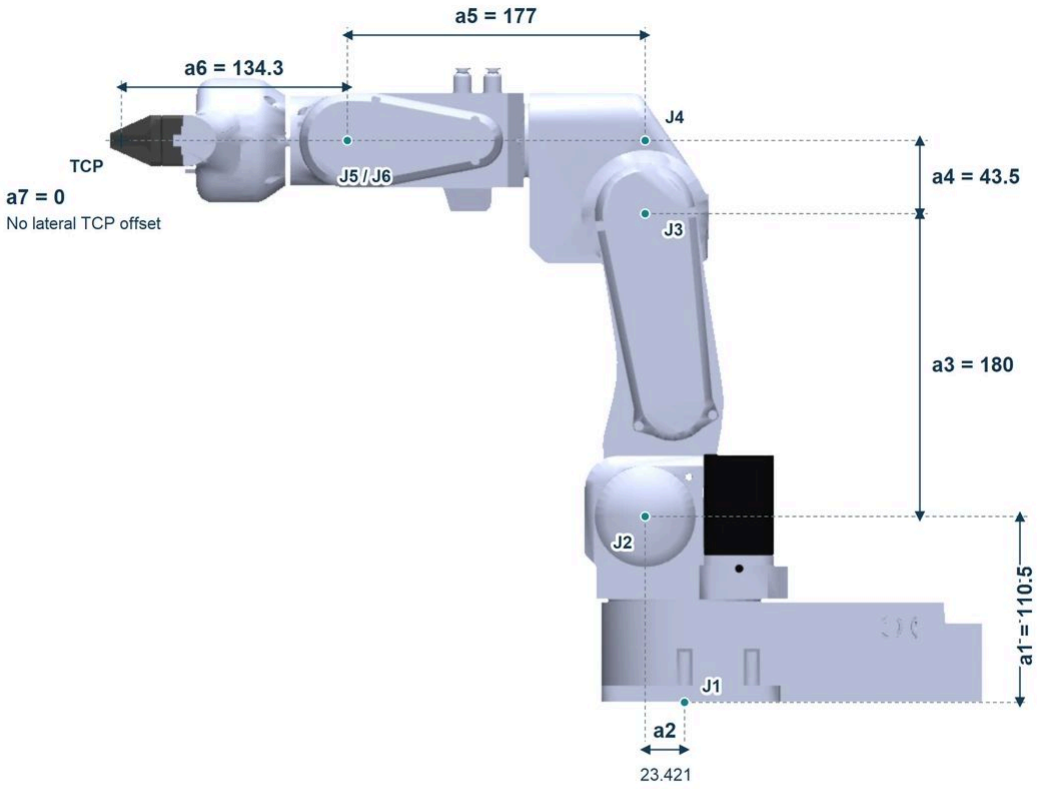
The tool centre point (TCP) is a6 = 134.3 mm from the joint 6 centre along the tool axis and a7 = 0 mm to the side. The TCP lies on the joint 6 axis, with no sideways offset, for both jaw variants. The sheet is the robot's model from robot_descriptions (version 2.15.1) with every joint at 0. Tell the software about the tool (Tools and the TCP); every tool is compared on Dimensions and kinematics.

Solid jaws

PAROL6 / SSG48 / Solid jaws

Kinematic review | Standard DH to the configured TCP

Pose: q1 ... q6 = 0 rad; jaws open. Orthographic side view; dimensions in mm.



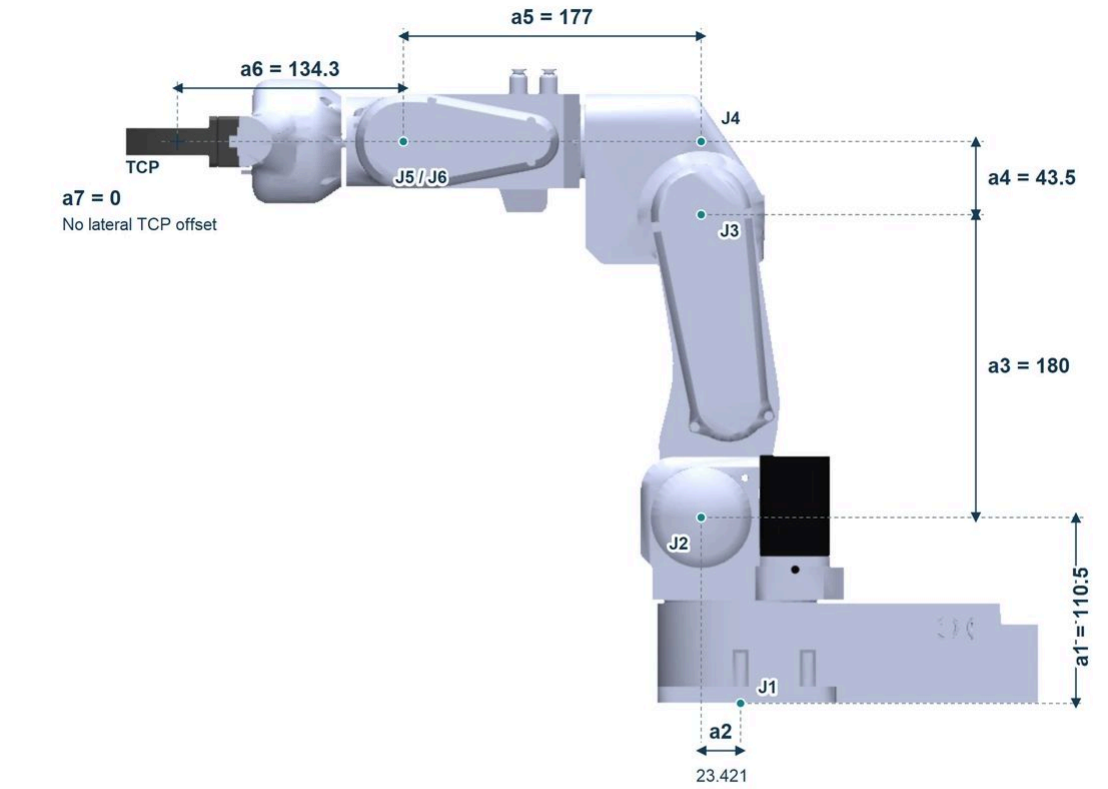
a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

Joint	Link	theta [rad]	alpha [deg]	a [mm]	d [mm]
1	Base	q1	-90	a2 = 23.421	a1 = 110.5
2	Shoulder	q2 - pi/2	180	a3 = 180	0
3	Elbow	q3 + pi	90	-a4 = -43.5	0
4	Wrist 1	q4	-90	0	-a5 = -177
5	Wrist 2	q5	90	0	0
6	Wrist 3 / TCP	q6 + pi	180	-a7 = 0	-a6 = -134.3

Figure 2: PAROL6 with the SSG-48, solid jaws: dimensions to the TCP and the DH table.

Fin-ray jaws



a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

Joint	Link	theta [rad]	alpha [deg]	a [mm]	d [mm]
1	Base	q1	-90	a2 = 23.421	a1 = 110.5
2	Shoulder	q2 - pi/2	180	a3 = 180	0
3	Elbow	q3 + pi	90	-a4 = -43.5	0
4	Wrist 1	q4	-90	0	-a5 = -177
5	Wrist 2	q5	90	0	0
6	Wrist 3 / TCP	q6 + pi	180	-a7 = 0	-a6 = -134.3

Figure 3: PAROL6 with the SSG-48, fin-ray jaws: dimensions to the TCP and the DH table.

The joint 5 homing offset

Joint 5 homes against the tool, and the SSG-48's coupler sits differently from the pneumatic gripper holder. Set `j5_homing_offset` to **8035** in `main.cpp` of the control board firmware (the default, 8900, is for the pneumatic gripper) and upload (Firmware). Check with the pin through the joint 5 witness marks after the first homing, and master if needed (Mastering).

Driving it

The gripper must be calibrated after every power-up: it opens fully and closes fully to find its end stops, and does not move before that. Clear any error first, then calibrate, then wait until the gripper reports itself calibrated before the first move: about 3 s.

Commander software

The **Gripper** tab: after homing the robot press **Clear gripper error**, then **Calibrate gripper**, and wait until the tab shows `Calibration status is: 1`. Then set position, speed and current with the sliders and press **Move GoTo**. The feedback section shows the current, the position and the status (The GUI). In RBTscript, after clearing the error in the tab: `Gripper_cal()`, `Delay(4)`, then `Gripper(position, speed, current)` (RBTscript).

Python API

```
rbt.select_tool("SSG-48")                                  # or add variant_key="pinch"
rbt.tool.calibrate()                                        # queued; the API counts it done after 2 s
rbt.delay(2)                                                # the rest of the calibration, with a
margin
rbt.tool.set_position(0.8, speed=0.5, current=0.4)          # fractions of the ranges; 0.4 is 580 mA
rbt.tool.close(current=0.6)
rbt.tool.open()
rbt.tool.release()                                          # drop the grip, the jaws stay
rbt.tool.stop()                                             # immediate: halt where the jaws are
```

Tool actions are queued with the motion, so a move after `close()` waits for the grip; a `move` on a gripper that was never calibrated is refused when its turn comes. The API does not read the calibrated bit and has no clear-error action at this version: clear an error in the commander's Gripper tab or with Motor GUI (Tools and the TCP, Tools).

Current and force

The current limit is the grip force, about 54 N per ampere: 300 mA is about 16 N, 1 A about 54 N. Start low and raise it for a firmer grip; a high current with a high speed slams the jaws into the part. The gripper manual's Force and speed has the curve and the heat limits.

The commander takes the current in milliamps, 100 to 1000 mA, so its strongest grip is about 54 N. The Python API takes it as a fraction of 100 to 1300 mA: `current=1.0` is 1.3 A, about 70 N.

Status

The robot passes the gripper's status byte through: calibrated, in motion, object detected (while closing, while opening, or none), and its error bits. The commander's Gripper tab shows the calibrated and E-stop error bits, and the whole status byte as a number (labelled *Gripper object detection is*). The Python API's `tool.status()` gives the jaw position and the current; its `engaged` and `part_detected` do not follow the status byte's bits at this version (Tool status). The gripper manual's errors page says what each error needs: Errors and recovery.

MSG compliant AI stepper gripper



Source Robotics' modular stepper gripper: a STEPFOC driver in gripper mode, three rail lengths, three stepper lengths and a camera mount. It plugs into the M8 connector at the forearm and the control board drives it over CAN with the same commands as the SSG-48. Its own manual, MSG-gripper-docs, covers the gripper; this page covers PAROL6.

Stroke	55, 105 or 155 mm, with the 100, 150 or 200 mm rail
Grip force	about 20, 25 or 36 N per ampere with the 21.5, 40 or 60 mm stepper; up to about 31, 50 or 76 N
Mass	about 380 g plus the camera: mind the payload at full reach
Shop	MSG gripper

Mounting

1. Replace the gripper's `ring_cover` with `ring_cover-PAROL6` from the repository's `Common parts`: it bolts to the robot's last joint, and the coupler plate's shaft coupler (8 mm bore, two M4 clamp screws) takes the joint 6 shaft (Mounting on a robot).
2. Slide the gripper onto the shaft, orient it with the camera looking down between the jaws, and tighten the two clamp screws with blue threadlocker (Couplers and threadlocker).
3. Connect the PAROL6 gripper cable between the gripper and the M8 connector at the forearm.
4. The camera is a plain USB camera with its own cable: route it outside the arm, with slack for joints 4, 5 and 6, and fix it so the arm cannot catch it.

Only the PAROL6 cable

PAROL6 and PAR6 put the four signals on different pins. The PAR6 cable reverses the gripper's supply, and a self-made cable with a wrong pin can put 24 V on a CAN pin; either destroys the gripper's driver. The socket's pins are in The gripper socket, the cable's in the gripper manual's cable page.

Leave the gripper's CAN node ID at 0 and its termination switch, reachable through the stepper cover, on: it is the last node on the robot's bus (CAN bus).

Tool centre point

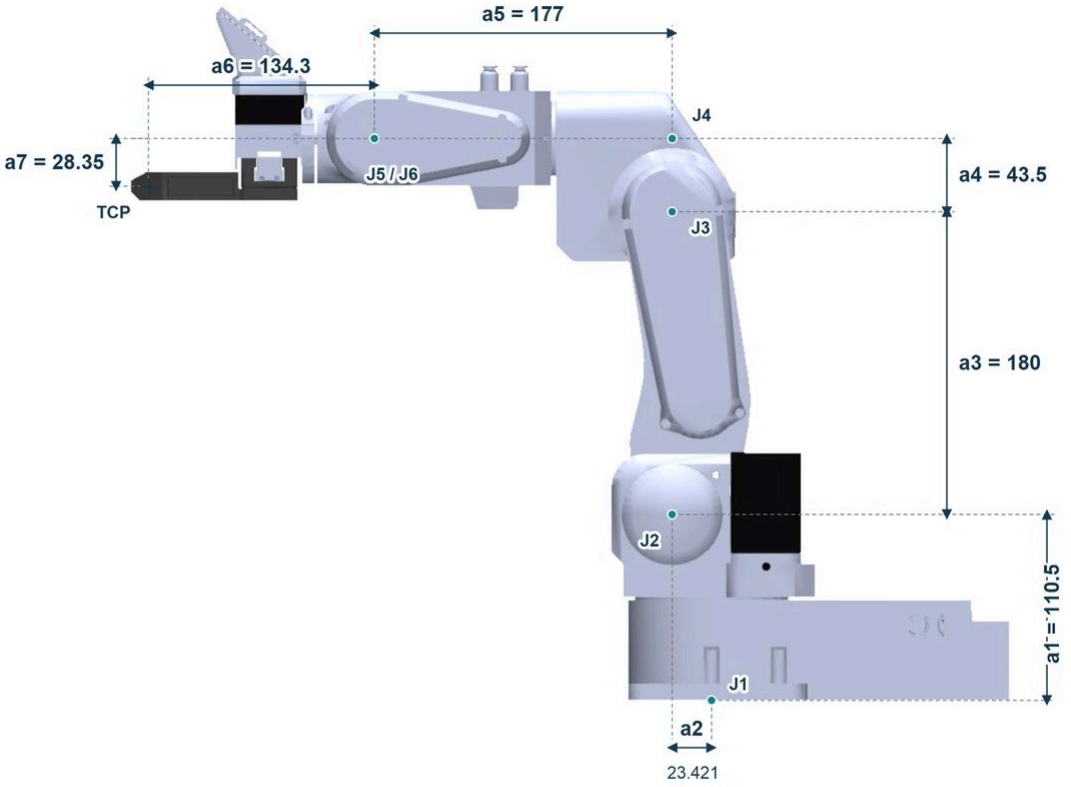
The tool centre point (TCP) is a6 = 134.3 mm from the joint 6 centre along the tool axis and a7 = 28.35 mm to the side. Both jaw variants share one TCP: an operating point, not a measured grasp centre. The sheet is the robot's model from robot_descriptions (version 2.15.1) with every joint at 0. Tell the software about the tool (Tools and the TCP); every tool is compared on Dimensions and kinematics.

PAROL6 / MSG / Solid jaws

Kinematic review | Standard DH to the configured TCP

08 / 12

Pose: q1 ... q6 = 0 rad; jaws open. Orthographic side view; dimensions in mm.



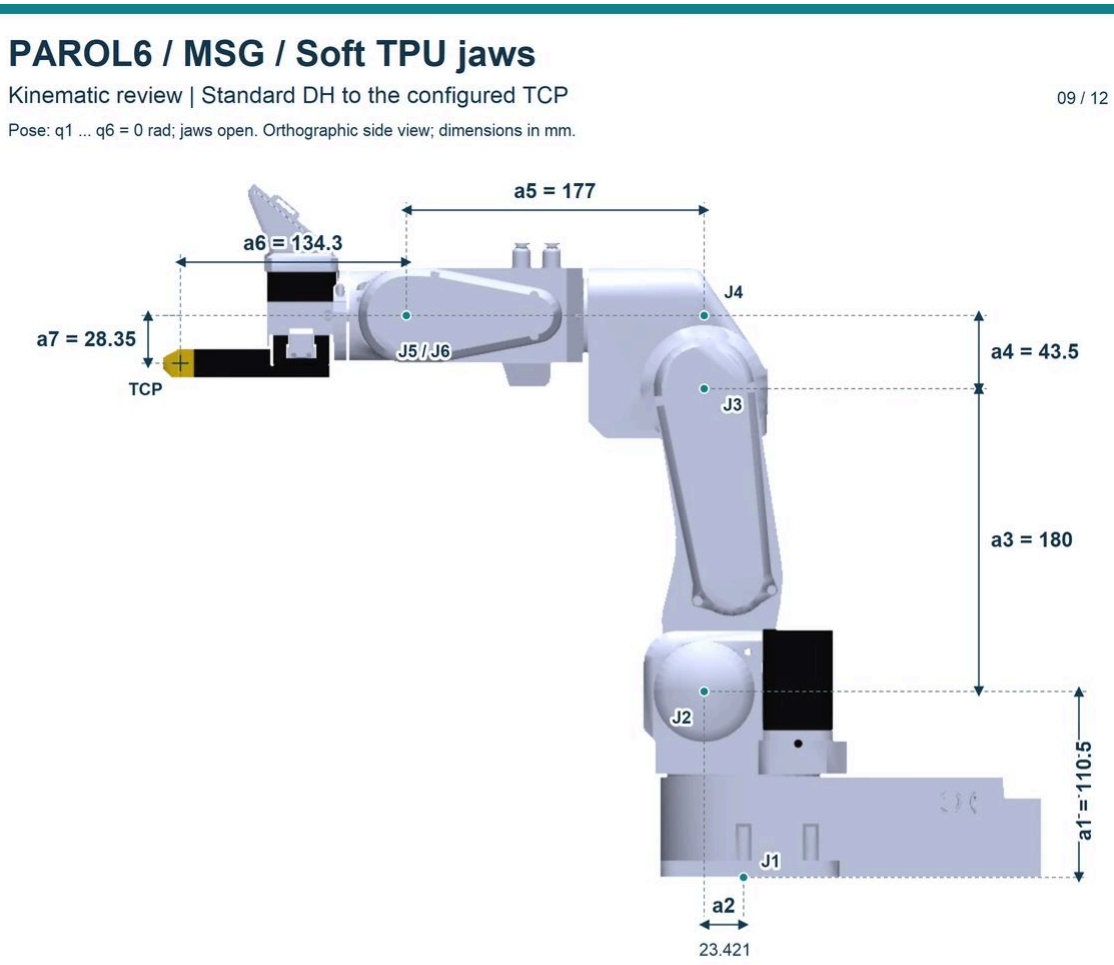
a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

Joint	Link	theta [rad]	alpha [deg]	a [mm]	d [mm]
1	Base	q1	-90	a2 = 23.421	a1 = 110.5
2	Shoulder	q2 - pi/2	180	a3 = 180	0
3	Elbow	q3 + pi	90	-a4 = -43.5	0
4	Wrist 1	q4	-90	0	-a5 = -177
5	Wrist 2	q5	90	0	0
6	Wrist 3 / TCP	q6 + pi	180	-a7 = -28.35	-a6 = -134.3

Figure 1: PAROL6 with the MSG, solid jaws: dimensions to the TCP and the DH table.

Soft TPU jaws



Joint	Link	theta [rad]	alpha [deg]	a [mm]	d [mm]
1	Base	q1	-90	a2 = 23.421	a1 = 110.5
2	Shoulder	q2 - pi/2	180	a3 = 180	0
3	Elbow	q3 + pi	90	-a4 = -43.5	0
4	Wrist 1	q4	-90	0	-a5 = -177
5	Wrist 2	q5	90	0	0
6	Wrist 3 / TCP	q6 + pi	180	-a7 = -28.35	-a6 = -134.3

Figure 2: PAROL6 with the MSG, soft TPU jaws: dimensions to the TCP and the DH table.

The joint 5 homing offset

Joint 5 homes against the tool, and the MSG's ring cover is not the pneumatic gripper holder that the firmware's default offset (8900) assumes. After the first homing, check joint 5 with the pin through the witness marks; if it is off, measure the offset and set `j5_homing_offset` in `main.cpp` as Mastering describes.

Driving it

The gripper must be calibrated after every power-up: it opens fully and closes fully to find its end stops, and does not move before that. Clear any error first, then calibrate, then wait until the gripper reports itself calibrated before the first move: about 5 to 6 s on the longer rails.

Commander software

The **Gripper** tab: after homing the robot press **Clear gripper error**, then **Calibrate gripper**, and wait until the tab shows `Calibration status is: 1`. Then set position, speed and current with the sliders and press **Move GoTo**. The feedback section shows the current, the position and the status (The GUI). In RBTscript, after clearing the error in the tab: `Gripper_cal()`, `Delay(7)`, then `Gripper(position, speed, current)` (RBTscript).

Python API

```
rbt.select_tool("MSG", variant_key="150mm")           # 100mm, 150mm or 200mm: the rail
rbt.tool.calibrate()                                   # queued; the API counts it done after 2 s
rbt.delay(5)                                           # the rest of the calibration, with a
margin
rbt.tool.set_position(0.8, speed=0.5, current=0.4)     # fractions of the ranges; 0.4 is 1180 mA
rbt.tool.close(current=0.4)
rbt.tool.open()
rbt.tool.release()                                     # drop the grip, the jaws stay
rbt.tool.stop()                                        # immediate: halt where the jaws are
```

Tool actions are queued with the motion, so a move after `close()` waits for the grip; a `move` on a gripper that was never calibrated is refused when its turn comes. The API does not read the calibrated bit and has no clear-error action at this version: clear an error in the commander's Gripper tab or with Motor GUI (Tools and the TCP, Tools).

Current and force

The current limit is the grip force, and the slope depends on the stepper: about 20 N per ampere with the 21.5 mm stepper, 25 with the 40 mm, 36 with the 60 mm. Stay within the stepper's maximum, 1.5 A, 2.0 A or 2.1 A. The gripper manual's Force and speed has the curves.

The commander takes the current in milliamps, 100 to 1000 mA, which is within every stepper's maximum. The Python API takes it as a fraction of 100 to 2800 mA, a range above every stepper's maximum, so keep the fraction at or below the value for your stepper:

Stepper	Maximum current	Highest `current=` fraction
21.5 mm	1.5 A	0.51
40 mm	2.0 A	0.70

Stepper	Maximum current	Highest `current=` fraction
60 mm	2.1 A	0.74

Status

The robot passes the gripper's status byte through: calibrated, in motion, object detected (while closing, while opening, or none), and its error bits. The commander's Gripper tab shows the calibrated and E-stop error bits, and the whole status byte as a number (labelled *Gripper object detection is*). The Python API's `tool.status()` gives the jaw position and the current; its `engaged` and `part_detected` do not follow the status byte's bits at this version (Tool status). The gripper manual's errors page says what each error needs: Errors and recovery.

Your own tool

Mounting

Clamp onto the 8 mm shaft of joint 6 with a shaft coupler from the BOM, or print a holder with a coupler connector: the SSG-48 repository has a template, `STEPS/coupler_connector_generic_STEP.STEP` (and `Gripper STL parts/connector_coupler_generic.STL`), and the `STL/WRIST/wrist.STL` in the robot's files shows the mating geometry. Keep the tool within the payload and short: every millimetre past the flange is leverage against joints 4 and 5.

The joint 5 homing feature

Joint 5 finds its reference by driving the forearm's limit switch against a feature of the tool holder. Your holder needs a surface in the same place as the pneumatic gripper holder's; compare with `Pneumatic_gripper_holder.STL`. Then measure the homing offset: home, jog joint 5 until the pin fits the witness marks, and set `j5_homing_offset` in `main.cpp` to the steps that gives (Mastering).

Connections

Up to 24 V and CAN at the M8 connector (its pins: The gripper socket), two air lines, and the control board's two inputs and two outputs at the base. A tool that needs more, a camera's USB for instance, runs its cable outside the arm, with enough slack for joints 4, 5 and 6 and fixed so the arm cannot catch it.

The tool frame

The software positions the tool centre point. In the Python API, register the tool once with its transform from the flange, or set an offset at run time:

```
rbt.select_tool("NONE")
rbt.set_tcp_offset(0, 0, -120)                # the tip, 120 mm out from the flange: along the
tool's -Z
index = rbt.set_tcp_transform(0, 0, -120, 0, 15, 0)  # with a rotation, intrinsic XYZ degrees
rbt.wait_command(index)
```

The Python API's tool frame has Y and Z the other way

In the Python API (version 0.9.1) the flange frame's Z points back into the wrist, the other way from the commander's and these pages' (The flange frame), so a tip in front of the flange has a negative z there (known issue KI-11).

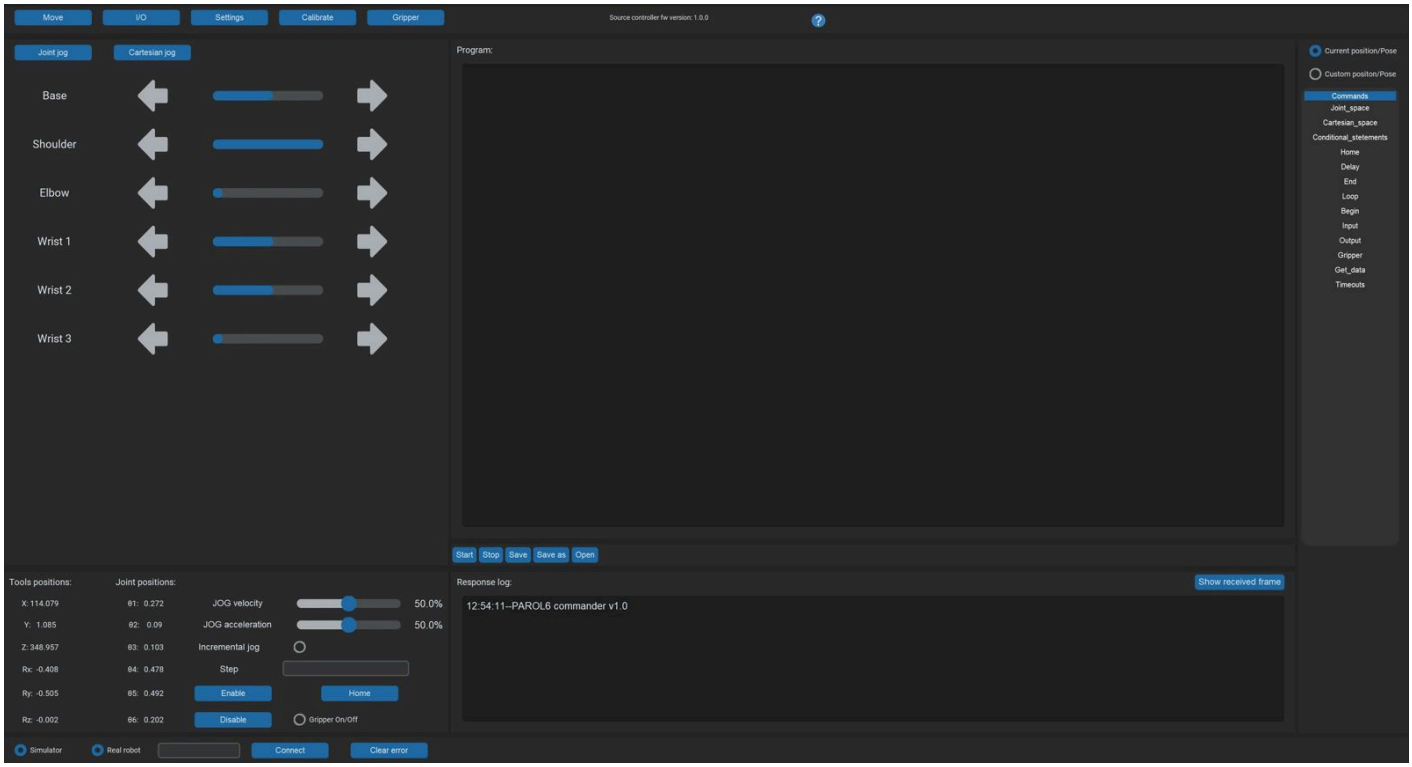
A permanent tool goes into the API's tool registry as a `ToolConfig` with its collision mesh, so that the collision checker knows its shape (Tools and the TCP). The commander software has no tool setting: its poses are for the standard pneumatic gripper, and a different tool means editing its robot model.

Check before you run

Jog joint 5 and joint 6 through their ranges slowly with the tool mounted and watch for contact with the forearm: the software's joint limits do not know the tool's shape.

GUI interfaces

PAROL6 GUI Guide



Document number: SR-4005

PAROL6 release: 1.1

PAROL6 commander software: main

Revision: 2026.B

Published: 2026-11-05

Applies to: PAROL6 release 1.1, PAROL6 commander software main

Original instructions

The graphical programs for PAROL6 and its grippers: which one to open for which job, how to install it, and what is on its screen. Programming the robot, in RBTscript or Python, is in the Programming Guide.

	Commander software	Waldo Commander	Motor GUI
What it is	Source Robotics' desktop program: jogging, a 3D simulator, the RBTscript	A community browser interface on the Python API: a simulator, Python programs, teaching by	Source Robotics' browser tool for the

	Commander software	Waldo Commander	Motor GUI
	editor, the I/O and gripper tabs	demonstration, an MCP server for LLMs	drivers inside the electric grippers
For	the first day, teaching poses, demos, simple sequences	remote use, programs in the browser, AI-driven control	setting up, testing and updating an SSG-48 or MSG gripper
Runs	on the PC at the robot, Python 3.10 or 3.11	a server at the robot, any browser	any Chromium browser, with the CANvas adapter
Talks to	the control board over USB	the Python API's controller	the gripper's driver over CAN

In this guide

- The commander software: Installation, The GUI
- Waldo Commander
- Motor GUI

One program owns the control board at a time: the commander and the Python API's controller cannot be connected together (Control ownership).

The commander software

The commander software

The PAROL6 commander software is the desktop program for using the robot by hand: a GUI with joint and Cartesian jogging, a 3D simulator that mirrors the robot, a response log, a program editor and interpreter for RBTscript, the I/O tab and the gripper tab. It is written in Python with customtkinter and the Robotics Toolbox, runs a 100 Hz loop to the control board, and is the program to start with.

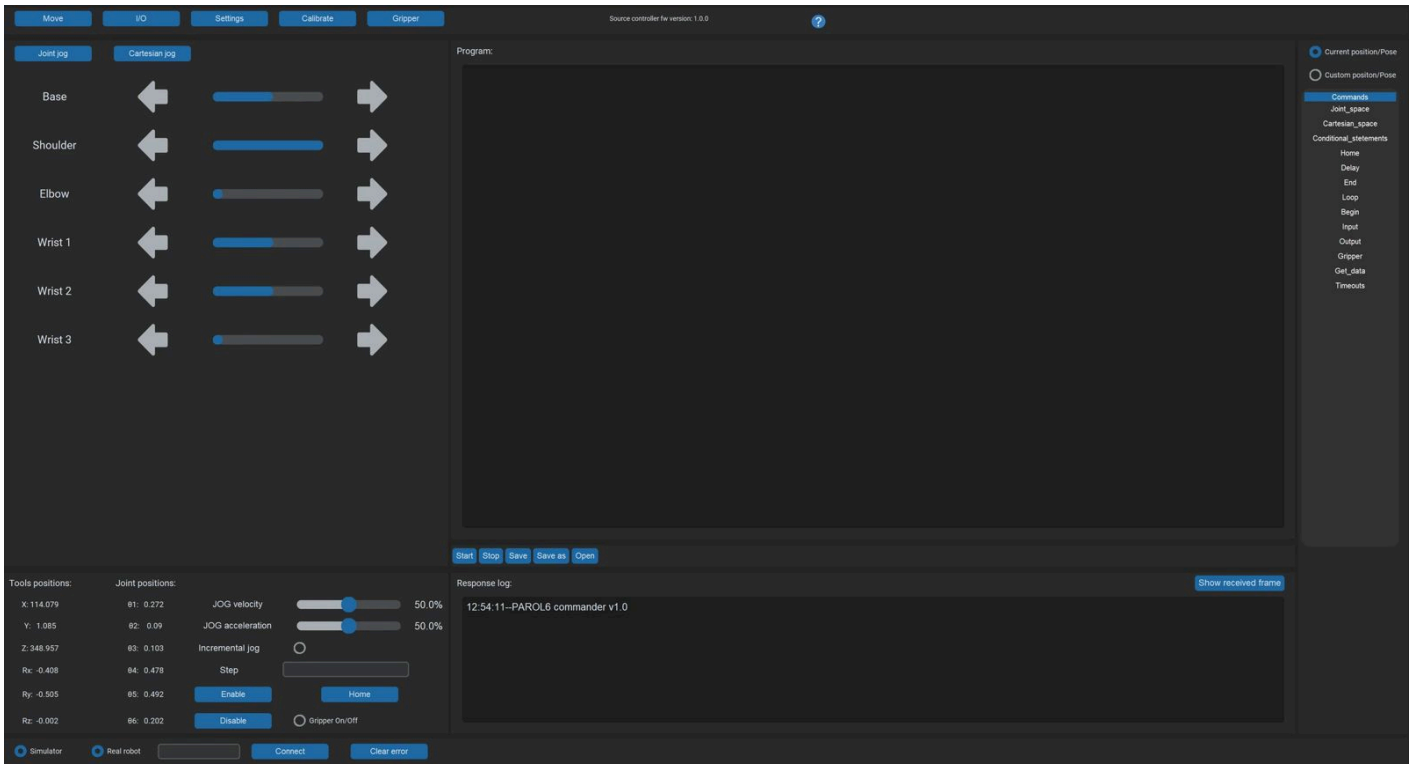


Figure 1: The commander window.

In this guide

- Installation
- The GUI: the windows, the menus, connecting, jogging, the tabs

Its program language, RBTscript, is in the Programming Guide: RBTscript, Examples.

What it is and is not

It is the tool for the first day, for teaching poses, for demos and for simple sequences. It has no blending, no collision checking, no network interface and no tool selection: its kinematic model is the one with the pneumatic gripper. For programs that need those, the Python API. The two cannot be connected to the robot at the same time.

The experimental kinematics branch

The repository's `experimental_kinematics` branch resolves most Cartesian jogging errors but can command dangerous motion. The `main` branch is the safe one; use the other at your own risk.

Installation

Requirements

- Python 3.10 or 3.11 (tested on 3.10.12 and 3.11.0; newer versions break some pinned dependencies).
- Windows, Linux or macOS. On macOS, follow the Linux steps, as the repository's README says.
- A PC that keeps the 100 Hz loop: What you need.

Install

The commander goes into its own folder with its own virtual environment (`.venv`), on every operating system. It pins numpy 1.23.4 and the Python API needs numpy 2, so the two cannot share one Python: give each its own environment, as the Quick start does.

Windows

```
git clone https://github.com/Source-Robotics/PAROL-commander-software.git
cd PAROL-commander-software
py -3.11 -m venv .venv
.venv\Scripts\activate
python -m pip install -r requirements.txt
```

`py -3.11` picks Python 3.11 when several are installed; `py -3.10` works too. The robot's serial port is `COM` and a number: Device Manager lists it under **Ports (COM & LPT)**.

Linux

```
sudo apt install python3.11 python3.11-venv python3.11-tk
git clone https://github.com/Source-Robotics/PAROL-commander-software.git
cd PAROL-commander-software
python3.11 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.txt
sudo usermod -aG dialout $USER     # access to the serial port; log out and in again once
```

Ubuntu's own Python is 3.10 on 22.04 and 3.12 on 24.04; where Python 3.11 is not in the distribution, the deadsnakes PPA has it (`sudo add-apt-repository ppa:deadsnakes/ppa` first). On 22.04 the system Python 3.10 with `python3-venv` and `python3-tk` works as well. The robot's serial port is `ttyACM0` (`ls /dev/ttyACM*` lists it).

macOS

```
git clone https://github.com/Source-Robotics/PAROL-commander-software.git
cd PAROL-commander-software
python3.11 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.txt
```

With Python 3.11 from python.org; the repository's README says the Linux steps work on macOS.

After the activate line the prompt starts with `(.venv)` . Activate the environment again in every new terminal before you start the program.

The repository's Windows and Linux guides list the packages one by one if `requirements.txt` fails: the robotics toolbox, swift, numpy and scipy are pinned to the versions that work together, and the `Working dependency` folder has the full module lists of PCs on which the software runs.

Run

For the first run, write the robot's port into the program: open `GUI/files/Serial_sender_good_latest.py` in a text editor and find these lines near the top:

```
if my_os == "Windows":
    STARTING_PORT = 58 # COM3
else:
    STARTING_PORT = 0
```

On Windows, replace `58` with the number of the robot's port (`5` for `COM5` ; Device Manager, **Ports (COM & LPT)**, shows it). On Linux, `0` means `/dev/ttyACM0` . Save, then start the program from the activated virtual environment:

```
cd GUI/files
python Serial_sender_good_latest.py
```

Two windows open: the commander and the simulator. Enter the serial port at the bottom of the commander (`COM5` , or `ttyACM0` on Linux) and press **Connect**, more than once if needed. On Linux, a permission error on the port is fixed with `sudo chmod 666 /dev/ttyACM0` , or by joining the `dialout` group.

Check the loop

The terminal the program runs in prints the robot's data; `Time between 2 commands in ms` should sit near 0.010 while you jog. The value is in seconds, whatever the label says: 0.010 is the 10 ms cycle. If it is often much higher, the PC is too slow for the loop, and the robot will stutter (Communication).

> ✓ **Check**
>
> The commander shows the robot's data after **Connect**, and the simulator moves with the robot when you jog. If not: Communication.

The GUI

Two windows: the **commander**, where everything is done, and the **simulator**, a 3D view of the robot that follows it once the robot is connected and homed.

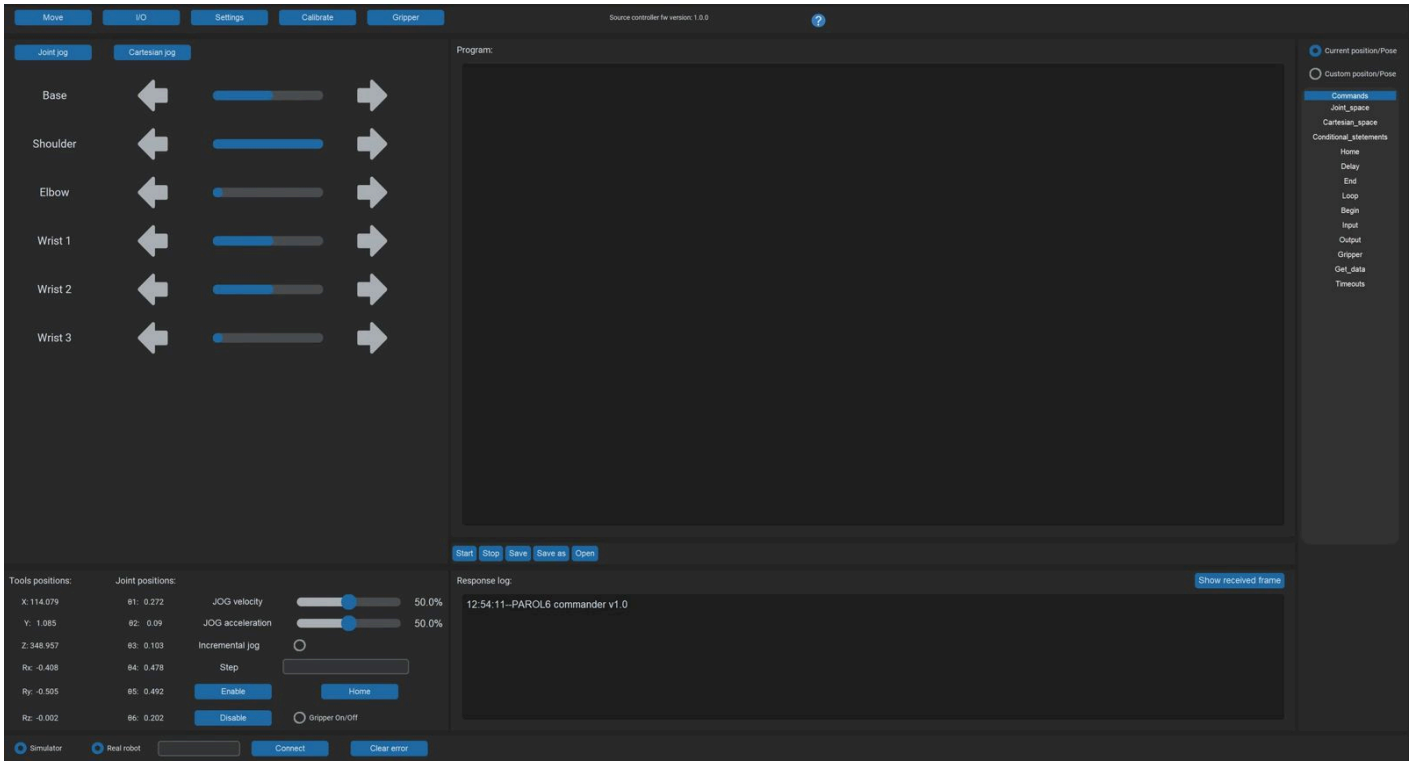


Figure 1: The commander window, with the joint jog menu.

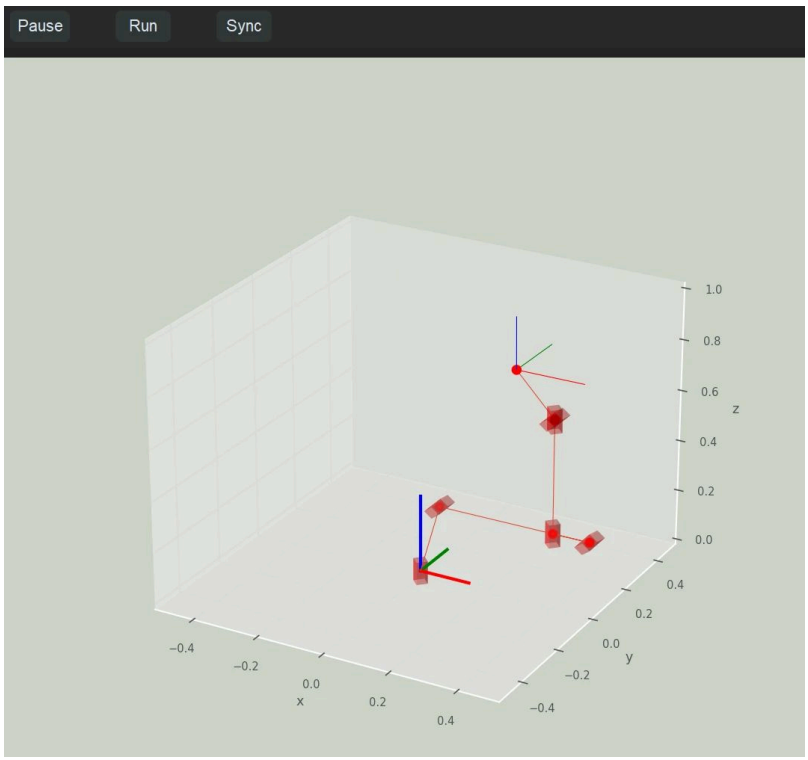


Figure 2: The simulator window.

Connecting



Figure 3: The connection bar.

At the bottom: the serial port field and **Connect**. `COMx` on Windows, `ttyACMx` on Linux. The response log starts showing the robot's data when the connection is up; press Connect again if it does not. The **Simulator** and **Real robot** buttons have no function at present. **Clear error** resets the homing state; unlike **Enable** it does not re-enable a disabled robot.

The Move menu

The left part of the window is the jog menu, with two variants:

Joint jog: six pairs of arrows, left for the positive direction and right for the negative. The speed slider below sets the jog speed as a percentage; it affects jogging only, not programs. Works before homing.

Cartesian jog: X, Y, Z and the rotations, in the **WRF** (the base) or the **TRF** (the tool), chosen with the switch; **TRF** is the default. The rotation buttons only work in the TRF: in the WRF they do nothing. Home the robot first (the software does not check), and the jog stops at a singularity (Configurations and singularities).

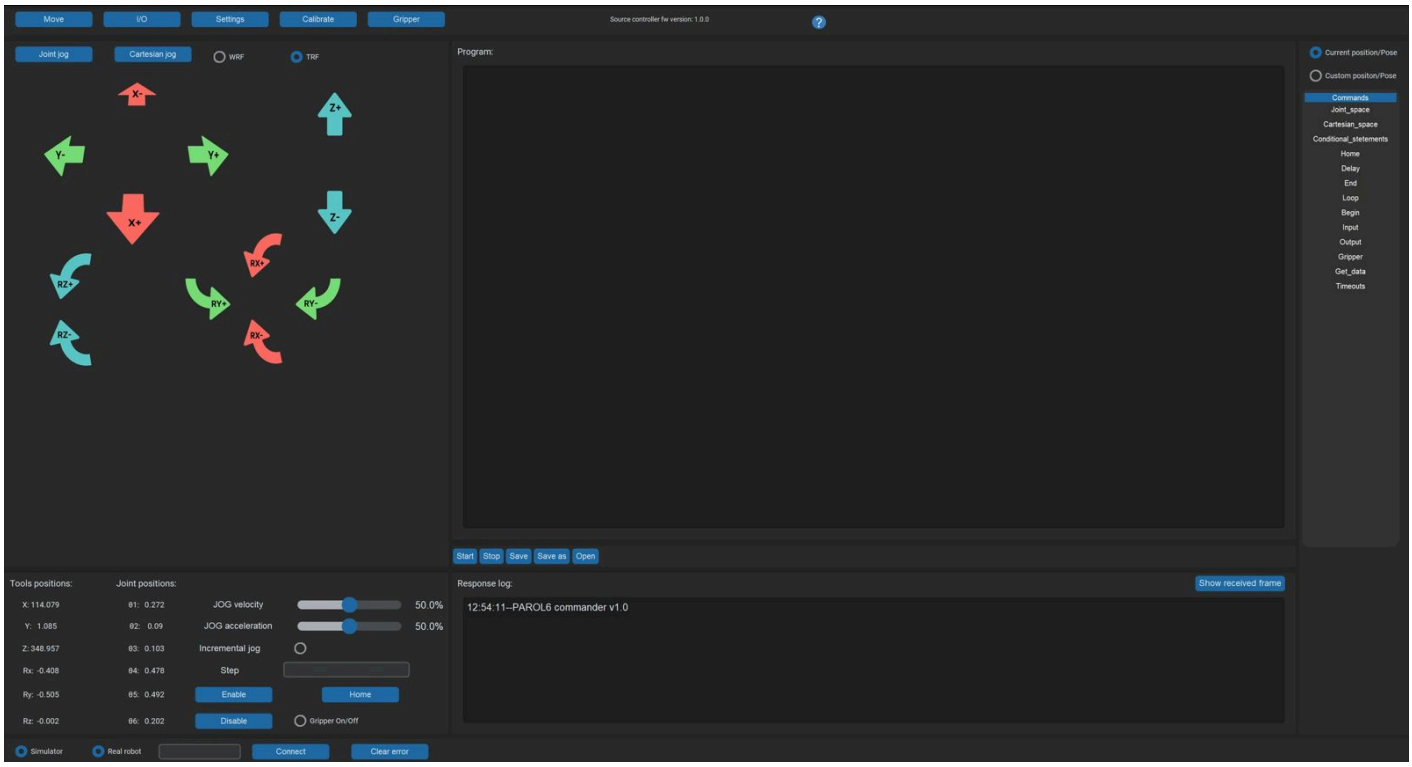


Figure 4: The Cartesian jog menu.

The robot position



Figure 5: Joint angles, the tool pose and the speed slider.

The six joint angles and the tool pose in the WRF, live. **Home** starts the homing sequence; **Enable** clears errors and enables the robot; **Disable** disables it, the motors still holding. **Park** does not move the robot: it inserts a three-line program at the cursor of the program window, `Begin()`, a `MoveJoint` to the park pose with `t=4`, and `End()`, which you then run with **Start**.

The program window

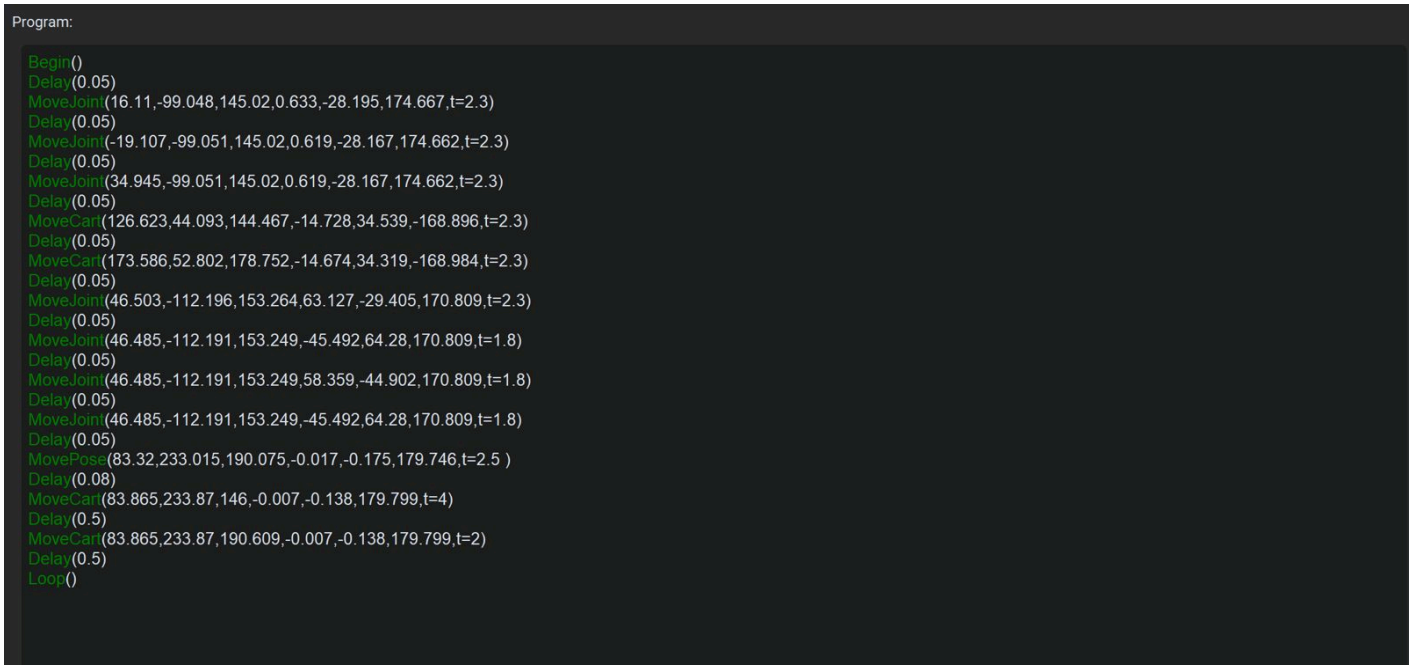


Figure 6: The program editor.

An RBTscript program, one command per line (RBTscript). **Start** runs it from `Begin()`, **Stop** halts it. **Start** first saves the editor's text: into the file you opened, if any, overwriting it, and into `Programs/execute_script.txt`.

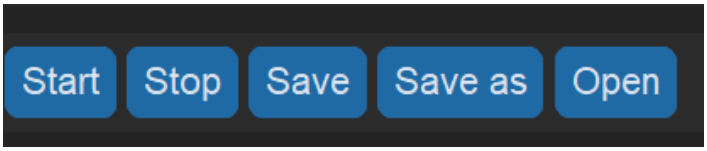


Figure 7: Start and stop.

The command list inserts a command at the cursor: with *Current* selected, a motion command is inserted with the robot's current joint angles or pose as its arguments, which is how you teach a pose; with *Custom* the arguments are left for you.

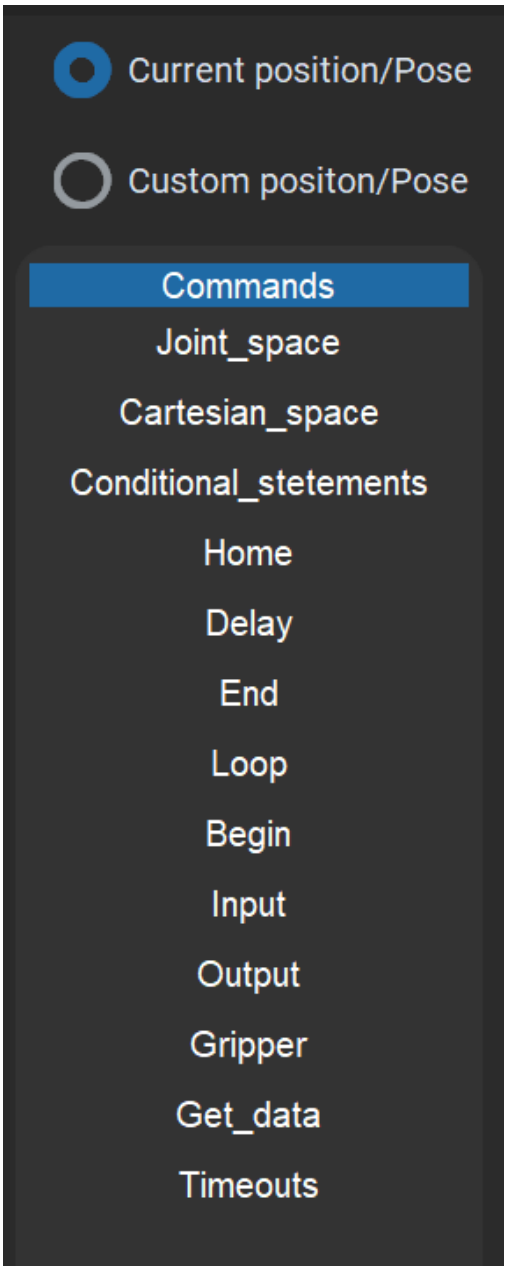


Figure 8: The command list.

The response log

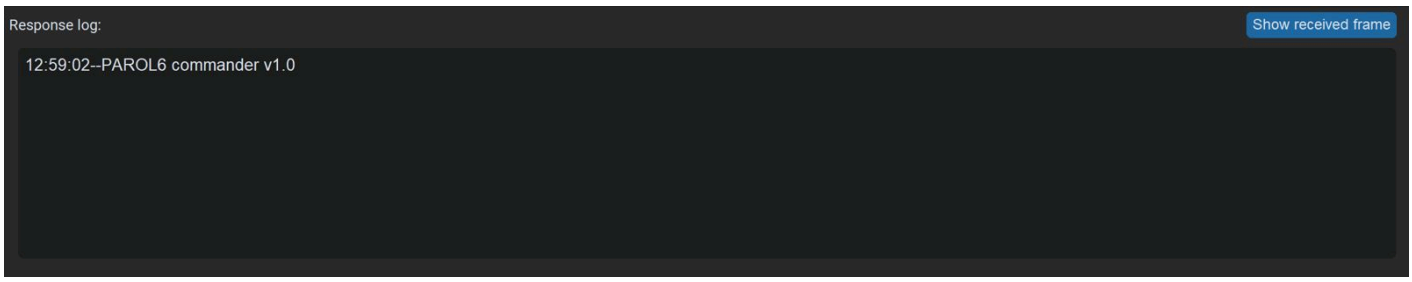


Figure 9: The response log.

What the robot and the program are doing, and why a command was refused: the first place to look when something does not happen (Errors and recovery).

The IO tab

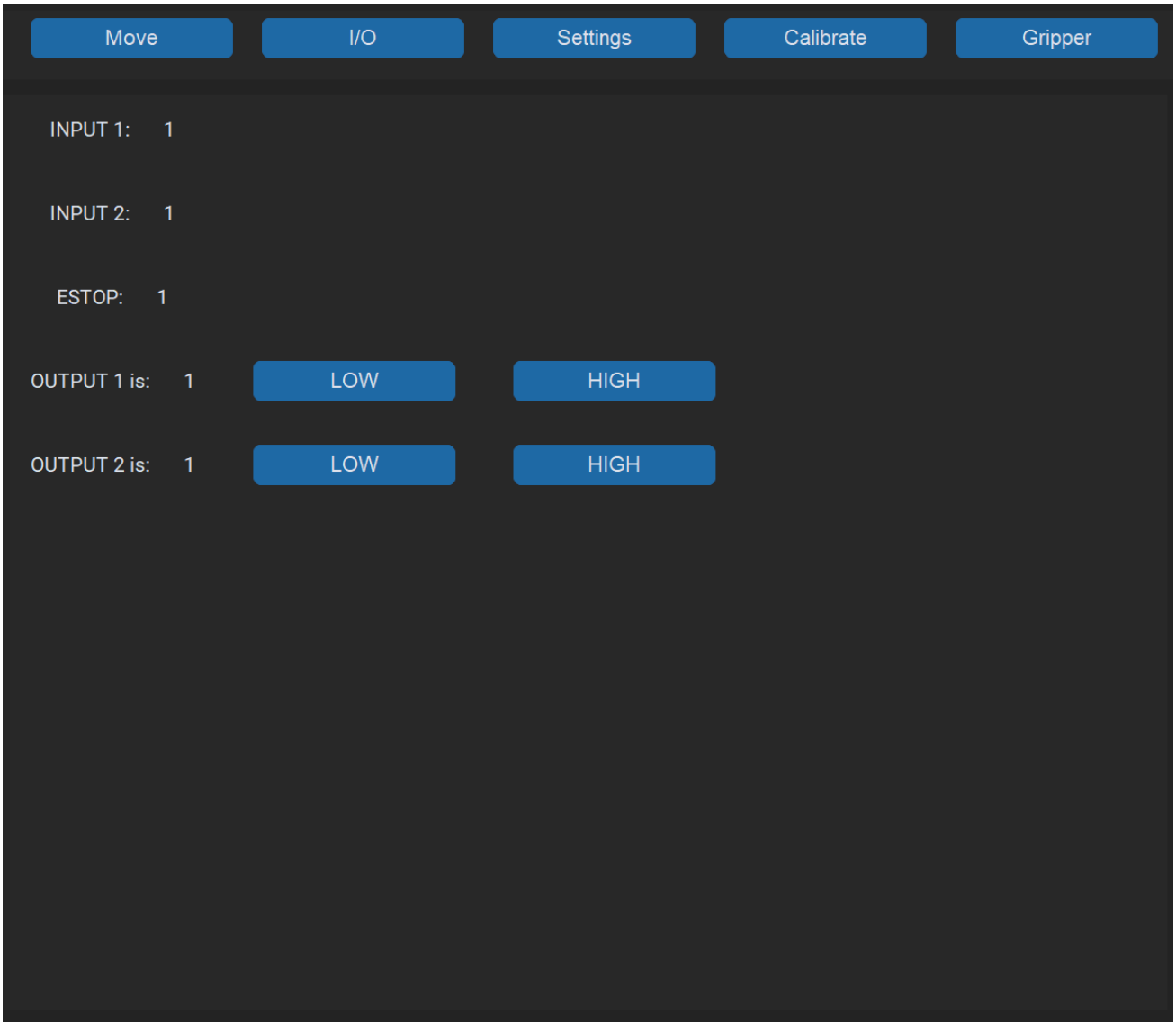


Figure 10: The IO tab.

The state of the two inputs, and buttons for the two outputs.

The Gripper tab

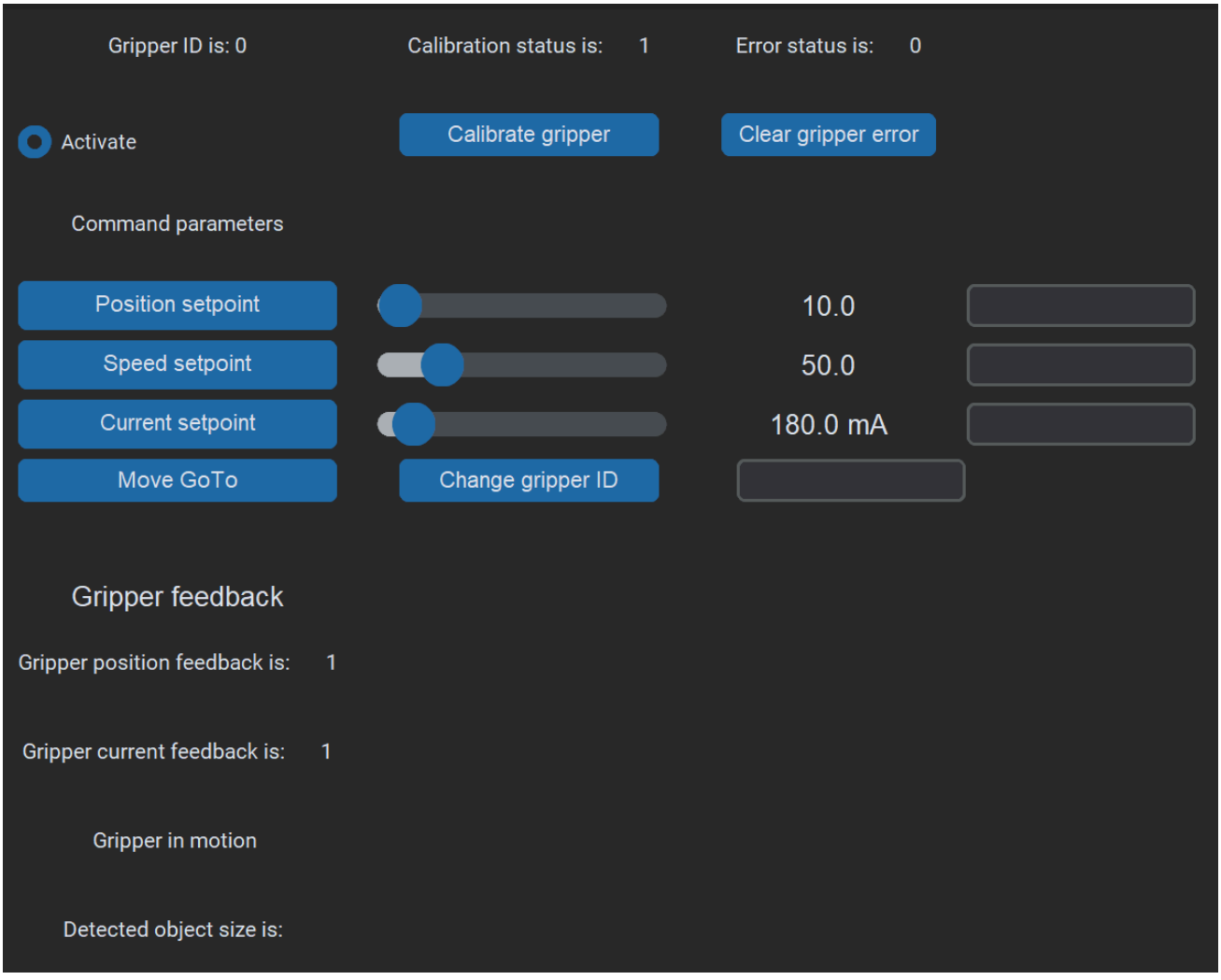


Figure 11: The gripper tab.

For the SSG-48 and MSG grippers. After homing: **Clear gripper error**, then **Calibrate gripper**, then wait until the tab shows `Calibration status is: 1` (about 3 s for the SSG-48, 5 to 6 s for the MSG on its longer rails). Then set the position, speed and current with the sliders and press **Move GoTo**. The feedback section shows the gripper's position and current, its calibrated and E-stop error bits, and its whole status byte as a number (labelled *Gripper object detection is*). **Change gripper ID** sets the CAN ID the board addresses, 0 as the grippers ship.

The Calibrate and Plot buttons

The menu also has **Calibrate** and **Plot**. Both open the same panel, with **Disable motor**, **Enable motor**, **Go to limit** and a joint menu; none of them does anything in the current version.

The Settings tab

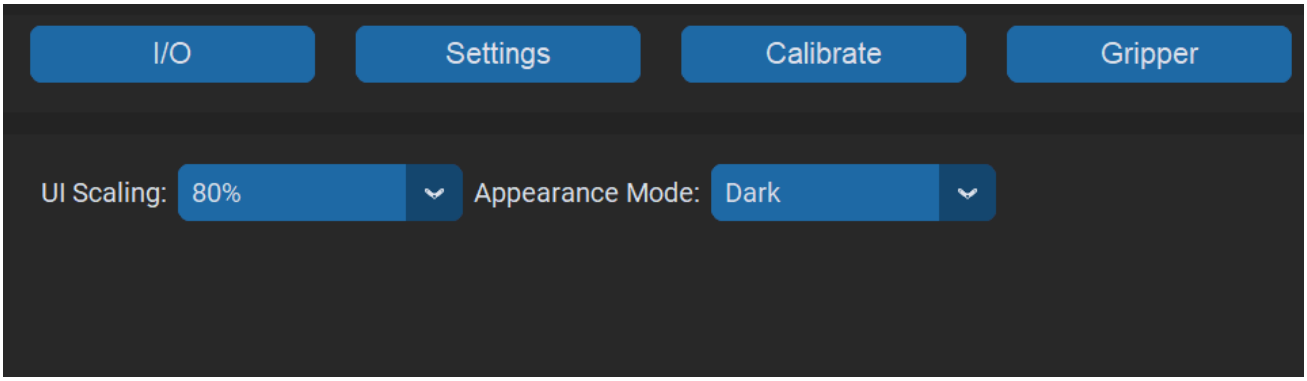


Figure 12: The settings tab.

Dark or light mode, and the scaling of the GUI for your display.

The terminal

The terminal the program was started from prints a block of data every few seconds: the positions and speeds from the robot, the homed, I/O and error bytes, the timing, the gripper data, the commanded data and the GUI's state. `Time between 2 commands in ms` is the one to watch: near 0.010 is right. The value is in seconds, whatever the label says: 0.010 is the 10 ms cycle.

Waldo Commander (web)

Waldo Commander, by a member of the community, is a browser interface on top of the Python API: control from any device on the network, Python programs with an editor and step-through debugging, a 3D simulation to preview paths, teaching by demonstration recorded as Python, and a built-in MCP server through which an LLM can read the robot's status, write and run programs and drive the arm with your approval. It runs in simulator mode without a robot.

```
git clone https://github.com/Jepson2k/Waldo-Commander.git
cd Waldo-Commander
pip install -e ".[paro16]"
waldo-commander
```

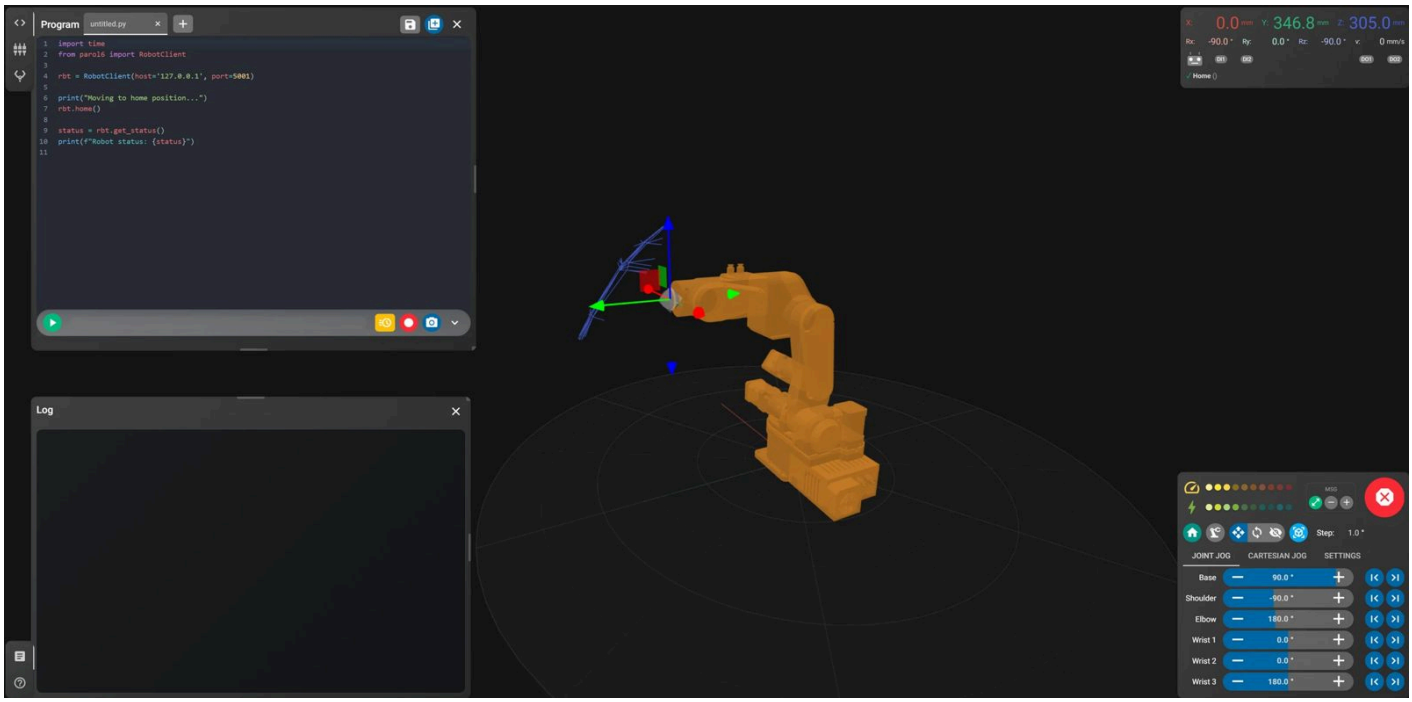


Figure 1: The web commander.

It is built on waldoctl, the abstraction layer the Python API implements, so what it offers tracks the API's features. Its documentation is at jepson2k.github.io/Waldo-Commander.

Motor GUI

Motor GUI is Source Robotics' browser tool for its motor drivers, and so for the SSG-48 and MSG grippers, whose driver sits inside. It reaches the hardware through the browser's serial port access: nothing to install, a Chromium-based browser and the CANvas USB-to-CAN adapter.

On PAROL6 the control board drives the gripper, so Motor GUI is not part of everyday use. It is the tool when you:

- **update the gripper's firmware**, over CAN, from the driver's GitHub releases, through the gripper's bootloader;
- **set up a self-built gripper**, or restore a gripper's configuration: the setup page has presets for the SSG-48 and for the three MSG steppers. They are applied over UART, with the USB-to-serial adapter on the driver's UART port, not over CAN. A preset writes CAN ID 6, the PAR6 gripper's: change it to 0 before you apply it for PAROL6;
- **test a gripper off the robot**: calibrate, open and close it, watch its position and current, run a cycle test, in the gripper panel.

The control board polls the gripper continuously while the robot is on, so use Motor GUI with the gripper off the robot's bus: the gripper on the CANvas adapter's CAN terminals and on a 24 V supply. Then follow the gripper manual's Motor GUI page, SSG-48 or MSG; the firmware update is on their Firmware pages (SSG-48, MSG).

Programming

PAROL6 Programming Guide

Document number: SR-4004

PAROL6 release: 1.1

Python API (parol6): v0.9.1

PAROL6 commander software: main

Revision: 2026.A

Published: 2026-11-03

Applies to: PAROL6 release 1.1, Python API (parol6) v0.9.1, PAROL6 commander software main

Original instructions

This guide is for everyone who makes PAROL6 do something: from a first RBTscript program in the commander software to a Python application on the API, a ROS 2 pipeline, or a program that an AI agent writes. It starts with the interfaces and how to choose, walks through the Python API and the commander software, and ends with ROS 2 and the other interfaces. The ideas the software is built on, the units, the frames, the kinematics, the configurations, the motion profiles, are in the Theory corner, a tab of their own. The API reference is where every method and command is listed, read from the API's source at the version named above.

Start here

The interfaces at a glance: which program for which job, and what they have in common.

Theory corner

Units, joints and poses, frames, kinematics, singularities, motion profiles, blending, control ownership, and more.

The Python API

Installation, the controller, clients, motion, jogging and servoing, tools, status, errors, examples.

RBTscript

RBTscript, the commander software's program language, and examples; the commander itself is in the GUI Guide.

ROS 2 and others

API reference

ROS 2 and MoveIt, other interfaces: the serial protocol directly, the legacy API, AI agents.

Every Python method, RBTscript, the serial protocol, firmware parameters.

Guide

1. The interfaces at a glance

Every way of programming PAROL6 ends at the same place: a packet to the control board every 10 ms with the joint speeds and positions to execute. What differs is who computes those packets and how you tell it what you want.

	Commander software	**Python API**	**ROS 2 + MoveIt**	**Waldo Commander (community)**
What it is	A desktop GUI with a simulator and a program editor	A controller process and a Python client library	A MoveIt configuration and a `ros2_control` interface on top of the Python API's controller	A browser GUI on top of the Python API, with an MCP server for LLMs
You write	RBTscript, one command per line	Python	ROS 2 nodes, MoveIt plans	Python, in the browser, or prompts
Motion	Joint and Cartesian moves, trapezoidal or quintic profiles, teach by jogging	Planned moves (joint, linear, circular, spline, process), streamed jogs and servo targets, blending, five profiles, collision checking	MoveIt planning	What the API offers, plus teach by demonstration
Tools	SSG-48 and MSG gripper tab, outputs	Registered tools with variants, TCP transforms, tool actions in the motion queue	Through the bridge	Through the API
Simulator	3D window, needs the robot connected	Simulator mode and a dry-run client, no robot needed	RViz	3D preview, no robot needed
Runs on	The PC at the robot	The controller at the robot; clients anywhere on the network	Ubuntu 22.04 with ROS 2 Humble	Any browser; the server at the robot
Python	3.10 or 3.11	3.11 to 3.14		3.11 or newer

	Commander software	Python API	ROS 2 + MoveIt	Waldo Commander (community)
Maturity	Stable, no releases, main branch	Pre-1.0, released often, interfaces can change between minors	Demo and real-robot launch files	Community project
Best for	The first day; teaching points; demos; simple sequences	Everything else	ROS users	Remote use; AI-driven control

Which one

- **You have never run the robot**: the commander software. The GUI shows the robot, the jog buttons check the wiring, and a three-line program is a minute's work.
- **You want to write programs**: the Python API. Its controller runs the robot; your script, or several, talk to it; the dry-run client tests the script without a robot.
- **You have a ROS 2 stack**: the MoveIt configuration, with the Python API's controller underneath.
- **You want to drive the arm from a browser or an LLM**: Waldo Commander, a community project on the API.

What they share

- The serial protocol to the control board, at 100 Hz.
- The robot model: the joint limits, the gear ratios, the standby pose, the kinematics (Theory corner).
- The rule that **one program owns the board**: the commander and the API's controller cannot be connected at the same time (Control ownership).
- Homing after every power-up, and the E-stop handled by the PC software (Emergency stop).

What they do not share

Poses and programs do not transfer between the commander and the API without care: the commander's kinematic model assumes the pneumatic gripper, the API's assumes the tool you select, and the two have different speed limits and profiles. A pose taught in one is a starting point in the other, not a result.

2. The Python API

2. The Python API

The Python API, package `parol6`, is two things: a **controller** process that owns the robot, talks to the control board over serial at 100 Hz, plans the motion and checks collisions; and a **client library** that sends it commands over UDP, from the same PC or any other on the network, with a blocking interface for scripts and an asynchronous one for applications. A **dry-run client** runs the same planning without a robot, for tests.

This chapter walks through it in the order you use it. The reference lists every method: Python API reference, read from the API's source at the version this site describes.

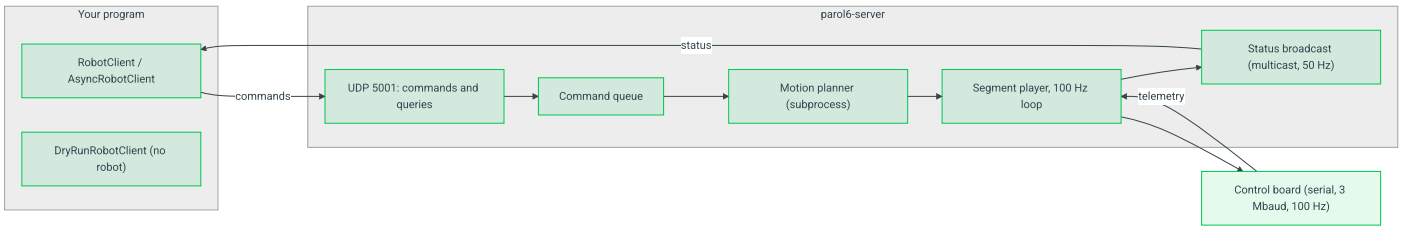
In this chapter

- Installation
- The controller: starting it, the serial port, simulator mode, the environment variables
- Clients: the sync client, the async client, the dry-run client, the `Robot` lifecycle helper
- Motion: joint, linear, circular, spline and process moves, speed and duration, blending, profiles
- Jogging and servoing: streamed velocity and position targets
- Tools and the TCP: selecting a tool, gripper actions, the TCP transform
- I/O, delays and checkpoints
- Status and waiting: queries, the status stream, waiting for commands
- Errors
- Examples

In three lines

```
from parol6 import RobotClient
with RobotClient() as rbt:            # the controller runs already: parol6-server --serial COM5
    rbt.home(wait=True); rbt.move_j([90, -90, 180, 0, 0, 180], speed=0.5)
```

Architecture



The controller is the only process that touches the serial port. The planner runs in a subprocess so that a long plan never delays the 100 Hz loop; the loop plays the planned segments and streams the jog and servo targets. Status goes out as multicast, so any number of listeners see the robot's state without asking.

Installation

Requirements

- Python 3.11 to 3.14.
- Windows (x86-64), Linux (x86-64, ARM64) or macOS (ARM64): the kinematics backend, pinokin, ships wheels for these, for Python 3.11 to 3.14. Its Linux wheels need glibc 2.39 or newer: Ubuntu 24.04 or newer, not 22.04. Its macOS wheels need macOS 15 or newer.
- The PC at the robot: the minimum PC of the Robot Manual. A client that only sends commands runs anywhere.

Install

From a clone of PAROL6-python-API, in a virtual environment:

Windows

```
git clone https://github.com/PCrnjak/PAROL6-python-API.git
cd PAROL6-python-API
git checkout v0.9.1
py -3.11 -m venv .venv
.venv\Scripts\activate
pip install .
```

The robot's serial port is `COM` and a number: Device Manager lists it under **Ports (COM & LPT)**.

Linux

```
git clone https://github.com/PCrnjak/PAROL6-python-API.git
cd PAROL6-python-API
git checkout v0.9.1
python3 -m venv .venv
source .venv/bin/activate
pip install .
sudo usermod -aG dialout $USER     # access to the serial port; log out and in again once
```

The robot's serial port is `/dev/ttyACM0` (`ls /dev/ttyACM*` lists it).

macOS

```
git clone https://github.com/PCrnjak/PAROL6-python-API.git
cd PAROL6-python-API
git checkout v0.9.1
python3 -m venv .venv
source .venv/bin/activate
pip install .
```

Apple silicon (ARM64) only. The robot's serial port is `/dev/cu.usbmodem` and a number (`ls /dev/cu.usbmodem*` lists it).

The package pulls pinokin, numpy, scipy, Ruckig, TOPP-RA, numba, pyserial, msgspec, ormsgpack, psutil and the waldoctl interface layer. `pip install -e .` installs it editable for development.

Check

```
parol6-server --log-level=INFO
```

starts the controller in simulator mode when no serial port is given. In another terminal:

```
from parol6 import RobotClient
with RobotClient() as rbt:
    print(rbt.ping())
```

Check

It prints a `PingResult` with `hardware_connected=False`. Stop the server with Ctrl-C.

Versions

The API is pre-1.0 and released often; a minor version can change an interface, and the controller and the client must be the same version, since they share the wire protocol. The release notes list the changes; this site describes v0.9.1.

The controller

`parol6-server` is the process that owns the robot. Start it on the PC the control board is plugged into, and leave it running; clients connect and disconnect as they like.

Starting it

```
parol6-server --serial COM5                # Windows
parol6-server --serial /dev/ttyACM0        # Linux
parol6-server --serial /dev/cu.usbmodem1   # macOS: ls /dev/cu.usbmodem* gives the name
parol6-server                              # no serial port: simulator mode
parol6-server --serial COM5 --log-level INFO
```

The options, from the source:

Option	Default	Meaning
`--host`	`0.0.0.0`	UDP host address
`--port`	`5001`	UDP port
`--serial`	—	Serial port (e.g., /dev/ttyUSB0 or COM3)
`--baudrate`	`3000000`	Serial baudrate
`-v`, `--verbose`	`0`	Increase verbosity; -v=INFO, -vv=DEBUG, -vvv=TRACE
`-q`, `--quiet`	flag	Enable quiet logging (WARNING level)
`--log-level`	—	Set specific log level (one of `TRACE`, `DEBUG`, `INFO`, `WARNING`, `ERROR`, `CRITICAL`)

The serial port can also come from the environment (`PAROL6_COM_PORT`) or from the file the controller saves the last port to, so that a second start needs no option. Once running, `connect_hardware("COM5")` from a client connects or re-connects the port, and `simulator(True)` switches to the virtual arm.

From Python

The `Robot` class starts and stops the controller as a subprocess, for scripts that should be self-contained:

```python
from parol6 import Robot, RobotClient

with Robot(host="127.0.0.1", port=5001) as robot:      # starts parol6-server, waits until it answers
    with RobotClient() as rbt:
        rbt.wait_ready(timeout=5)
        rbt.simulator(True)
        rbt.home(wait=True)
```

`Robot` also carries the robot model (kinematics, limits, tools) that the clients use for previews.

Simulator mode

Without a serial port, or after `simulator(True)`, the controller runs the same planner against a virtual arm and broadcasts its state. Everything works except the hardware: homing is instant, `teleport()` sets the pose, collisions are checked. Develop in the simulator and switch to the hardware with `connect_hardware()` or a restart with `--serial`.

The loop

The controller sends a packet to the board every 10 ms (100 Hz) and broadcasts status at 50 Hz. If the PC cannot keep up it logs `loop overbudget by +N%`, as a warning when it happens more than three times a minute (at DEBUG level otherwise); `loop_stats()` shows the figures. Lower `PAROL6_CONTROL_RATE_HZ`, close other programs, and keep logging at INFO or above: TRACE and DEBUG cost time.

Environment variables

Everything configurable without code, read at start-up:

Variable	Default
`PAROL6_BLEND_HOLD_S`	`0.1`
`PAROL6_BUSY_THRESHOLD_MS`	`1.0`
`PAROL6_COLLISION_CHECK`	`1`
`PAROL6_COLLISION_CLEARANCE_M`	`0.005`
`PAROL6_COLLISION_JOG_LOOKAHEAD_S`	`0.15`
`PAROL6_COLLISION_PATH_SAMPLES`	`16`

Variable	Default
`PAROL6_COLLISION_SRDF`	`computed`
`PAROL6_COM_FILE`	`computed`
`PAROL6_COM_PORT`	—
`PAROL6_CONTROL_RATE_HZ`	`100`
`PAROL6_EXECUTION_OVERRIDE_TRANSITION_S`	`1.0`
`PAROL6_MAX_BACKLOG_COUNT`	`500`
`PAROL6_MAX_BLEND_LOOKAHEAD`	`100`
`PAROL6_MCAST_GROUP`	`239.255.0.101`
`PAROL6_MCAST_IF`	`127.0.0.1`
`PAROL6_MCAST_PORT`	`50510`
`PAROL6_MCAST_TTL`	`1`
`PAROL6_PATH_SAMPLES`	`200`
`PAROL6_SERIAL`	—
`PAROL6_SETTLE_MAX_TICKS`	`20`
`PAROL6_STATUS_RATE_HZ`	`50`
`PAROL6_STATUS_STALE_S`	`0.5`
`PAROL6_STATUS_TRANSPORT`	`MULTICAST`
`PAROL6_STATUS_UNICAST_HOST`	`127.0.0.1`
`PAROL_TRACE`	`0`

The ones you are most likely to touch: `PAROL6_COM_PORT` (the serial port), `PAROL6_CONTROL_RATE_HZ` and `PAROL6_STATUS_RATE_HZ` (the loop and broadcast rates), `PAROL6_MCAST_GROUP` and `PAROL6_MCAST_PORT` (where the status goes; `PAROL6_STATUS_TRANSPORT=UNICAST` with `PAROL6_STATUS_UNICAST_HOST` for networks without multicast), and `PAROL6_COLLISION_CHECK` (0 disables collision checking). A path or an expression as the default is marked *computed*: see `config.py`.

Security

The controller accepts any well-formed command on its UDP port from anyone who can reach it. That is by design, so that a GUI and a script can share it, and it means: run it on a trusted network, or bind `--host 127.0.0.1`.

Clients

Three clients, one interface.

	`RobotClient`	`AsyncRobotClient`	`DryRunRobotClient`
Style	Blocking calls	`async` / `await`, status as an async generator	Blocking, same methods
Needs	A running controller	A running controller	Nothing: plans in-process
For	Scripts, programs, tests on the robot	GUIs, teleoperation, anything that streams	Unit tests, previews, CI

All three are created with the controller's host and port (`127.0.0.1`, `5001` by default), work as context managers, and are closed with `close()`.

The sync client

```
from parol6 import RobotClient

with RobotClient(host="127.0.0.1", port=5001, timeout=2.0) as rbt:
    if not rbt.wait_ready(timeout=5):
        raise SystemExit("no controller")
    rbt.home(wait=True)
    index = rbt.move_j([90, -90, 180, 0, 0, 180], speed=0.5)    # waits by default (wait=True)
    print(rbt.pose(), rbt.angles())
```

A motion call returns the **command index** the controller assigned, or −1 if no acknowledgement arrived; a command the controller refuses raises `MotionError`. With `wait=True` the call blocks until that command completes and raises `MotionError` if it failed; with `wait=False` the call returns at once and `wait_command(index)` waits later. Queries return values or `None` on a timeout.

The defaults differ between the calls and between the clients:

Call	`RobotClient`	`AsyncRobotClient`	`timeout` **(used when waiting)**
`move_j`, `move_l`, `move_c`, `move_s`, `move_p`	`wait=True`	`wait=False`	10 s

Call	`RobotClient`	`AsyncRobotClient`	`timeout` (used when waiting)
`home`	`wait=False`	`wait=False`	60 s
`tool_action` and the `rbt.tool` methods	`wait=False`	`wait=False`	10 s

A move that runs longer than its `timeout` raises `TimeoutError` while it goes on running: give long moves a larger `timeout`.

The async client

```
import asyncio
from parol6 import AsyncRobotClient

async def main():
    async with AsyncRobotClient() as rbt:
        await rbt.wait_ready(timeout=3)
        await rbt.home(wait=True)
        async for status in rbt.stream_status():
            print(status.angles, status.action_state)
            if status.completed_index >= 0:
                break

asyncio.run(main())
```

The same methods, awaited, with one difference: the moves return at once unless you pass `wait=True` (the table above). Plus `stream_status()`, which yields a copy of every status broadcast, and `stream_status_shared()`, which yields the same buffer each time for zero-copy hot paths. Slow consumers skip to the latest status.

The dry-run client

```
from parol6 import DryRunRobotClient

with DryRunRobotClient() as rbt:
    rbt.home(wait=True)
    rbt.move_l([0, 263, 242, 90, 0, 90], speed=0.5)     # planned, collision-checked, not
executed anywhere
```

The same calls run through the planner in your process and report what the controller would do: refused targets, collisions, durations. Its status and timing are a preview, not a measurement; the API marks that by advertising `execution.preview`.

The bound tool

`rbt.tool` is the tool object of the selected tool, with `open()`, `close()`, `set_position()`, `calibrate()`, `release()`, `stop()` and `status()` as the tool type has them (Tools and the TCP). It raises if no tool is selected.

Several clients

The controller serves any number of clients at once; the queue and the streamed motion are shared (Control ownership). A client that must know its command finished waits on its own index, never on "the robot stopped".

Motion

Planned motion: the client sends a target, the controller plans the trajectory in its planner subprocess, checks it for collisions and limits, queues it, and plays it. Every planned command returns a command index and runs in queue order.

The five moves

Call	Path	Target
`move_j(angles)`	Straight in joint space: every joint moves proportionally, all finish together. The tool's path curves	Six joint angles, or `pose=` a Cartesian pose solved once at the target
`move_l(pose)`	A straight line of the tool, orientation interpolated	A pose in the WRF, or in the TRF with `frame="TRF"`
`move_c(via, end)`	A circular arc from the current pose through `via` to `end`	Two poses
`move_s(waypoints)`	A cubic spline through the waypoints	A list of poses
`move_p(waypoints)`	A process path: the waypoints joined by lines, run at constant tool speed with automatic blending at the corners	A list of poses

`rel=True` on `move_j` and `move_l` makes the target relative to the current angles or pose; with `frame="TRF"` a relative `move_l` moves along the tool's own axes.

```
rbt.move_j([90, -90, 180, 0, 0, 180], speed=0.5)                # joint move
rbt.move_j(pose=[100, 340, 334, 90, 0, 90], speed=0.5)          # IK at the target, then a joint
move
rbt.move_l([-50, 340, 334, 90, 0, 90], speed=0.3)               # straight line
rbt.move_l([0, 0, -20, 0, 0, 0], frame="TRF", rel=True, speed=0.2)   # 20 mm out of the tool: the
API's -Z
rbt.move_c(via=[0, 250, 250, 90, 0, 90], end=[0, 263, 242, 90, 0, 90], speed=0.5)
rbt.move_s([[0, 263, 242, 90, 0, 90], [50, 250, 200, 90, 0, 90]], speed=0.5)
rbt.move_p(waypoints, speed=0.8)
```

Speed, acceleration, duration

`speed` and `accel` are fractions of the robot model's maxima in (0, 1], default 0.5 each. For a joint move they scale the leading joint; for a Cartesian move they scale the tool's linear and angular limits, with the joint limits as hard bounds. A `duration` in seconds times the move instead and `speed` is ignored; `duration` and `speed` are not combined. The move's actual speed is also capped by what the profile and the limits allow: a short move never reaches the asked speed.

```
rbt.move_l(target, speed=0.25, accel=1.0)      # a quarter of the Cartesian speed limit, full
acceleration
rbt.move_l(target, duration=2.0)               # two seconds, whatever that takes
```

Blending

`r`, in millimetres, on `move_j`, `move_l` and `move_c`: the move does not stop at its target but blends into the next queued move within that radius (Blending). `move_p` blends on its own.

Profiles

`select_profile("TOPPRA")` and the others set how the speed curve is shaped for all planned moves (Motion profiles); `profile()` reads it. The profiles of this version: `TOPPRA`, `RUCKIG`, `QUINTIC`, `TRAPEZOID`, `LINEAR`.

Waiting

`wait=True`, the default, blocks until the move completes and raises `MotionError` if it was cancelled or failed; `TimeoutError` if the `timeout` runs out first, in which case the move is still queued and `stop()` cancels it. `wait=False` returns the index for a later `wait_command()`; `wait_motion()` waits for the whole queue and for the arm to come to rest (Status and waiting).

Execution speed and pause

`set_execution_speed(0.5)` runs the queued trajectory at half its planned speed from now on, 10 to 100 %; `pause()` decelerates the queue to a hold and `resume()` continues it. Both keep the queue. A paused robot's `execution_speed().paused` is true once it has actually stopped.

Refusals

A target outside the joint limits, a pose the IK cannot solve, a path that collides, an unhomed robot, a latched protective stop: the command is refused before anything moves, with a `MotionError` whose code says why (Errors). A refused command cancels nothing already queued; a command that **fails while running** cancels everything queued behind it.

Jogging and servoing

Streamed motion is for a hand on a button or a loop that computes the next target: a jog is a velocity held for a short time, a servo target is a position to go to now. Both bypass the queue and take the arm at once (the queue is cancelled). They end differently:

- **A jog** runs for its `duration` (0.1 s by default), then the arm brakes to rest and holds. A held button sends a new jog before the last one runs out.
- **A servo stream** ends about 0.25 s after the last target arrives, when the arm brakes to rest and holds.

Jogging

```
rbt.jog_j(0, speed=0.3, duration=0.5)                              # joint 1 (index 0), 30 %, 0.5
s
rbt.jog_j(joints=[1, 2], speeds=[0.2, -0.2], duration=1.0)         # two joints
rbt.jog_l("WRF", "X", speed=0.2, duration=0.5)                     # the tool along world X
rbt.jog_l("TRF", axes=["Z", "RZ"], speeds_list=[0.2, 0.1], duration=0.5)
```

Speeds are signed fractions of the maximum in [−1, 1]; the controller clamps jogs to 80 % of the limits. A GUI sends a jog every few tens of milliseconds while the button is held; the controller de-duplicates repeated targets and extends the motion. A jog stops at a joint limit, and a Cartesian jog stops before a singularity.

Servoing

```
rbt.servo_j([90, -90, 180, 0, 0, 180], speed=0.5, accel=0.5)       # go to these angles, now
rbt.servo_j(pose=[0, 263, 242, 90, 0, 90])                          # via IK
rbt.servo_l([0, 263, 242, 90, 0, 90], speed=0.5)                    # a straight line to the pose
```

A servo target is fire-and-forget: the controller plans a jerk-limited motion from where the arm is to the target and follows it, re-planning as new targets arrive. A stream of targets at 20 to 100 Hz gives smooth tracking of a moving goal, which is how teleoperation and following a camera are built. The stream ends a quarter second after the last target.

Rules

- Cartesian jogs (`jog_l`) and servo targets need a homed robot, like planned motion; they are refused with `MOTN_NOT_HOMED` before it. Joint jogs (`jog_j`) work before homing, so that an arm can be jogged clear of an obstruction first.
- It cancels the planned queue: every queued command fails with `MOTN_CANCELLED` . Do not interleave a servo stream with planned moves you still want.
- Keep the targets reachable and the steps between them small; a target far from the current pose is a fast move at the asked speed.

- Collision checking applies to jogs with a look-ahead (`PAROL6_COLLISION_JOG_LOOKAHEAD_S`).

Status while streaming

The status stream reports `action_state` and the actual angles at 50 Hz; `stream_status()` on the async client is the natural partner of a servo loop (Status and waiting).

Tools and the TCP

The controller needs to know the tool to compute the pose of its tip, to check it for collisions and to drive it. Tools are registered in the API with a transform, a collision mesh, variants and the actions they support.

Registered tools

Key	Tool	Variants (default first)
`NONE`	Bare flange - no tool attached	—
`PNEUMATIC`	Pneumatic gripper assembly (vertical/horizontal mounting)	`vertical`, `horizontal`
`SSG-48`	SSG-48 adaptive electric gripper (Spectral micro BLDC)	`finger`, `pinch`
`MSG`	MSG compliant AI stepper gripper (StepFOC)	`100mm`, `150mm`, `200mm`
`VACUUM`	Vacuum gripper (pneumatic valve I/O)	—

`select_tool(key, variant_key)` sets the active tool for everyone, queued with the motion; `tools()` lists them; `rbt.tool.status()` and the status stream's `tool_status` report the applied variant and the tool's state. Changing the tool or the variant clears any TCP correction.

```
rbt.select_tool("PNEUMATIC")                        # the default variant
rbt.select_tool("PNEUMATIC", variant_key="horizontal")
rbt.select_tool("SSG-48", variant_key="pinch")
rbt.select_tool("MSG", variant_key="150mm")
```

Driving the tool

`rbt.tool` is the selected tool's object. The electric grippers (SSG-48, MSG) have position, speed and current as fractions of their ranges, and the actions calibrate, stop and release; the pneumatic and vacuum grippers have open and close on an output.

```
rbt.select_tool("SSG-48")
rbt.tool.calibrate()                                    # queued; the API gives it 2 s
rbt.delay(2)                                            # the rest of the SSG-48's
calibration
rbt.tool.set_position(0.7, speed=0.5, current=0.4)      # 0 open, 1 closed
rbt.tool.close(speed=0.5, current=0.6)
rbt.tool.open()
rbt.tool.release()                                      # drop the grip, the jaws stay
rbt.tool.stop()                                         # immediate
print(rbt.tool.status())                                # ToolStatus: state, engaged,
part_detected, positions
```

The current fraction spans the tool's current range: 100 to 1300 mA for the SSG-48 and 100 to 2800 mA for the MSG, so `current=0.4` is 580 mA on the SSG-48 and 1180 mA on the MSG. Keep an MSG within its stepper's rating (MSG).

Calibrating

The gripper has to be calibrated after every power-up: it opens fully and closes fully to find its end stops. `calibrate()` sends the calibration and counts it done after a fixed 2 s (200 control cycles); it does not read the gripper's calibrated bit. The gripper takes longer: about 3 s for the SSG-48, and about 5 to 6 s for the MSG on the longer rails. Queue a `delay()` after `calibrate()` to cover the rest with a margin, 2 s for the SSG-48 and 5 s for the MSG, before the first move. A gripper that has latched an error must be cleared first: the API has no clear-error action at this version, so use the commander's **Clear gripper error** or Motor GUI, then calibrate.

Tool status

`rbt.tool.status()` returns a `ToolStatus`. For the electric grippers, `positions` is the jaw position (0 open, 1 closed) and `channels` the current in mA. `engaged` is taken from the gripper-speed field of the telemetry, which firmware 1 always reports as 1, so it reads true. `part_detected` is taken from bits 3 and 2 of the gripper's status byte, where the gripper manuals put the temperature and timeout errors (they put the object detection in bits 5 and 4). The calibrated and error bits are not in `ToolStatus` at this version. To know that a part is held, check the position the jaws stopped at, or use a sensor.

Tool actions are **queued work**: an action waits for the motion queued before it, the arm holds still while it runs, and the motion queued after it waits for it. A `move` on a gripper that was never calibrated is refused when its turn comes and, like any failing command, cancels what is queued behind it. `stop()` is the exception: it is immediate, halts the running action, and keeps the queue. `tool_action(key, action, params)` is the generic form the tool methods are built on.

The TCP

The tool centre point is the point the robot positions: the registered tool's tip. Two corrections on top of it:

```
index = rbt.set_tcp_offset(0, 0, -190)           # a translation in the tool frame, mm: 190 mm
out of the tool
index = rbt.set_tcp_transform(0, 0, 25, 0, 90, 0) # translation and intrinsic XYZ rotation,
degrees
rbt.wait_command(index)                           # queued: wait before relying on it
print(rbt.tcp_transform())                        # six values; tcp_offset() the three
translations
```

Both are queued with the motion so that moves planned before the change keep their frame. `set_tcp_offset` clears a rotation; a tool change clears both. The correction applies to live FK, Cartesian planning, TRF motion and dry-run previews alike; the collision meshes stay on their links regardless.

> **The Python API's tool frame has Y and Z the other way**
>
> In the Python API (version 0.9.1) the tool frame is the commander's turned 180° about X: its Z points back into the wrist. An offset or a TRF move out of the tool is negative z, as in `set_tcp_offset(0, 0, -190)` (Reference frames, known issue KI-11).

Your own tool

Add a `ToolConfig` and `register_tool("KEY", config)` in `parol6/tools.py`: the transform from the flange, the collision mesh, the variants, and the actions if it is driven through the robot (Your own tool). The controller validates every tool action against the config before it is acknowledged.

Held objects

A program can declare the shape of what the gripper holds, attached to the flange, so that the collision checker accounts for it: `shape.attach(...)` on a shape from `rbt.shapes()`, applied with `set_shapes()`. The declaration is tied to the tool, the world and a fresh position reference; a change in any of them invalidates it and blocks motion until it is reconciled. The README of the API has the rules in full.

I/O, delays and checkpoints

Digital I/O

The control board's two inputs, two outputs and the E-stop, as the firmware reports them:

```
state = rbt.io()                # [in1, in2, out1, out2, estop], or None on a timeout
rbt.write_io(0, 1)              # output 1 high; logical index 0 = the first output
rbt.write_io(1, 0, timeout=1.0)  # raises TimeoutError if not confirmed in time
```

The firmware's I/O byte is `[in1, in2, out1, out2, estop, ...]`, so logical output 0 is bit 2 of the byte; `write_io` does the mapping. `is_estop_pressed()` reads the E-stop alone. A per-call `timeout` bounds the setup, the retries and the reply; without one the client's defaults apply.

A write is queued with the motion: it takes effect when its turn in the queue comes, and it returns a command index that `wait_command()` waits on. The pneumatic gripper's `open()` and `close()` are output writes of the same kind. While the controller is disabled, after `estop()` or the E-stop until `reset()`, a write is refused with `SYS_CONTROLLER_DISABLED`, like a move.

Switching an output at a point of a program

Queue the write between the two moves it belongs between. The output changes when the first move has finished, and the second move starts after it:

```
rbt.move_l([0, 263, 200, 90, 0, 90], speed=0.5, wait=False)   # to the part
rbt.write_io(0, 1)                                              # output 1 high, once the arm is
there
rbt.delay(0.5)                                                  # let the valve act
rbt.move_l([0, 263, 242, 90, 0, 90], speed=0.5, wait=False)   # away again
rbt.write_io(0, 0)                                              # output 1 low at the end
rbt.wait_motion()
```

A `checkpoint()` is for something outside the robot that must know when the queue reaches a point; the outputs do not need one.

Delays

```
rbt.delay(1.0)                  # a second of nothing, in the queue
```

A queued delay holds the arm for its duration at its place in the queue; it is not `time.sleep()`, which would hold your script and not the robot. A paused queue keeps the delay's remaining time.

Checkpoints

```
index = rbt.checkpoint("pick_done")
...
rbt.wait_checkpoint("pick_done", timeout=30)
```

A checkpoint is a labelled marker in the queue; `wait_checkpoint()` blocks until the queue reaches it, and the status stream reports `last_checkpoint`. It is how a program synchronises something outside the robot, a conveyor or a camera, with a point in the motion.

Status and waiting

Queries

Every query is a request to the controller with a reply, returning `None` on a timeout:

```
rbt.ping()            # PingResult: hardware_connected
rbt.angles()          # [J1..J6] in degrees
rbt.pose()            # [x, y, z, rx, ry, rz], mm and degrees; pose("TRF") in the tool frame
rbt.joint_speeds()    # deg/s
rbt.tcp_speed()       # mm/s
rbt.io()              # [in1, in2, out1, out2, estop]
rbt.status()          # StatusSnapshot: pose, angles, speeds, io, tool_status
rbt.activity()        # what the robot does now: state, command, parameters, error
rbt.queue()           # the queued commands
rbt.error()           # the current RobotError, or None
rbt.reachable()       # remaining travel per joint and axis
rbt.tools(); rbt.profile(); rbt.tcp_offset(); rbt.tcp_transform(); rbt.shapes()
rbt.loop_stats(); rbt.status_rate(); rbt.execution_speed(); rbt.is_simulator()
```

The status stream

The controller broadcasts its state at 50 Hz (`set_status_rate()` changes it) as multicast; the async client yields it as a stream:

```
async for s in rbt.stream_status():
    s.angles, s.speeds, s.pose, s.io, s.tool_status
    s.action_state, s.action_current, s.executing_index, s.completed_index, s.accepted_index
    s.error, s.homed, s.enabled, s.simulator_active, s.collision_active
    s.session_id, s.seq, s.mono_time_ns          # for recording: a session, a sequence, a
timestamp
```

The fields are listed on Status fields. `seq` gaps mean missed broadcasts; `session_id` changes when the controller restarts; `mono_time_ns` is the time of the controller's snapshot, not of the sensor reading. The sync client's `wait_status(predicate)` waits for a status that satisfies a function.

Waiting for a command

```
index = rbt.move_l(target, wait=False)
ok = rbt.wait_command(index, timeout=10)        # True when completed; raises MotionError if it
failed
```

`wait_command` asks the controller for that command's exact result. Queued commands run in index order, so a wait on one covers everything queued before it; a jog or a `tool.stop()` can finish earlier than something queued before it, which is why a wait never infers success from the highest completed index. A command discarded by `stop()`, an E-stop, `reset_state()`, a teleport or a failure ahead of it raises `MotionError` with `MOTN_CANCELLED` at once; an unknown or expired result stays unconfirmed and the wait returns `False` at its timeout. The controller keeps its last 1024 completions and 1024 failures.

`wait_motion()` waits for the whole queue to run out and the arm to hold still for a settle window; `wait_ready()` polls `ping()` until the controller answers; `is_robot_stopped()` looks at the joint speeds only.

What "done" means

A blocking move returns when the controller reports the command complete: the trajectory played to its end and the arm at the commanded position, in counted steps. For a tool action, when the gripper's estimated travel time or its reported state says so. Neither is a measurement of the world: a program that needs to know the part is in the gripper checks where the jaws stopped (`tool.status().positions`) or reads a sensor. For the electric grippers, the tool status's `part_detected` does not follow the gripper's object-detection bits at this version (Tool status).

Errors

MotionError

A command the controller refuses or fails raises `MotionError`, carrying a `RobotError` with a code, a title, a cause, an effect and a remedy from the controller's catalog:

```
from parol6 import RobotClient, MotionError

with RobotClient() as rbt:
    try:
        rbt.move_l([900, 0, 0, 0, 0, 0], speed=0.5)
    except MotionError as e:
        print(e)            # IK: target unreachable — ... — Verify target is within workspace...
```

The codes are on Errors. The ones you meet first:

Code	When
`MOTN_NOT_HOMED`	Planned motion before the robot is referenced: `home()` first
`IK_TARGET_UNREACHABLE`, `IK_PARTIAL_PATH`	The pose, or part of a Cartesian path, has no solution
`SYS_ESTOP_ACTIVE`	The E-stop is pressed or a protective stop is latched: release, then `reset()`
`SYS_CONTROLLER_DISABLED`	The controller is disabled
`SYS_SELF_COLLISION`	The path collides with the robot itself or with a declared shape
`MOTN_CANCELLED`	The command was discarded by `stop()`, an E-stop, a stream taking the arm, a teleport, or a failure ahead of it in the queue
`COMM_VALIDATION_ERROR`, `COMM_DECODE_ERROR`	A malformed command; usually a version mismatch between client and controller

`error()` returns the current error, `activity()` the state with the error description, and the status stream carries `error`.

ValueError and TimeoutError

The client checks parameters before sending: a wrong number of angles, a speed outside (0, 1], a non-finite value, an unknown tool action raise `ValueError` locally. A blocking call whose completion is not confirmed in its `timeout` raises `TimeoutError`; the command stays queued, and `stop()` cancels it. A query that gets no reply returns `None`, except the TCP queries and bounded I/O writes, which raise rather than report a misleading value.

ConnectionError

A controller restart during a wait changes the session, and the wait raises `ConnectionError`: the index you were waiting for belongs to a controller that no longer exists.

Recovering

After	Do
`stop()`	Nothing: the controller is enabled and holding; queue again
`estop()` or the E-stop button	Release the button; `reset()`
`reset_state()`	The shapes, tool, errors, pause, profile and speed are back to defaults; select the tool again; a protective stop stays until `reset()`
A collision or stall on the hardware	Stop; move the arm clear; `home(calibrate=True)`
A failed command in the queue	Everything behind it was cancelled: re-queue from the failed one

Examples

The API's repository has runnable scripts in `examples/`; these are the shapes of program you will write most.

A self-contained script

Starts the controller, runs in the simulator, moves, stops the controller.

```
from parol6 import Robot, RobotClient

with Robot(host="127.0.0.1", port=5001, normalize_logs=True):
    with RobotClient(timeout=2.0) as rbt:
        if not rbt.wait_ready(timeout=5.0):
            raise SystemExit("controller did not start")
        rbt.simulator(True)
        rbt.home(wait=True)                    # planned motion is refused until the robot is
referenced
        print("pose:", rbt.pose()[:3], "angles:", rbt.angles())
        rbt.move_l([0, 0, 5, 0, 0, 0], rel=True, duration=1.0)
```

Pick and place with an electric gripper

```
from parol6 import RobotClient

ABOVE_PICK  = [100, 300, 200, 90, 0, 90]
PICK        = [100, 300, 120, 90, 0, 90]
ABOVE_PLACE = [-100, 300, 200, 90, 0, 90]
PLACE       = [-100, 300, 120, 90, 0, 90]

with RobotClient() as rbt:
    rbt.select_tool("SSG-48")
    rbt.home(wait=True)
    rbt.tool.calibrate()                                # the API gives it 2 s
    rbt.delay(2)                                        # the rest of the SSG-48's calibration
    rbt.tool.open()
    for _ in range(3):
        rbt.move_j(pose=ABOVE_PICK, speed=0.5)
        rbt.move_l(PICK, speed=0.3)
        rbt.tool.close(speed=0.5, current=0.5)          # queued: the next move waits for the
grip
        rbt.move_l(ABOVE_PICK, speed=0.3)
        rbt.move_j(pose=ABOVE_PLACE, speed=0.5)
        rbt.move_l(PLACE, speed=0.3)
        rbt.tool.open()
        rbt.move_l(ABOVE_PLACE, speed=0.3)
    rbt.home()
```

A circle, a spline, a raster

From the API's showcase: three ways to draw a curve, in the XZ plane at a fixed Y.

```
import math
from parol6 import RobotClient

rbt = RobotClient()
rbt.home()
R, Y, ORI = 30, 340, [90, 0, 90]

def pt(cx, cz, deg):
    a = math.radians(deg)
    return [cx + R * math.cos(a), Y, cz + R * math.sin(a)] + ORI

rbt.move_j(pose=pt(0, 280, 0), speed=0.5)
rbt.move_c(via=pt(0, 280, 180), end=pt(0, 280, 0), speed=0.8)            # a full circle in one
arc

waypoints = [pt(0, 140, i * 30) for i in range(12)] + [pt(0, 140, 0)]
rbt.move_l(waypoints[0], speed=0.8)
rbt.move_p(waypoints, speed=0.8)                                         # the same circle as a
process path

spline = [[R * math.cos(t * 6 * math.pi), Y, 140 + t * 140] + ORI for t in (i / 36 for i in
range(37))]
rbt.move_s(spline, speed=0.8)                                            # a sine wave up the
three circles

for i in range(6):                                                       # a raster with blended
corners
    y0, y1 = (0, 160) if i % 2 == 0 else (160, 0)
    z = 200 + i * 20
    rbt.move_l([280, y0, z, -180, -90, -180], speed=0.5, r=15)
    rbt.move_l([280, y1, z, -180, -90, -180], speed=0.5, r=15)
rbt.close()
```

Following a target with a servo stream

```
import asyncio, math, time
from parol6 import AsyncRobotClient

async def main():
    async with AsyncRobotClient() as rbt:
        await rbt.home(wait=True)
        await rbt.move_l([0, 263, 272, 90, 0, 90], speed=0.5, wait=True)   # to the circle's
start
        t0 = time.monotonic()
        while time.monotonic() - t0 < 10:
            t = time.monotonic() - t0
            await rbt.servo_l([30 * math.sin(t), 263, 242 + 30 * math.cos(t), 90, 0, 90],
speed=0.5)
            await asyncio.sleep(0.02)                    # 50 targets a second
        # the stream ends by itself a quarter second after the last target

asyncio.run(main())
```

The async client's moves return at once unless `wait=True` is given (the sync client waits by default). Without it, the first `servo_l` would take the arm from the `move_l` and cancel it.

Switching an output between moves, and a checkpoint for the script

```
with RobotClient() as rbt:
    rbt.home(wait=True)
    rbt.move_l([0, 263, 200, 90, 0, 90], speed=0.5, wait=False)
    rbt.write_io(0, 1)                                  # the valve, queued: it switches when the
arm is at the part
    rbt.checkpoint("at_part")
    rbt.delay(0.5)
    rbt.move_l([0, 263, 242, 90, 0, 90], speed=0.5, wait=False)
    rbt.write_io(0, 0)                                  # queued after the second move
    rbt.wait_checkpoint("at_part", timeout=20)
    print("at the part")                                # the script learns it here, e.g. to
trigger a camera
    rbt.wait_motion()
```

`write_io` is queued with the motion, so the valve switches between the two moves without the script's help; the checkpoint only tells the script when the queue got there.

3. RBTscript

3. RBTscript

RBTscript is the program language of the commander software: one command per line, run by the commander's interpreter while its GUI shows the robot. It is the quickest way from a taught pose to a repeating sequence, and it is the limit of what the commander does: for blending, collision checking, tool selection and a network interface, the Python API.

- RBTscript: the language, its units, every command
- Examples

The commander itself, its installation and its windows, is in the GUI Guide.

RBTscript

RBTscript is the commander's program language: one command per line, executed in order, synchronised to the 100 Hz control loop. It has motion in joint and Cartesian space, delays, outputs and the gripper, and a loop. There are no variables, conditions or arithmetic: a program is a sequence.

```
Begin()
MoveJoint(90,-90,180,0,0,180,t=3)
MoveCart(0,263,242,90,0,90)
Delay(1)
Loop()
```

Units

Angles in degrees, positions in millimetres, speed and acceleration in percent of the robot's maximum, time in seconds, gripper position and speed 0 to 255, gripper current in milliamps (Units).

Motion

MoveJoint

```
MoveJoint(j1, j2, j3, j4, j5, j6, v=0, a=0, t=0, trap|poly, speed)
```

Moves all joints to the given angles; the path is a straight line in joint space, all joints stopping together. The tool's path curves.

Argument	Meaning	Unit
`j1` … `j6`	The target joint angles	degrees
`v`	Speed of the leading joint, the one with the longest way	% (0 to 100)
`a`	Acceleration and deceleration of the leading joint	%
`t`	Duration of the move; overrides `v` and `a`	seconds
`trap` , `poly`	The profile, as a bare word: `poly` (quintic polynomial) or `trap` (trapezoidal). `"trap"` in quotes or `func=trap` is ignored without a message and gives the default	

Argument	Meaning	Unit
`speed`	The word `speed` as the last argument: track the speed curve only	

The arguments must be in this order. The rules:

- With `t`, the move takes that long; the profile is polynomial unless the word `trap` follows, and a trapezoid with `t` spends a third of the time in each phase.
- With `v` and `a`, the profile is trapezoidal, for the leading joint; the other joints get what makes them finish together.
- With neither, a conservative trapezoid.
- With `speed` last, only the speed curve is tracked: smoother and quieter, and the robot may stop a few steps short of the target (Target, commanded and actual).

```
MoveJoint(0,-90,180,0,0,180,t=4)
MoveJoint(0,-90,180,0,0,180,t=4,trap)
MoveJoint(0,-90,180,0,0,180,v=50,a=50)
MoveJoint(0,-90,180,0,0,180)
MoveJoint(0,-90,180,0,0,180,speed)
```

MovePose

```
MovePose(x, y, z, Rx, Ry, Rz, v=0, a=0, t=0, trap|poly, speed)
```

Like `MoveJoint`, but the target is a pose in the WRF: the inverse kinematics is solved once at the target and the robot moves in joint space to the configuration it found, which may not be the one you expected. The same arguments and rules as `MoveJoint`.

MoveCart

```
MoveCart(x, y, z, Rx, Ry, Rz, v=0, a=0, t=0, trap|poly, speed)
```

Moves the tool in a straight line to the pose, in the WRF. The joints' motion is whatever the straight line requires, and the move stops if it meets a singularity; the robot does not deviate from the line to avoid one. The timing rules differ from `MoveJoint`:

- With `t`, the move takes that long and the profile is **trapezoidal** unless the word `poly` follows, the opposite of `MoveJoint`.
- Without `t`, **every move takes 3.6 s**, however long it is, so a long move without `t` is fast. `v` and `a` are accepted but only checked for range. Give `t` for every Cartesian move.

MoveCartRelTRF

```
MoveCartRelTRF(x, y, z, Rx, Ry, Rz, v=0, a=0, t=0, trap|poly, speed)
```

A straight-line move relative to the current tool frame: `MoveCartRelTRF(-80,0,0,0,0,0)` moves 80 mm along the tool's −X. For rotations about more than one axis, use two commands, one axis each. The timing rules are those of `MoveCart`.

SpeedJoint, SpeedCart

Listed in the command menu, but neither works. `SpeedJoint()` passes the program check and has no action: the program stops on that line until you press **Stop**. `SpeedCart()` fails the check, and a program that contains it is refused; so are the menu's `Get_data` and `Timeouts`.

Timing and flow

Command	Meaning
`Begin()`	Must be the first line; a program that does not start with it is refused
`End()`	Stops the program; the last line must be `End()` or `Loop()`, or the program is refused
`Loop()`	Restarts at the line after `Begin()`
`Delay(t)`	Waits `t` seconds; `t` must be more than 0.01, and `Delay(0.01)` is refused and stops the program

I/O

```
Output(1,HIGH)
Output(1,LOW)
```

Sets output 1, with no space after the comma. In the current version `Output(2,HIGH)` and `Output(2,LOW)` are refused by a bug in the check (`Invalid Output() command`) and stop the program, so only output 1 can be switched from a program. `Input()` appears in the menu but has no action: a program stops on it until you press **Stop**; `Home()` behaves the same.

Gripper

```
Gripper_cal()
Delay(4)
Gripper(position, speed, current)
Delay(2)
```

`Gripper_cal()` calibrates the SSG-48 or MSG gripper. Clear the gripper's error first, with the Gripper tab's **Clear gripper error** (RBTscript has no command for it), and add a delay for the calibration to finish: `Delay(4)` for the SSG-48 (about 3 s) and `Delay(7)` for the MSG (5 to 6 s on the longer rails). Activating it is the Gripper tab's **Activate** option, which is on by default. `Gripper()` moves it, with whole numbers only: position 0 (open) to 255 (closed), speed 0 to 255, current 100 to 1000 mA, the current being the grip force; below about 150 mA the gripper does not move. A value outside these ranges is logged as `Gripper() invalid input value`, the line is skipped and the program goes on. The program continues at once, so add a `Delay()` for the gripper to arrive.

Danger

A current above about 700 mA with a high speed slams the jaws into the part and can damage the gripper, the robot and the part. Start low.

A gripper error (not calibrated, E-stop, timeout) stops the gripper until it is cleared in the Gripper tab.

Reference

The commands on one page: RBTscript commands.

Examples

Programs from the repository's `GUI/files/Programs` folder and the documentation. Paste one into the program window, read it against the simulator, and press Start. The speed slider does not slow a program: for a first run, lower `v=` or raise `t=` in the moves.

A joint-space loop

Two poses, back and forth, with a timed first move.

```
Begin()
Delay(1)
MoveJoint(85.078,-111.195,143.513,-32.92,18.084,129.448,t=3)
Delay(1)
MoveJoint(66.129,-117.368,136.77,46.28,-29.588,149.293)
Delay(1)
Loop()
```

Cartesian moves in the world frame

A joint move to a start pose, then straight-line moves: translations first, then changes of orientation. Watch the simulator to see the tool follow.

```
Begin()
Delay(1)
MoveJoint(90.0,-90.0,180.001,0.0,0.0,180.0,t=4)
Delay(1)
MoveCart(0.0,263.23,161.241,-90.0,0.0,-89.499)
Delay(1)
MoveCart(81.565,262.804,161.047,-90.135,-0.029,-89.578)
Delay(1)
MoveCart(81.649,194.981,160.077,-90.14,-0.043,-89.593)
Delay(1)
MoveCart(83.598,194.356,161.493,-91.155,1.306,-80.28)
Delay(1)
MoveCart(84.56,194.131,162.347,-48.387,1.261,-80.188)
Delay(1)
MoveCart(82.89,191.698,162.369,-40.587,26.026,-107.028)
Delay(1)
End()
```

The `MoveCart` lines have no `t=`, so each takes 3.6 s whatever its length (MoveCart). Give the long ones a `t=` for a slower first run.

A move relative to the tool

80 mm back along the tool's X axis, from the start pose the program moves to first.

```
Begin()
Delay(1)
MoveJoint(90.0,-90.0,180.001,0.0,0.0,180.0,t=4)
Delay(1)
MoveCartRelTRF(-80,0,0,0,0,0)
Delay(1)
End()
```

Pick and place with the gripper

From the repository: a joint move over the part, a Cartesian descent, a grip, a transfer and a release. Note the delays after each gripper command.

```
Begin()
Delay(1)
MoveJoint(89.209,-35.021,174.356,-1.463,-28.645,85.972,t=4)
Delay(1)
Gripper(255,20,120)
Delay(2)
MoveCart(-40.499,298.455,131.712,6.081,-0.739,-179.276)
MoveJoint(15.478,-56.126,178.673,-0.928,-54.084,85.5)
Delay(1)
Gripper(0,100,120)
MoveJoint(89.033,-49.402,174.486,-1.027,-43.341,85.596)
MoveCart(-41.535,300.316,17.635,6.141,-0.686,-179.224)
Delay(1)
Gripper(255,100,1000)
Delay(1)
MoveCart(-40.499,298.455,131.712,6.081,-0.739,-179.276)
MoveJoint(15.478,-56.126,178.673,-0.928,-54.084,85.5)
Gripper(0,100,100)
Delay(1)
End()
```

Before you run it:

- It does not calibrate the gripper: clear the gripper's error and calibrate it in the Gripper tab first.
- `Gripper(255,20,120)`, `Gripper(0,100,120)` and `Gripper(0,100,100)` ask for 120 or 100 mA, below the about 150 mA the gripper needs to move: raise the current to at least 150 mA.
- `Gripper(255,100,1000)` grips at 1000 mA and speed 100. Above about 700 mA, keep the speed low, or lower the current (Gripper).
- The `MoveCart` lines have no `t=` and take 3.6 s each.

Gripper test

Calibration, a few open and close cycles, then a small joint move and a Cartesian move.

```
Begin()
Gripper_cal()
Delay(3)
Gripper(100,60,600)
Delay(2)
Gripper(200,60,600)
Delay(2)
MoveJoint(85.192,-104.451,158.769,-1.983,-46.491,179.899,t=6)
Delay(1)
MoveCart(4.317,103.384,192.319,-84.465,1.447,-143.154)
Gripper(100,60,600)
Delay(2)
End()
```

Clear the gripper's error in the Gripper tab before you start it. `Delay(3)` is just the SSG-48's calibration time; `Delay(4)` leaves a margin, and the MSG needs `Delay(7)`. The `MoveCart` has no `t=` and takes 3.6 s.

Toggling an output

A pneumatic gripper, or a lamp, once a second, forever.

```
Begin()
Output(1,HIGH)
Delay(1)
Output(1,LOW)
Delay(1)
Loop()
```

A dance

Timed joint and Cartesian moves in quick succession: `example_dance_2.txt` in the repository, for a demo once the robot is known to run every pose in it.

4. ROS 2 and MoveIt

PAROL6-ROS2-MOVEIT is a MoveIt 2 configuration for PAROL6 with the URDF, a simulation demo and launch files, tested on Ubuntu 22.04 with ROS 2 Humble.

Simulation

```
sudo apt install ros-humble-desktop ros-humble-moveit
mkdir -p ~/ros2_ws/src && cd ~/ros2_ws/src
git clone https://github.com/Source-Robotics/PAROL6-ROS2-MOVEIT.git
cd ~/ros2_ws && colcon build
source /opt/ros/humble/setup.bash && source install/setup.bash
LC_NUMERIC=en_US.UTF-8 ros2 launch parol6_moveit demo.launch.py
```

RViz opens with the robot, and MoveIt plans and executes on the mock hardware.

The real robot

`real_robot.launch.py` brings up the same stack on `ros2_control`, with a hardware interface and a bridge that talk to the Python API's controller, which must be running (`parol6-server --serial ...`, or its simulator mode). The hardware interface lives in parol6_ros2_control, a community repository, cloned next to the MoveIt package:

```
cd ~/ros2_ws/src
git clone https://github.com/grahas/parol6_ros2_control.git
cd ~/ros2_ws && colcon build
ros2 launch parol6_moveit real_robot.launch.py bridge_python:=/path/to/venv/bin/python3
```

The `bridge_python` must have the `parol6` package installed. On Ubuntu 22.04 it does not install: the package needs Python 3.11 or newer, and on Linux its kinematics backend ships wheels only for glibc 2.39 or newer (Ubuntu 24.04), while 22.04 has Python 3.10 and glibc 2.35 (Installation). The interface's README gives the way round it: run `parol6-server` and the bridge on another machine, or under WSL2, where `parol6` installs, and point the hardware interface's `bridge_host` parameter at it over the network.

The URDF

The robot description, URDF and MuJoCo with meshes and a version per tool, is in robot_descriptions, and an earlier URDF is inside the Python API (`parol6/urdf_model`). The gripper repositories have URDF and MJCF descriptions of the SSG-48 and MSG for attaching to it.

Other ROS 2 projects

parol6_commander, a web GUI on ROS 2 and MoveIt 2, and Parol6-firmware, alternative firmware with micro-ROS, are community projects (Community projects).

5. Other interfaces

Waldo Commander (web)

A community browser interface on the Python API, with a simulator, programs in the browser, teaching by demonstration and an MCP server for LLMs: Waldo Commander in the GUI Guide.

The serial protocol directly

Anything that frames the packets of the serial protocol and keeps the 100 Hz cadence can drive the control board from any language. The Python API's serial transport and the commander's `Serial_sender_good_latest.py` are the two reference implementations.

The legacy Python API

PAROL6-python-API-Alvar was the first Python API: a headless commander (`headless_commander.py`) on the PC at the robot, listening for text commands over UDP on port 5001, and a thin client (`robot_api.py`) with functions like `move_robot_joints` , `move_robot_pose` , `control_electric_gripper` . It is frozen and kept for the projects built on it, including the Gemini vision demo. New work belongs on the current API.

AI agents

The current API's typed client, its dry-run preview and the MCP server in Waldo Commander are the paths for LLM-driven control: an agent writes Python against the client, previews it dry, and runs it with a human at the E-stop. The community list has examples (Community projects).

6. Troubleshooting

The Python API

Symptom	Cause	Remedy
`ping()` returns `None`	No controller, wrong host or port, a firewall	Start `parol6-server`; check `--host` / `--port`; allow UDP 5001
`ping()` says `hardware_connected=False`	No serial port given, or it failed to open	`--serial COM5`, or `connect_hardware("COM5")`; check the port is free
No status in `stream_status()`	Multicast blocked, or a different group	`PAROL6_STATUS_TRANSPORT=UNICAST` with `PAROL6_STATUS_UNICAST_HOST`; check `PAROL6_MCAST_IF`
`MotionError: MOTN_NOT_HOMED`	The robot is not referenced	`home(wait=True)`
`MotionError: SYS_ESTOP_ACTIVE`	The E-stop is pressed or a protective stop is latched	Release the button; `reset()`
`IK_TARGET_UNREACHABLE` on a pose that looks fine	Numerical IK, often around joint 4	Change the orientation slightly; start from another posture; a joint move first
`COMM_VALIDATION_ERROR` or decode errors	Client and controller of different versions	The same version on both sides
`TimeoutError` on a blocking move	The move did not complete in `timeout`	Longer `timeout`; `stop()` to cancel the queued move
`loop overbudget by +N%` in the log	The PC cannot keep 100 Hz	Lower `PAROL6_CONTROL_RATE_HZ`; no DEBUG or TRACE logging; close other programs
Jittery motion	The same, or a long serial cable	As above; a short shielded USB cable
Motion refused with `SYS_SELF_COLLISION` where there is nothing	A declared shape, a held object, the tool's mesh	`shapes()`; the tool's variant; `PAROL6_COLLISION_CHECK=0` to confirm, then fix the world

Symptom	Cause	Remedy
A tool `move` fails with `MOTN_GRIPPER_*`	Not calibrated; no gripper answering on CAN	`tool.calibrate()`; the gripper's cable and manual

The commander software

Symptom	Cause	Remedy
Connect does nothing	Wrong port; the port in use; Linux permissions	Try again; `COMx` / `ttyACMx`; `chmod 666`
The simulator shows a different pose	Not homed	Home
A command in the program does nothing	A typo in the name or the argument order; `End()` before it	The response log; the exact syntax on RBTscript
`MoveCart` stops half way	A singularity	A joint move, or another orientation
`Gripper()` does nothing	Not calibrated; an error latched; a value out of range	Gripper tab: **Clear gripper error**, then **Calibrate gripper**; whole numbers, current 150 to 1000 mA
The program runs once and stops	No `Loop()`	Add it, or `End()` is where you want it
Import errors at start	Wrong Python or package versions	Python 3.10 or 3.11 and the pinned versions, in the commander's own virtual environment, activated (Installation)
`Time between 2 commands` far above 0.010 (seconds, despite the label)	The PC	Communication

Both

The control board answers in text when you expected packets: it runs the test firmware; upload the main one (Firmware). Two programs cannot share the port (Control ownership).

Reference

API reference

Where you look things up. The Python API pages are generated from the API's source at v0.9.1: the signature and the docstring of every public method of `RobotClient`, the tool methods, the registered tools, the error codes, the environment variables and the controller's options, with notes written around them. The build fails when a page names a method the API no longer has, and the tests fail when a method is on no page.

Python API

Every method by category: motion, jogging and servoing, control, queries, synchronization, tools, I/O, configuration, the status fields, the errors, the controller.

RBTscript

Every command of the commander's program language, with its arguments.

The serial protocol

The packets between the PC and the control board, byte by byte, and the command codes.

Firmware and test firmware

The firmware's parameters per joint, and the test firmware's commands.

Python API

Python API reference

The public methods of `RobotClient` (the sync client; the async client has the same methods, awaited, plus the two status streams), generated from the API's source at v0.9.1. `rbt` in the signatures is a client, `rbt = RobotClient()`.

The conventions, once: angles in degrees in order J1 to J6, poses as `[x, y, z, rx, ry, rz]` in millimetres and degrees, `speed` and `accel` as fractions in (0, 1], `duration` in seconds, `r` a blend radius in millimetres, joints indexed from 0 in jog calls. A motion call returns the command index (≥ 0), or −1 if no acknowledgement arrived; a query returns `None` on a timeout; a refused or failed command raises `MotionError` (Errors).

Method	Category	Documented on
`connect_hardware`	Configuration	configuration
`select_profile`	Configuration	configuration
`select_tool`	Configuration	tools
`set_shapes`	Configuration	configuration
`set_status_rate`	Configuration	configuration
`set_tcp_offset`	Configuration	configuration
`tcp_offset`	Configuration	configuration
`estop`	Control	control
`pause`	Control	control
`reset`	Control	control
`reset_state`	Control	control
`resume`	Control	control
`set_execution_speed`	Control	control
`simulator`	Control	control
`stop`	Control	control

Method	Category	Documented on
`teleport`	Control	control
`tool_action`	I/O	tools
`write_io`	I/O	io
`jog_j`	Jog	streaming
`jog_l`	Jog	streaming
`home`	Motion	motion
`move_c`	Motion	motion
`move_j`	Motion	motion
`move_l`	Motion	motion
`move_p`	Motion	motion
`move_s`	Motion	motion
`close`	Other	control
`run_skill`	Other	control
`set_tcp_transform`	Other	configuration
`stream_status`	Other	streaming
`stream_status_shared`	Other	streaming
`tcp_transform`	Other	configuration
`wait_command`	Other	synchronization
`wait_ready`	Other	synchronization
`wait_status`	Other	synchronization

Method	Category	Documented on
`host`	Property	queries
`port`	Property	queries
`robot`	Property	queries
`tool`	Property	queries
`activity`	Query	queries
`angles`	Query	queries
`error`	Query	queries
`execution_speed`	Query	queries
`io`	Query	queries
`is_estop_pressed`	Query	queries
`is_robot_stopped`	Query	queries
`is_simulator`	Query	queries
`joint_speeds`	Query	queries
`loop_stats`	Query	queries
`ping`	Query	queries
`pose`	Query	queries
`profile`	Query	queries
`queue`	Query	queries
`reachable`	Query	queries
`reset_loop_stats`	Query	queries

Method	Category	Documented on
`shapes`	Query	queries
`status`	Query	queries
`status_rate`	Query	queries
`tcp_speed`	Query	queries
`tools`	Query	queries
`servo_j`	Streaming	streaming
`servo_l`	Streaming	streaming
`checkpoint`	Synchronization	synchronization
`delay`	Synchronization	synchronization
`wait_checkpoint`	Synchronization	synchronization
`wait_motion`	Synchronization	synchronization

Motion

Planned moves: queued, run in order, each returning its command index. `wait=True` (the default) blocks until the command completes and raises `MotionError` if it failed or was cancelled; `wait=False` returns at once for a later `wait_command()`. Either `duration` or `speed` times a move, not both. Motion in the guide explains the choices.

`home`

```
rbt.home(wait: bool = False, calibrate: bool = False, timeout: float = 60.0) -> int
```

Home the robot to its home position.

Uncalibrated (first home after power-on), this runs the full referencing sequence: each joint seeks its limit switch, then the robot moves to standby. Calibrated, it returns to standby with a normal planned, collision-checked joint move — unless `calibrate=True`, which re-runs the referencing sequence. The referencing sequence is firmware-driven and ignores the collision world, so clear keep-out geometry from the joints' sweep first.

Returns the command index (≥ 0) on success, -1 on failure.

```
rbt.home()
```

Parameter	Meaning
`wait`	If True, block until motion completes
`calibrate`	If True, always run the referencing sequence
`timeout`	Maximum time to wait in seconds (only used when wait=True)

Note

The referencing sequence is the firmware's (Homing): it ignores the collision world, and it takes about half a minute. A later `home()` is a planned joint move to the API's standby, the after-homing pose.

`move_j`

```
rbt.move_j(
    angles: list[float] | None = None,
    *,
    pose: list[float] | None = None,
    duration: float = 0.0,
    speed: float = 0.5,
    accel: float = 0.5,
    r: float = 0.0,
    rel: bool = False,
    wait: bool = True,
    timeout: float = 10.0,
) -> int
```

Joint-space move. Positional arg = joint angles; pose= kwarg = Cartesian target with IK.

Returns the command index (≥ 0) on success, -1 on failure.

```
rbt.move_j([90, -90, 180, 0, 0, 180], speed=0.5)
```

Parameter	Meaning
`angles`	6 joint angles in degrees (ignored if pose= is set)
`pose`	If set, Cartesian target [x,y,z,rx,ry,rz] — dispatches to MOVEJ_POSE
`duration`	Motion duration in seconds; > 0 times the move and speed is unused
`speed`	Speed fraction in (0, 1], used when duration is 0
`accel`	Acceleration fraction in (0, 1]
`r`	Blend radius in mm (0 = stop at target)
`rel`	If True, angles are relative to current position
`wait`	If True, block until motion completes

`move_l`

```
rbt.move_l(
    pose: list[float],
    *,
    frame: Frame = 'WRF',
    duration: float = 0.0,
    speed: float = 0.5,
    accel: float = 0.5,
    r: float = 0.0,
    rel: bool = False,
    wait: bool = True,
    timeout: float = 10.0,
) -> int
```

Linear Cartesian move to target pose.

Returns the command index (≥ 0) on success, -1 on failure.

```
rbt.move_l([0, 263, 242, 90, 0, 90], speed=0.5)
```

Parameter	Meaning
`pose`	Target [x,y,z,rx,ry,rz] in mm and degrees
`frame`	Reference frame ("WRF" or "TRF")
`duration`	Motion duration in seconds; > 0 times the move and speed is unused
`speed`	Speed fraction in (0, 1], used when duration is 0
`accel`	Acceleration fraction in (0, 1]
`r`	Blend radius in mm
`rel`	If True, pose is relative delta
`wait`	If True, block until motion completes

`move_c`

```
rbt.move_c(
    via: list[float],
    end: list[float],
    *,
    frame: Frame = 'WRF',
    duration: float = 0.0,
    speed: float = 0.5,
    accel: float = 0.5,
    r: float = 0.0,
    wait: bool = True,
    timeout: float = 10.0,
) -> int
```

Circular arc through current position -> via -> end.

Returns the command index (≥ 0) on success, -1 on failure.

```
rbt.move_c(via=[0, 250, 250, 90, 0, 90], end=[0, 263, 242, 90, 0, 90], speed=0.5)
```

Parameter	Meaning
`via`	Via-point pose [x,y,z,rx,ry,rz]
`end`	End-point pose [x,y,z,rx,ry,rz]
`frame`	Reference frame
`duration`	Motion duration in seconds; > 0 times the move and speed is unused
`speed`	Speed fraction in (0, 1], used when duration is 0
`accel`	Acceleration fraction in (0, 1]
`r`	Blend radius in mm
`wait`	If True, block until motion completes

`move_s`

```
rbt.move_s(
    waypoints: list[list[float]],
    *,
    frame: Frame = 'WRF',
    duration: float = 0.0,
    speed: float = 0.5,
    accel: float = 0.5,
    wait: bool = True,
    timeout: float = 10.0,
) -> int
```

Cubic spline through waypoints.

Returns the command index (≥ 0) on success, -1 on failure.

```
rbt.move_s([[0, 263, 242, 90, 0, 90], [50, 250, 200, 90, 0, 90]], speed=0.5)
```

Parameter	Meaning
`waypoints`	List of poses [[x,y,z,rx,ry,rz], ...]
`frame`	Reference frame
`duration`	Motion duration in seconds; > 0 times the move and speed is unused
`speed`	Speed fraction in (0, 1], used when duration is 0
`accel`	Acceleration fraction in (0, 1]
`wait`	If True, block until motion completes

`move_p`

```
rbt.move_p(
    waypoints: list[list[float]],
    *,
    frame: Frame = 'WRF',
    duration: float = 0.0,
    speed: float = 0.5,
    accel: float = 0.5,
    wait: bool = True,
    timeout: float = 10.0,
) -> int
```

Process move — constant TCP speed with auto-blending at corners.

Returns the command index (≥ 0) on success, -1 on failure.

```
rbt.move_p([[0, 263, 242, 90, 0, 90], [50, 250, 200, 90, 0, 90]], speed=0.5)
```

Parameter	Meaning
`waypoints`	List of poses [[x,y,z,rx,ry,rz], ...]
`frame`	Reference frame
`duration`	Motion duration in seconds; > 0 times the move and speed is unused
`speed`	Speed fraction in (0, 1], used when duration is 0
`accel`	Acceleration fraction in (0, 1]
`wait`	If True, block until motion completes

Jogging and servoing

Streamed motion: takes the arm at once and cancels the planned queue (`MOTN_CANCELLED` for every discarded command). A jog ends when its `duration` runs out; a servo stream ends about 0.25 s after the last target. Then the arm brakes to rest. Speeds are signed fractions in [−1, 1] for jogs and fractions in (0, 1] for servo targets. `jog_j` works before homing; `jog_l` and the servo targets need a homed robot. Jogging and servoing in the guide.

`jog_j`

```
rbt.jog_j(
    joint: int | None = None,
    speed: float = 0.0,
    duration: float = 0.1,
    *,
    joints: list[int] | None = None,
    speeds: list[float] | None = None,
    accel: float = 0.5,
) -> int
```

Joint velocity jog. Single-joint or multi-joint.

Single joint: jog_j(0, 0.5, 1.0) Multi joint: jog_j(joints=[0,1], speeds=[0.5, -0.3], duration=1.0)

```
rbt.jog_j(0, speed=0.5, duration=1.0)
```

Parameter	Meaning
`joint`	Joint index (0-5) for single-joint jog
`speed`	Signed speed fraction for single-joint jog
`duration`	Jog duration in seconds
`joints`	List of joint indices for multi-joint jog
`speeds`	List of signed speed fractions for multi-joint jog
`accel`	Acceleration fraction 0-1

`jog_l`

```
rbt.jog_l(
    frame: Frame,
    axis: Axis | None = None,
    speed: float = 0.0,
    duration: float = 0.1,
    *,
    axes: list[Axis] | None = None,
    speeds_list: list[float] | None = None,
    accel: float = 0.5,
) -> int
```

Cartesian velocity jog. Single-axis or multi-axis.

Single axis: jog_l("WRF", "X", 0.5, 1.0) Multi axis: jog_l("WRF", axes=["X","Y"], speeds_list=[0.5, -0.3], duration=1.0)

```
rbt.jog_l("WRF", "X", speed=0.5, duration=1.0)
```

Parameter	Meaning
`frame`	Reference frame ("WRF" or "TRF")
`axis`	Axis name for single-axis jog
`speed`	Signed speed fraction for single-axis jog
`duration`	Jog duration in seconds
`axes`	List of axis names for multi-axis jog
`speeds_list`	List of signed speed fractions for multi-axis jog
`accel`	Acceleration fraction 0-1

Note

Axis names: `X`, `Y`, `Z`, `RX`, `RY`, `RZ`; frames `WRF` and `TRF`. Jogs are clamped to 80 % of the limits.

`servo_j`

```
rbt.servo_j(
    angles: list[float] | None = None,
    *,
    pose: list[float] | None = None,
    speed: float = 0.5,
    accel: float = 0.5,
) -> int
```

Streaming joint position target. Fire-and-forget.

The stream stops about 0.25 s after the last target arrives: the arm brakes to rest and holds there.

```
rbt.servo_j([90, -90, 180, 0, 0, 180])
```

Parameter	Meaning
`angles`	6 joint angles in degrees (ignored if pose= is set)
`pose`	If set, Cartesian target — dispatches to SERVOJ_POSE
`speed`	Speed fraction 0-1
`accel`	Acceleration fraction 0-1

`servo_l`

```
rbt.servo_l(pose: list[float], *, speed: float = 0.5, accel: float = 0.5) -> int
```

Streaming linear Cartesian position target. Fire-and-forget.

The stream stops about 0.25 s after the last target arrives: the tool brakes along its line to rest and holds there.

```
rbt.servo_l([0, 263, 242, 90, 0, 90])
```

Parameter	Meaning
`pose`	Target [x,y,z,rx,ry,rz] in mm and degrees
`speed`	Speed fraction 0-1
`accel`	Acceleration fraction 0-1

`stream_status` (async client only)

```
rbt.stream_status() -> AsyncIterator[StatusBuffer]
```

Async generator that yields status updates from multicast broadcasts.

```
async for status in client.stream_status():
    print(f"Angles: {status.angles}")
```

This generator terminates automatically when `close` is called on the client, so callers do not need to manually cancel their consumer tasks.

Each yielded StatusBuffer is a copy - safe to store or process async. For zero-copy hot paths, use `stream_status_shared` instead.

Slow consumers automatically skip to the latest state (desired for real-time).

`stream_status_shared` (async client only)

```
rbt.stream_status_shared() -> AsyncIterator[StatusBuffer]
```

Zero-copy async generator that yields the shared status buffer.

```
async for status in client.stream_status_shared():
    # Process immediately - don't store references
    print(f"Angles: {status.angles}")
```

WARNING: The same buffer instance is yielded on every iteration. Do not store references to the yielded object - data will be overwritten on the next iteration. For safe storage, use `stream_status` or call status.copy().

This generator terminates automatically when `close` is called on the client.

Slow consumers automatically skip to the latest state (desired for real-time).

Control

Commands that change what the controller is doing: stops, the protective stop and its reset, pause and execution speed, simulator mode, teleport, and the state reset. They are system commands: immediate, not queued, and accepted also while the controller is disabled, except `teleport`, which is refused then. Control ownership.

`stop`

```
rbt.stop() -> int
```

Stop all motion — cancel the active move and clear the queue.

The controller stays enabled and holding position; the next motion command is accepted immediately.

```
rbt.stop()
```

Returns: 1 if acknowledged, 0 on failure.

`estop`

```
rbt.estop() -> int
```

Protective stop: stop all motion and latch the controller disabled until `reset()`.

```
rbt.estop()
```

Returns: 1 if acknowledged, 0 on failure.

`reset`

```
rbt.reset() -> int
```

Clear a latched protective stop, re-enabling motion.

```
rbt.reset()
```

Returns: 1 if acknowledged, 0 on failure.

`reset_state`

```
rbt.reset_state() -> int
```

Reset program state — world shapes, tool selection, errors, pause, motion profile and execution speed — and discard the queue, each discarded command failing with `MOTN_CANCELLED`.

The arm holds where it is. A protective stop stays latched (only `reset()` clears it), and homed state, digital outputs and the serial connection are kept.

```
rbt.reset_state()
```

`pause`

```
rbt.pause(*, timeout: float = 3.0) -> int
```

Request a controlled hold, retaining queued trajectory progress.

A confirmed request returns 1. Read `execution_speed().paused` to confirm the hold. Standalone Python completion timeouts continue.

```
rbt.pause()
```

`resume`

```
rbt.resume(*, timeout: float = 3.0) -> int
```

Resume the retained queue at its selected positive speed.

```
rbt.resume()
```

`set_execution_speed`

```
rbt.set_execution_speed(scale: float, *, timeout: float = 3.0) -> int
```

Select 10–100% of planned queued-motion speed, preserving pause.

```
rbt.set_execution_speed(0.5)
```

`simulator`

```
rbt.simulator(enabled: bool) -> int
```

Enable or disable simulator mode.

```
rbt.simulator(True)
```

`teleport`

```
rbt.teleport(angles_deg: list[float], tool_positions: list[float] | None = None) -> int
```

Instantly set joint angles and optional tool positions (simulator only).

The pose is exact, so the arm counts as homed afterwards and planned motion may follow. Whatever was driving the arm is cancelled with `MOTN_CANCELLED`.

```
rbt.teleport([90, -90, 180, 0, 0, 180])
rbt.teleport([90, -90, 180, 0, 0, 180], tool_positions=[1.0])
```

Returns: 1 once the pose is applied, 0 when no reply arrives.

Raises:

- `ValueError`: for a non-finite angle, one outside the hard joint limits, or a tool position outside `[0, 1]`.
- `MotionError`: off the simulator (`SYS_NOT_SIMULATOR`), when the tool positions do not match the fitted tool's DOF count, or while the controller is disabled.

`run_skill`

```
rbt.run_skill(invoke: Callable[[AsyncRobotClient], Coroutine[Any, Any, T]]) -> T
```

Execute a Python skill using this connection and its existing loop.

> **Note**
>
> Skills are decorated Python functions from the waldoctl skill runtime, run on the connected client; Waldo Commander uses them. A plain script does not need them.

`close`

```
rbt.close() -> None
```

Release UDP transport and background tasks.

Safe to call multiple times.

Queries

Requests with a reply, immediate, returning `None` when no reply arrives in the client's timeout (the TCP queries raise instead, see Configuration). Status and waiting in the guide.

`ping`

```
rbt.ping() -> PingResult | None
```

Return parsed ping result with hardware_connected status.

```
rbt.ping()
```

`angles`

```
rbt.angles() -> list[float] | None
```

Current joint angles in degrees [J1, J2, J3, J4, J5, J6].

```
angles = rbt.angles()
```

`pose`

```
rbt.pose(frame: str = 'WRF') -> list[float] | None
```

Current TCP pose as [x, y, z, rx, ry, rz] in mm and degrees.

```
pose = rbt.pose()
```

`joint_speeds`

```
rbt.joint_speeds() -> list[float] | None
```

Current joint velocities in deg/s [J1, J2, J3, J4, J5, J6], the units of `StatusBuffer.speeds`.

```
speeds = rbt.joint_speeds()
```

`tcp_speed`

```
rbt.tcp_speed() -> float | None
```

TCP linear velocity in mm/s.

```
speed = rbt.tcp_speed()
```

`io`

```
rbt.io(*, timeout: float | None = None) -> list[int] | None
```

Digital I/O status [in1, in2, out1, out2, estop].

`timeout` bounds setup, retries, and the reply; None uses client defaults.

```
io = rbt.io()
```

`status`

```
rbt.status() -> StatusSnapshot | None
```

Aggregate status snapshot (pose, angles, speeds, io, tool_status). Its `tool_status` is always a `ToolStatus`, key `"NONE"` when no tool is fitted.

```
status = rbt.status()
```

`activity`

```
rbt.activity() -> ActivityResult | None
```

What the robot is currently doing.

Returns state (idle/executing/error), current command name, parameters, and error description if applicable.

```
act = rbt.activity()
```

`queue`

```
rbt.queue() -> list[str] | None
```

Queued command list.

```
queue = rbt.queue()
```

`error`

```
rbt.error() -> RobotError | None
```

Current error state, or None if no error.

```
err = rbt.error()
```

`reachable`

```
rbt.reachable() -> EnablementResultStruct | None
```

Remaining freedom of movement per joint/axis before hitting limits.

```
en = rbt.reachable()
```

`is_estop_pressed`

```
rbt.is_estop_pressed() -> bool
```

Check if E-stop is pressed. Returns True if pressed.

```
pressed = rbt.is_estop_pressed()
```

`is_robot_stopped`

```
rbt.is_robot_stopped(threshold_speed: float = 0.5) -> bool
```

Check if robot has stopped moving.

Prefer `wait_command()` for waiting on specific commands. This method polls raw joint speeds and is mainly useful for diagnostics or manual stopping logic.

```
stopped = rbt.is_robot_stopped()
```

Parameter	Meaning
`threshold_speed`	Speed threshold in deg/s

Returns: True if all joints below threshold

`is_simulator`

```
rbt.is_simulator() -> bool
```

Query whether simulator mode is active.

```
active = rbt.is_simulator()
```

`tools`

```
rbt.tools() -> ToolResult | None
```

Current tool and available tools.

```
tools = rbt.tools()
```

`profile`

```
rbt.profile() -> str | None
```

Current motion profile name.

```
profile = rbt.profile()
```

`shapes`

```
rbt.shapes()
```

The collision world the controller is currently enforcing, by layer.

Readback truth for displays; re-query when `StatusBuffer.scene_epoch` changes. Returns None if the controller is unreachable.

```
world = rbt.shapes()
```

`execution_speed`

```
rbt.execution_speed(*, timeout: float = 3.0) -> ExecutionSpeed
```

Read fresh requested, applied, and retained execution scales.

```
speed = rbt.execution_speed()
```

`status_rate`

```
rbt.status_rate() -> StatusRate | None
```

Current broadcast rate and the control rate it divides.

```
rate = rbt.status_rate()
```

`loop_stats`

```
rbt.loop_stats() -> LoopStatsResult | None
```

Fetch control-loop runtime metrics.

```
stats = rbt.loop_stats()
```

`reset_loop_stats`

```
rbt.reset_loop_stats() -> int
```

Reset control-loop min/max metrics and overrun count.

```
rbt.reset_loop_stats()
```

Properties

`host` (property)

```
rbt.host
```

No docstring.

`port` (property)

```
rbt.port
```

No docstring.

`robot` (property)

```
rbt.robot
```

The backend this client drives, built on first read when a bare client (what a user script constructs) supplied none.

`tool` (property)

```
rbt.tool
```

Active bound tool. Raises if no tool has been set.

Synchronization

Waiting for the controller, for a command, for a checkpoint or for the arm to come to rest; and the two queued markers, delays and checkpoints. Status and waiting in the guide says what "done" means.

`wait_ready`

```
rbt.wait_ready(timeout: float = 5.0, interval: float = 0.05) -> bool
```

Poll ping() until server responds or timeout.

Parameter	Meaning
`timeout`	Maximum time to wait for server to respond
`interval`	Polling interval between ping attempts

Returns: True if server responded to PING, False on timeout

`wait_command`

```
rbt.wait_command(command_index: int, timeout: float = 10.0) -> bool
```

Wait until a specific command index has been completed.

Queries exact success in the controller's last 1024 completions: a later command that finishes first — a jog, a `tool.stop()` — does not complete an unfinished one queued before it. Queued commands, tool actions among them, run in index order, so a wait on one covers everything queued ahead of it. Unknown or expired results are never inferred successful from the status high-water mark. A command that ended as a failure — cancelled by `stop()` / `estop()` (`MOTN_CANCELLED`) or failed by the pipeline — raises MotionError.

Parameter	Meaning
`command_index`	The command index to wait for (returned by motion commands).
`timeout`	Maximum time to wait in seconds.

Returns: True if the command completed within timeout, False otherwise.

Raises:

- `MotionError`: If the pipeline errored at or before command_index.

`wait_motion`

```
rbt.wait_motion(
    timeout: float = 10.0,
    settle_window: float = 0.25,
    speed_threshold: float = 0.5,
    angle_threshold: float = 0.5,
) -> bool
```

Wait until the queue has run out and the arm has come to rest.

First waits for the latest command the controller has accepted — every command this client queued, and each newer one the status stream reports accepted — to end, then for the arm to hold still for `settle_window`. A command that failed or was cancelled ends the wait like one that completed; `wait_command()` tells which.

```
rbt.wait_motion()
```

Parameter	Meaning
`timeout`	Maximum time to wait in seconds
`settle_window`	How long robot must be stable to be considered stopped
`speed_threshold`	Max joint speed to be considered stopped (deg/s)
`angle_threshold`	Max angle change to be considered stopped (degrees)

Returns: True if robot stopped, False if timeout

Raises:

- `ConnectionError`: If the controller session changes while a command is still being waited on.

`wait_status`

```
rbt.wait_status(predicate: Callable[[StatusBuffer], bool], timeout: float = 5.0) -> bool
```

Wait until a multicast status satisfies predicate(status) within timeout.

`checkpoint`

```
rbt.checkpoint(label: str) -> int
```

Insert a checkpoint marker in the motion queue.

Returns the command index (≥ 0) on success, -1 on failure.

```
rbt.checkpoint("pick_done")
```

Parameter	Meaning
`label`	Checkpoint label for progress tracking

`wait_checkpoint`

```
rbt.wait_checkpoint(label: str, timeout: float = 30.0) -> bool
```

Wait until a checkpoint with the given label has been reached.

```
rbt.wait_checkpoint("pick_done")
```

Parameter	Meaning
`label`	Checkpoint label to wait for
`timeout`	Maximum wait time in seconds

`delay`

```
rbt.delay(seconds: float) -> int
```

Insert a non-blocking delay in the motion queue.

Returns the command index (≥ 0) on success, -1 on failure.

```
rbt.delay(1.0)
```

Tools

Selecting a tool, the generic tool action, and the methods of the bound tool object `rbt.tool`. Tool actions are queued with the motion; `tool.stop()` is immediate. Tools and the TCP in the guide.

Registered tools

Key	Tool	Variants (default first)
`NONE`	Bare flange - no tool attached	—
`PNEUMATIC`	Pneumatic gripper assembly (vertical/horizontal mounting)	`vertical`, `horizontal`
`SSG-48`	SSG-48 adaptive electric gripper (Spectral micro BLDC)	`finger`, `pinch`
`MSG`	MSG compliant AI stepper gripper (StepFOC)	`100mm`, `150mm`, `200mm`
`VACUUM`	Vacuum gripper (pneumatic valve I/O)	—

Selecting and acting

`select_tool`

```
rbt.select_tool(tool_name: str, variant_key: str = '') -> int
```

Set the active end-effector tool on the controller.

```
rbt.select_tool("PNEUMATIC")
```

Parameter	Meaning
`tool_name`	Name of the tool ('NONE', 'PNEUMATIC', 'SSG-48', 'MSG', 'VACUUM')
`variant_key`	Optional variant within the tool type.

Returns: Command index (>= 0) if queued, 0 on failure.

`tool_action`

```
rbt.tool_action(
    tool_key: str,
    action: str,
    params: list | None = None,
    *,
    wait: bool = False,
    timeout: float = 10.0,
) -> int
```

Send a generic tool action command.

Returns the command index (>= 0) on success, -1 on failure. The action and its parameters are checked before anything is sent: a malformed one raises `ValueError`. A key naming a tool other than the selected one is refused by the controller.

A tool action is queued work: it runs in queue order with motion, the arm holding still while it does, so a wait on it covers the motion queued ahead of it, and one sent after a blended move ends the blend there. A `move` before a completed `calibrate` is refused when its turn comes, failing its own index; a tool action that fails cancels what is queued behind it (`MOTN_CANCELLED`). A pause holds the tool actions still queued. `stop()`, `estop()`, `reset_state()`, a teleport, or a jog or servo stream taking the arm discards them and halts the one running where the jaws are, keeping the grip.

`stop` alone acts at once, on the tool fitted now: it halts the action running, failing it as cancelled by a tool stop, keeps what is queued behind it, and holds the next queued tool action until the jaws are still.

Electric grippers take `move [position, speed, current]` (exactly three fractions in `[0, 1]`; current spans the tool's `current_range`), `calibrate`, `stop` (halt in place, keep grip) and `idle` (release, the jaws left where they are); `set_position` / `open` / `close` map onto `move`. Pneumatic grippers take `open`, `close`, and `move` / `set_position [position]`.

```
rbt.tool_action("PNEUMATIC", "open")
```

Parameter	Meaning
`tool_key`	Tool registry key (e.g. "PNEUMATIC", "SSG-48", "MSG")
`action`	Action name (e.g. "open", "close", "move", "calibrate")
`params`	Positional parameters (meaning defined by tool config)
`wait`	If True, block until action completes
`timeout`	Maximum time to wait in seconds (only used when wait=True)

Note

The actions of the electric grippers: `move` with `[position, speed, current]` as fractions, `calibrate`, `stop`, `idle` (release). Of the pneumatic and vacuum grippers: `open`, `close`, and `move` (or `set_position`) with `[position]`, below 0.5 opening and from 0.5 closing. The tool methods below call these.

The electric gripper tool (SSG-48, MSG)

`tool.set_position` (electric)

```
rbt.tool.set_position(
    position: float,
    *,
    speed: float = 0.5,
    current: float = 0.5,
    **wait_kwargs: Any,
) -> int
```

Moves the jaws to `position`, 0 open and 1 closed, at `speed` (a fraction of the gripper's speed range) and with `current` as the current limit, the grip force. `current` is a fraction of the tool's current range: 100 to 1300 mA for the SSG-48, 100 to 2800 mA for the MSG, so `current=0.5` is 700 mA or 1450 mA. Queued: it waits for the motion before it, and the motion after it waits for the jaws to arrive, stop on a part or stall. Refused when its turn comes on a gripper that was never calibrated.

`tool.open` (electric)

```
rbt.tool.open(*, speed: float = 0.5, current: float = 0.5, **wait_kwargs: Any) -> int
```

`set_position(0.0, ...)`: the jaws fully open.

`tool.close` (electric)

```
rbt.tool.close(*, speed: float = 0.5, current: float = 0.5, **wait_kwargs: Any) -> int
```

`set_position(1.0, ...)`: the jaws close until they reach the end or stop on a part.

`tool.calibrate` (electric)

```
rbt.tool.calibrate(**kwargs: object) -> int
```

Sends the gripper its calibration: it opens fully and closes fully to find its end stops. Queued. The API counts it done after a fixed 2 s (200 control cycles) and does not read the gripper's calibrated bit; the SSG-48 needs about 3 s and the MSG 5 to 6 s, so queue a `delay()` after it (Calibrating). There is no clear-error action: a latched gripper error is cleared from the commander or Motor GUI.

`tool.release` (electric)

```
rbt.tool.release(**kwargs: object) -> int
```

Drop the grip, freeing the jaws for manual handling, once the commands queued ahead of it have run. The jaws stay where they are.

`tool.stop` (electric)

```
rbt.tool.stop(**kwargs: object) -> int
```

Halt the jaws where they are, keeping the grip, ahead of anything still queued: the action running fails as cancelled, and the queue behind it is kept. On an uncalibrated gripper, which has no position to hold, this releases instead.

The pneumatic gripper tool (PNEUMATIC, VACUUM)

`tool.set_position` (pneumatic)

```
rbt.tool.set_position(position: float, **kwargs: float | int) -> int
```

Binary position: < 0.5 opens, >= 0.5 closes.

`tool.open` (pneumatic)

```
rbt.tool.open(**kwargs: float | int) -> int
```

Switches output 1 high, queued with the motion; the action lasts the tool's estimated stroke (a single control cycle for the vacuum gripper). Which way the gripper moves depends on the tubing.

`tool.close` (pneumatic)

```
rbt.tool.close(**kwargs: float | int) -> int
```

Switches output 1 low, queued with the motion; otherwise as `open()`.

Every tool object also has `status()`, returning the `ToolStatus` of the selected tool: `key`, `state`, `engaged`, `part_detected`, `fault_code`, `positions`, `channels` and `variant_key`. For the electric grippers, Tool status says where each field comes from.

I/O

The control board's digital inputs and outputs. The firmware's I/O byte is `[in1, in2, out1, out2, estop, ...]`; reads return the first five, writes take the logical output index 0 or 1. I/O, delays and checkpoints in the guide.

`write_io`

```
rbt.write_io(index: int, value: int, *, timeout: float | None = None) -> int
```

Set digital output by logical index (0 = first output pin).

The firmware I/O byte layout is `[in0, in1, out0, out1, estop, ...]` so logical output index 0 maps to bit position 2.

Returns the command index (≥ 0) on success, -1 on failure.

`timeout` bounds command acceptance. TimeoutError leaves application unconfirmed; None uses the client defaults.

```
rbt.write_io(0, 1)   # Set first output HIGH
```

The read side is `io()` under Queries, and `is_estop_pressed()`.

Configuration

The hardware connection, the motion profile, the TCP corrections, the collision world and the status rate. Tools and the TCP and The controller in the guide.

`connect_hardware`

```
rbt.connect_hardware(port_str: str) -> int
```

Connect to robot hardware via serial port.

```
rbt.connect_hardware("/dev/ttyUSB0")
```

Parameter	Meaning
`port_str`	Serial port path (e.g., '/dev/ttyUSB0' or 'COM3').

Returns: 1 if acknowledged, 0 on failure.

Raises:

- `ValueError`: If port_str is empty.

`select_profile`

```
rbt.select_profile(profile: str) -> int
```

Set the motion profile (e.g. `"TOPPRA"`).

```
rbt.select_profile("TOPPRA")
```

Parameter	Meaning
`profile`	Motion profile type ('TOPPRA', 'RUCKIG', 'QUINTIC', 'TRAPEZOID', 'LINEAR')
`Note`	RUCKIG is point-to-point only; Cartesian moves will use TOPPRA.

Returns: 1 once the profile is selected, 0 when no reply arrives.

Raises:

- `MotionError`: when `profile` is not a profile name.

> **Note**
>
> The profiles of this version: `TOPPRA`, `RUCKIG`, `QUINTIC`, `TRAPEZOID`, `LINEAR`. Motion profiles says what each does.

`set_tcp_offset`

```
rbt.set_tcp_offset(x: float = 0, y: float = 0, z: float = 0) -> int
```

Set TCP offset in mm, composed on top of the current tool transform.

The offset shifts the effective TCP point in the tool's local frame. Subsequent motion (especially TRF relative moves) will use the new TCP. Returns the queued index; await `wait_command(index)` to confirm application. This translation-only setter clears user TCP rotation. Call with (0, 0, 0) to reset. Changing tools resets the offset.

```
rbt.set_tcp_offset(0, 0, -190)  # pencil tip 190mm from gripper
```

`set_tcp_transform`

```
rbt.set_tcp_transform(
    x: float = 0,
    y: float = 0,
    z: float = 0,
    roll: float = 0,
    pitch: float = 0,
    yaw: float = 0,
) -> int
```

No docstring.

`tcp_offset`

```
rbt.tcp_offset() -> list[float]
```

Query current TCP offset in mm [x, y, z].

```
offset = rbt.tcp_offset()
```

`tcp_transform`

```
rbt.tcp_transform() -> list[float]
```

No docstring.

Note

`set_tcp_transform(x, y, z, roll, pitch, yaw)` : millimetres and intrinsic XYZ degrees relative to the registered tool. Both setters return a queued index; wait for it before relying on the correction. The TCP queries raise `TimeoutError` instead of returning a misleading zero when the controller does not answer.

`set_shapes`

```
rbt.set_shapes(shapes: list) -> int
```

Replace the program-layer collision-world shapes (keep-out barriers).

Collision-enabled shapes are added to the controller's checkers so motion is blocked against them; an empty list clears the program layer. Installation-layer shapes (robot config) are unaffected.

Acknowledged: returns 1 only after the controller confirms the world was applied; 0 on timeout. Raises MotionError if the controller rejects the shapes.

```
rbt.set_shapes([Box(name="table", x=0.6, y=0.4, z=0.02,
                    pose=(0.3, 0, -0.01, 0, 0, 0))])
```

Note

Shapes are the program layer of the collision world: boxes and other primitives with a pose in metres and extrinsic XYZ radians, collision-enabled or visual only, attachable to the flange as held objects. The installation layer, from the robot's configuration, is not affected.

`set_status_rate`

```
rbt.set_status_rate(hz: float) -> int
```

Set the rate the controller broadcasts status at.

```
rbt.set_status_rate(100)
```

Status fields

The fields of the status buffer the controller broadcasts, as `stream_status()` yields it and `status()` summarises it, from the source at v0.9.1. Arrays are numpy arrays.

Field	Type
`session_id`	`int`
`seq`	`int`
`mono_time_ns`	`int`
`pose`	`np.ndarray`
`angles`	`np.ndarray`
`speeds`	`np.ndarray`
`io`	`np.ndarray`
`tool_status`	`ToolStatus`
`joint_en`	`np.ndarray`
`cart_en_wrf`	`np.ndarray`
`cart_en_trf`	`np.ndarray`
`action_current`	`str`
`action_params`	`str`
`action_state`	`ActionState`
`executing_index`	`int`
`completed_index`	`int`
`last_checkpoint`	`str`

Field	Type
`error`	`RobotError | None`
`queued_segments`	`int`
`queued_duration`	`float`
`tcp_speed`	`float`
`simulator_active`	`bool`
`collision_active`	`bool`
`collision_pairs`	`list[tuple[str, str]]`
`scene_epoch`	`int`
`accepted_index`	`int`
`homed`	`bool`
`enabled`	`bool`
`torques`	`np.ndarray`
`torques_ext`	`np.ndarray`
`warnings`	`list[tuple]`
`link_health`	`dict`
`drive_health`	`dict`
`loop_health`	`dict`
`homing`	`dict`
`cart_en`	`dict[str, np.ndarray]`

Meaning

Field	Meaning
`session_id`, `seq`, `mono_time_ns`	The status publisher's lifetime (changes on restart), a sequence number (gaps mean missed broadcasts), and the monotonic time of the controller's snapshot
`pose`	The tool pose as a 4×4 matrix, flattened (16 values), in the WRF
`angles`, `speeds`	Joint angles in degrees and speeds in deg/s, J1 to J6
`io`	`[in1, in2, out1, out2, estop]`
`tool_status`	The selected tool's `ToolStatus`; key `NONE` when no tool is fitted
`joint_en`, `cart_en_wrf`, `cart_en_trf`, `cart_en`	Enablement: whether each joint, and each Cartesian axis in each frame, can still move in the positive and negative direction (12 values: six axes, two directions)
`action_current`, `action_params`, `action_state`	What the controller is executing, its parameters, and the state: idle, executing, error
`executing_index`, `completed_index`, `accepted_index`	The index of the command running, the highest completed, and the latest accepted
`last_checkpoint`	The label of the last checkpoint reached
`error`	The current `RobotError`, or `None`
`queued_segments`, `queued_duration`	How much motion is queued, in segments and in seconds of estimated duration (tool actions included)
`tcp_speed`	The tool's linear speed in mm/s
`simulator_active`	Simulator mode
`collision_active`, `collision_pairs`	Whether the collision checker is on, and the pairs it currently reports
`scene_epoch`	Changes when the collision world changes: re-query `shapes()`
`homed`, `enabled`	Whether the robot is referenced, and whether motion is enabled

Field	Meaning
`torques`, `torques_ext`	Reserved for arms with torque sensing; zero on PAROL6
`warnings`, `link_health`, `drive_health`, `loop_health`	Diagnostics: warnings, the serial link, the drives, the control loop timing
`homing`	The progress of a homing sequence

The buffer's `freedrive` property is always false: PAROL6's steppers cannot be back-driven. `mode` is an alias of `action_state`.

Errors

The error codes of the controller, with the catalog text a `MotionError` carries, from the source at v0.9.1. The number ranges: 10 to 19 inverse kinematics, 20 to 29 trajectory planning, 30 to 39 motion execution, 40 to 49 communication, 50 to 59 system and safety. `MOTN_CANCELLED` uses the number shared with the waldoctl interface, so that a client sees one code for a discarded command whichever arm raised it. Errors in the guide says how to recover.

Code	Number	Title	Cause	Effect	Remedy
`IK_TARGET_UNREACHABLE`	10	IK: target unreachable	Target pose has no valid IK solution.	Motion command rejected. Pipeline halted.	Verify target is within workspace. Try different orientation.
`IK_PARTIAL_PATH`	11	IK: partial path failure	Only {valid}/{total} poses along the path are reachable.	Motion command rejected. Pipeline halted.	Shorten the move, add intermediate waypoints, or adjust orientation.
`TRAJ_EMPTY_RESULT`	20	Trajectory: empty result	Trajectory generation returned no waypoints.	Motion command rejected.	Check motion parameters. Start and end may be too close.
`TRAJ_NO_STEPS`	21	Trajectory: no steps	Trajectory calculation produced zero steps.	Motion command rejected.	Increase duration or reduce speed fraction.
`MOTN_HOME_TIMEOUT`	30	Homing timeout	Robot did not start homing sequence within timeout.	Home command aborted.	Check serial connection and robot power. Ensure E-stop is released.
`MOTN_GRIPPER_TIMEOUT`	31	Gripper timeout	Gripper command timed out in state {state}.	Gripper command aborted.	Check gripper connection and calibration.

Code	Number	Title	Cause	Effect	Remedy
`MOTN_GRIPPER_UNKNOWN`	32	Gripper: unknown state	Gripper entered an unknown internal state.	Gripper command aborted.	Reset the controller and recalibrate the gripper.
`MOTN_SETUP_FAILED`	33	Command setup failed	Command could not be initialized.	Command rejected. Pipeline halted.	Check command parameters and robot state.
`MOTN_TICK_FAILED`	34	Command execution error	Unexpected error during execution.	Command aborted. Robot stopped.	Check robot state. May need to re-home.
`MOTN_NOT_HOMED`	35	Robot not homed			
`MOTN_CANCELLED`	shared	Command cancelled	The command was cancelled by {scope} before it finished.	The motion did not run to completion.	Re-issue the command if the motion is still wanted.
`COMM_QUEUE_FULL`	40	Command queue full	Motion queue at maximum capacity.	Command rejected.	Wait for current motions to complete.
`COMM_UNKNOWN_COMMAND`	41	Unknown command	No handler for received command type.	Command ignored.	Check client library version matches server.
`COMM_DECODE_ERROR`	42	Command decode error	Failed to decode command.	Command ignored.	Check command encoding. Possible version mismatch.

Code	Number	Title	Cause	Effect	Remedy
`COMM_VALIDATION_ERROR`	43	Command validation error	Invalid parameters.	Command rejected.	Check parameter ranges and types.
`SYS_STATUS_RATE_INVALID`	55	Status rate not achievable	Status is broadcast every Nth control tick, so {requested} Hz does not divide the {control} Hz control loop evenly.	Broadcast rate unchanged.	Pick a rate that divides {control} Hz: {allowed}.
`SYS_CONTROLLER_DISABLED`	50	Controller disabled	Motion command sent while controller is disabled.	Command rejected.	Call reset() to re-enable the controller.
`SYS_ESTOP_ACTIVE`	51	E-stop active	Emergency stop is currently engaged.	All motion stopped. Queue cleared.	Release the E-stop button and call reset().
`SYS_PORT_SAVE_FAILED`	52	Serial port save failed	Could not save serial port configuration.	Port may not persist across restarts.	Check file permissions and disk space.
`SYS_PROFILE_INVALID`	53	Invalid motion profile	Unrecognized motion profile:	Profile not changed.	Use one of: TOPPRA, RUCKIG, QUINTIC, TRAPEZOID, LINEAR.
`SYS_NOT_SIMULATOR`	56	Simulator-only command	{detail} is only available on the simulator.	Command rejected; the arm is unchanged.	Switch to the simulator with simulator(True), or drive the arm with a planned move.

Code	Number	Title	Cause	Effect	Remedy
`SYS_SELF_COLLISION`	54	Self-collision predicted	Planned configuration would self-collide at sample {sample} of {total}: {pairs}	Motion command rejected before dispatch.	Choose a different target, add intermediate waypoints, or disable via PAROL6_COLLISION_CHECK=0.

A `RobotError` has the fields `code`, `title`, `cause`, `effect` and `remedy`, and `str(error)` joins them.

The controller and its environment

The command line

`parol6-server`, from the source at v0.9.1:

Option	Default	Meaning
`--host`	`0.0.0.0`	UDP host address
`--port`	`5001`	UDP port
`--serial`	—	Serial port (e.g., /dev/ttyUSB0 or COM3)
`--baudrate`	`3000000`	Serial baudrate
`-v`, `--verbose`	`0`	Increase verbosity; -v=INFO, -vv=DEBUG, -vvv=TRACE
`-q`, `--quiet`	flag	Enable quiet logging (WARNING level)
`--log-level`	—	Set specific log level (one of `TRACE`, `DEBUG`, `INFO`, `WARNING`, `ERROR`, `CRITICAL`)

Without `--serial` the controller starts in simulator mode. The port can also come from `PAROL6_COM_PORT` or from the file the last successful port was saved to.

Environment variables

Read at the controller's start:

Variable	Default
`PAROL6_BLEND_HOLD_S`	`0.1`
`PAROL6_BUSY_THRESHOLD_MS`	`1.0`
`PAROL6_COLLISION_CHECK`	`1`
`PAROL6_COLLISION_CLEARANCE_M`	`0.005`

Variable	Default
`PAROL6_COLLISION_JOG_LOOKAHEAD_S`	`0.15`
`PAROL6_COLLISION_PATH_SAMPLES`	`16`
`PAROL6_COLLISION_SRDF`	`computed`
`PAROL6_COM_FILE`	`computed`
`PAROL6_COM_PORT`	—
`PAROL6_CONTROL_RATE_HZ`	`100`
`PAROL6_EXECUTION_OVERRIDE_TRANSITION_S`	`1.0`
`PAROL6_MAX_BACKLOG_COUNT`	`500`
`PAROL6_MAX_BLEND_LOOKAHEAD`	`100`
`PAROL6_MCAST_GROUP`	`239.255.0.101`
`PAROL6_MCAST_IF`	`127.0.0.1`
`PAROL6_MCAST_PORT`	`50510`
`PAROL6_MCAST_TTL`	`1`
`PAROL6_PATH_SAMPLES`	`200`
`PAROL6_SERIAL`	—
`PAROL6_SETTLE_MAX_TICKS`	`20`
`PAROL6_STATUS_RATE_HZ`	`50`
`PAROL6_STATUS_STALE_S`	`0.5`
`PAROL6_STATUS_TRANSPORT`	`MULTICAST`
`PAROL6_STATUS_UNICAST_HOST`	`127.0.0.1`

Variable	Default
`PAROL_TRACE`	`0`

A default marked *computed* is a path or an expression; see `parol6/config.py`. The controller in the guide says which ones matter.

The wire protocol

Clients and the controller exchange msgpack messages over UDP: a command or query with a request ID, and an OK, an error, a response or a status. The protocol version is checked in every status broadcast, and a client that speaks another version raises `ProtocolVersionError`. The command and query types of this version:

Commands: `CmdType.PING` and the others, 54 in all, the queries among them; **queries:** `PING`, `STATUS`, `ANGLES`, `POSE`, `IO`, `SPEEDS`, `TOOL`, `QUEUE`, `CURRENT_ACTION`, `LOOP_STATS`, `PROFILE`, `TOOL_STATUS`, `ENABLEMENT`, `ERROR`, `TCP_SPEED`, `IS_SIMULATOR`, `TCP_OFFSET`, `SHAPES`, `STATUS_RATE`, `TCP_TRANSFORM`, `EXECUTION_SPEED`, `COMMAND_COMPLETION`. The definitions are `parol6/protocol/wire.py`; a program in another language can speak the protocol from them, though the Python client is the supported path.

Python

The package needs Python >=3.11.

RBTscript commands

The commands of the commander software's program language, as its interpreter accepts them. One command per line; arguments in the order given; angles in degrees, positions in millimetres, speed and acceleration in percent, time in seconds. RBTscript in the guide has the rules and examples.

Command	Arguments	What it does
`Begin()`		Must be the first line, or the program is refused
`End()`		Stops the program; the last line must be `End()` or `Loop()`
`Loop()`		Restart at the line after `Begin()`
`Delay(t)`	`t` seconds, more than 0.01	Wait; `Delay(0.01)` is refused
`MoveJoint(j1,j2,j3,j4,j5,j6, v=, a=, t=, trap|poly, speed)`	six joint angles; `v`, `a` in % for the leading joint; `t` duration; the bare word `trap` or `poly`; the word `speed`	A move to joint angles, straight in joint space
`MovePose(x,y,z,Rx,Ry,Rz, v=, a=, t=, trap|poly, speed)`	a WRF pose, then as `MoveJoint`	IK at the target, then a joint-space move
`MoveCart(x,y,z,Rx,Ry,Rz, v=, a=, t=, trap|poly, speed)`	a WRF pose; `t`; `poly` or `trap`; `speed`	A straight-line move of the tool in the WRF; stops at a singularity. With `t`: trapezoidal unless `poly`. Without `t`: always 3.6 s
`MoveCartRelTRF(x,y,z,Rx,Ry,Rz, v=, a=, t=, trap|poly, speed)`	a pose relative to the tool frame	A straight-line move relative to the tool; timing as `MoveCart`
`SpeedJoint()`		No action: the program stops on it until Stop
`SpeedCart()`		Not accepted: a program containing it is refused
`Output(n,HIGH|LOW)`	output 1; `HIGH` or `LOW`, no space	Sets output 1; `Output(2,...)` is refused in the current version and stops the program

Command	Arguments	What it does
`Input()`, `Home()`		No action: the program stops on it until Stop
`Gripper_cal()`		Calibrates the electric gripper (clear its error in the Gripper tab first; activation is the tab's Activate option); follow with `Delay(4)` for the SSG-48, `Delay(7)` for the MSG
`Gripper(position, speed, current)`	whole numbers: 0 to 255; 0 to 255; mA, 100 to 1000	Moves the electric gripper; continues at once, so follow with a `Delay()`

Rules for `MoveJoint` and `MovePose`: `t` overrides `v` and `a` and makes the profile polynomial unless the word `trap` follows; `v` with `a` makes it trapezoidal; neither gives a conservative trapezoid; `speed` as the last argument tracks the speed curve only and may stop short of the target. `MoveCart` and `MoveCartRelTRF` are trapezoidal with `t` unless `poly` follows, and take 3.6 s for any move without `t`. The list is in the commander's `PAROL6_ROBOT.py` as `Commands_list`.

The serial protocol

The packets between the PC and the control board, from the firmware's `main.cpp` (`Unpack_data`, `Pack_data`). Serial over USB at 3 000 000 baud; the PC sends one command packet per 10 ms cycle and the board answers each valid packet with one telemetry packet. Multi-byte values are big-endian; positions and speeds are signed 24-bit integers in motor microsteps and microsteps per second. The serial protocol in the Control Board Manual explains the cycle.

PC → board: the command packet

Offset after the length byte	Size	Field	Meaning
—	3	Start bytes	`0xFF 0xFF 0xFF`
—	1	Length	52: the number of bytes that follow, CRC and end bytes included
0	18	Joints[6]	Commanded position of each joint, 3 bytes each, signed, microsteps
18	18	Speed[6]	Commanded speed of each joint, 3 bytes each, signed, microsteps/s
36	1	Command	The command byte, below
37	1	Affected joint	Bit per joint; reserved
38	1	InOut	Bits from the MSB: bit 2 = output 1, bit 3 = output 2; the rest reserved
39	1	Timeout	Reserved
40	2	Gripper position	0 to 255
42	2	Gripper speed	0 to 255
44	2	Gripper current	mA

Offset after the length byte	Size	Field	Meaning
46	1	Gripper command	Bits from the MSB: 0 activate, 1 action, 2 E-stop, 3 release direction
47	1	Gripper mode	0 operate, 1 calibrate, 2 clear error
48	1	Gripper ID	The gripper's CAN ID
49	1	CRC	Reserved: carried, not checked by firmware 1
50	2	End bytes	`0x01 0x02`

Command bytes

Value	Name	Effect
255	Idle (dummy)	The joints stand still (speeds are set, no joint is stepped). Continues a homing in progress
123	Jog	Runs each joint at its commanded speed
156	Go to position with speed	Runs each joint at the average of the commanded speed and the speed that closes the position error in one cycle
100	Home	Starts the homing sequence
101	Enable	Enables motion, resets the homing state
102	Disable	Disables motion; the motors hold
103	Clear error	Resets the homing state
69	Repeatability test	A built-in two-pose cycle

The gripper fields are forwarded over CAN every cycle: a move frame when any of them changed, a calibrate for mode 1, a clear error for mode 2, an empty status poll otherwise.

Board → PC: the telemetry packet

Offset after the length byte	Size	Field	Meaning
—	3	Start bytes	`0xFF 0xFF 0xFF`
—	1	Length	56
0	18	Position[6]	Actual position of each joint, 3 bytes each, signed, microsteps
18	18	Speed[6]	Actual speed of each joint
36	1	Homed	Bits from the MSB: joint 1 to joint 6 homed; bits 6 and 7 set
37	1	IO	Bits from the MSB: input 1, input 2, output 1, output 2, E-stop (1 = closed, not pressed); bits 5 to 7 set
38	1	Temperature error	Bits from the MSB: joint 1 to joint 6, reserved, reported cleared by firmware 1; bits 6 and 7 set
39	1	Position error	Bits from the MSB: joint 1 to joint 6, reserved; bits 6 and 7 set
40	2	Timing data	Ticks of 1.422 µs between the last two valid packets (90 MHz / 128); 7031 = 10 ms
42	1	Timeout error	Reserved
43	1	XTR2	The last command byte received
44	1	Gripper ID	Always 0 in firmware 1 (not reported by the gripper)
45	2	Gripper position	0 to 255
47	2	Gripper speed	Always 1 in firmware 1 (not reported by the gripper)
49	2	Gripper current	mA

Offset after the length byte	Size	Field	Meaning
51	1	Gripper status	The gripper's status byte, bits from the MSB: activated, moving to a position, object detection (bits 5 and 4, bit 5 the low bit: 0 moving, 1 object while closing, 2 object while opening, 3 at position), temperature error, timeout error, E-stop error, calibrated
52	1	Object detection	Always 1 in firmware 1; the real value is in the status byte's object-detection bits
53	1	CRC	Reserved
54	2	End bytes	`0x01 0x02`

The gripper's status byte and object detection are as the gripper manuals define them.

Firmware parameters

The per-joint parameters of the control board's main firmware (`constants.h`, `motor_init.cpp`, `main.cpp` at release V1): what a builder edits when mastering, fitting a different tool, reducing the currents, or fixing a joint's direction. Steps are 32-microstep motor steps. Firmware says how to upload a change.

Currents (`constants.h`)

Define	Value	Note
`MICROSTEP`	32	
`MOTOR1_MAX_CURRENT`	2000 mA	motor rated 2100
`MOTOR2_MAX_CURRENT`	2000 mA	motor rated 2100
`MOTOR3_MAX_CURRENT`	1900 mA	motor rated 2000
`MOTOR4_MAX_CURRENT`	1700 mA	reduced: the motor is enclosed in the forearm
`MOTOR5_MAX_CURRENT`	1700 mA	reduced: enclosed
`MOTOR6_MAX_CURRENT`	965 mA	motor rated 1000
`VERSION`	1	

The driver is set to `MOTORx_MAX_CURRENT × 0.85` as the running current and `hold_multiplier` of that for holding (the per-joint table below has both); the driver rounds them to its own current steps.

Per joint (`motor_init.cpp`)

Parameter	J1	J2	J3	J4	J5	J6
`reduction_ratio`	6.4	20	18.0952	4	4	10
`standby_position`	10240	−32000	57905	0	0	32000

Parameter (steps at standby)	J1	J2	J3	J4	J5	J6
`homed_position` (steps from the switch to standby)	13500	19588	23020	−10200	8900 (`j5_homing_offset`)	15900
`joint_range_positive_steps`	14000	−1200	92605	7500	6400	64000
`joint_range_negative_steps`	−14000	−51587	34700	−7500	−6400	0
`limit_switch_trigger` (input level when triggered)	0 (inductive)	1 (switch)	1 (switch)	0 (inductive)	1 (switch)	0 (inductive)
`direction_reversed`	1	0	1	0	0	1
`hold_multiplier`	0.7	0.8	0.8	0.8	0.8	0.88
Running current, `MOTORx_MAX_CURRENT × 0.85`	1700 mA	1700 mA	1615 mA	1445 mA	1445 mA	820 mA
Holding current, the running current × `hold_multiplier`	1190 mA	1360 mA	1292 mA	1156 mA	1156 mA	722 mA

Parameter	J1	J2	J3	J4	J5	J6
Driver position / LIMIT input	1 / 6	6 / 2	5 / 3	4 / 4	2 / 5	3 / 1

Joint 1's standby of 10240 steps is +90° (90 / 0.05625 × 6.4): the firmware homes to the pose the software reports as J1 = 90°. The firmware's "standby" is what these pages call the after-homing pose: the standby pose turned 90° about joint 1 (Joint limits and directions).

Joint 5 and the tool (`main.cpp`)

Variable	Value	Note
`j5_homing_offset`	8900	the pneumatic gripper holder; **8035** for the SSG-48

Homing speeds and distances (`main.cpp`, `home_all`)

The approach speeds in steps/s: J1 −2050, J2 −2550, J3 −2550, J4 5550, J5 −5050, J6 −9550; the back-off distances 850 (J1), 1050 (J2, J3), 1550 (J4, J5, J6) steps; the slow re-approach −750 steps/s for joints 1 to 3, ±4050 for 4 and 5, −8050 for 6. The firmware uses the second, slow trigger as the reference.

The serial link

`Serial.begin(3000000)`; the timing counter runs at 90 MHz / 128 = 703 125 Hz.

Test firmware commands

The commands the test firmware (`PAROL6 control board test code`) acts on: `#` followed by the name, case as written, and a line feed (a terminal that sends only a carriage return gets no answer). None takes an argument. The test firmware in the Control Board Manual shows the output.

The motors turn from power-on

The test firmware turns all six motors slowly from the moment it boots until the power is switched off; no command stops them.

Command	What it does
`#IO`	Prints, once a second, the E-stop input, input 1, input 2 and the supply voltage in mV
`#LIMIT`	Prints, once a second, `Joint N is pressed` for every triggered limit switch
`#STOP`	Ends the `#IO` or `#LIMIT` printout; the motors keep turning
`#start`	Toggles the two LEDs

The parser (`SerialPacketParser.cpp`) also recognises `STATUS`, `STEP1` to `STEP6`, `INPUT`, `ESTOP`, `OUT1TOGGLE`, `OUT2TOGGLE`, `LED1TOGGLE`, `LED2TOGGLE`, `SUPPLYV`, `ONOFF`, `CAN` and `FLASH`, but `main.cpp` has no action for them (the `FLASH` test is commented out): each is answered with `Unknown command`. The driver report (`joint N Connection is: GOOD` or `BAD`, the currents and fault flags) is printed once, about three seconds after power-on, not on a command.

Theory corner

Theory corner

The theory behind a six-axis robot arm and its software, explained in the order you meet it: what a joint angle and a pose are, which frames the numbers are in, how the software goes from one to the other, why some poses cannot be reached and some moves cannot be made, what shape a move has in time, and who is in charge of the robot at any moment. Each page says how the idea shows up in PAROL6's software. None of it needs maths beyond what is on the page, and none of it is needed to run the robot: read it when you want to know why the robot does what it does.

The pages

- Units
- Joints, directions and the standby pose
- Reference frames
- Pose and orientation
- Forward and inverse kinematics
- Configurations and singularities
- Workspace and reach
- Homing and mastering
- Motion profiles
- Blending
- Target, commanded and actual
- Control ownership
- Steppers, open loop and backlash
- Further reading

Robot types

Industrial robots are sorted by their joints and how they are arranged: vertically articulated arms, Cartesian gantries, cylindrical and polar arms, SCARA arms, delta robots. PAROL6 is a **vertically articulated** arm with six revolute joints, the most common industrial type, and the ideas on these pages are the ones every such arm uses. The last three joints meet in a point, a **spherical wrist**, which the pages on kinematics return to.

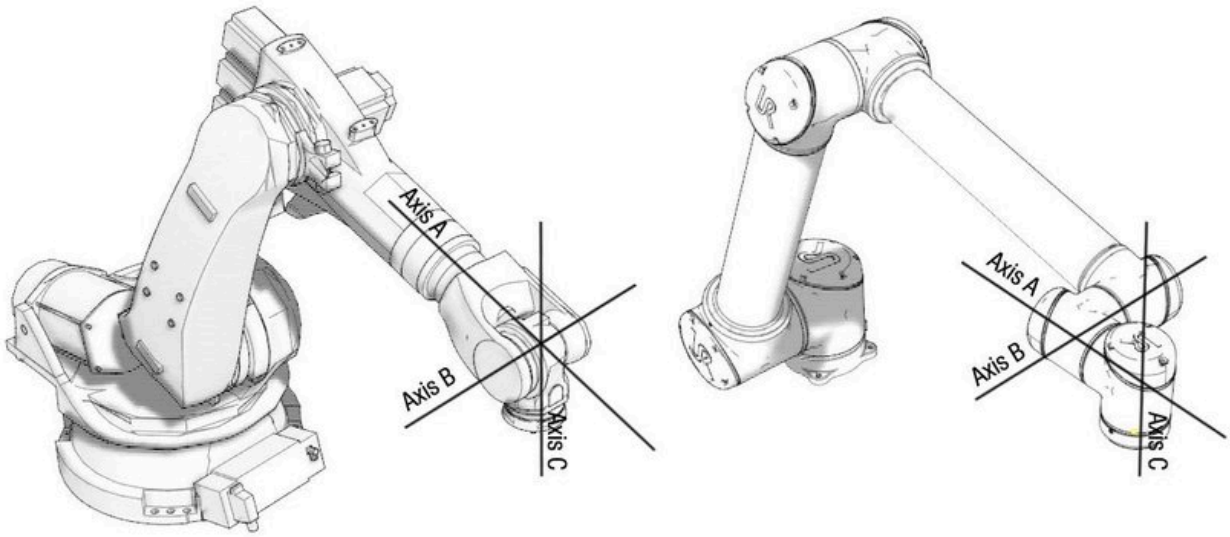


Figure 1: A robot with a spherical wrist (left) and one without (right).

Units

Robotics mixes units, and PAROL6's software converts between them at several places. The rule is: the interfaces speak degrees and millimetres, the firmware speaks motor steps, and the maths in between is in radians and metres.

Quantity	At the interface (commander GUI, RBTscript, Python API)	Inside the software	At the firmware
Joint angle	degrees	radians	motor steps (microsteps)
Position	millimetres	metres	—
Orientation	degrees (three angles)	rotation matrices, radians	—
Joint speed	commander: percent of the maximum; API: a fraction of the maximum in (0, 1]	rad/s	steps per second
Tool speed	commander: percent; API: a fraction	m/s, rad/s	—
Acceleration	percent, or a fraction	rad/s², m/s²	steps/s²
Time	seconds	seconds	10 ms cycles; the timing counter in 1.422 µs ticks
Gripper position	commander: 0 to 255; API: a fraction 0 to 1		0 to 255
Gripper current	commander: milliamps; API: a fraction of the tool's current range		milliamps

Steps

The firmware knows nothing but steps: 200 full steps per motor turn, 32 microsteps each, 6400 per turn, times the joint's gear ratio for a turn of the joint (Drives, ratios and resolution). The PC software converts with the two constants it keeps in its robot model:

```
degrees per step = 360 / (200 × 32) / ratio          # 0.05625° / ratio
steps = degrees / 0.05625 × ratio
```

Speed fractions

The Python API's `speed` and `accel` are fractions of the robot model's maxima, in (0, 1], not percentages: a `speed=0.5` joint move runs the leading joint at half its maximum speed, and the others proportionally so that all finish together. A `duration` above zero overrides the speed: the move takes that long, whatever the speed fraction says. The commander's speed slider is a percentage applied to its own, lower, maxima.

Angles wrap

Joint 3 reports about 105° to 295° and joint 6 0° to 360°, both with the standby at 180°; joints 1, 2, 4 and 5 report signed angles. Orientation angles wrap at ±180°: a pose with Rz = 179° and one with Rz = −181° are the same, and a program that compares angles must take that into account.

Joints, directions and the standby pose

Numbering

Joints are numbered 1 to 6 from the base up: J1 turns the whole arm about the vertical axis, J2 is the shoulder, J3 the elbow, J4 rotates the forearm, J5 pitches the wrist, J6 rotates the tool. The Python API indexes joints from 0 in its jog calls (`jog_j(0, ...)` is J1) and lists angles in order J1 to J6 everywhere.

Joints 4, 5 and 6 are also called wrist 1, wrist 2 and wrist 3, as in the robot model and ROS. The forearm is also called the lower arm: both names mean the link from the elbow to the wrist.



Figure 1: Each joint turned 30 degrees in its positive direction, in colour, over the pose in grey.

Positive direction

Each joint has a positive direction, the one in which its angle increases, drawn as the arrows in the figure on Joint limits and directions. The commander's left arrows jog positive, the right arrows negative. The firmware has a per-joint flag that inverts the motor so that the hardware agrees with this convention; a new build sets it per joint on the first run.

Joint space and the standby pose

A set of six joint angles is a point in **joint space**: it describes the arm's configuration completely, and a joint move is a straight line in it, every joint moving proportionally and finishing at the same time.

The reference point is the **standby pose**, the pose of the Denavit-Hartenberg table; after homing the robot stands in it turned 90° about joint 1:

Pose	J1	J2	J3	J4	J5	J6
Standby	0°	−90°	180°	0°	0°	180°
After homing	90°	−90°	180°	0°	0°	180°

Every joint angle the software reports is measured in this convention. At J1 = 0° the arm points along the base's X axis, away from the electronics box; homing ends with J1 a quarter turn further, at 90°, and the software reports 90° there. The Python API calls that after-homing pose its standby.

Limits

Each joint has a software limit in both directions, enforced by the planner and by the jog, and a mechanical stop a little past it (Joint limits and directions). The limits are in the robot model (`PAROL6_ROBOT.py` in both programs); a tool or a base that changes the real clearance does not change them.

Reference frames

A frame is a coordinate system: an origin and three axes, X, Y and Z, at right angles. A position is three numbers in a frame; the same point has different numbers in different frames. PAROL6's software uses two frames you choose between, the world frame and the tool frame, and several it keeps to itself, the flange frame among them.

The right-hand rule

All frames are right-handed: with the right hand's thumb along X and the index finger along Y, the middle finger points along Z. For rotations, the thumb along the axis, the fingers curl in the positive direction. Every orientation angle in the software follows this rule.

The frames on the robot

Flange only

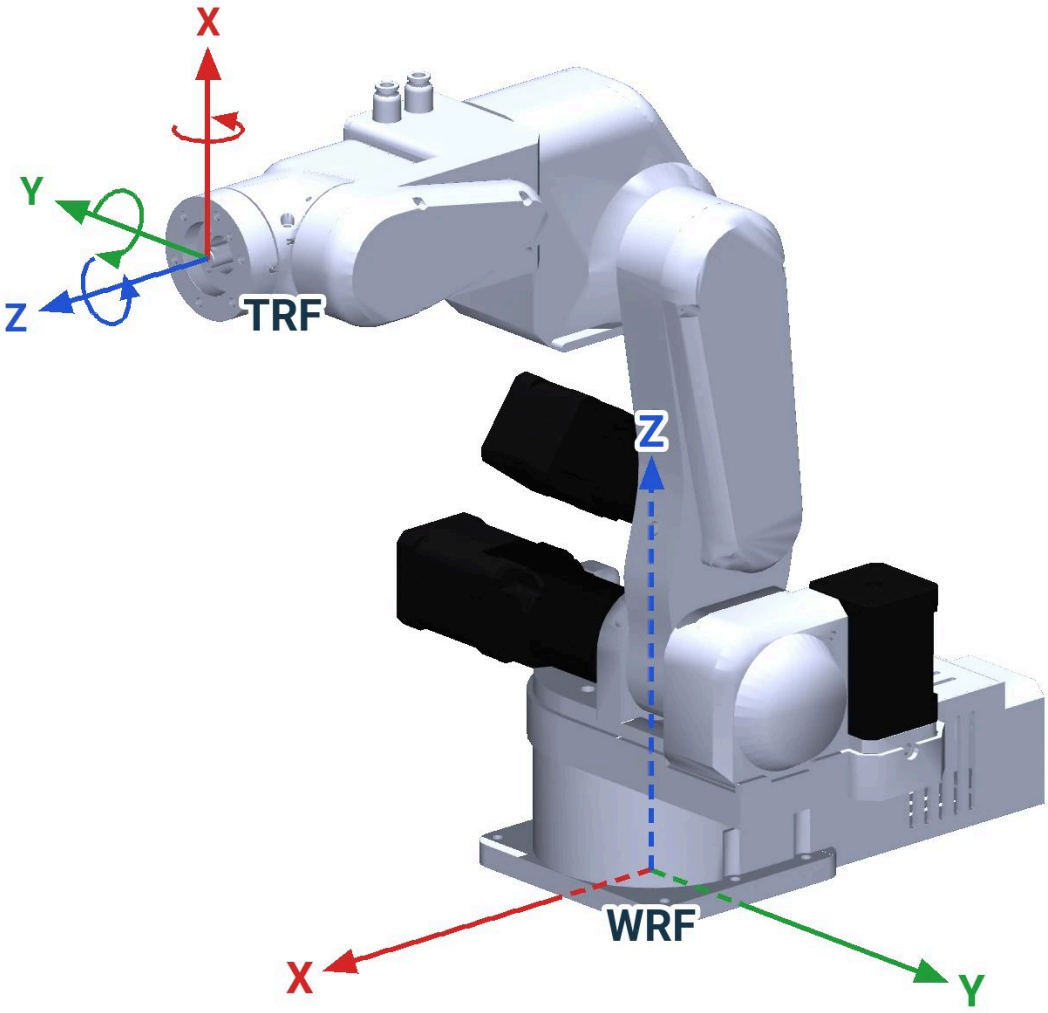


Figure 1: The world frame (WRF) at the base and the tool frame (TRF) at the flange, in the standby pose; with nothing on the flange the TRF is the flange frame.

Pneumatic gripper

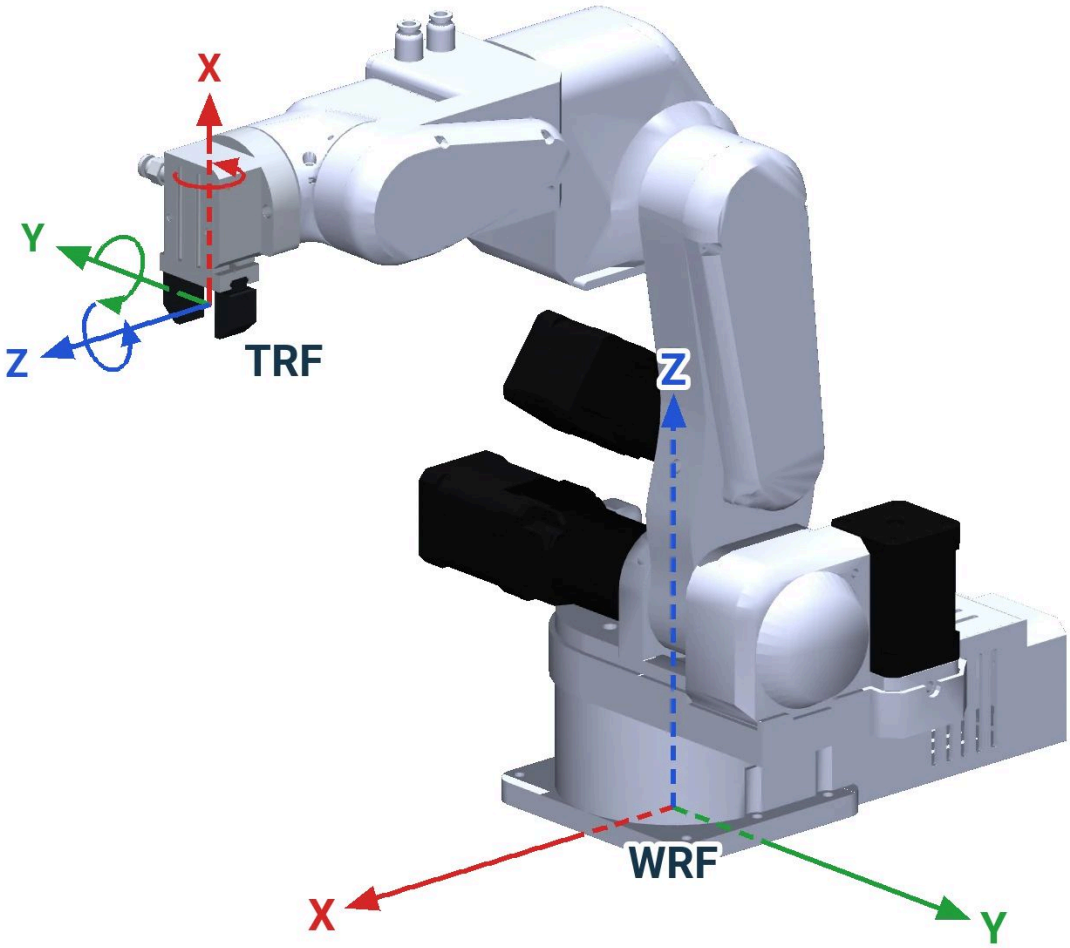


Figure 2: The world frame (WRF) at the base and the tool frame (TRF) at the pneumatic gripper's TCP, in the standby pose.

SSG-48

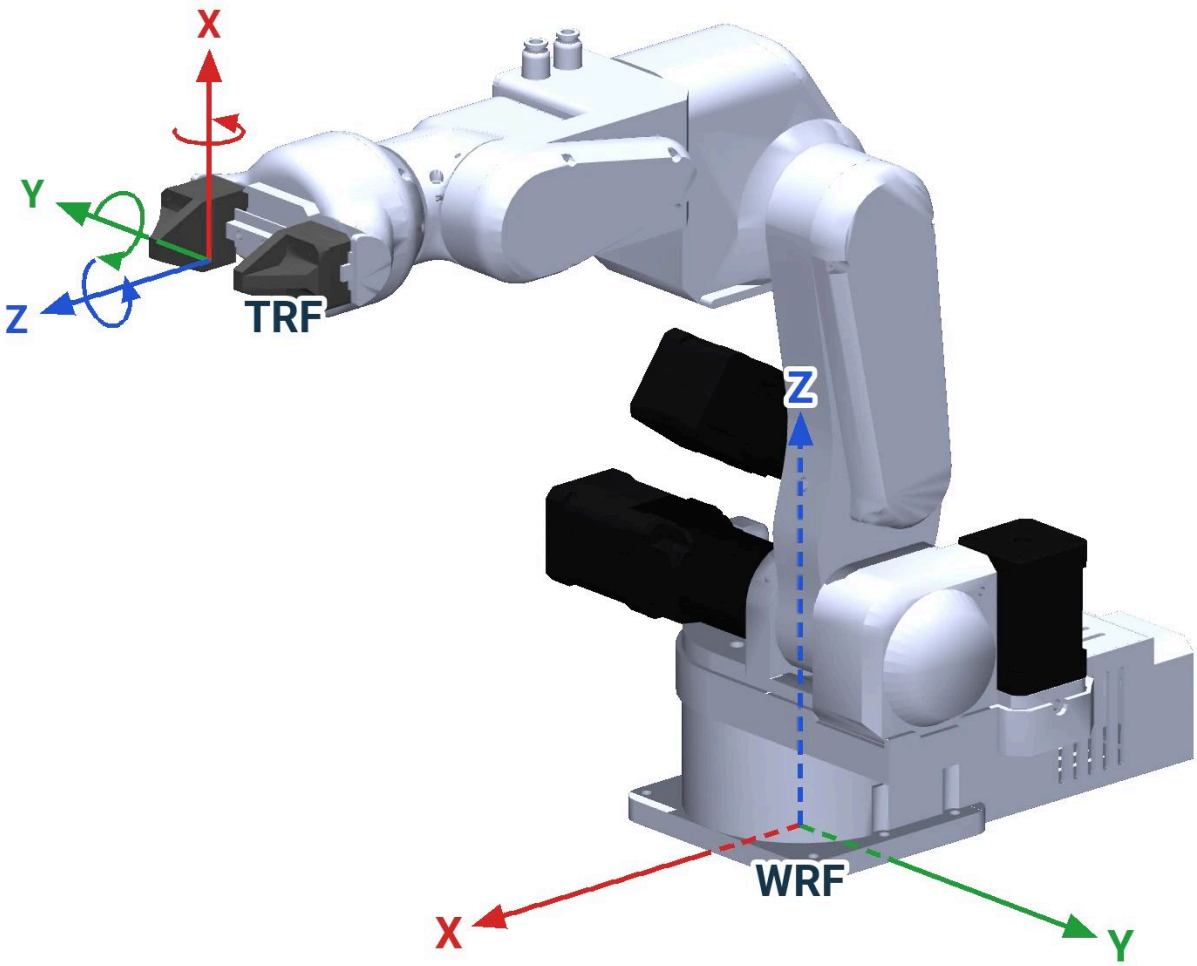


Figure 3: The world frame (WRF) at the base and the tool frame (TRF) at the SSG-48's TCP, in the standby pose.

MSG

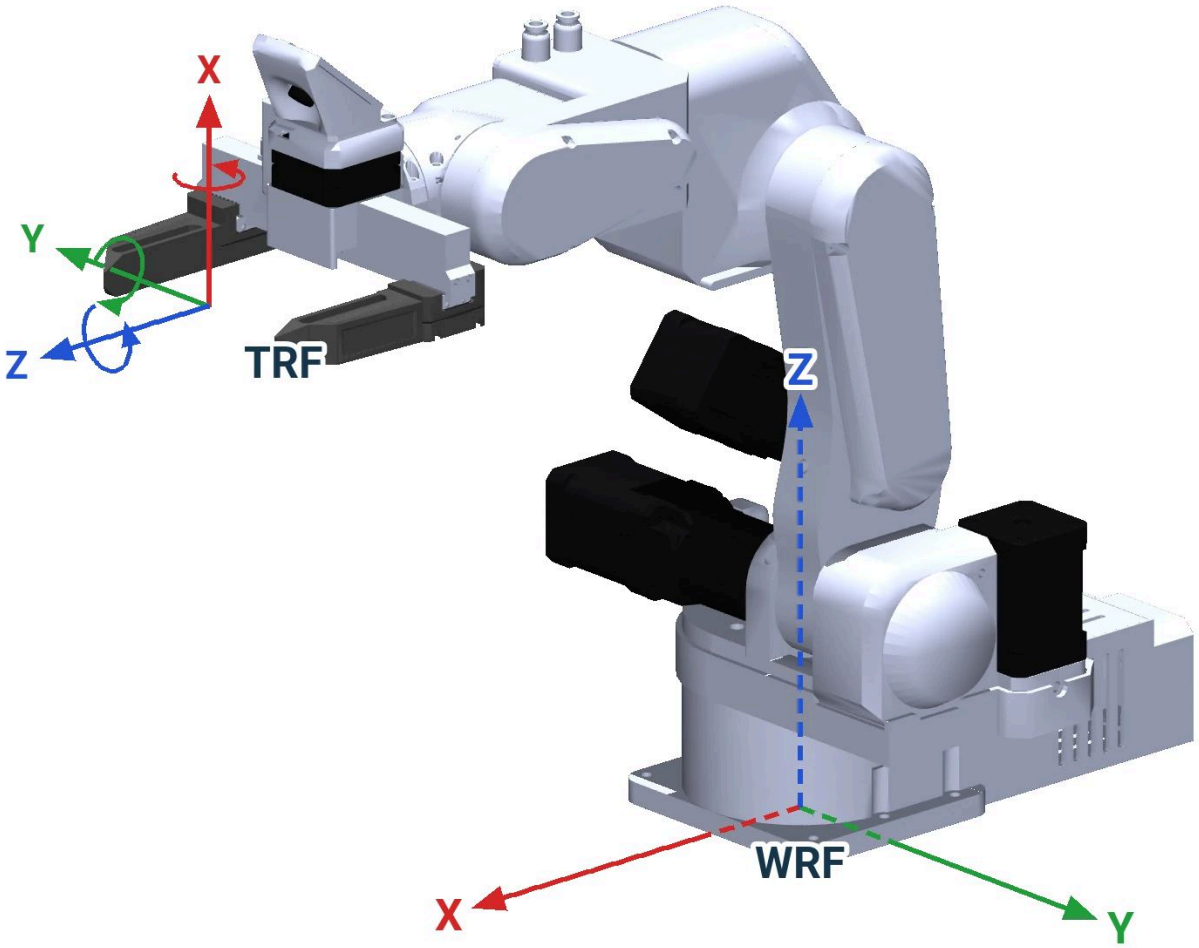


Figure 4: The world frame (WRF) at the base and the tool frame (TRF) at the MSG's TCP, in the standby pose.

The pictures show the frames on the robot's model, in the commander software's convention, which these pages use: X red, Y green, Z blue. The arcs at the tool frame are the positive rotations RX, RY and RZ. An axis is dashed where it runs inside the robot. The directions in the table are those of the standby pose.

In the Python API, the tool frame's Y and Z point the other way

In the Python API (version 0.9.1) the flange and tool frames are these frames turned 180° about X: X is the same, Y and Z point the other way, so the API's Z points back into the wrist. A TRF move or jog, or a TCP offset, "out of the tool" is negative z there: `move_l([0, 0, -20, 0, 0, 0], frame="TRF", rel=True)` moves the tool 20 mm out along the joint 6 axis, and `set_tcp_offset(0, 0, -120)` puts the TCP 120 mm in front of the flange. The code examples on these pages are written for the API as it is (known issue KI-11).

Frame	Origin	X	Y	Z
WRF	On the joint 1 axis, at the base's mounting surface	Forward: the way the arm points in the standby pose, away from the electronics case	To the left, seen from behind the robot: the way the arm points after homing	Up, along the joint 1 axis
Flange frame	On the joint 6 axis, at the flange face	Up; it turns with joint 6	To the robot's right	Out of the flange, along the joint 6 axis
TRF	At the tool centre point (TCP)	As the flange frame	As the flange frame	As the flange frame: out of the tool, along the joint 6 axis

WRF, the world reference frame

Fixed to the robot's base, as in the table: X the way the arm points in the standby pose, Y the way it points after homing, Z up. Everything that does not move with the arm is described in it: the table, the parts, the positions a program teaches. `MoveCart` in RBTscript and `move_l` in the API take WRF poses by default. The commander software and the Python API agree on it.

TRF, the tool reference frame

Fixed to the tool: the origin at the tool centre point, the axes those of the flange frame. It moves with the arm. Z points out of the tool along the joint 6 axis, the way the SSG-48's and the MSG's jaws point; the pneumatic gripper's fingers hang beside the axis and point along −X. A move **in** the TRF is relative to where the tool is now: "100 mm out along the tool" is `MoveCartRelTRF(0,0,100,0,0,0)` in the commander, and `move_l([0, 0, -100, 0, 0, 0], frame="TRF", rel=True)` in the Python API, whose Z points the other way (above). It goes in a different world direction every time, depending on the tool's orientation. Approaching a part along the gripper's axis, retreating after a grip, and rotating the tool about its own axis are TRF moves. Jog slowly in the TRF before you run a TRF move written for the other program.

The flange frame

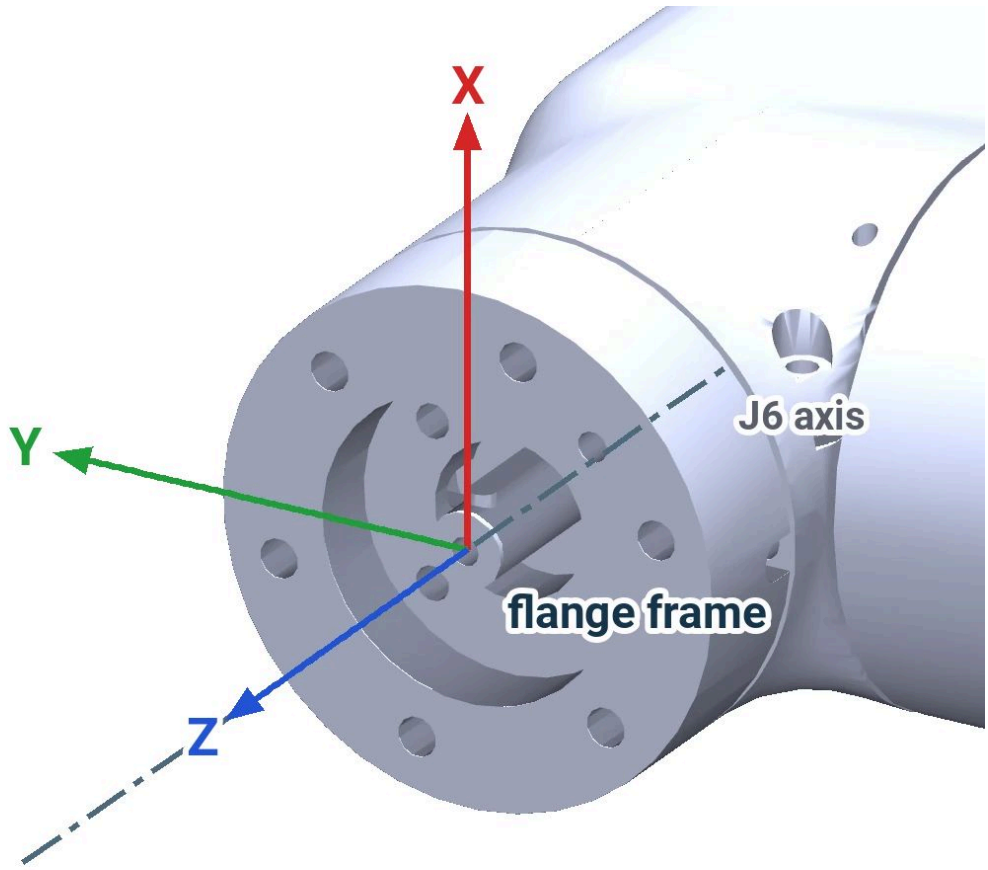


Figure 5: The flange frame on the flange face, in the standby pose: X up, Y to the robot's right, Z out of the flange along the joint 6 axis.

Every tool is measured from the flange frame. Its origin is on the joint 6 axis at the flange face, 50.5 mm from the wrist centre (the point where the axes of joints 4, 5 and 6 meet), and its Z axis points out of the flange along that axis. X and Y turn with joint 6: in the standby pose, with joint 6 at 180°, X points up. The Python API's built-in model puts the origin 37 mm from the wrist centre, 13.5 mm short of the face, measures its tool offsets from there, and has Y and Z the other way (above).

The TCP of each tool

Tool	`select_tool` key, variants	The API's offset, in its own axes: x, y, z	From the wrist centre, along the tool and beside it: API	The same in the model: a6, a7	Fingers point along; jaws close along
Bare flange	`NONE`	0, 0, 0 mm	37 mm, 0	50.5 mm, 0	

Tool	`select_tool` key, variants	The API's offset, in its own axes: x, y, z	From the wrist centre, along the tool and beside it: API	The same in the model: a6, a7	Fingers point along; jaws close along
Pneumatic gripper	`PNEUMATIC`, `vertical` (the default)	−55, 0, −27 mm	64 mm, 55 mm	62.3 mm, 45.25 mm	−X; Y
Pneumatic gripper	`PNEUMATIC`, `horizontal`	0, 0, −82 mm	119 mm, 0	not in the model	Z; X
SSG-48	`SSG-48`, `finger`, `pinch`	0, 0, −105 mm	142 mm, 0	134.3 mm, 0	Z; Y
MSG	`MSG`, `100mm`, `150mm`, `200mm`	−29, 0, −109.5 mm	146.5 mm, 29 mm	134.3 mm, 28.35 mm	Z; Y
Vacuum gripper	`VACUUM`	0, 0, −37 mm	74 mm, 0	not in the model	

Every tool frame has the flange frame's axes: the Python API's tool definitions are offsets, with no rotation, and their negative z is out of the tool. A TCP beside the tool axis lies along the flange frame's −X, below the joint 6 axis in the standby pose. The model's TCP sits between the fingers, near their tips; the API's offsets come from the earlier model and differ from the model's by up to about 14 mm. The pictures above are drawn at the model's TCP. `set_tcp_offset()` and `set_tcp_transform()` move the TCP for a correction or a tool of your own (Tools and the TCP).

Jogging in a frame

The commander's Cartesian jog and the API's `jog_l` take the frame: WRF jogs move the tool along the base's axes, TRF jogs along the tool's own. Switch to TRF to go straight in or out of a part whatever its angle.

In the TRF column, +Z is out of the tool in the commander and into the wrist in the Python API.

Jog	In the WRF	In the TRF
X, Y, Z	The TCP moves along the base's X, Y or Z; the tool keeps its orientation	The TCP moves along the tool's own X, Y or Z, as the tool stands; the tool keeps its orientation

Jog	In the WRF	In the TRF
RX, RY, RZ	The tool turns about a line through the TCP, parallel to the base's X, Y or Z; the TCP stays where it is. The Python API only: the commander's WRF rotation buttons do nothing	The tool turns about its own X, Y or Z, through the TCP; the TCP stays where it is

Orientation

A pose gives the tool frame's orientation in the world frame as three angles in degrees. The Python API reads them as intrinsic XYZ rotations, R = Rx · Ry · Rz, of its own tool frame: after homing, with the tool horizontal and pointing away from the base, an API pose ends in 90, 0, 90. Pose and orientation explains the angles.

The frames inside

The kinematic model keeps a frame at every joint, each defined relative to the previous one by the Denavit-Hartenberg parameters, and the flange frame. The tool frame is the flange frame moved by the tool's transform; the Python API's tool definitions and `set_tcp_transform()` set that transform (Tools and the TCP). The commander's model has the pneumatic gripper's transform built in.

Pose and orientation

A **pose** is a position and an orientation together: where the tool is and which way it points. Six numbers: x, y, z in millimetres and three angles in degrees, `[x, y, z, Rx, Ry, Rz]`, in the frame you name, the WRF unless you say TRF.

Position

The tool centre point's coordinates in the frame. For the standard pneumatic gripper the TCP is between the finger tips; for a selected tool in the API it is the tool's registered point, moved by any offset you set.

Orientation

Three angles describe how the tool frame is rotated relative to the reference frame. There are many conventions; PAROL6's software uses **roll, pitch and yaw as rotations about X, Y and Z**, which is what the commander shows as Rx, Ry, Rz and what the API's `pose()` returns and its motion commands take. The API's `set_tcp_transform()` takes its angles as intrinsic XYZ rotations, `Rx · Ry · Rz`; its collision shapes take extrinsic XYZ in radians. When a pose from one program goes into another, check the convention before trusting the numbers.

The same position can be reached with many orientations, and the orientation is what a program usually gets wrong first: a few degrees off puts a finger into the table. The Python API examples on this site use `[90, 0, 90]`, the orientation the tool has after homing: horizontal, pointing forward, away from the base. A tool pointing down at the table has other angles; jog it there and read them.

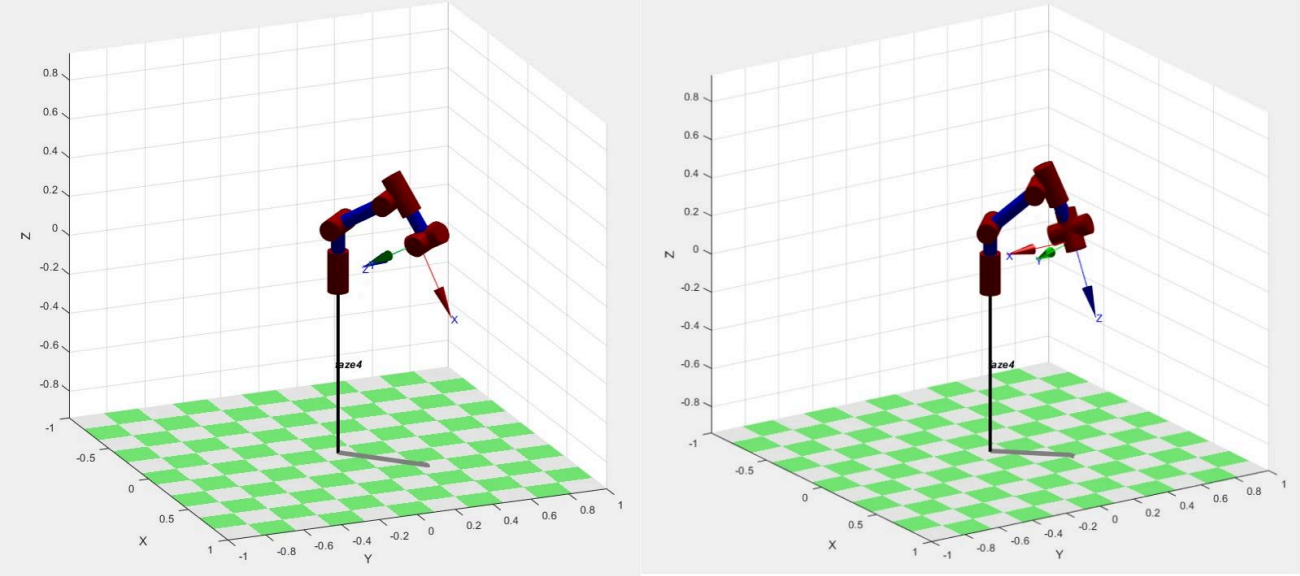


Figure 1: The same position with two orientations.

Reading a pose from the robot

Jog to where you want to be and read the pose: the commander's position display, or `rbt.pose()`. That is how poses are taught. A taught pose contains the tool's transform: taught with one tool, it is wrong for another.

Rotating about several axes

A rotation about X followed by one about Y is not the same as Y then X, and three angles at once are hard to predict. Rotate about one axis per command when you can, especially in the tool frame:

```
MoveCartRelTRF(0, 0, 0, 0, 45, 0)
MoveCartRelTRF(0, 0, 0, 0, 0, 45)
```

Forward and inverse kinematics

Kinematics is the geometry of the arm: how the joint angles decide where the tool is, and the other way round.

The kinematic diagram and the DH table

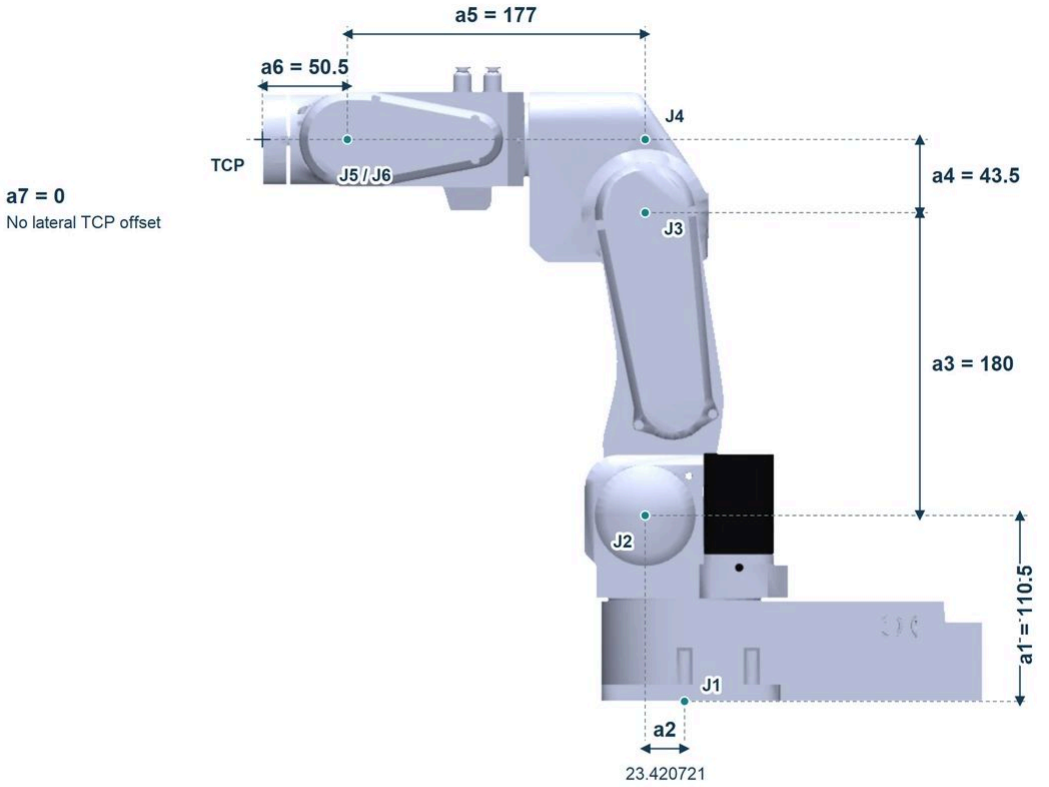
A kinematic diagram shows the links and joints without the shapes: six revolute joints, their axes, and the distances between them. The Denavit-Hartenberg convention writes that down as four numbers per joint, a link length, a link offset, a joint angle and a link twist, and PAROL6's table is in the Robot Manual (Dimensions and kinematics). The commander software's robot model is built from this table; the Python API loads the robot's URDF, which carries the same geometry, and computes the kinematics numerically with Pinocchio.

The sheet below shows both at once for PAROL6 with nothing on the flange: the robot's real outline with every joint at 0, the joint centres J1 to J6, the dimensions a1 to a7 between them, and under it the DH table those dimensions make. The Robot Manual has the same sheet for every gripper.

PAROL6 / Flange only

Kinematic review | Standard DH to the configured TCP 07 / 12

Pose: q1 ... q6 = 0 rad; no jaws. Orthographic side view; dimensions in mm.



a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

Joint	Link	theta [rad]	alpha [deg]	a [mm]	d [mm]
1	Base	q1	-90	a2 = 23.420721	a1 = 110.5
2	Shoulder	q2 - pi/2	180	a3 = 180	0
3	Elbow	q3 + pi	90	-a4 = -43.5	0
4	Wrist 1	q4	-90	0	-a5 = -177
5	Wrist 2	q5	90	0	0
6	Wrist 3 / TCP	q6 + pi	180	-a7 = 0	-a6 = -50.5

Figure 1: PAROL6, flange only: the dimensions a1 to a7 between the joint centres, and the DH table they make.

Forward kinematics

From six joint angles to the pose of the tool. It is a chain of transformations, one per joint, multiplied together, and it always has exactly one answer: the software runs it every cycle to show you where the tool is. `rbt.pose()` is forward kinematics of the current angles plus the tool's transform.

Inverse kinematics

From a pose of the tool to the joint angles that reach it. It is the hard direction. The spherical wrist helps: the first three joints decide where the wrist centre is, and the last three decide the orientation from there, so the problem splits in two. Even so:

- A pose can have **several solutions**: elbow up or down, wrist flipped or not, the base turned the other way. The software picks the one nearest the current configuration.
- A pose can have **no solution**: out of reach, or an orientation the wrist cannot make from that position. The move is refused (`IK_TARGET_UNREACHABLE` in the API, an inverse kinematics error in the commander).
- Near a **singularity** the solution exists but demands enormous joint speeds; the next page.

The Python API solves the inverse kinematics numerically, which is general but can fail on poses a closed-form solver would find, especially around joint 4; re-planning from a slightly different pose or starting posture usually helps. `MovePose` in RBTscript and `move_j(pose=...)` in the API solve the kinematics once, at the target, and move in joint space; `MoveCart` and `move_l` solve it all along a straight line, which is where the next page matters.

Configurations and singularities

Configurations

The same tool pose can be reached with the elbow up or down, the wrist flipped or not, the base facing the target or turned 180° with the arm reaching backwards. Each combination is a **configuration**. A joint move keeps the configuration it is in unless the target is in another; a Cartesian move cannot change configuration at all, because that would mean passing through a pose where two configurations meet, which is a singularity.

Three choices with two answers each make eight configurations. The figure shows all eight for one tool pose: PAROL6 with nothing on the flange, the tool centre point 250 mm in front of the base and 230 mm up, the tool tilted 30° from vertical. The flange is in exactly the same place and orientation in every picture; only the joints differ. The teal dot marks the centre of the flange and the arrow its axis, the same in every picture.

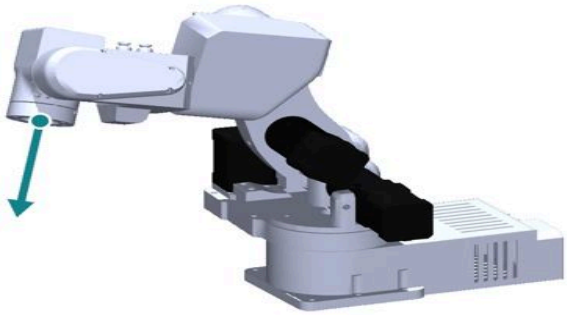
- **Shoulder right or left**: the wrist centre in front of the joint 1 axis, or behind it with joint 1 turned about 180° and the arm reaching back over itself.
- **Elbow up or down**: the elbow above or below the line from the shoulder to the wrist centre.
- **Wrist flipped or not**: joint 5 positive or negative; joints 4 and 6 turn by 180° to keep the tool where it is.

(a) elbow up, shoulder right
wrist not flipped

within the joint limits



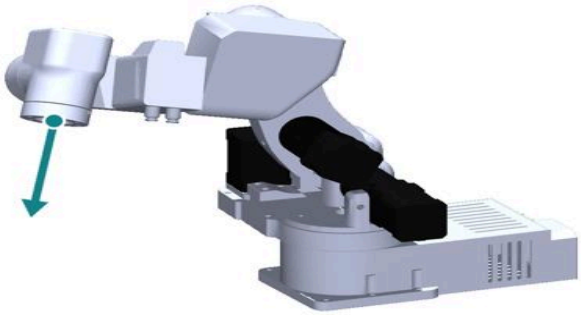
(b) elbow up, shoulder left
wrist not flipped



(c) elbow up, shoulder right
wrist flipped



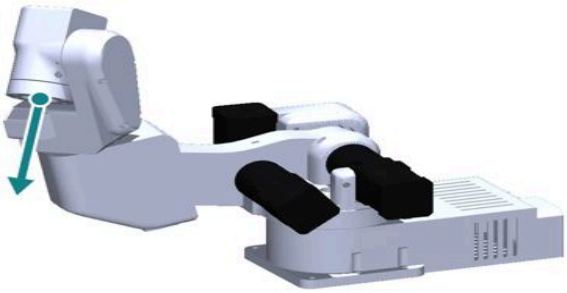
(d) elbow up, shoulder left
wrist flipped



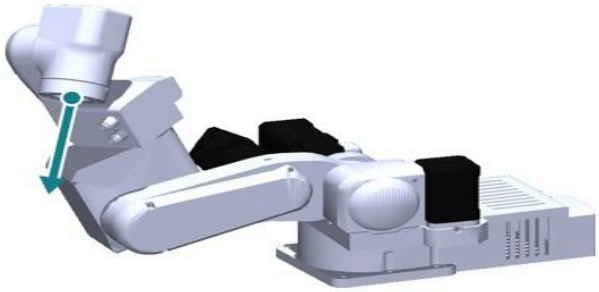
(e) elbow down, shoulder right
wrist not flipped



(f) elbow down, shoulder left
wrist not flipped



(g) elbow down, shoulder right
wrist flipped



(h) elbow down, shoulder left
wrist flipped

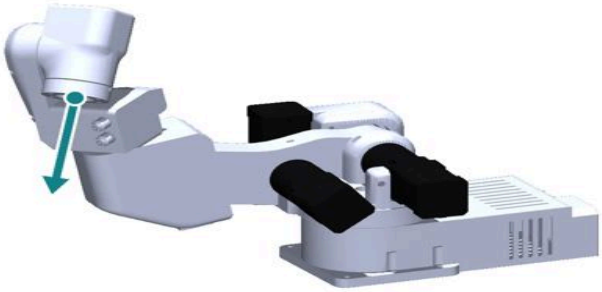


Figure 1: The eight configurations of PAROL6 for one tool pose: elbow up or down, shoulder right or left, wrist flipped or not; of these, only (a) is within the joint limits.

Mathematically all eight reach the pose. On the real robot, the joint limits rule most of them out: for this pose only (a) is within every joint limit (Joint limits and directions), and several of the others would also hit the robot itself. The poses were solved and drawn from PAROL6's model in robot_descriptions, the same model the limits come from.

The practical rule: teach the poses of a program in the configuration you want, and make the move between two configurations a joint move (`MoveJoint` , `move_j`), never a Cartesian one.

How the software chooses, when a target is given as a pose:

- **The Python API** solves every pose starting from the joints where the robot is, so it keeps the current configuration, in Cartesian moves and in joint moves to a pose alike.
- **The commander software** solves `MovePose` starting from the standby pose, so it takes the configuration nearest to standby, whichever the robot is in; its Cartesian moves keep the current one.

To be sure of the configuration, give the target as joint angles.

Singularities

When the arm moves in Cartesian space, the software converts a tool velocity into six joint velocities every cycle. In most poses that works. In some poses two joint axes line up, and a motion of the tool in a certain direction would need infinite joint speed, or the joints can turn without the tool moving: a **singularity**. The maths divides by something that goes to zero. Near it, small tool motions ask for huge joint speeds; at it, the move is impossible.

The three singularities of a six-axis arm with a spherical wrist:

	When	Looks like	You meet it
Wrist	Joint 5 at 0°: the axes of joints 4 and 6 in one line	The wrist straight; the standby pose is a wrist singularity	whenever the tool points the way the forearm does
Shoulder	The wrist centre on the joint 1 axis. The shoulder axis is 23.4 mm in front of it, so the arm leans back to get there: with joint 3 at 180°, at joint 2 = −133.1°	The wrist straight above the middle of the base	working close to the joint 1 axis, above the base
Elbow	Joint 3 at 256.2°: the upper arm and the line from the elbow to the wrist centre in one line,	The arm fully stretched	at full reach

When	Looks like	You meet it
the wrist centre 362.3 mm from the shoulder axis		

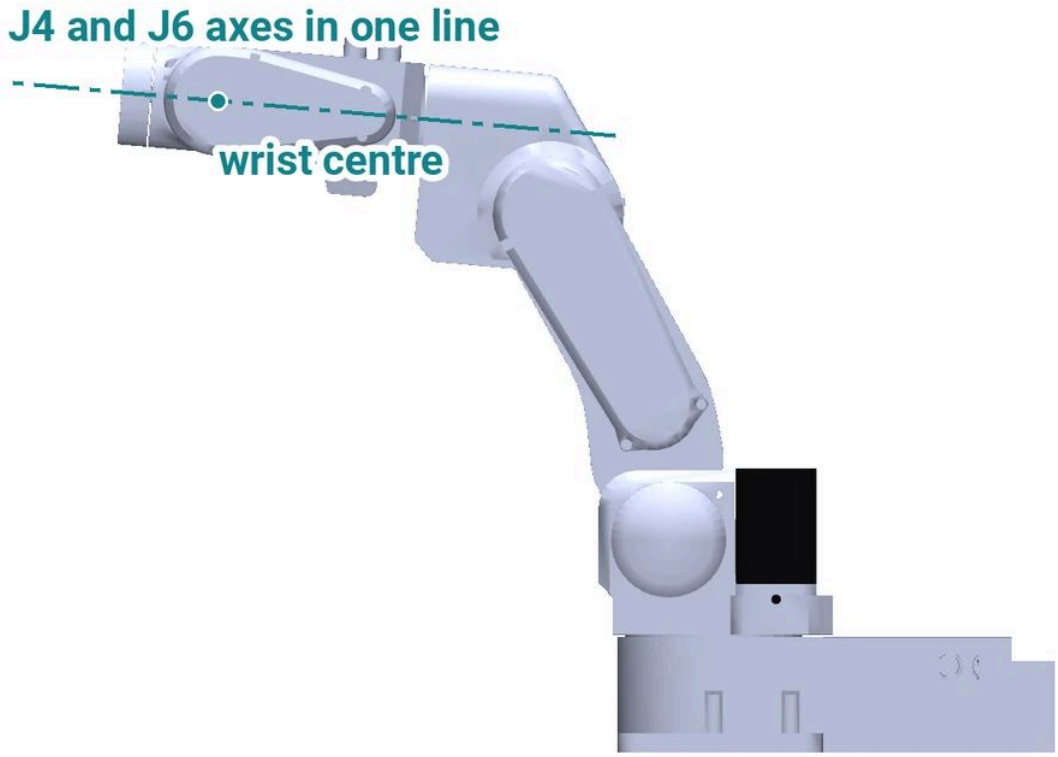


Figure 2: The wrist singularity: joint 5 at 0°, so the axes of joints 4 and 6 lie in one line.

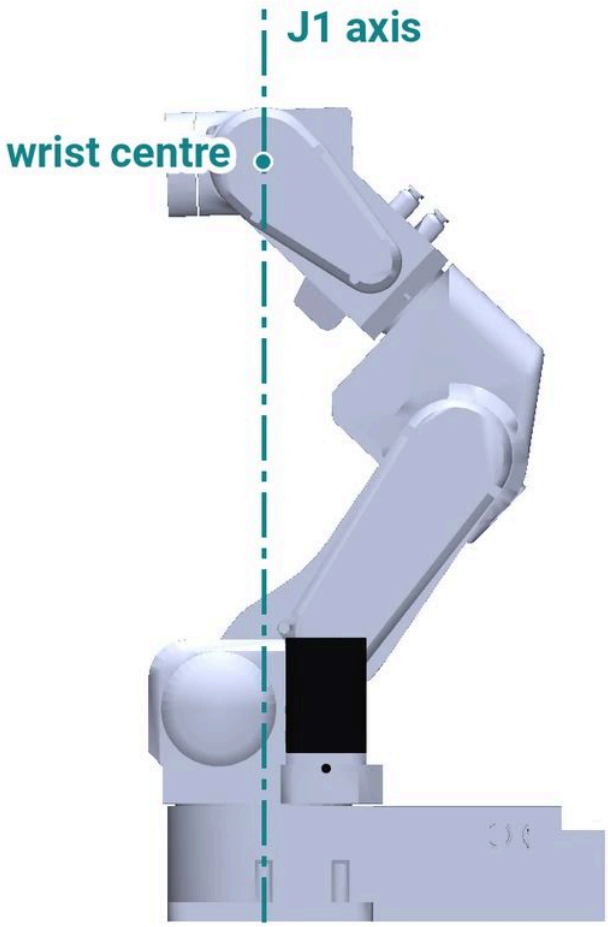


Figure 3: The shoulder singularity: the wrist centre on the joint 1 axis, here with joint 2 at −133.1° and joint 3 at 180°.

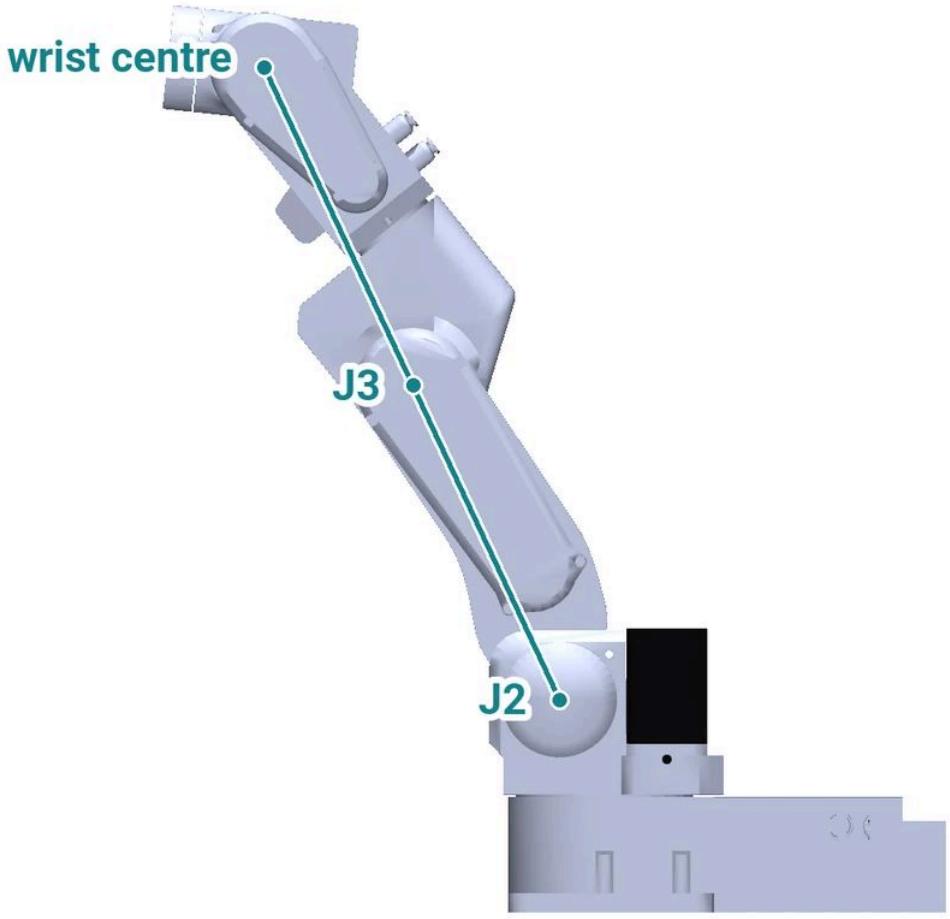


Figure 4: The elbow singularity: joint 3 at 256.2°, so joints 2 and 3 and the wrist centre lie in one line, the arm fully stretched.

The angles are the ones the software shows, from PAROL6's model. The wrist centre is where the axes of joints 4, 5 and 6 meet. The elbow singularity is the outer edge of the workspace (Workspace and reach): singularities are part of its boundary, with the joint limits.

What PAROL6 does

- **The commander software** stops a Cartesian move or jog when it reaches a singularity; the robot stays where it stopped. It does not deviate from the path to go around.
- **The Python API** refuses a Cartesian command whose path it cannot solve, before anything moves, and a jog that would enter a singular region stops.

Some industrial controllers can bend the path slightly to pass a singularity; PAROL6 keeps the path and stops instead. The demonstration video shows what singularities look like on an industrial arm.

Avoiding them

Singularities are a property of the geometry, so they are avoided by design, with trial and error:

- Keep the wrist bent: a few degrees of joint 5 away from 0° at the poses where you work.
- Place the parts so that the arm reaches them with a bent elbow, not at full stretch, and off the axis of joint 1.
- Mount the gripper at an angle, or on a plate that turns it, so that the working orientation needs a bent wrist.
- When a move keeps failing, make it a joint move, or add an intermediate pose.
- Run the program in the simulator: a stop or a refusal in the simulator is a singularity you can see.

Workspace and reach

Reach

The furthest the flange centre gets from the axis of joint 1, within the joint limits: 408 mm. The standard pneumatic gripper's tool centre point gets to 434 mm. A longer tool reaches further and weighs on joints 4 and 5.

Workspace

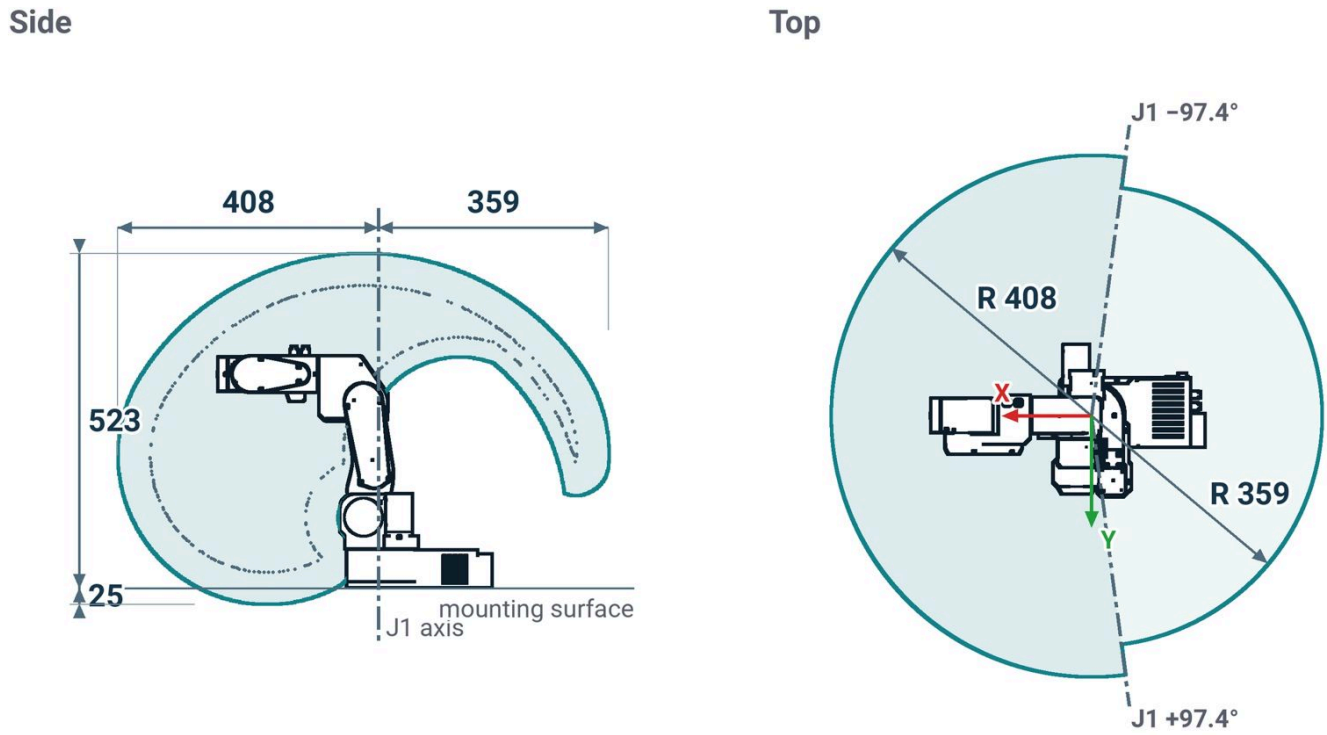


Figure 1: Where the flange centre can go within the joint limits, side and top view, in mm.

In the side view the shaded region is everywhere the centre of the flange can go in the plane of the arm, within the joint limits of the robot's model; the dotted line is the same for the wrist centre. It reaches 408 mm in front of the joint 1 axis and 359 mm behind it, 523 mm above the mounting surface and 25 mm below it. The top view sweeps that region about joint 1 through its ±97.4°: in front of the base the arm reaches out directly, behind it (lighter) only by folding back over the base. A tool adds its own length; the same figure for each gripper, with a table of the numbers, is on Dimensions and kinematics. The region is computed from the limits only: the arm's own parts, the tool and the table can block some of it.

The volume the tool can reach. The **reachable workspace** is every point it can touch with some orientation: a near-sphere of the reach, minus the base, minus the volume behind the elbow that the limits of joints 2 and 3 exclude. The **dexterous workspace** is the part of it where the tool can take any orientation, much smaller, around the middle of the reach and in front of the robot. A program that needs a particular orientation at a point needs that point in the dexterous workspace, and the way to find out is to jog there and try.

Payload across the workspace

1 kg close to the base, 0.5 kg at full reach: the torque on joints 2 and 3 grows with the distance, and the steppers' torque does not. A heavy tool shrinks the workspace in which the arm moves without losing steps.

Placing the task

Faster cycles and a larger usable workspace come from the layout, not from the program: put the work in front of the robot at mid reach, at a height that keeps the elbow bent, and away from the joint 1 axis. Move the robot or the fixture rather than fighting the kinematics. The Python API's `reachable()` reports how much travel each joint and axis has left from the current pose, which tells you how close to a limit the program works.

Homing and mastering

Stepper motors without encoders know nothing about where they are when the power comes on. Two procedures give PAROL6 its position: homing, every time, and mastering, once per robot.

Homing

Each joint drives to a switch or sensor whose position in the mechanics is known from the CAD model, then moves a fixed number of steps from there to the after-homing pose. After that the firmware's step counters hold the joint positions, and every move is counted from them. The process is the one 3D printers use. The sequence and what can go wrong are in the Robot Manual's Homing.

In the software: the commander's **Home** button, the API's `home()`. The API tracks whether the robot is referenced and refuses planned motion before it (`MOTN_NOT_HOMED`); a `home()` on a referenced robot is a planned joint move back to the after-homing pose, and `home(calibrate=True)` runs the firmware's sequence again.

Mastering

The fixed number of steps from the switch to the after-homing pose is measured on the reference robot. Printers and switches vary, so on another robot the homed pose is a fraction of a degree off per joint. Mastering measures the error with the witness marks, pins through holes in neighbouring links that line up in the after-homing pose, and corrects the number in the firmware (Mastering).

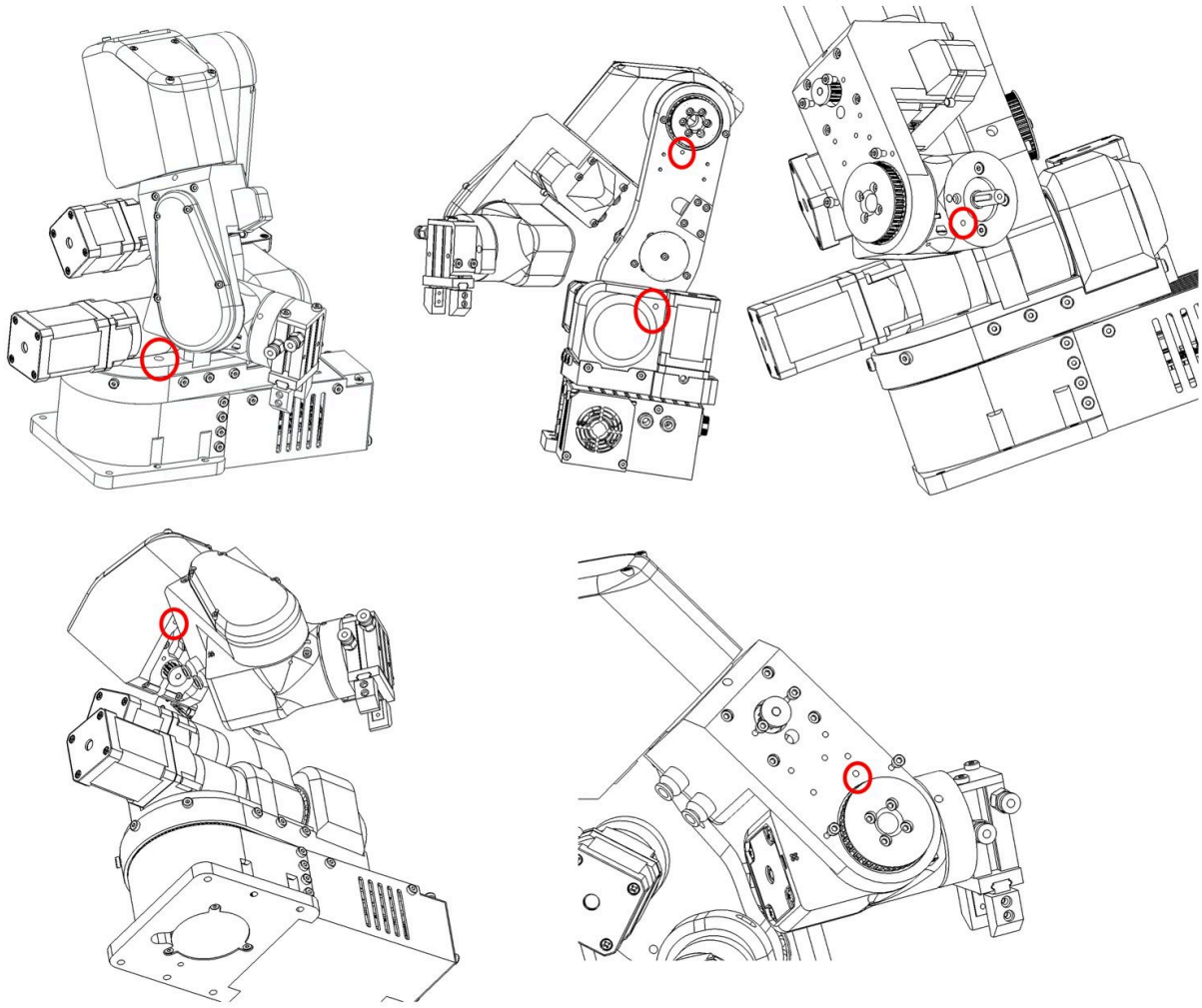


Figure 1: Witness marks on PAROL6.

Mastering affects the **absolute** accuracy, whether the pose the software reports is the true pose. It does not affect the **repeatability**, whether the robot returns to the same place every time, which comes from the mechanics and the switches alone. A program taught on an unmastered robot runs correctly on that robot; it runs offset on another.

When the position is lost

A collision, a stall, a move by hand or a power failure breaks the count, and nothing in the robot can tell. Home again. A program that must not run with a wrong position checks something external before it starts: a sensor on an input, a camera, a touch.

Motion profiles

A move has a start, an end and a time; the profile is how the speed changes in between. The profile decides how smooth the move is, how much it shakes the arm, and whether the steppers can follow it.

Trapezoidal

Accelerate at a constant rate, cruise at the maximum speed, decelerate at the same rate. The speed curve is a trapezoid; the acceleration jumps between +a, 0 and −a, which is felt as a jerk at the three corners. With the commander's `t=` argument, each third of the move is one phase.

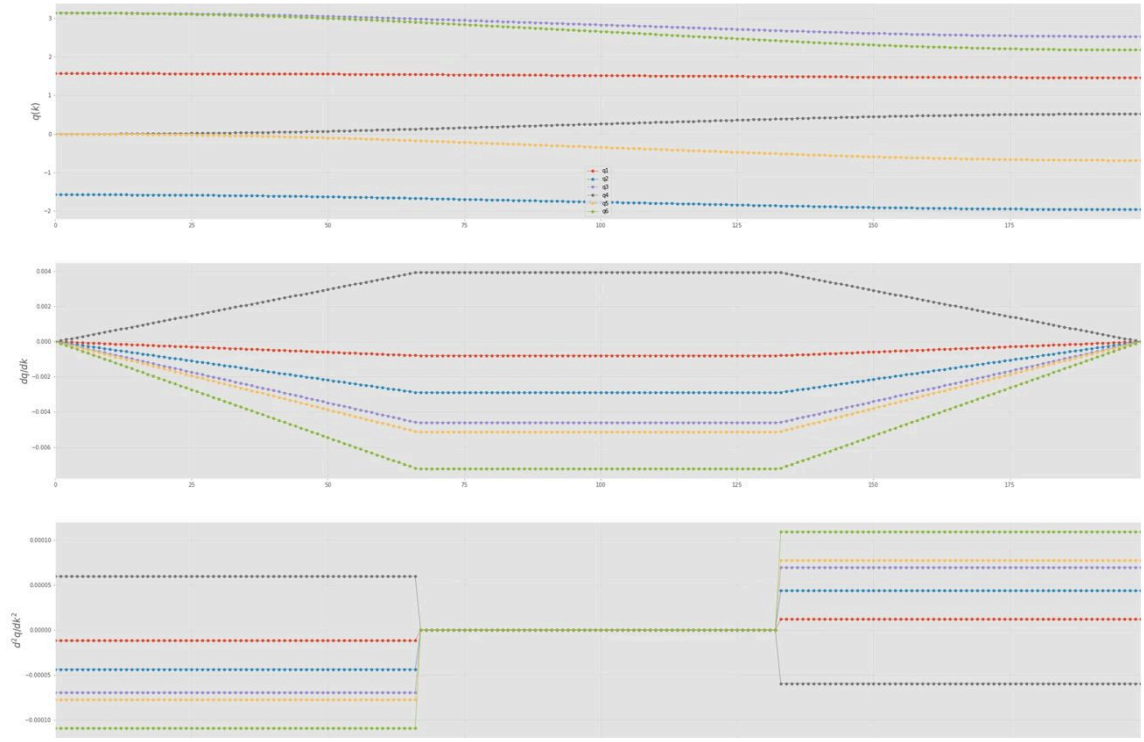


Figure 1: A trapezoidal velocity profile.

Polynomial

A quintic (fifth-order) polynomial through the start and end positions with zero speed and zero acceleration at both ends: the speed rises and falls as a smooth bell, the acceleration is continuous, and the move starts and ends from rest without a jolt. Slower in the middle for the same total time, and the arm likes it better.

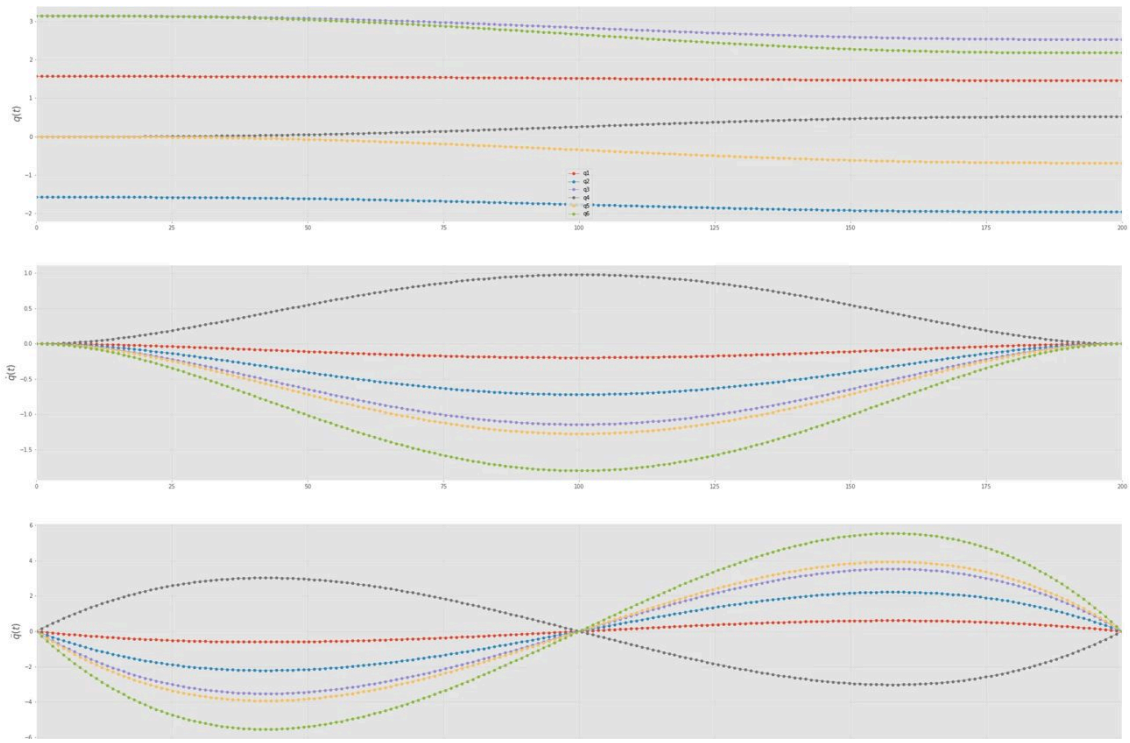


Figure 2: A quintic polynomial velocity profile.

In the commander software

`MoveJoint`, `MovePose`, `MoveCart` and `MoveCartRelTRF` take the bare word `trap` or `poly` after the other arguments (`func="trap"` is ignored). A `MoveJoint` or `MovePose` with `t=` defaults to polynomial, with `v=` and `a=` it is trapezoidal, and with neither it runs a conservative trapezoid. A `MoveCart` or `MoveCartRelTRF` with `t=` defaults to trapezoidal, and without `t=` it always takes 3.6 s (RBTscript).

In the Python API

`select_profile()` sets the profile for all planned moves. The profiles of the version this site describes: `TOPPRA`, `RUCKIG`, `QUINTIC`, `TRAPEZOID`, `LINEAR`.

Profile	What it is
`TOPPRA`	Time-optimal path following: the fastest run along the given path within the joint speed and acceleration limits. The default
`RUCKIG`	Jerk-limited point-to-point motion: acceleration ramps instead of jumping. Joint moves only; a Cartesian move under it runs TOPPRA
`QUINTIC`	The quintic polynomial above, C² smooth

Profile	What it is
`TRAPEZOID`	Trapezoidal, with parabolic blends at the corners
`LINEAR`	Straight interpolation, no smoothing: for tests

The move's `speed` and `accel` fractions scale the limits the profile works within; a `duration` fixes the time instead. Streamed jogs and servo targets use their own jerk-limited profile in the controller, not the selected one.

Why the profile matters for a stepper arm

A stepper follows a speed command exactly until the torque runs out; a sudden acceleration demand at a corner is where it runs out first. Smooth profiles, lower accelerations and lower currents go together: if a joint loses steps at the start of moves, a quintic or jerk-limited profile at the same speed often fixes it.

Blending

A program with many waypoints stops at each one unless told otherwise, and every stop costs time and a jolt. Blending lets the robot pass near a waypoint without stopping: when the tool comes within a radius of the waypoint, it starts the next move, and the path rounds the corner.

- The tool does not pass **through** a blended waypoint; it cuts the corner inside the radius. A waypoint where the tool must be exactly, a pick or a place, is not blended.
- The blend radius is in millimetres of the tool's path.
- A blend needs the next move to be known when the current one starts: in the Python API the moves are queued, so the controller looks ahead and blends.

In the Python API

`move_j`, `move_l` and `move_c` take `r`, the blend radius; `r=0` stops at the target. `move_p` is the process move: a list of waypoints run at constant tool speed with automatic blending at every corner, for scan and dispensing paths. A tool action or a `delay()` queued after a blended move ends the blend there: the arm stops for it. The zig-zag example in the API's repository shows a raster scan with blends.

```
rbt.move_l([200, 0, 250, 90, 0, 90], speed=0.5, r=15)
rbt.move_l([200, 160, 250, 90, 0, 90], speed=0.5, r=15)
rbt.move_l([250, 160, 250, 90, 0, 90], speed=0.5)          # stops here
```

In the commander software

RBTscript has no blending: every move ends at rest. Make the path with fewer, longer moves, or with `MoveCart` through points far enough apart that the stops do not matter.

Target, commanded and actual

Three positions exist at every moment, and they differ.

- The **target** is where you asked the robot to go: the end of the move.
- The **commanded** position is where the planner says the robot should be right now, on the trajectory it computed from the profile, sent to the control board in this cycle.
- The **actual** position is where the motors are, as the control board counts their steps: what `rbt.angles()` and the commander's display show.

Why commanded and actual differ

The board receives a speed and a position every 10 ms and runs the stepper at that speed until the next packet. Two things pull the actual position away from the commanded one:

- The PC is not a real-time machine; the packets are 10 ms apart on average, not exactly, and a late packet lets the motor run a little too long or short.
- A stepper cannot make the very small speed changes at the start and end of a smooth profile exactly; it rounds them.

If the board followed the speed alone, the errors would add up over a long move and the robot would miss the target by a few steps. So the planners use the go-to-position command (156), in which the board runs each joint at the average of the commanded speed and the speed that would close the position error within one cycle: the speed curve gives the smoothness, the position curve pulls the error back to zero every cycle, and the robot arrives exactly.

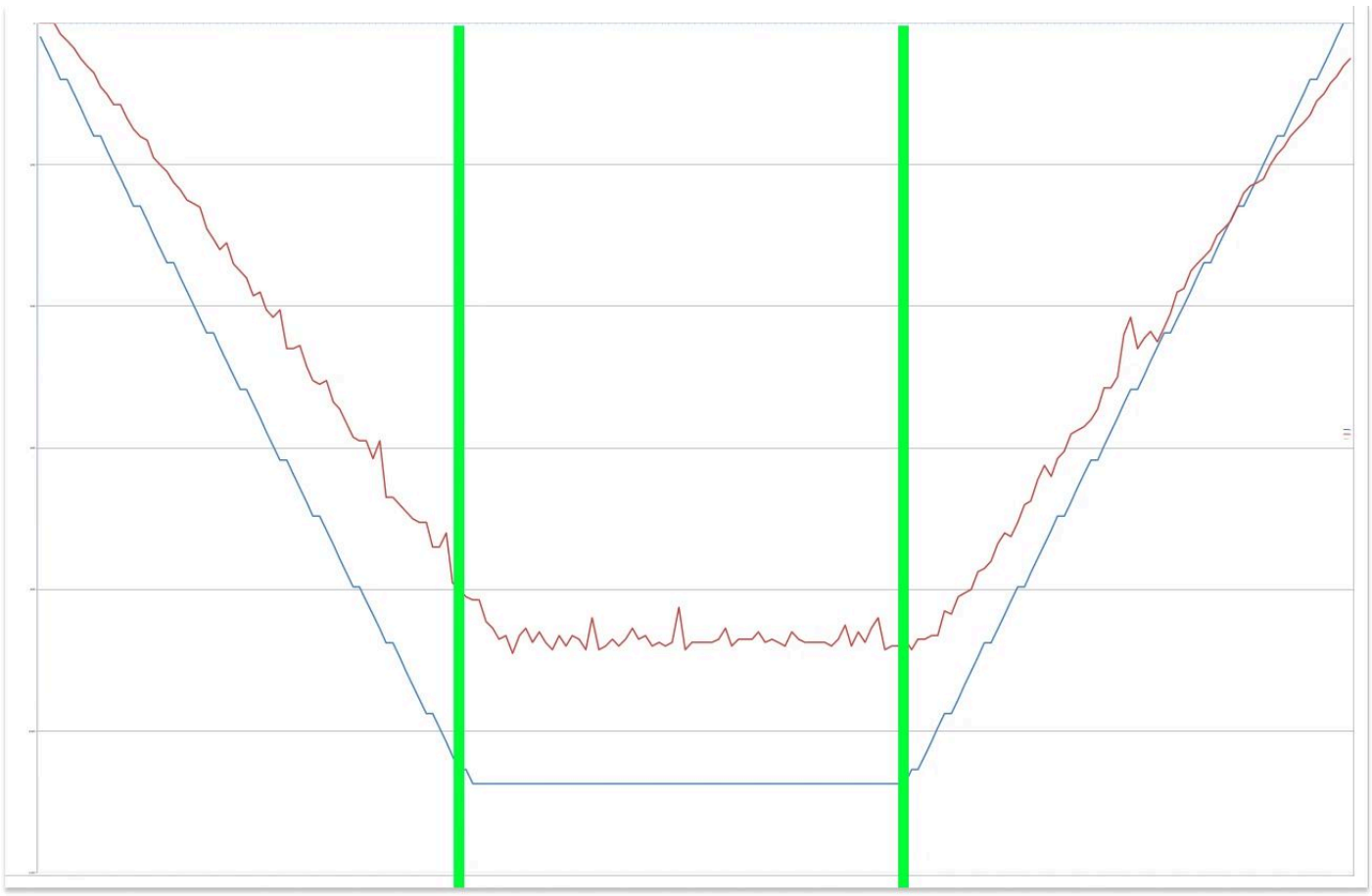


Figure 1: Commanded (blue) and actual (red) speed on a trapezoidal move; the robot spends a third of the time in each phase.

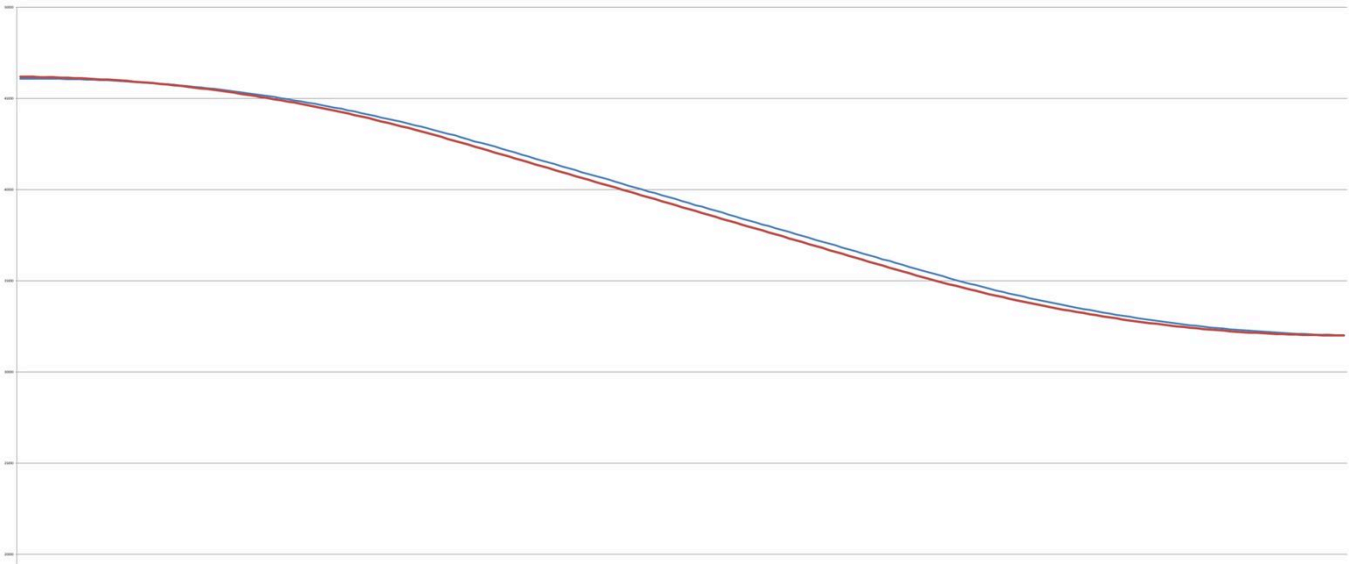


Figure 2: Position with speed-only tracking and with speed-and-position tracking: the second reaches the target exactly.

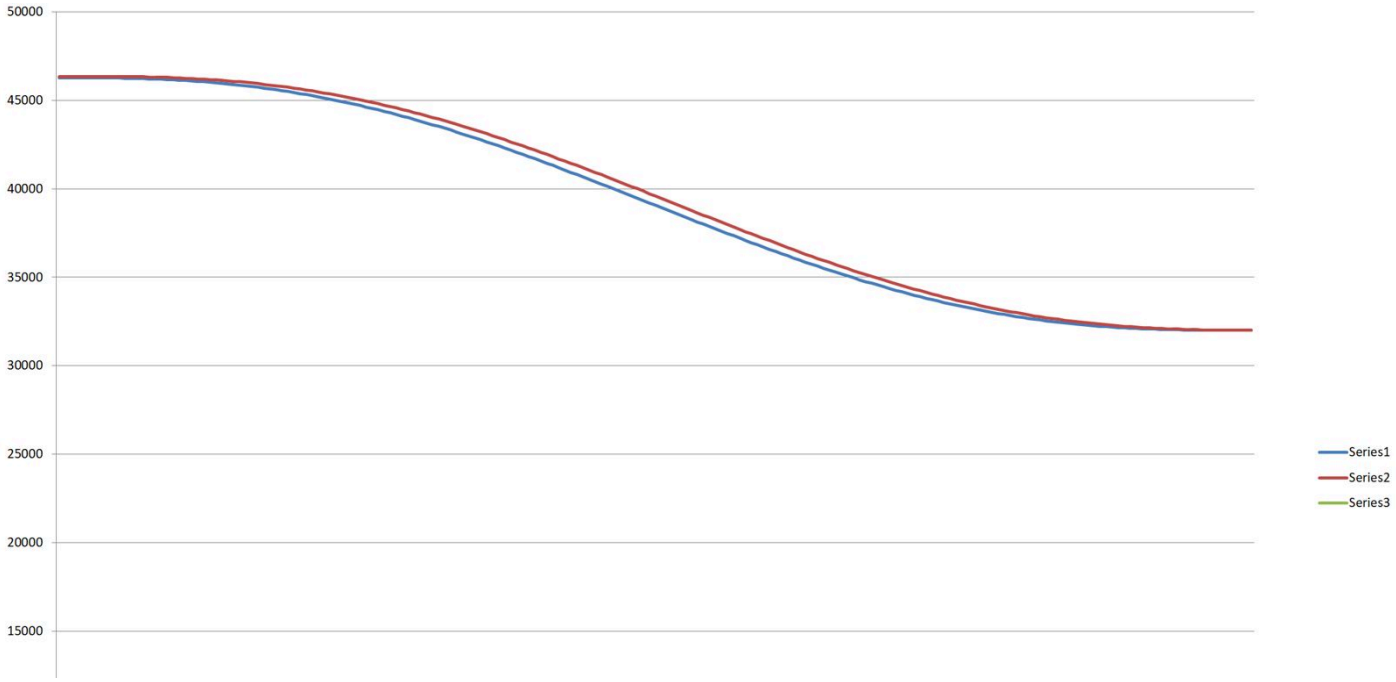


Figure 3: The same with a polynomial profile.

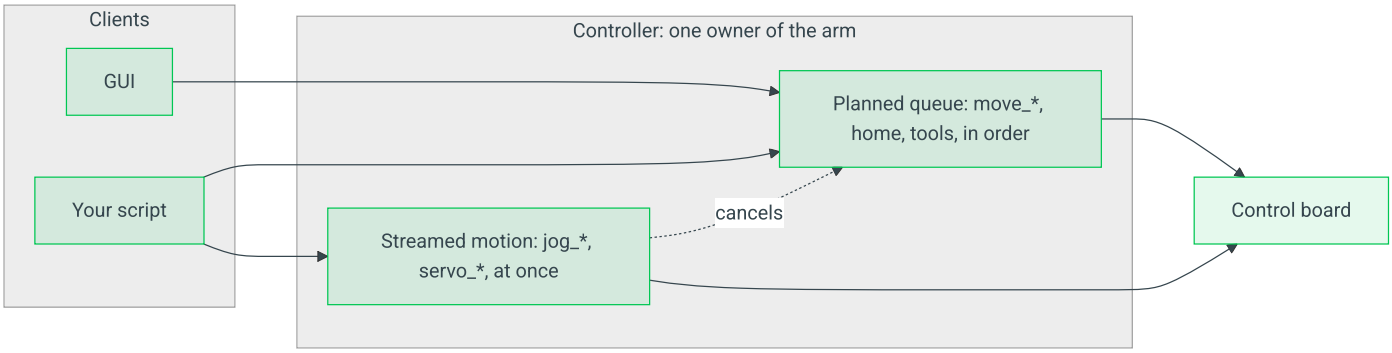
The commander's RBTscript exposes the choice: a move with `speed` as its last argument tracks the speed curve only, which is smoother and quieter and may stop short of the target; without it, speed and position are tracked.

What the software reports

The commander shows the actual joint angles and the actual tool pose; the Python API's status stream carries the actual angles and speeds, plus `executing_index` and `completed_index` for where the queue is. "Actual" means "counted steps": a lost step moves the real arm away from all three without any of them noticing (Homing and mastering).

Control ownership

Only one program can own the control board, and inside the Python API only one source can drive the arm at a time. Knowing who is in charge saves confusion and collisions.



One host on the serial port

The control board executes whatever packets arrive on its USB port. The commander software and the Python API's controller each take the port exclusively; the second one to try gets an error or silence. Stop one before you start the other.

Inside the Python API

The **controller** owns the arm. **Clients** send it commands; it accepts commands from any number of clients on the network, without authentication, so a GUI and a script can both be connected. What happens when they both command motion:

- **Planned motion** (`move_*`, `home`, `delay`, tool actions, checkpoints) goes into one **queue**, in the order it arrives, and runs in that order, together with the output writes (`write_io`), the tool selection (`select_tool`) and the TCP corrections. Every queued command gets an index; `wait_command(index)` waits for that one.
- **Streamed motion** (`jog_*`, `servo_*`) takes the arm at once: it cancels the planned queue (every discarded command fails with `MOTN_CANCELLED`). A jog runs for its duration; a servo stream drives the arm while targets keep coming and ends about a quarter of a second after the last one. Then the arm brakes to rest and holds.
- `stop()` cancels the active move and clears the queue, from any client. `estop()` does the same and latches a protective stop until `reset()`.
- `pause()` holds the queue where it is; `resume()` continues it; `set_execution_speed()` scales it between 10 and 100 %.
- **System commands** (`simulator`, `connect_hardware`, `select_profile`, `set_shapes`) take effect at once and change the controller's state for everyone. A queued `select_tool` does too, when its turn comes.

The rule for a program that shares the controller: own the arm for the duration of a task by queuing it, wait for your indices, and expect `MotionError` if someone else stopped you.

The firmware

The firmware's enable and disable commands (101, 102) decide whether it accepts motion at all; the E-stop makes the PC software send disable or latch a stop (Emergency stop). Between packets the firmware keeps the last speeds it was given, which is why a host must never stop sending while a joint is in motion.

Simulator versus hardware

The controller runs against the hardware or in **simulator mode**, where the same planner drives a virtual arm and the status stream reports it. `simulator(True)` switches; `is_simulator()` tells; `teleport()` sets the virtual arm's pose. The **dry-run client** goes further: it runs the planner in the client's own process, with no controller at all, for tests and previews (Clients).

Steppers, open loop and backlash

Actuators

Robots are moved by actuators: servo motors with encoders, stepper motors, brushless motors, pneumatic and hydraulic cylinders. PAROL6 uses **stepper motors**, which move in discrete steps on command, hold position with their current, and need no sensor to be positioned: the cheapest way to get six precise axes, and the reason the robot has no encoders and no brakes.

How a stepper works

A stepper has 50 pole pairs and two windings; energising the windings in sequence pulls the rotor one step, 1.8°, at a time, 200 steps per turn. The driver can set the currents to intermediate values and hold the rotor between full steps: **microstepping**, 32 positions per step on PAROL6, 6400 per turn. A microstep is a real position only while the torque against it is small; under load the rotor lags up to a full step behind the commanded microstep, which is why the resolution on paper is finer than the accuracy in practice.

A stepper's torque falls as its speed rises, because the windings' inductance limits how fast the current can reverse. Above a certain speed, at a certain load, it **stalls**: it stops following the steps, buzzes, and the position is lost. The speed and acceleration limits of the software exist to stay below that.

Open loop and closed loop

Open loop: the controller commands and assumes the result. Simple, cheap, no sensors, and no way to know when something went wrong. **Closed loop**: a sensor measures the result and the controller corrects the difference; accurate, robust against load, and more complex. PAROL6's joints are open loop: the step count is the position, and a lost step is an error nobody sees. The safeguards are the margins in the limits, the homing that resets the count, and the user's eyes.

The electric grippers are different: their drivers run the motor in closed loop with an encoder, a brushless motor on the SSG-48's Spectral Micro and a stepper on the MSG's STEPFOC. That is what makes a force-controlled gripper possible (End effectors).

Position, velocity and torque control

A joint can be controlled in **position** (go there and hold), **velocity** (turn at this speed) or **torque** (push this hard). PAROL6's firmware offers velocity (the jog command) and a velocity-plus-position tracking (the go-to command); the planner on the PC turns position targets into those. Torque control, which collaborative robots use to be safe around people, is not possible with open-loop steppers: the robot cannot feel.

Backlash

The play between gear teeth, in a belt's mesh and in a coupler: when a joint reverses, the motor turns a little before the output follows. PAROL6's planetary gearboxes have 15 to 20 arc-minutes of it, the belts less. It shows as a small position error that depends on the direction of the last move, and as a slight looseness at the tool. The cure is preload (tight belts), approaching a pose always from the same direction, and keeping the couplers tight (Couplers and threadlocker).

Encoders

An encoder measures a shaft's rotation: incremental ones count from a start, absolute ones know the angle at power-up. PAROL6 has none on its joints; adding them is a common modification, and the community has closed-loop driver projects. The gripper drivers have an absolute magnetic encoder on their motor.

Further reading

Good places to go deeper into the concepts of this chapter:

- Robot Academy: Peter Corke's video lectures on kinematics, frames and control, the basis of the Robotics Toolbox the commander software uses.
- Automatic Addison: practical tutorials, including how to assign DH frames and how to find the DH table.
- Robogrok: a robotics course with the maths worked out.
- Robotics coursework collection: a curated list of courses and books.
- Robotics Explained: short explanations of the vocabulary.
- How do robot manipulators move? on Robotic Sea Bass: motion planning from the ground up.
- How many axes does my robot need? on Robohub.
- Singularities demonstrated on an industrial arm.

The Python API's repository has a `docs` folder and a README that go into its architecture, the motion planner and the collision checker beyond what this guide covers.

Support

Getting help

- **Questions and discussion**: the Source Robotics Discord.
- **A problem with the robot's hardware, its files or the control board firmware**: an issue in the robot repository; see Report a problem for what to include.
- **A problem with the commander software**: an issue in PAROL-commander-software.
- **A problem with the Python API**: an issue in PAROL6-python-API.
- **A mistake on a page of this site**: every page ends with a link to edit it and one to report it; see Contributing.
- **Orders and hardware**: source-robotics.com.

Before you ask, Troubleshooting goes through the common symptoms, and the commander's terminal output or the API's `error()` tells most of what anyone will ask you for.

Report a problem

A report that can be acted on says what you did, what happened, and what the robot said. Include:

1. **The robot**: bought or self-built; control board version (v2.0 with terminal blocks, or v1.2); the firmware (main, from which commit, and whether you changed `constants.h` or `motor_init.cpp`).
2. **The software**: commander (which commit, which Python version, which OS) or Python API (`pip show parol6`), or something else.
3. **What you did**: the exact steps, the program or script, the jog, the command.
4. **What happened**: and what you expected instead. A short video helps for anything mechanical.
5. **What the robot said**: the commander's response log and terminal output, or the API's exception text and the controller's log.
6. **The PC**: processor and OS, for anything about timing or communication.

Where to file it is on Getting help. A problem with a gripper goes to the gripper's own repository, with the same facts.

Contributing

The documentation lives in Source-Robotics/PAROL-docs. A small fix, a typo, a wrong number, a missing link, can be made from the browser: every page ends with a link to edit it on GitHub, which opens a pull request. For anything larger, open an issue first and say what you want to change.

House rules

- Write for the reader who has the robot in front of them: what to do, in what order, and why when the why matters. Short sentences, one idea each.
- Facts come from the files: the firmware, the software's source, the BOM. When a page and the code disagree, the code is right and the page is fixed; a change to a command is made in the code first.
- Version facts live in `versions.yml`, never in a page. The API reference is generated from the Python API's source: edit the notes around the markers, not the markers' output.
- Link to a page of this site with its relative path, so that the build checks it; link to code on GitHub by its permanent URL.
- Images under 2400 px, with the exact file name (the live site is case-sensitive).
- Run `python tools/test_tools.py` and `mkdocs build --strict` before you open a pull request. Both run on every push.

The text and the images you contribute are published under CC BY-SA 4.0, the code examples under MIT.